

(19) **United States**
(12) **Patent Application Publication**
Birt
(10) **Pub. No.: US 2025/0276018 A1**
(43) **Pub. Date: Sep. 4, 2025**

(54) **CAR CONSTRUCTS AND METHODS OF TREATMENT**

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(21) Appl. No.: **19/002,602**
(22) Filed: **Dec. 26, 2024**

Related U.S. Application Data

(60) Provisional application No. 63/616,278, filed on Dec. 29, 2023.

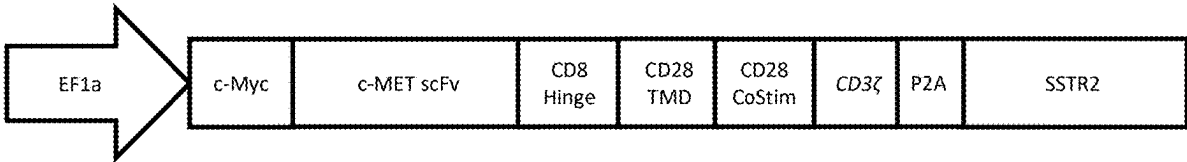
Publication Classification

(51) **Int. Cl.**
A61K 35/17 (2025.01)
A61K 31/506 (2006.01)
A61K 40/31 (2025.01)
A61K 40/42 (2025.01)
A61K 45/06 (2006.01)
A61K 51/12 (2006.01)
A61P 35/00 (2006.01)
C07K 14/655 (2006.01)
C07K 14/705 (2006.01)
C07K 14/725 (2006.01)

C07K 16/28 (2006.01)
C12N 15/86 (2006.01)
(52) **U.S. Cl.**
CPC *A61K 35/17* (2013.01); *A61K 31/506* (2013.01); *A61K 40/31* (2025.01); *A61K 40/421* (2025.01); *A61K 40/4237* (2025.01); *A61K 45/06* (2013.01); *A61K 51/1203* (2013.01); *A61P 35/00* (2018.01); *C07K 14/655* (2013.01); *C07K 14/7051* (2013.01); *C07K 14/70521* (2013.01); *C07K 14/70578* (2013.01); *C07K 16/2818* (2013.01); *C07K 16/2821* (2013.01); *C07K 16/2827* (2013.01); *C12N 15/86* (2013.01); *A61K 2123/00* (2013.01); *A61K 2239/13* (2023.05); *A61K 2239/17* (2023.05); *A61K 2239/21* (2023.05); *A61K 2239/31* (2023.05); *A61K 2239/38* (2023.05); *C07K 2317/565* (2013.01); *C07K 2317/622* (2013.01)

(57) **ABSTRACT**
The present disclosure relates to low affinity chimeric antigen receptors (CARs) and CAR-T cells, which provide cytotoxicity against tumors overexpressing the proto-oncogene c-MET and alleviate on-target, off-tumor toxicities. The CAR-T cells of the present disclosure comprise low affinity anti-c-MET scFvs, which facilitate enhanced anti-tumor activity and a reduced rate of tumor relapse.

Specification includes a Sequence Listing.



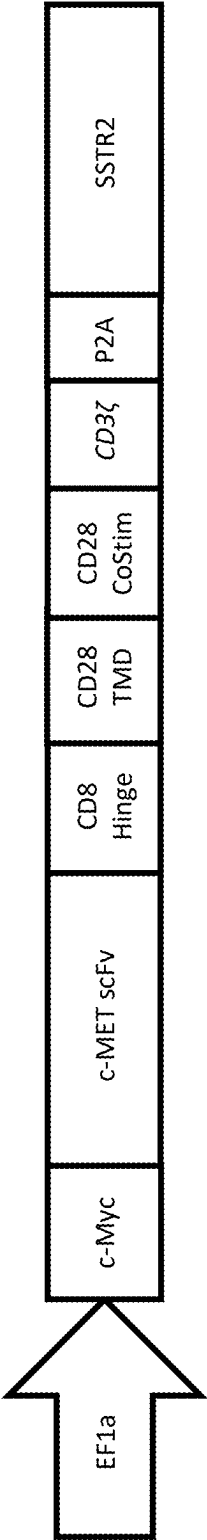
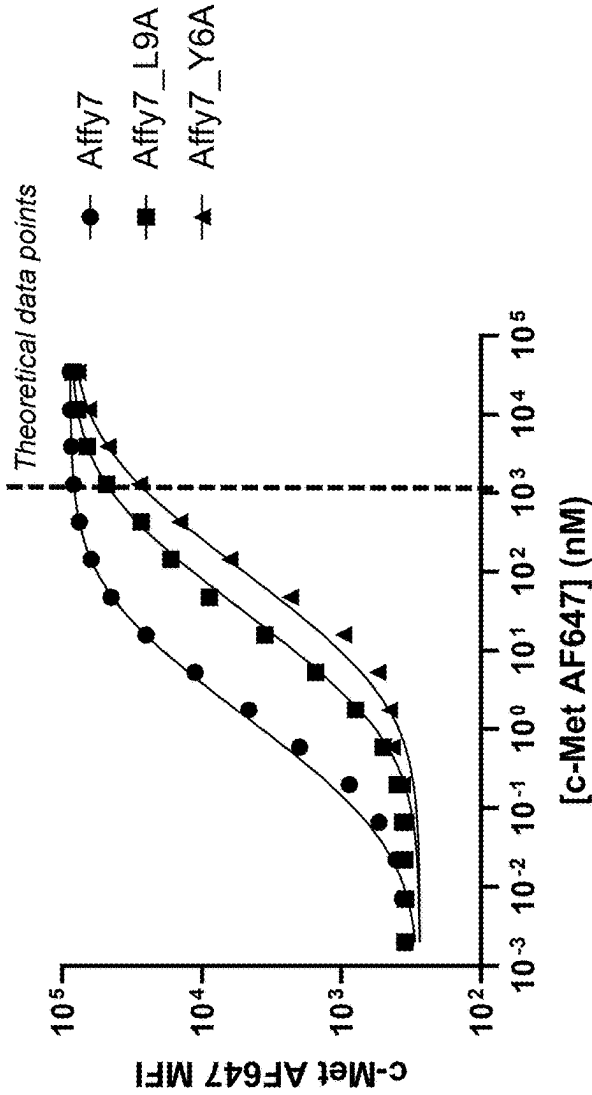


FIG. 1

Binding of c-MET to Affy7 Parent/Variant CAR-T Cells



*Fold change relative to "parent" Affy7

FIG. 2

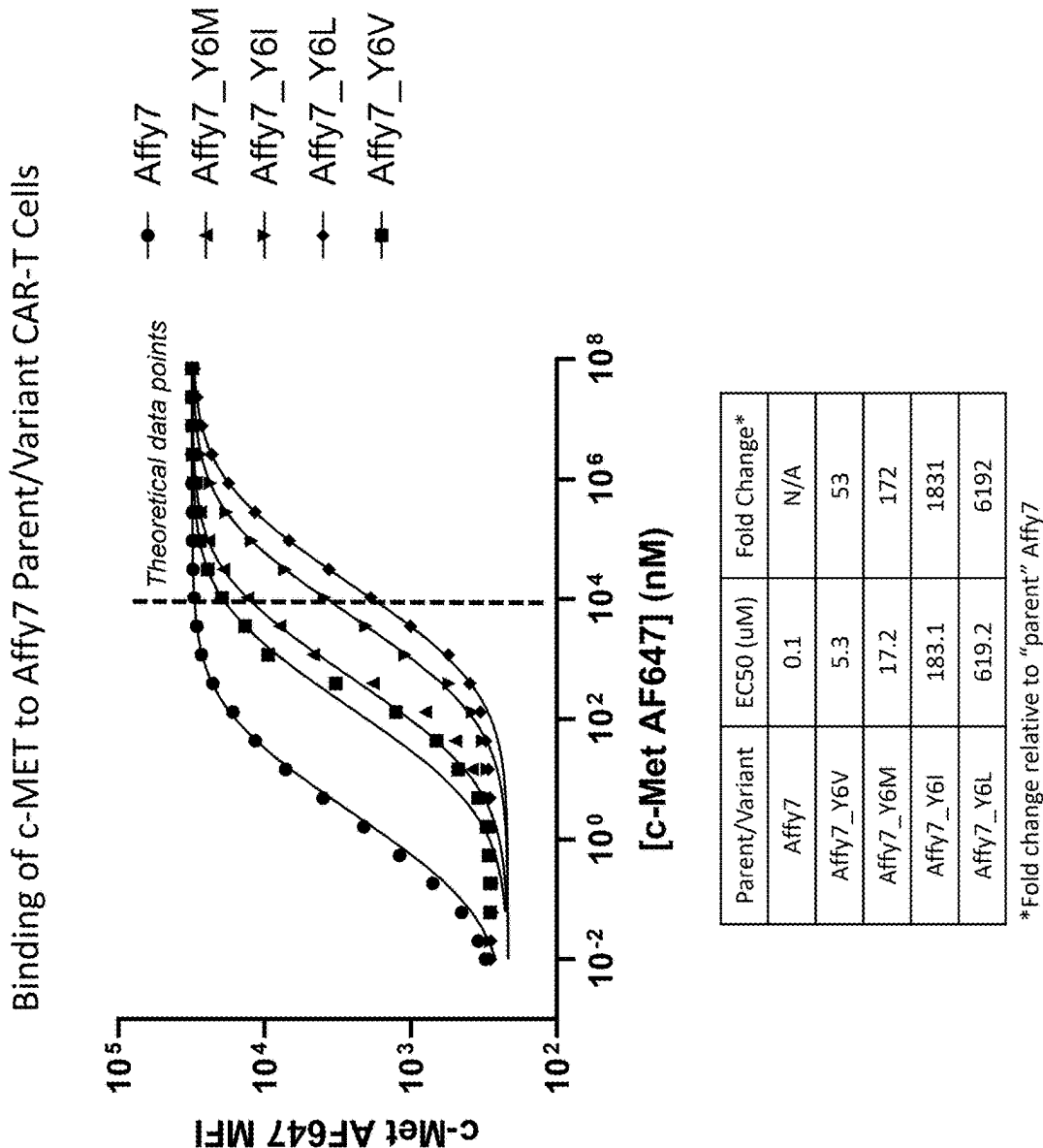
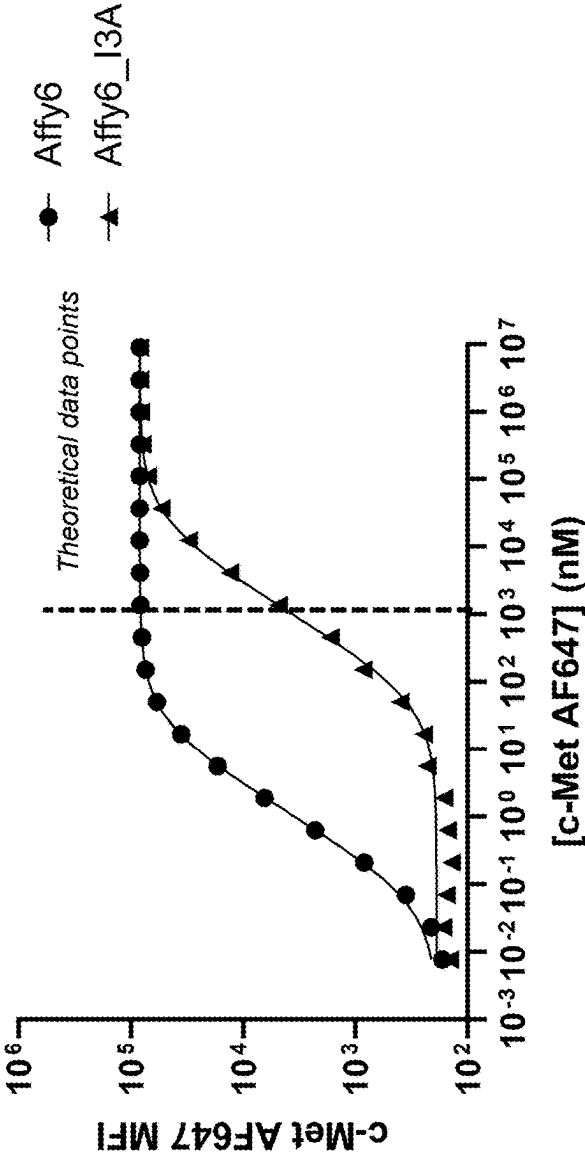


FIG. 3

Binding of c-MET to Affy6 Parent/Variant CAR-T Cells

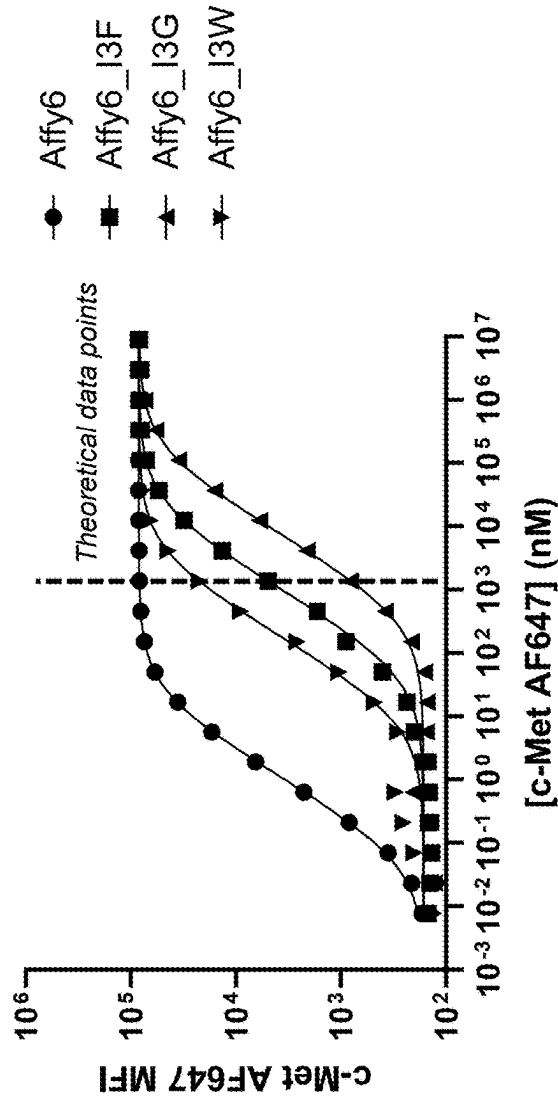


Parent/Variant	EC50 (uM)	Fold Change*
Affy6	0.02	N/A
Affy6_I3A	20.9	956

*Fold change relative to "parent" Affy6

FIG. 4

Binding of c-MET to Affy6 Parent/Variant Cells



Parent/Variant	EC50 (uM)	Fold Change*
Affy6	0.02	N/A
Affy6_I3W	3.6	167
Affy6_I3F	20.5	937
Affy6_I3G	150.8	6899

*Fold change relative to "parent" Affy6

FIG. 5

SNU-638 Human Stomach Carcinoma Cell Line Co-Cultured With Affy7 Parent/Variant CAR-T Cells

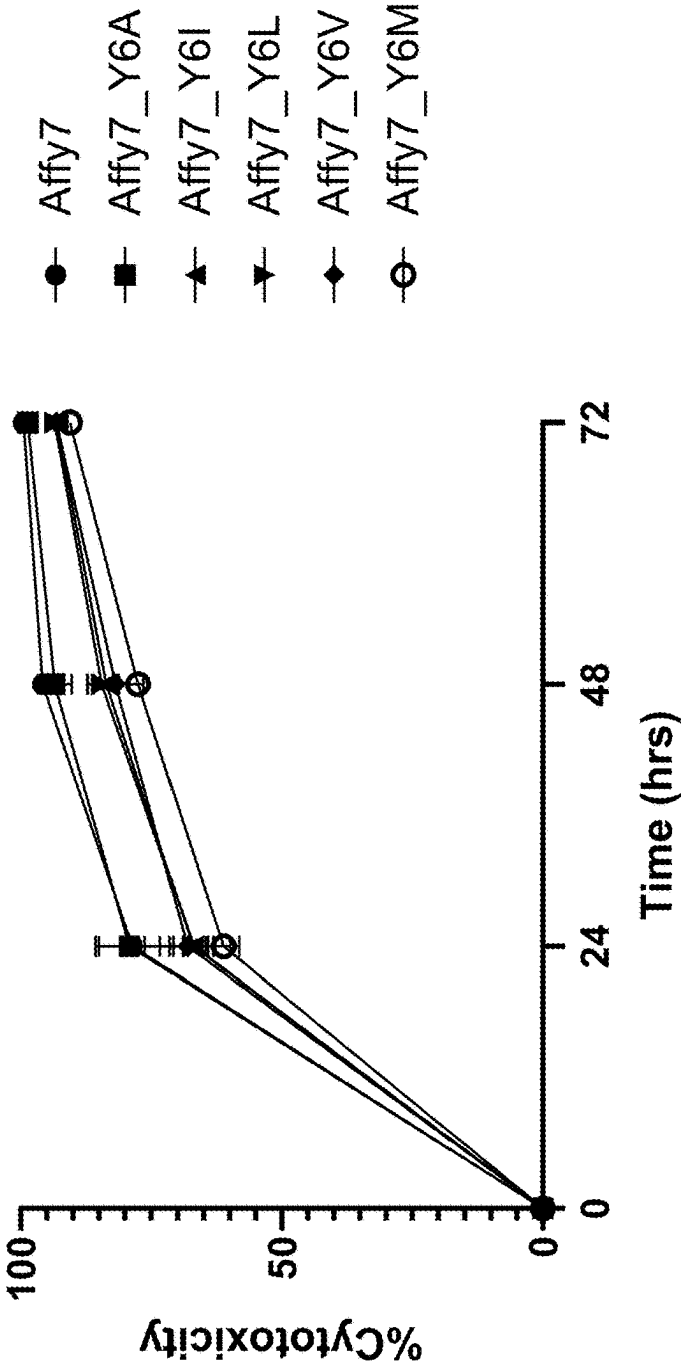
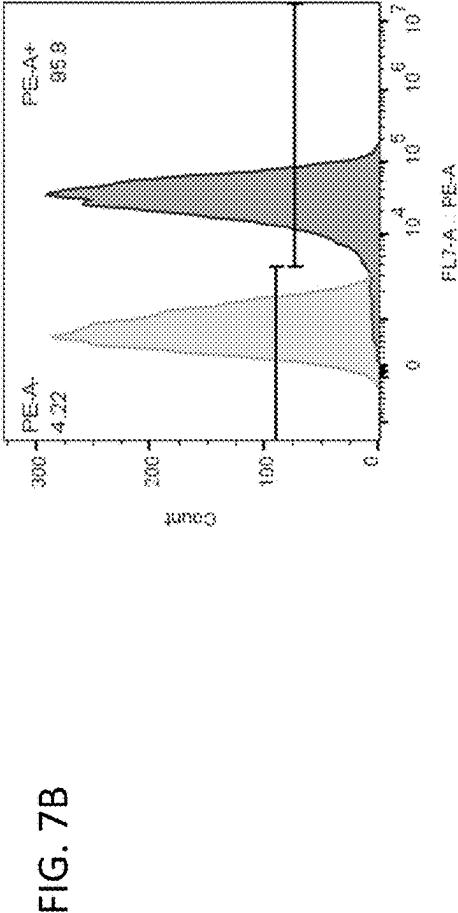
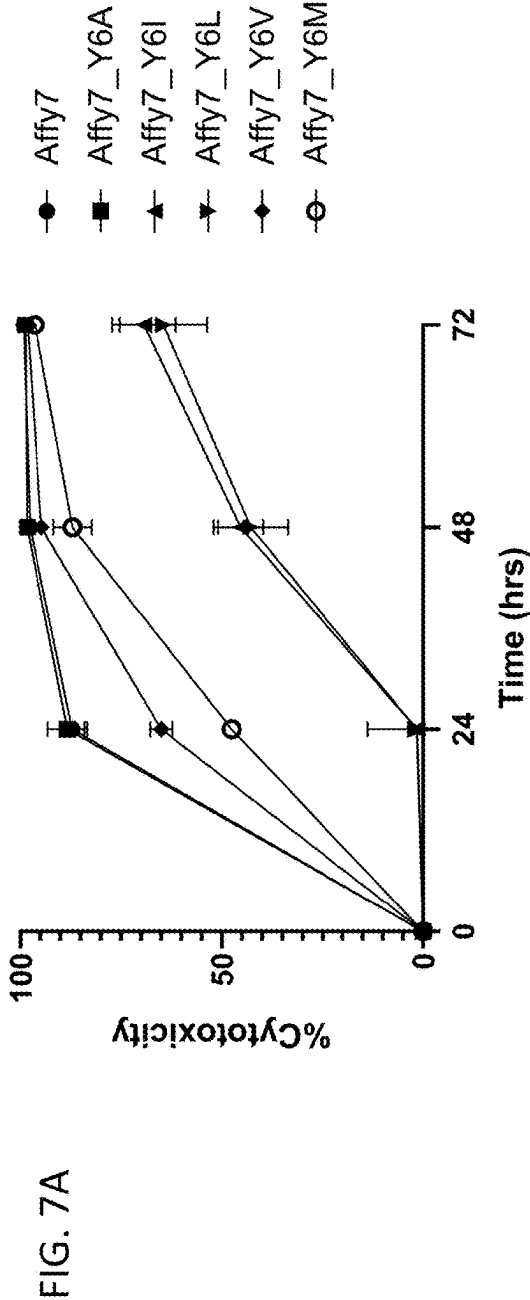


FIG. 6

Hs746T Human Stomach Carcinoma Cell Line Co-Cultured With Affy7 Parent/Variant CAR-T Cells



H1993 Human Non-Small Cell Lung Adenocarcinoma Cell Line
Co-Cultured With Affy7 Parent/Variant CAR-T Cells

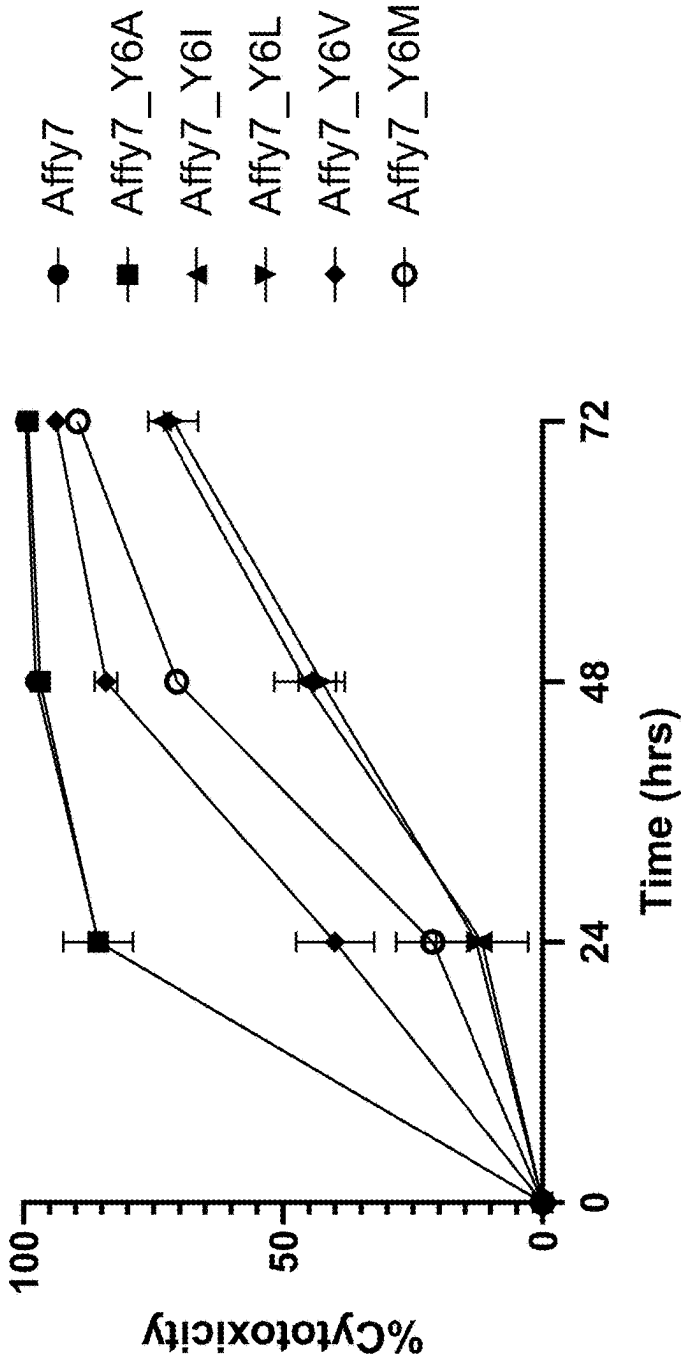


FIG. 8

SNU-638 Human Stomach Carcinoma Cell Line Co-Cultured With
Affy6 Parent/Variant CAR-T Cells

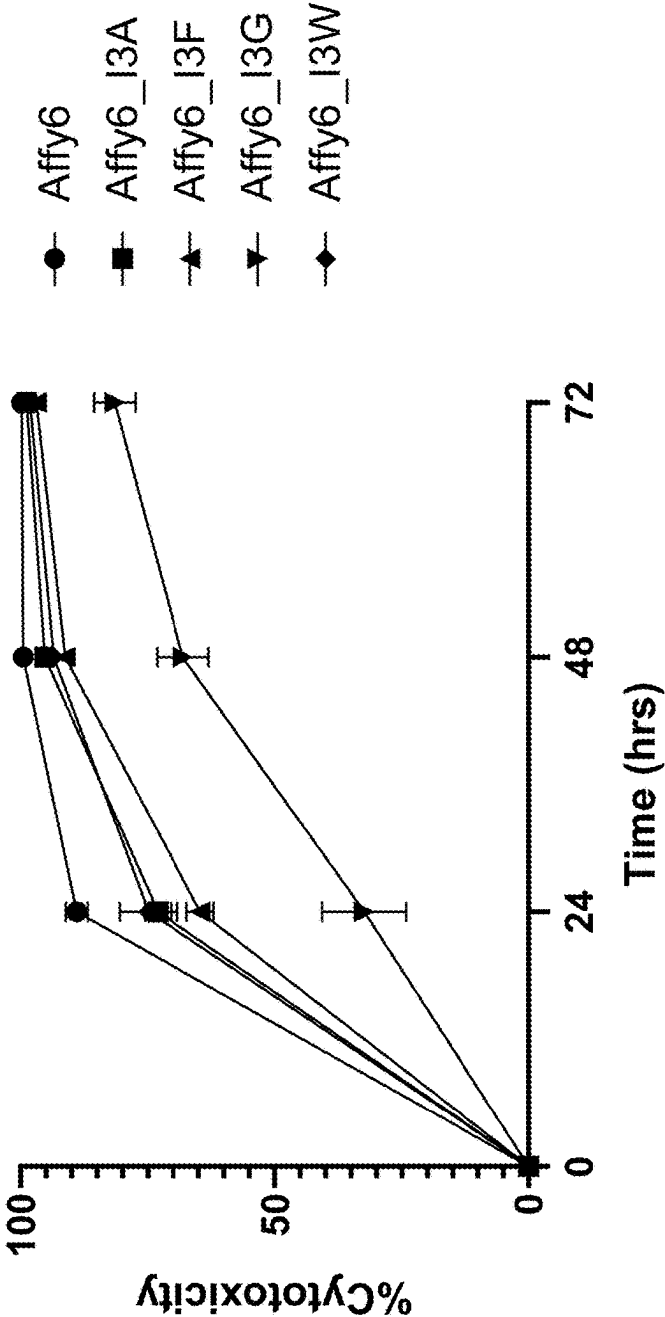


FIG. 9

c-MET Expression in Healthy Cells

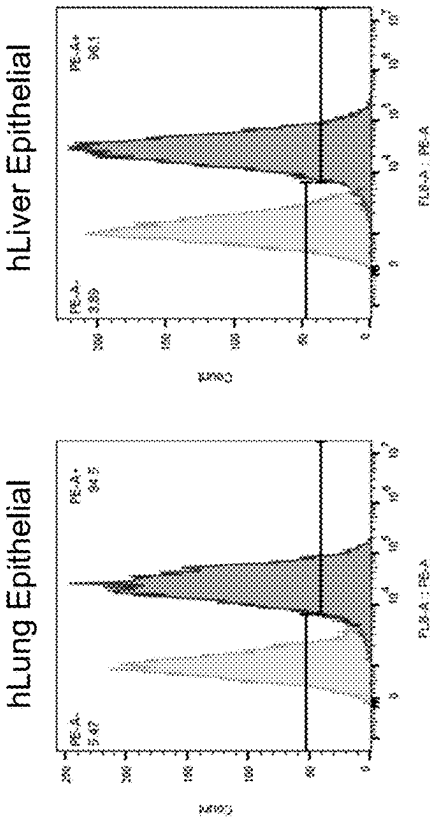


FIG. 10A

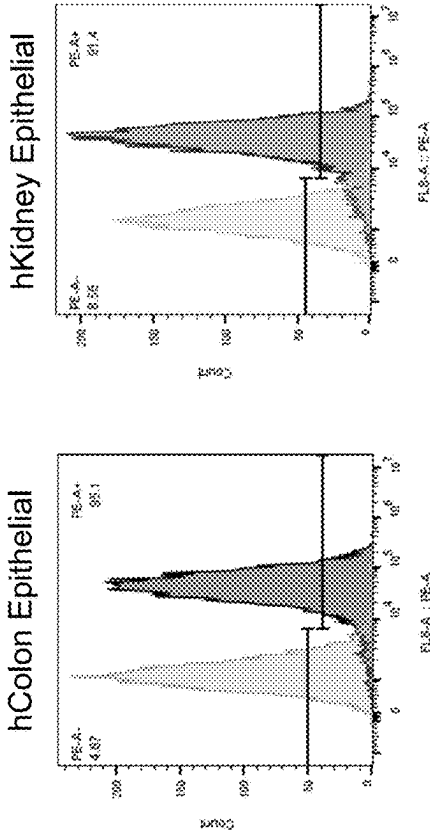


FIG. 10C

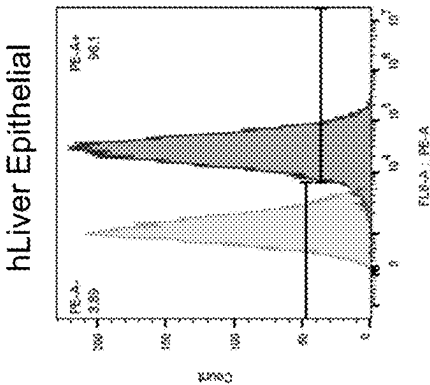


FIG. 10B

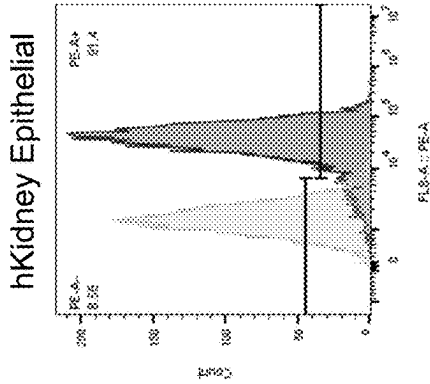


FIG. 10D

Primary Human Lung Cells Co-Cultured With Affy7 Parent/Variant CAR-T Cells

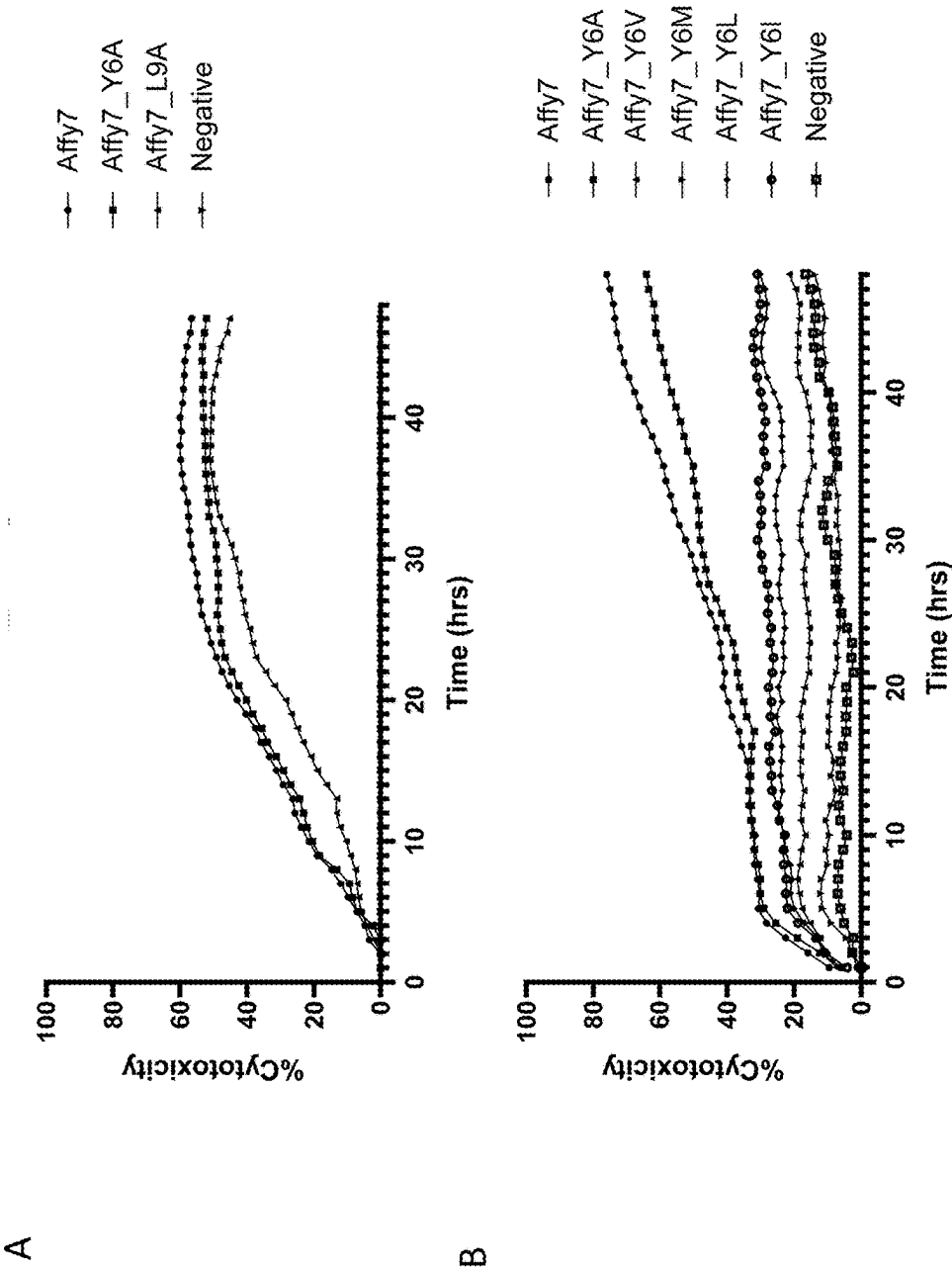


FIG. 11

Primary Human Lung Cells Co-Cultured With Affy6 Parent/Variant CAR-T Cells

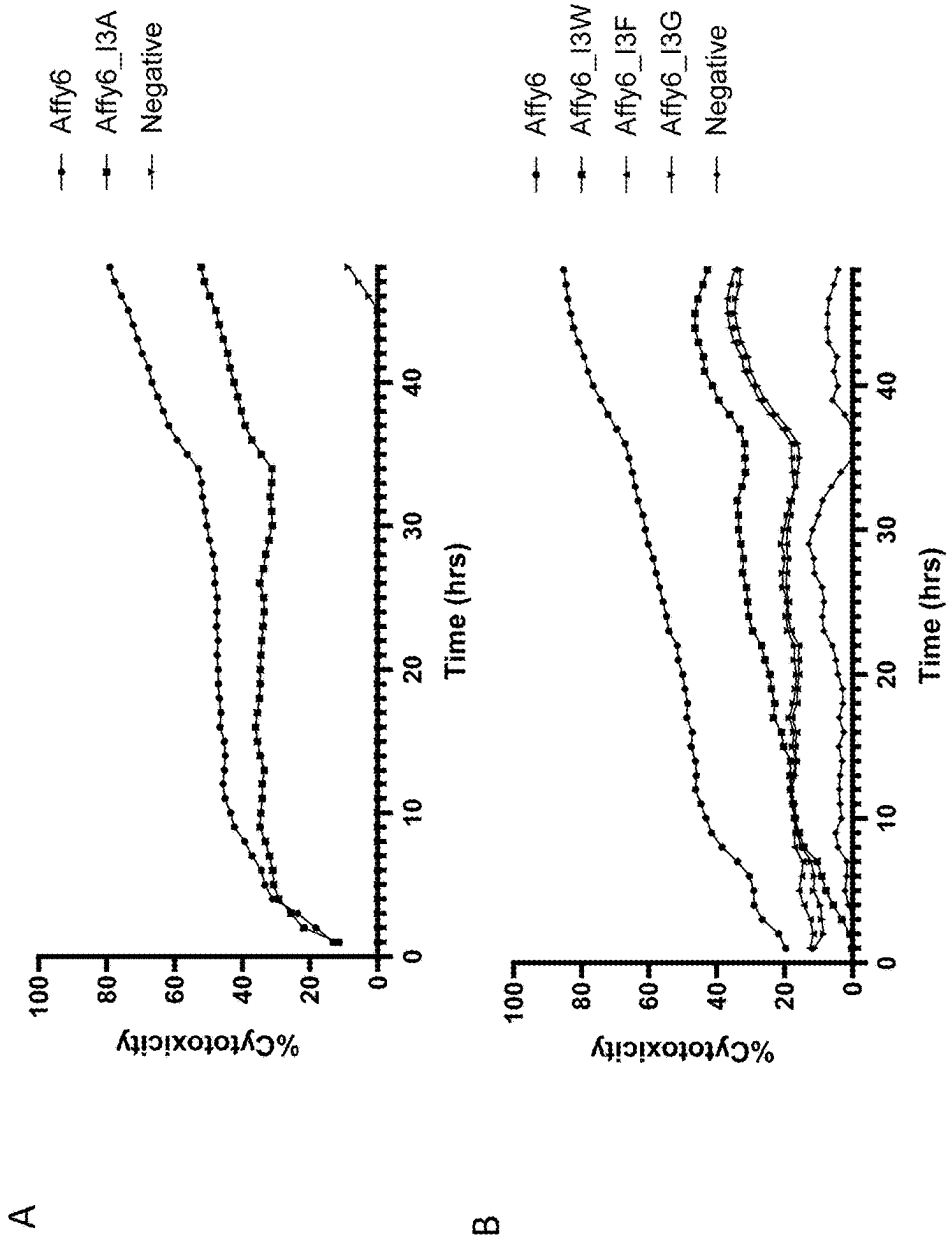


FIG. 12

Primary Human Liver Cells Co-Cultured With Affy7 Parent/Variant CAR-T Cells

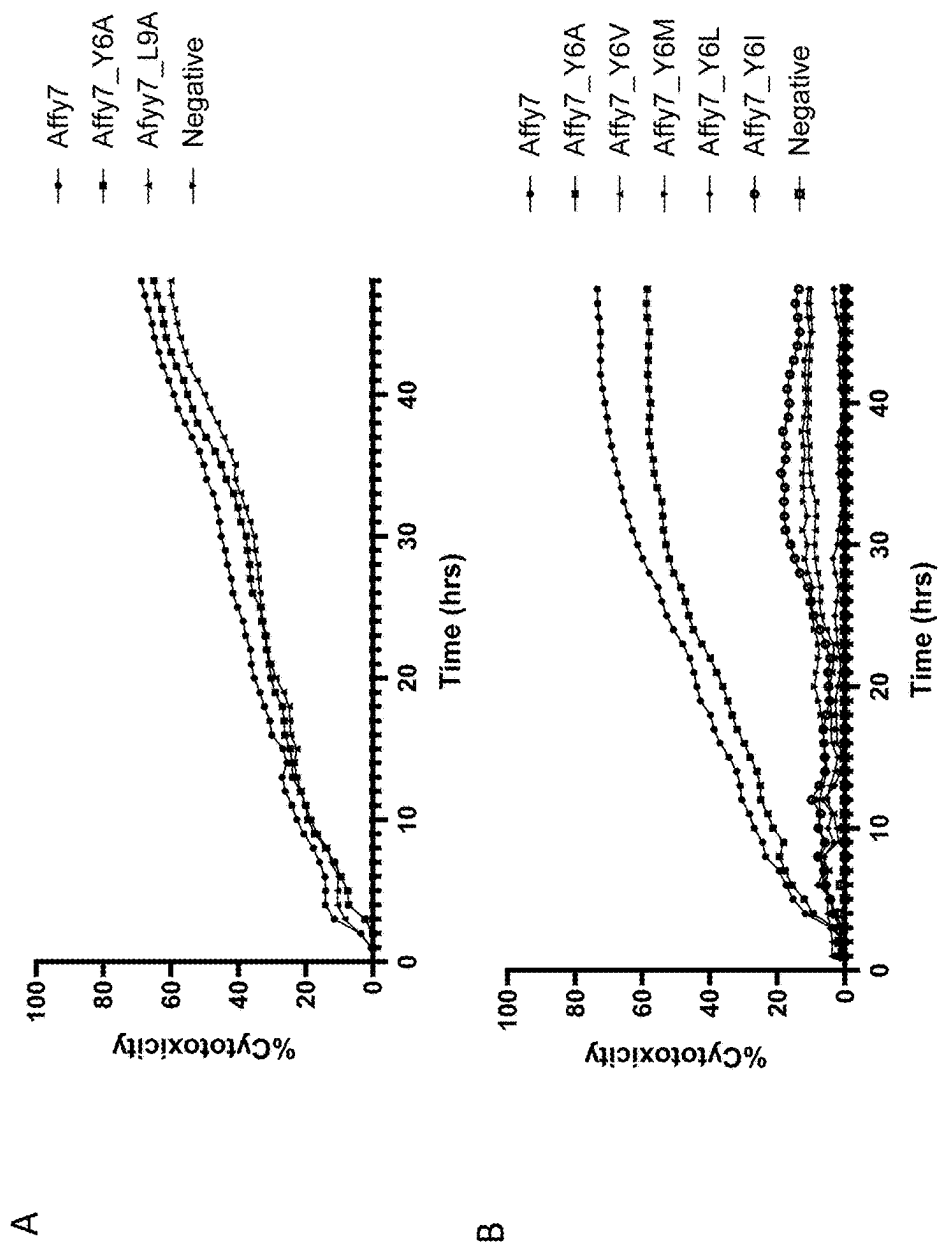


FIG. 13

Primary Human Liver Cells Co-Cultured With Affy6 Parent/Variant CAR-T Cells

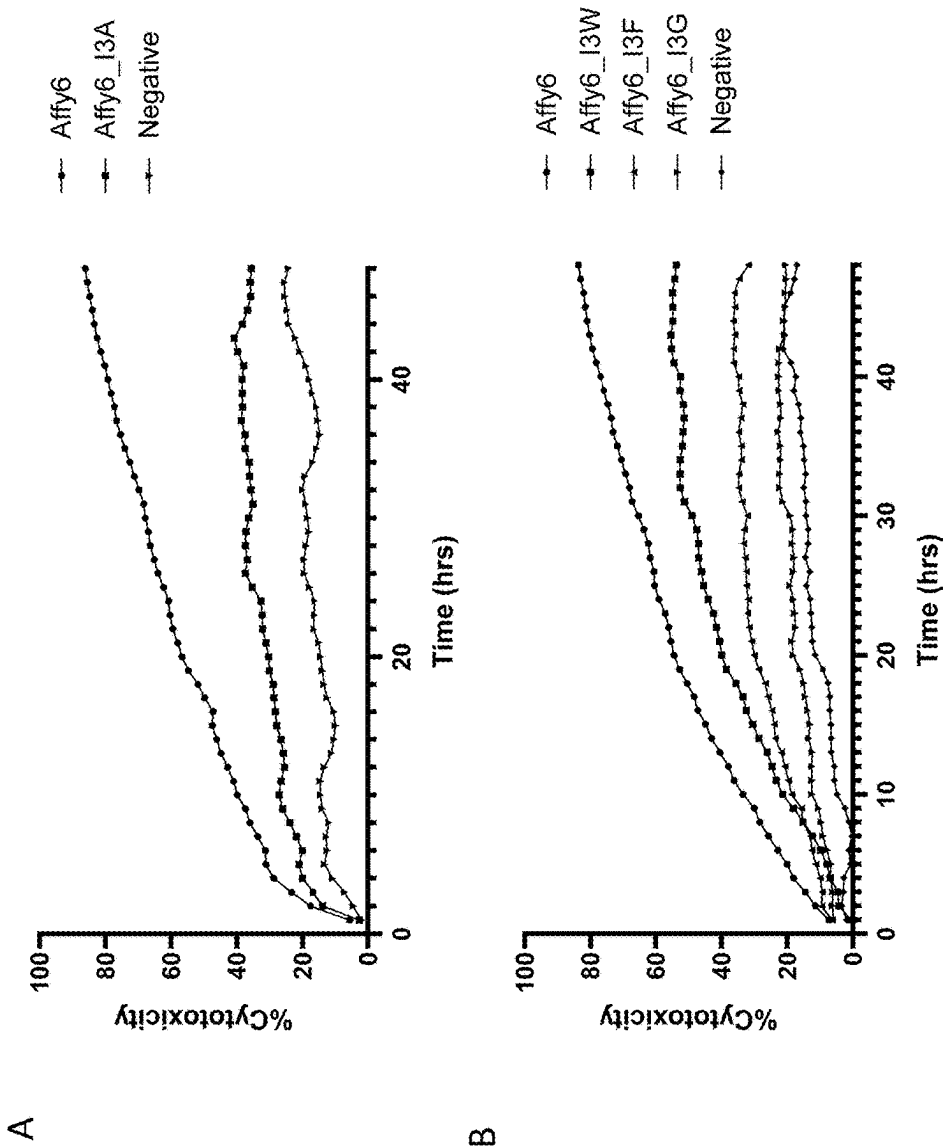


FIG. 14

Primary Human Kidney Cells Co-Cultured With Affy7 Parent/Variant CAR-T Cells

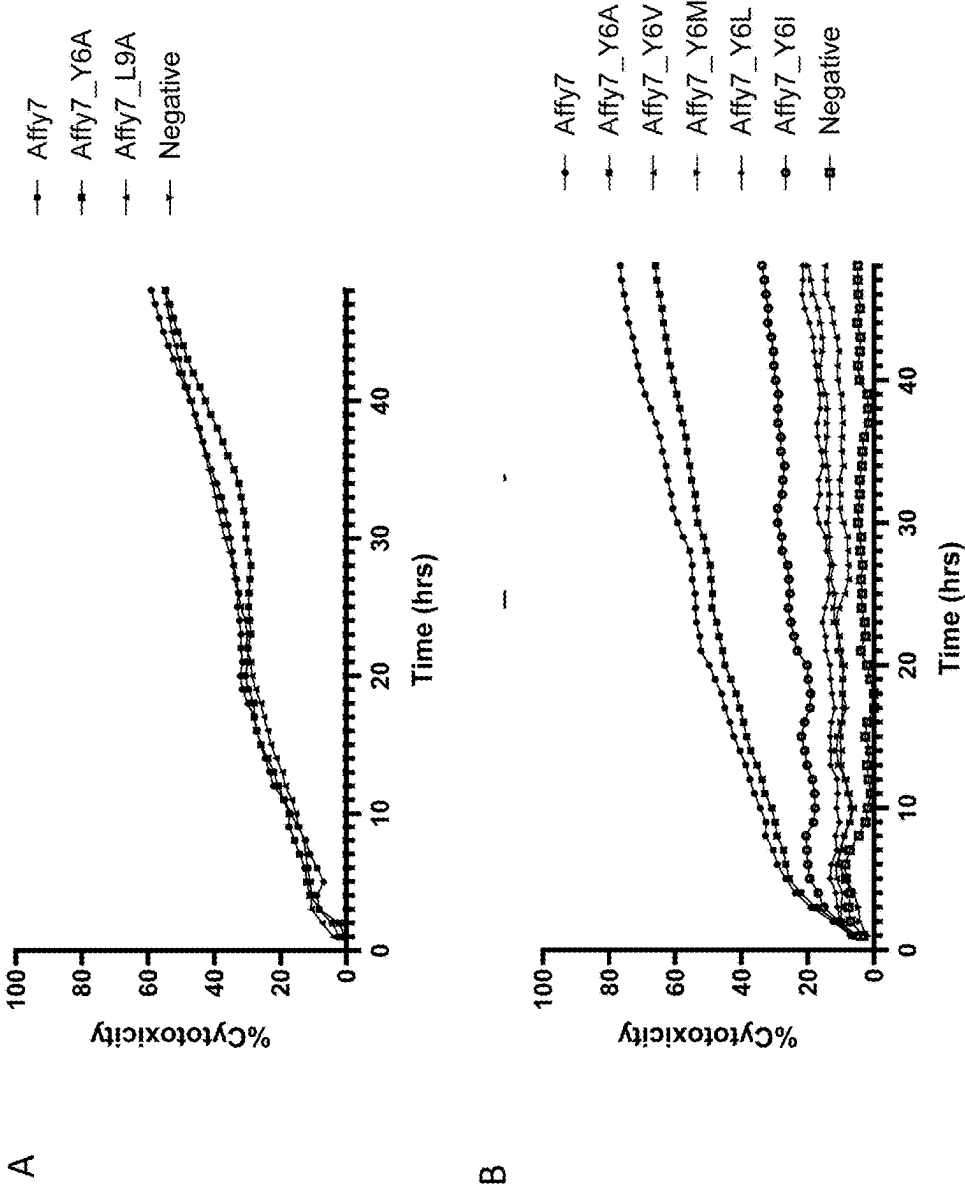


FIG. 15

Primary Human Kidney Cells Co-Cultured With Affy6 Parent/Variant CAR-T Cells

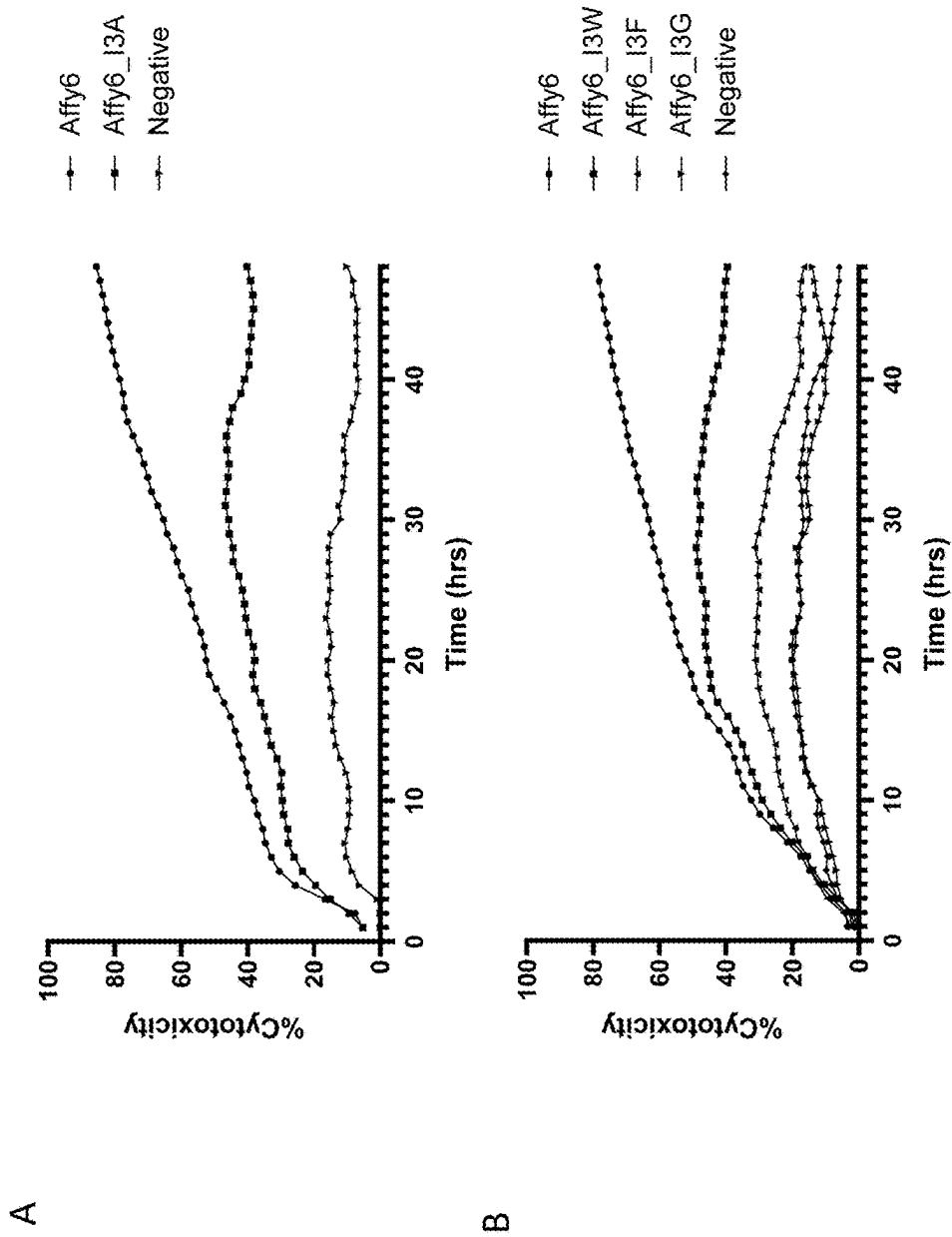


FIG. 16

Primary Human Colon Cells Co-Cultured With Affy7 Parent/Variant CAR-T Cells

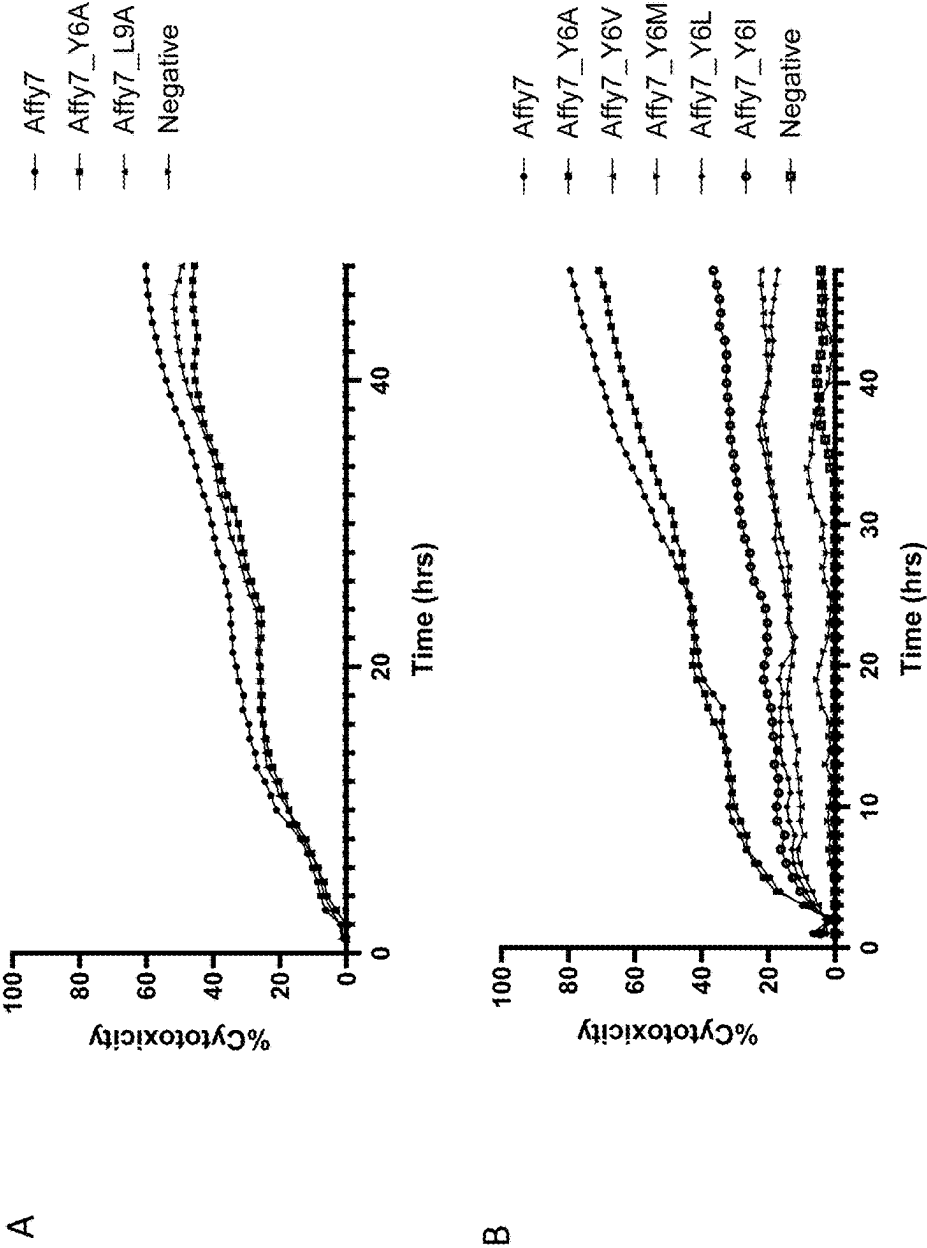


FIG. 17

Primary Human Colon Cells Co-Cultured With Affy6 Parent/Variant CAR-T Cells

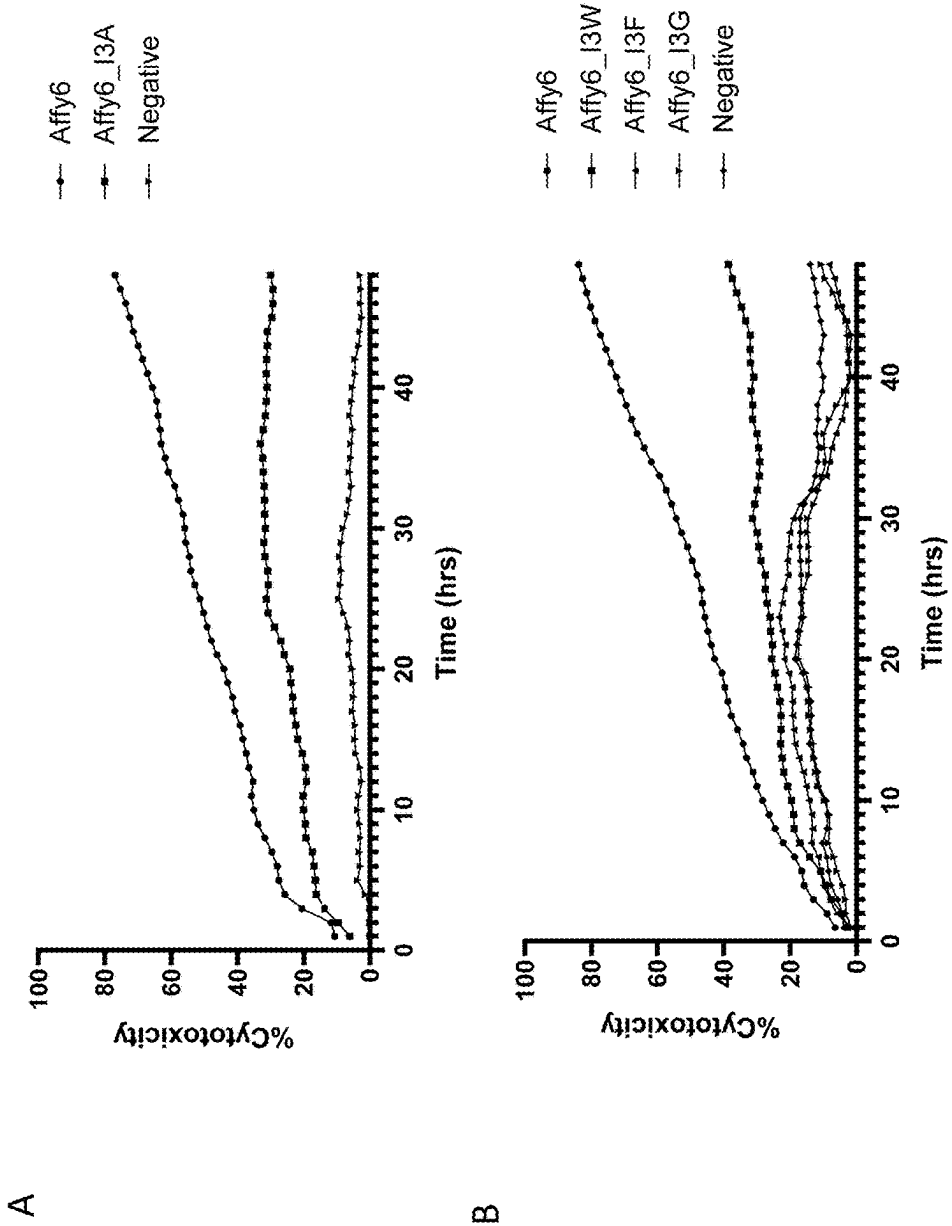


FIG. 18

CAR CONSTRUCTS AND METHODS OF TREATMENT

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application claims priority to U.S. Provisional Patent Application No. 63/616,278, filed on Dec. 29, 2023, the entire content of which is incorporated herein by reference.

SEQUENCE LISTING

[0002] The instant application contains a Sequence Listing which has been submitted electronically in XML file format and is hereby incorporated by reference in its entirety. Said XML copy, created on Jan. 20, 2025, is named 115568-0030_SL.xml and is 246,882 bytes in size.

FIELD

[0003] The present disclosure relates to novel c-MET CAR constructs, c-MET CAR-expressing cells and methods of their use in the treatment of cancer.

BACKGROUND

[0004] Adoptive cell transfer therapy is a type of immunotherapy involving ex vivo expansion of autologous or allogeneic immune cells and subsequent infusion into a patient. The immune cells may be modified ex vivo to specifically target malignant cells. Modifications include engineering of T cells to express chimeric antigen receptors (CARs). The advantage of CAR-T technology compared with chemotherapy or therapeutic antibodies is that reprogrammed engineered T cells can proliferate and persist in the patient and work like a living drug.

[0005] The ideal CAR target antigen would be a native, surface-exposed tumor neoantigen that is highly expressed and is undetectable in healthy tissues. However, many commonly targeted solid tumor antigens are also expressed by non-tumor tissues, albeit at lower levels. CAR molecules with high affinity for such antigens can lead to collateral targeting of healthy tissues resulting in on-target, off-tumor toxicity, a major limiting factor to the progress of CAR T cell therapy to date.

[0006] C-MET is a member of the receptor tyrosine kinase family of proteins, and the product of the proto-oncogene MET. See, e.g., <https://www.ncbi.nlm.nih.gov/gene/4233>. The encoded preproprotein is proteolytically processed to generate alpha and beta subunits that are linked via disulfide bonds to form the mature receptor. Binding of its ligand, hepatocyte growth factor (HGF), induces dimerization and activation of the receptor, which plays a role in cellular survival, embryogenesis, and cellular migration and invasion. Id. Amplification and overexpression of this gene are associated with multiple human cancers. As a proto-oncogene, abnormal activation of c-MET can promote the development and progression of multiple carcinomas such as liver, lung, colon, rectal, breast, pancreatic, ovarian, prostate, renal hepatocellular, gastric, and various head and neck carcinomas, in addition to cancers of the nervous system such as glioblastoma.

[0007] High affinity c-MET CAR-T cells recognize c-MET-expressing cells, including both normal epithelial tissues with low levels of c-MET and carcinomas with considerably higher c-MET expression levels. The recogni-

tion of antigen both on normal, non-target cells as well as on cancer cells can lead to severe toxicities and T cell exhaustion. CAR-T cell exhaustion has been recognized as a major cause of nonresponse and relapse associated with CAR-T cell therapy. T cells expressing chimeric antigen receptors (CARs) at high levels undergo tonic, antigen independent signaling due to receptor clustering, and function poorly as a result of T cell exhaustion, as evidenced by high levels of PD-1, TIM-3, LAG-3 expression, diminished antigen induced cytokine production, poor CAR-T cell persistence, and excessive programmed cell death. These problems are particularly acute in CAR-T treatments in patients with solid tumors, particularly carcinomas.

[0008] It is therefore of great interest to develop CARs and CAR-T expressing cells facilitating more selective killing of tumor cells, reduced T cell exhaustion and T cell activation, improved in vivo persistence, and reduced systemic toxicities.

SUMMARY OF THE INVENTION

[0009] Previously, it was observed that Met is expressed in normal human tissue (liver, gastric, thyroid and kidney) and neoplastic human tissues (Di Renzio et al. (1991) 6(11): 1997-2003). The inventors observed that CAR-T cells expressing anti-c-MET scFvs have the undesirable effect of killing healthy cells and tissue. Accordingly, it has now been unexpectedly found that affinity tuned anti-c-MET scFvs described herein avoid this problem and exhibit reduced killing of healthy cells (lung, liver, colon, and kidney epithelial) while maintaining efficacy of high c-met expressing cancer cell lines. The present disclosure is based, at least in part, on the development of affinity tuned chimeric antigen receptors (CARs) with variant heavy chain variable (V_H) region CDR3 domains providing reduced binding affinity to c-MET as compared to a corresponding binding domain from the parent antibody.

[0010] Accordingly, one aspect of the present disclosure features a chimeric antigen receptor (CAR) comprising: (a) an affinity-tuned extracellular antigen binding domain; (b) a co-stimulatory domain; and an (c) activation domain. The affinity-tuned extracellular antigen binding domain comprises an anti-c-MET antibody. In some embodiments, the anti-c-MET antibody is a single-chain antibody fragment (scFv).

[0011] In some embodiments, the extracellular binding domain comprises a heavy chain variable (V_H) region comprising complementary determining 1 (HC CDR1), complementary determining 2 (HC CDR2), and complementary determining 3 (HC CDR3) regions; and a light chain variable (V_L) region comprising complementary determining 1 (LC CDR1), complementary determining 2 (LC CDR2), and complementary determining 3 (LC CDR3) regions, wherein: the V_H region comprises HC CDR1 and HC CDR2 regions comprising amino acid sequences set forth SEQ ID NOs: 1 and 2, respectively, and an HC CDR3 region comprising an amino acid sequence set forth in any one of SEQ ID NOs: 3-8; and the V_L region comprises LC CDR1, LC CDR2, and LC CDR3 regions comprising amino acid sequences set forth in SEQ ID NOs: 9-11, respectively. In certain embodiments, the HC CDR3 comprises an amino acid sequence set forth in SEQ ID NO: 5 or SEQ ID NO: 8.

[0012] In some embodiments, the V_H and V_L region amino acid sequences are selected from: (a) SEQ ID NOs: 13 and 19, respectively; (b) SEQ ID NOs: 14 and 19, respectively;

(c) SEQ ID NOs: 15 and 19, respectively; (d) SEQ ID NOs: 16 and 19, respectively; (e) SEQ ID NOs: 17 and 19, respectively; or (f) SEQ ID NOs: 18 and 19, respectively. In some embodiments, the extracellular binding domain comprises an anti-c-MET scFv comprising an amino acid sequence set forth in any one of SEQ ID NOs: 29-34.

[0013] In some embodiments, the V_H region in the extracellular binding domain (e.g., anti-c-MET single-chain antibody) comprises HC CDR1 and HC CDR2 regions comprising the amino acid sequences set forth in SEQ ID NOs: 70 and 71, respectively, and an HC CDR3 region comprising an amino acid sequence set forth in any one of SEQ ID NOs: 72-75; and the V_L region comprises LC CDR1, LC CDR2, and LC CDR3 regions comprising amino acid sequences set forth in SEQ ID NOs: 76-78, respectively.

[0014] In some embodiments, the extracellular binding domain comprises V_H and V_L region amino acid sequences selected from: (a) SEQ ID NOs: 80 and 84, respectively; (b) SEQ ID NOs: 81 and 84, respectively; (c) SEQ ID NOs: 82 and 84, respectively; or (d) SEQ ID NOs: 83 and 84, respectively. In some embodiments, extracellular binding domain comprises an anti-c-MET scFv comprising an amino acid sequence set forth in any one of SEQ ID NOs: 92-95.

[0015] In some embodiments, the anti-c-MET scFv comprises a linker between the V_H and V_L regions. In some embodiments, the linker comprises the amino acid sequence (GxS)_n in which x is an integer between 1-6, inclusive, and n is an integer between 1-10, inclusive. In some embodiments, the linker comprises the amino acid sequence (GGGS)_n in which n is an integer between 1-10, inclusive (SEQ ID NO: 143). In one embodiment, the linker comprises the amino acid sequence multimer GGGSGGGSGGGSGGS (SEQ ID NO: 121).

[0016] In some embodiments, the costimulatory domain is from 4-1BB (CD 137), CD28, OX40, ICOS, GITR, CD27, CD30, CD40, DAP10, DAP12, BAFR, HVEM, ICAM-1, lymphocyte function-associated antigen-1 (LFA-1), CD2, CDS, CD7, CD287, LIGHT, NKG2C, NKG2D, SLAMF7, NKp80, NKp30, NKp44, NKp46, CD 160, B7-H3, a ligand that specifically binds with CD83, or a combination thereof. In one embodiment, the extracellular antigen binding domain (or scFv) comprises a costimulatory domain from 4-1BB (SEQ ID NO: 122), CD28 (SEQ ID NO: 123) or OX-40 (SEQ ID NO: 124). In another embodiment, the extracellular binding domain (or scFv) comprises costimulatory domains from 4-1BB (SEQ ID NO: 122) and CD28 (SEQ ID NO: 123). In some embodiments, the costimulatory domain comprises a mutant CD28 costimulatory domain as described in U.S. Patent Application No. 2020/0129554.

[0017] In some embodiments, the activation domain comprises one or more immunoreceptor tyrosine-based activation motifs (ITAMs). Examples of ITAM containing primary activation domain include, but are not limited to, those derived from CD3 zeta, common FcR gamma (FCER1G), Fc gamma RIIa, FcR beta (Fc epsilon 1b), CD3 gamma, CD3 delta, CD3 epsilon, CD5, CD22, CD79a, CD79b, CD278 (ICOS), FcεRI, and CD66d. In one embodiment, the activation domain is from CD3 zeta (SEQ ID NO: 125).

[0018] In some embodiments, the CAR further comprises a hinge domain, a transmembrane domain, or a combination thereof, which is located between the extracellular binding domain and the costimulatory domain. In some embodiments, the CAR disclosed herein may further comprise a hinge domain, which may be linked to the C-terminus of the

extracellular antigen binding domain and to the N-terminus of the transmembrane domain. In some embodiments, the CAR comprises a hinge domain from CD8 alpha (SEQ ID NO: 126), CD28 (SEQ ID NO: 127), or IgG4 (SEQ ID NO: 128).

[0019] In some embodiments, the CAR comprises a transmembrane domain from a cell surface receptor, which is C-terminal to the extracellular antigen binding domain and N-terminal to the costimulatory domain. Exemplary transmembrane domains, include but are not limited to those derived from an alpha, beta or zeta chain of a T cell receptor, CD28, CD3 epsilon, CD45, CD4, CD5, CD8, CD9, CD16, CD22, CD33, CD37, CD64, CD80, CD86, CD134, CD137, CD154, CD271, CD278 (ICOS), TNFRSF18 (GITR), TNFRSF19, or killer cell immunoglobulin-like receptor (KIR). In certain embodiments, the transmembrane domain is from CD8 alpha (SEQ ID NO: 129), CD28 (SEQ ID NO: 130), ICOS (SEQ ID NO: 131), or GITR (SEQ ID NO: 132).

[0020] In exemplary embodiments, the c-MET CAR comprises an amino acid sequence set forth in any one of SEQ ID NOs: 43-48 or 102-105.

[0021] In some embodiments, the c-MET CAR is a bispecific CAR or a tandem CAR (tanCAR), which further comprises a second extracellular antigen binding domain. In some embodiments, the second extracellular antigen binding domain is a second scFv. In some embodiments, the second extracellular binding domain is a binding domain of a protein ligand. In some embodiments, the second extracellular antigen binding domain comprises an αL subunit I domain of human lymphocyte function-associated antigen-1 (LFA-1) targeting ICAM-1. In one embodiment, the binding domain comprises an amino acid sequence set forth in SEQ ID NO: 134 or a mutant thereof. In some embodiments, the second extracellular binding domain targets a tumor-associated antigen (TAA), particularly one that is overexpressed in solid tumors and metastatic tumors.

[0022] Any of the CARs disclosed herein may further comprise a signal peptide located at the N-terminus of the CAR precursor protein. This may include, for example, a CD8 alpha chain signal peptide (SEQ ID NO: 162).

[0023] In another aspect, the present disclosure provides nucleic acids encoding the c-MET scFvs and CARs described herein. In some embodiments, the nucleic acid encoding an anti-c-MET scFv comprises a nucleotide sequence set forth in any one of SEQ ID NOs: 36-41 or 92-95. In some embodiments, the nucleic acid encoding a c-MET CAR comprises a nucleotide sequence set forth in any one of SEQ ID NOs: 57-62, 64-69, 112-115, or 117-120. In some embodiments, the nucleic acid is a plasmid. In some embodiments, the nucleic acid is a vector, such as a lentivirus vector. In some embodiments, the nucleic acid is an RNA.

[0024] In some embodiments, the nucleic acid further comprises a human somatostatin receptor 2 (SSTR2) coding sequence. In some embodiments, the nucleic acid encodes a full length SSTR2 polypeptide (aa 1-381; SEQ ID NO: 137). In other embodiments, the nucleic acid encodes a truncated SSTR2 polypeptide (aa 1-314; SEQ ID NO: 136). In certain embodiments, the nucleic acid encodes an SSTR2-CAR fusion protein comprising an amino acid cleavage sequence between the SSTR2 and CAR coding sequences. In some embodiments, the SSTR2 coding sequence is 5' of the CAR coding sequence. In other embodiments, the CAR coding sequence is 5' of the SSTR2 coding sequence. In some

embodiments, the amino acid cleavage sequence is encoded by a self-cleaving 2A peptide sequence from porcine teschovirus-1 (P2A), equine rhinitis A virus (E2A), *Thosea asigna* virus (T2A), or foot-and-mouth disease virus (F2A), or a combination thereof. In certain embodiments, the amino acid cleavage sequence comprises an amino acid sequence set forth in any one of SEQ ID NOs: 139-142.

[0025] In another aspect, the present disclosure provides a population of immune cells expressing one or more of the c-MET CARs disclosed herein. In some embodiments, the population of cells comprises one or more of the nucleic acids encoding an anti-c-MET CAR and/or an additional nucleic acid described herein. In some embodiments, the population of c-MET expressing cells further expresses a human somatostatin receptor 2 (SSTR2) marker protein. In some embodiments, the population of cells are immune cells. In some embodiments, the immune cells are T cells, natural killer (NK) cells, tumor infiltrating lymphocytes, dendritic cells, macrophages, B cells, neutrophils, eosinophils, basophils, mast cells, myeloid-derived suppressor cells, stem cells, precursors thereof, subtypes thereof, or a combination thereof; optionally wherein the immune cell is a human immune cell. In some embodiments, the immune cells are T cells. In some embodiments, the T cells additionally express an SSTR2 marker. In some embodiments, the population of cells further comprises a second population of immune cells. In some embodiments, the second population of immune cells expresses a second CAR or other polypeptide of interest.

[0026] In another aspect, the present disclosure provides a cell therapy-based method of treating cancer comprising administering to a subject in need thereof a population of immune cells described herein. In some embodiments, the subject is a human patient. In some embodiments, the cancer is a solid tumor. In some embodiments, the solid tumor is a carcinoma. In some embodiments, the carcinoma is a small cell lung carcinoma, non-small cell lung carcinoma, squamous cell lung carcinoma, large cell lung carcinoma, pancreatic carcinoma, pancreatic ductal carcinoma, prostate carcinoma, esophageal carcinoma, breast carcinoma, ovarian carcinoma, prostate carcinoma, colorectal carcinoma, bladder carcinoma, cervical carcinoma, hepatocellular carcinoma, renal hepatocellular carcinoma, gastric carcinoma, papillary carcinoma, adrenocortical carcinoma, pituitary carcinoma, a head and neck carcinoma, an adenocarcinoma thereof, or a squamous cell carcinoma thereof. In some embodiments, the cancer is a glioblastoma or mesothelioma. In some embodiments, the cancer is metastatic.

[0027] In some embodiments, the method further comprises administering the cell therapy to a human patient prior to, concurrently, or after a therapy against the cancer to reduce tumor burden. In some embodiments, the therapy is a chemotherapy, an immunotherapy, a radiotherapy, or a surgery. In some embodiments, prior to the cell therapy, the subject has been administered a lymphodepleting treatment to condition the subject for the cell therapy. In some embodiments, the lymphodepleting treatment comprises administering to the subject fludarabine and/or cyclophosphamide.

[0028] In some embodiments, the therapy comprises administration of an immune checkpoint inhibitor, such as pembrolizumab (Keytruda™), ipilimumab (Yervoy™), nivolumab (Opdivo™), or atezolizumab (Tecentriq™). In some embodiments, the cell therapy comprises administration of a therapeutic antibody selected from the group

consisting of abagovomab, adecatumumab, afutuzumab, alemtuzumab, altumomab, amatuximab, anatumomab, arcitumomab, bavituximab, bectumomab, bevacizumab, bivatuzumab, blinatumomab, brentuximab, cantuzumab, catumaxomab, cetuximab, citatuzumab, cixutumumab, clivatuzumab, conatumumab, daratumumab, drozitumab, duligotumab, dusigitumab, detumomab, dacetuzumab, dalotuzumab, ecomeximab, elotuzumab, ensituximab, ertumaxomab, etaracizumab, farietuzumab, ficlatuzumab, figitumumab, flanvotumab, futuximab, ganitumab, gemtuzumab, girentuximab, glembatumumab, ibritumomab, igovomab, imgatuzumab, indatuximab, inotuzumab, intetumumab, ipilimumab, iratumumab, labetuzumab, lexatumumab, lintuzumab, lorvotuzumab, lucatumumab, mapatumumab, matuzumab, milatuzumab, minretumomab, mitumomab, moxetumomab, namatumab, naptumomab, necitumumab, nimotuzumab, nofetumomab, ocaratuzumab, ofatumumab, obinutuzumab, olaratumab, onartuzumab, oportuzumab, oregovomab, panitumumab, parsatuzumab, patritumab, pemtumomab, pertuzumab, pintumomab, pritumumab, racotumomab, radretumab, rilotumumab, rituximab, robatumumab, satumomab, sibrotuzumab, siltuximab, simtuzumab, solitumab, tacatuzumab, taplitumomab, tenatumomab, teprotumumab, tigatuzumab, tositumomab, trastuzumab, tucotuzumab, ublituximab, veltuzumab, vorsetuzumab, votumumab, zalutumumab, CC49 and 3F8.

[0029] In some embodiments, the method of cell therapy further comprises administration of a tyrosine kinase inhibitor capable of inhibiting TCR signaling and/or CAR signaling, such as dasatinib, ponatinib, saracatinib, bosutinib, nilotinib, or a combination thereof. In a particular embodiment, the inhibitor is dasatinib. In some embodiments, the tyrosine kinase inhibitor is administered for a period of time sufficient to restore at least partial T cell function and then discontinued. In some embodiments, the tyrosine kinase inhibitor is administered continuously. In other embodiments, the tyrosine kinase inhibitor is administered intermittently. In some embodiments, the tyrosine kinase inhibitor is administered intermittently to facilitate periods of T cell inactivation (e.g., during pharmaceutical composition administration) and periods of T cell activation (e.g., during absence of pharmaceutical composition administration). In some embodiments, the tyrosine kinase inhibitor is administered intermittently so that the concentration is maintained below a threshold level required to block CAR-T cell function.

[0030] In another aspect, the present disclosure provides a method for treating a cancer and monitoring CAR-T cell distribution in a patient. The method comprises the steps of: (a) incubating a population of CAR-T cells expressing SSTR2 with a radioactive label that binds to SSTR2; (b) intravenously infusing the labeled CAR-T cells into a patient in an amount of 10^6 - 10^8 cells/kg patient; and detecting the labeled CAR-T cell distribution by PET/CT imaging, wherein the labeled CAR-T cells are infiltrated into cancer cells to kill the cancer cells. In some embodiments, the label is radioactively labeled DOTATOC or radioactively labeled DOTATATE. In certain embodiments, DOTATOC or DOTATATE is radiolabeled with ^{68}Ga . In certain embodiments, the cancer is thyroid cancer, gastric cancer, pancreatic cancer, or breast cancer.

[0031] In another embodiment, a method for treating cancer and monitoring CAR-T cell distribution in a patient, comprises the steps of: (a) intravenously a population of

CAR-T cells expressing SSTR2 into a patient, where the CAR-T cells have been transduced to express at least 100,000 molecules of SSTR2 per T cell; (b) injecting into the patient a radioactive label that binds to SSTR2 at least one hour prior to PET/CT imaging, and (c) detecting the labeled CAR-T cell distribution by PET/CT imaging, wherein the labeled CAR-T cells are infiltrated into cancer cells to kill the cancer cells. In certain embodiments, the cancer is thyroid cancer, gastric cancer, pancreatic cancer, or breast cancer.

[0032] In another aspect, the present disclosure provides method of producing a population of genetically engineered immune cells. The method comprises introducing into a population of immune cells a nucleic acid coding for a CAR disclosed herein, and optionally one or more exogenous nucleic acids. In some embodiments, the immune cells include T cells, natural killer (NK) cells, tumor infiltrating lymphocytes, dendritic cells, macrophages, B cells, neutrophils, eosinophils, basophils, mast cells, myeloid-derived suppressor cells, stem cells, precursors thereof, or a combination thereof; optionally wherein the immune cell is a human immune cell. In some embodiments, the population of genetically engineered immune cells are CAR-T cells. In some embodiments, the method further comprises the step of expanding a population of CAR-T cells in the presence of a tyrosine kinase inhibitor capable of inhibiting TCR signaling and/or CAR signaling. In some embodiments, the tyrosine kinase is dasatinib. In other embodiments, the tyrosine kinase is ponatinib, saracatinib, bosutinib, nilotinib, or a combination thereof.

[0033] The details of one or more embodiments of the invention are set forth in the description below. Other features or advantages of the present invention will be apparent from the following drawings and detailed description of several embodiments, and also from the appended claims.

BRIEF DESCRIPTION OF THE DRAWINGS

[0034] FIG. 1 is a schematic depiction of a CAR expression construct used in the Examples described herein.

[0035] FIG. 2 depicts binding curves reflecting the binding of c-MET to parent Affy7 CAR-T-cells and to two affinity-reduced Affy7 variant CAR-T cells, including the results of this analysis expressed in EC50 and fold change in binding relative to the parent Affy7 CAR-T cells.

[0036] FIG. 3 depicts binding curves reflecting the binding of c-MET to parent Affy7 CAR-T-cells and to four affinity-reduced Affy7 variant CAR-T cells, including the results of this analysis expressed in EC50 and fold change in binding relative to the parent Affy7 CAR-T cells.

[0037] FIG. 4 depicts binding curves reflecting the binding of c-MET to parent Affy6 CAR-T-cells and to two affinity-reduced Affy6 variant CAR-T cells, including the results of this analysis expressed in EC50 and fold change in binding relative to the parent Affy6 CAR-T cells.

[0038] FIG. 5 depicts binding curves reflecting the binding of c-MET to parent Affy6 CAR-T-cells and to four affinity-reduced Affy6 variant CAR-T cells, including the results of this analysis expressed in EC50 and fold change in binding relative to the parent Affy6 CAR-T cells.

[0039] FIG. 6 depicts in vitro cytotoxicity of Affy7 parent CAR-T cells and five affinity-reduced variant Affy7 CAR-T cell populations against an SNU-638 human stomach carcinoma cell line.

[0040] FIG. 7A depicts in vitro cytotoxicity of Affy7 parent CAR-T cells and five affinity-reduced Affy7 variant CAR-T cell populations against an Hs746T human stomach carcinoma cell line. FIG. 7B shows expression of c-met on the cell surface of HS746T cells.

[0041] FIG. 8 depicts in vitro cytotoxicity of Affy7 parent CAR-T cells and five affinity-reduced Affy7 variant CAR-T cell populations against an H1993 human non-small cell lung adenocarcinoma cell line.

[0042] FIG. 9 depicts in vitro cytotoxicity of Affy6 parent CAR-T cells and four affinity-reduced Affy6 variant CAR-T cell populations against an SNU-638 human stomach carcinoma cell line.

[0043] FIGS. 10A-10D show expression levels of c-MET as determined by flow cytometry in normal, healthy human epithelial cells from lung (10A), liver (10B), colon (10C), and kidney (10D).

[0044] FIG. 11, panels A and B depict in vitro cytotoxicity of Affy7 parent CAR-T cells and seven affinity-reduced Affy7 variant CAR-T cell populations against primary human lung cells.

[0045] FIG. 12, panels A and B depict in vitro cytotoxicity of Affy6 parent CAR-T cells and five affinity-reduced Affy6 variant CAR-T cell populations against primary human lung cells.

[0046] FIG. 13, panels A and B depict in vitro cytotoxicity of Affy7 parent CAR-T cells and seven affinity-reduced Affy7 variant CAR-T cell populations against primary human liver cells.

[0047] FIG. 14, panels A and B depict in vitro cytotoxicity of Affy6 parent CAR-T cells and five affinity-reduced Affy6 variant CAR-T cell populations against primary human liver cells.

[0048] FIG. 15, panels A and B depict in vitro cytotoxicity of Affy7 parent CAR-T cells and seven affinity-reduced Affy7 variant CAR-T cell populations against primary human kidney cells.

[0049] FIG. 16, panels A and B depict in vitro cytotoxicity of Affy6 parent CAR-T cells and five affinity-reduced Affy6 variant CAR-T cell populations against primary human kidney cells.

[0050] FIG. 17, panels A and B depict in vitro cytotoxicity of Affy7 parent CAR-T cells and seven affinity-reduced Affy7 variant CAR-T cell populations against primary human colon cells.

[0051] FIG. 18, panels A and B depict in vitro cytotoxicity of Affy6 parent CAR-T cells and five affinity-reduced Affy6 variant CAR-T cell populations against primary human colon cells.

DETAILED DESCRIPTION OF THE INVENTION

[0052] The present disclosure is based on the unexpected discovery that chimeric antigen receptor (CAR)-T cells expressing an affinity tuned anti-c-MET scFv with reduced affinity for c-MET can effectively kill c-MET overexpressing cancer cells, while avoiding the undesirable effect of killing healthy cells and tissue.

I. Chimeric Antigen Receptor with Reduced Affinity Anti-c-MET Binding Domain

[0053] One aspect of the present disclosure provides a chimeric antigen receptor (CAR) comprising an affinity-

tuned anti-c-MET scFv with reduced target affinities to avoid targeting of healthy tissue with basal c-MET expression while maintaining sufficient avidity for targeting tumor tissues with high c-MET expression. The c-MET CARs disclosed herein comprise an artificial (non-naturally occurring) receptor having a binding specificity for the c-MET proto-oncogene, which is capable of triggering immune responses in immune cells upon binding to c-MET, particularly cells overexpressing c-MET.

[0054] The CARs disclosed herein comprise the c-MET binding domain, one or more costimulatory domains and an activation domain comprising a plurality of immunoreceptor tyrosine-based activation motifs (ITAMs), such as a CD3 ζ signaling domain (also referred to as CD3 zeta). The CAR may also have a hinge domain, a transmembrane domain, or a combination thereof. In some embodiments, the transmembrane domain is located between the extracellular antigen binding domain and the costimulatory domain and the hinge domain may be located between the transmembrane domain and the costimulatory domain.

[0055] In one embodiment, provided herein is a CAR comprising an anti-c-MET scFv, one or more costimulatory domains from 4-1BB, CD28, and/or OX-40, and an ITAM-containing activation domain, such as a CD3 ζ signaling domain.

Extracellular Antigen Binding Domain

[0056] The CAR constructs disclosed herein comprise an affinity tuned extracellular antigen binding domain with reduced affinity for the c-MET proto-oncogene. The affinity tuned antigen binding domain is derived from an antibody and comprises a heavy chain variable (V_H) region and a light chain variable (V_L) region. The V_H and V_L regions herein are further subdivided into regions of hypervariability, also known as “complementarity determining regions” (“CDR”), interspersed with regions that are more conserved, which are known as “framework regions” (“FR”). Each V_H and V_L is typically composed of three CDRs and four FRs, arranged from amino-terminus to carboxy-terminus in the following order: FR1, CDR1, FR2, CDR2, FR3, CDR3, FR4. The extent of the framework region and CDRs can be precisely identified using methodology known in the art, for example, by the Kabat definition, the Chothia definition, the AbM definition, and/or the contact definition, all of which are well known in the art. See, e.g., Kabat, E. A., et al. (1991) *Sequences of Proteins of Immunological Interest, Fifth Edition*, U.S. Department of Health and Human Services, NIH Publication No. 91-3242, Chothia et al., (1989) *Nature* 342:877; Chothia, C. et al. (1987) *J. Mol. Biol.* 196:901-917, Al-lazikani et al (1997) *J. Molec. Biol.* 273:927-948; and Almagro, J. *Mol. Recognit.* 17:132-143 (2004). See also the Human Genome Mapping Project Resources at the Medical Research Council in the United Kingdom and the antibody rules described at the Bioinformatics and Computational Biology group website at University College London. Amino acid and nucleotide sequences corresponding to the V_H , V_L and CDR sequences are shown in Table 18. The CDR regions in Table 18 were identified through NCBI Blast.

[0057] In some embodiments, the V_H and V_L regions are “humanized.” Humanized variable regions (or antibodies) are derived from chimeric immunoglobulins, human immunoglobulins, immunoglobulin chains, or antigen-binding fragments thereof in which minimal sequences are derived

from a non-human immunoglobulin. In some embodiments, the humanized variable regions (or antibodies) comprise residues from CDRs of a non-human species (donor antibody) such as mouse, rat, or rabbit having the desired specificity, affinity, and capacity, where the framework region (FR) residues from the non-human species are replaced by corresponding human residues. Furthermore, the humanized binding region may comprise residues that are found neither in the recipient antibody nor in the imported CDR or framework sequences, but are included to further refine and optimize antibody performance. In general, the humanized antibody will comprise substantially all of at least one, and typically two, variable domains, in which all or substantially all of the CDR regions correspond to those of a non-human immunoglobulin and all or substantially all of the FR regions are those of human immunoglobulin consensus sequences.

[0058] In some embodiments, the EC50 of the c-MET binding domain in the c-MET CAR to c-MET is greater than 5 μ M, greater than 15 μ M, greater than 30 μ M, greater than 60 μ M, greater than 150 μ M, greater than 300 μ M, greater than 600 μ M but less than 800 μ M, or any range thereof. In other embodiments, the fold change in EC50 relative to a corresponding wild-type c-MET binding domain in the c-MET CAR (i.e. from which the reduced affinity binding domain is derived) is greater than 50-fold, greater than 150-fold, greater than 500-fold, greater than 1000-fold, greater than 2000-fold, greater than 6000-fold but less than 8000-fold, or any range thereof.

[0059] In some embodiments, the extracellular binding domain (e.g., anti-c-MET scFv) comprises a V_H region having HC CDR1 and HC CDR2 regions comprising amino acid sequences set forth SEQ ID NOs: 1 and 2, respectively, and an HC CDR3 region comprising an amino acid sequence set forth in any one of SEQ ID NOs: 3-8; and a V_L region having LC CDR1, LC CDR2, and LC CDR3 regions comprising amino acid sequences set forth in SEQ ID NOs: 9-11, respectively. In certain embodiments, the HC CDR3 comprises an amino acid sequence set forth in SEQ ID NO: 5 or SEQ ID NO: 8. In some embodiments, the V_H and V_L region amino acid sequences are selected from: (a) SEQ ID NOs: 13 and 19, respectively; (b) SEQ ID NOs: 14 and 19, respectively; (c) SEQ ID NOs: 15 and 19, respectively; (d) SEQ ID NOs: 16 and 19, respectively; (e) SEQ ID NOs: 17 and 19, respectively; or (f) SEQ ID NOs: 18 and 19, respectively. In some embodiments, the extracellular binding domain comprises an anti-c-MET scFv comprising an amino acid sequence set forth in any one of SEQ ID NOs: 29-34. In some embodiments, the extracellular binding domain comprises an anti-c-MET scFv encoded by a nucleotide sequence set forth in any one of SEQ ID NOs: 36-41.

[0060] In some embodiments, the extracellular binding domain (e.g., anti-c-MET single-chain antibody) comprises a V_H region having HC CDR1 and HC CDR2 regions comprising amino acid sequences set forth in SEQ ID NOs: 70 and 71, respectively, and an HC CDR3 region comprising an amino acid sequence set forth in any one of SEQ ID NOs: 72-75; and a V_L region having LC CDR1, LC CDR2, and LC CDR3 regions comprising amino acid sequences set forth in SEQ ID NOs: 76-78, respectively. In some embodiments, the V_H and V_L region amino acid sequences are selected from: (a) SEQ ID NOs: 80 and 84, respectively; (b) SEQ ID NOs: 81 and 84, respectively; (c) SEQ ID NOs: 82 and 84, respectively; or (d) SEQ ID NOs: 83 and 84, respectively. In

some embodiments, extracellular binding domain comprises an anti-c-MET scFv comprising an amino acid sequence set forth in any one of SEQ ID NOs: 92-95.

[0061] In some instances, the extracellular antigen binding domain is a single-chain antibody fragment (scFv) comprising V_H and V_L regions connected by a peptide linker. In some embodiments, the scFv has a $V_H \rightarrow V_L$ orientation. In other embodiments, the scFv has a $V_L \rightarrow V_H$ orientation. In some embodiments, the anti-c-MET scFv comprises a linker between the V_H and V_L regions. Peptide linkers used in scFv constructs are well known in the art and include, for example, the amino acid sequence (GxS) $_n$ in which x is an integer between 1-6, inclusive, and n is an integer between 1-10, inclusive. In some embodiments, the linker comprises the amino acid sequence (GGGS) $_n$ in which n is an integer between 1-10, inclusive (SEQ ID NO: 143). In one embodiment, the linker comprises the amino acid sequence multi-mer GGGSGGGSGGGGS (SEQ ID NO: 121).

[0062] In some embodiments, the affinity tuned extracellular binding domain against c-MET comprises a single domain antibody (sdAb), such as a VHH fragment, a single chain Fab fragment, a single chain Fab' fragment, or a c-MET binding peptide.

Other CAR Components

[0063] In addition to the affinity tuned extracellular antigen binding domain disclosed above, the CAR constructs disclosed herein further comprise one or more costimulatory domains and an activation domain comprising one or more ITAMs (as described below), such as the CD3 ζ signaling domain (which contains 3), or a combination thereof.

[0064] The c-MET CARs disclosed herein comprise one or more costimulatory domains. Costimulatory domains typically enhance and/or alter the nature of the response to activation of the activation domain. Co-stimulatory domains suitable for use in the CARs of the present disclosure are typically receptor-derived polypeptides. In some embodiments, the co-stimulatory domains homodimerize. In some embodiments, the co-stimulatory domain may be the intracellular portion of a transmembrane protein (i.e., the co-stimulatory domain may be derived from the transmembrane protein). In exemplary embodiments, the costimulatory domains are from 4-1BB (CD 137), CD28, OX40, ICOS, GITR, DAP 10, DAP12, CD27, CD30, CD40, BAFFR, HVEM, ICAM-1, LFA-1 (CD11a/CD18), CD2, CDS, CD7, CD287, LIGHT, NKG2C, NKG2D, SLAMF7, NKp80, NKp30, NKp44, NKp46, CD 160, B7-H3, and a ligand that specifically binds with CD83.

[0065] The c-MET CARs disclosed herein further comprises an activation domain. As used herein, the term "activation domain" refers to an intracellular signaling domain that can trigger the production of one or more cytokines upon activation; increases antibody-dependent cellular cytotoxicity (ADCC) and cell death; and/or increased activation and/or proliferation of CD8 $^+$ T cells, CD4 $^+$ T cells, natural killer T cells, $\gamma\delta$ T cells, and/or neutrophil proliferation. In some embodiments, the activation domain comprises at least one (e.g., 1, 2, 3, 4, 5, 6, etc.) immunoreceptor tyrosine-based activation motif (ITAM or ITAMa) with the sequence Yxx[L/I]x $_{(6-8)}$ Yxx[L/I] present in the cytoplasmic tail of an immune signaling receptor or associated subunit to induce cell signaling. The ITAM domains for use in the CARs disclosed herein can include signaling domains from several types of immune signaling receptors, including CD3, B7

family costimulatory intracellular signaling proteins such as molecules and tumor necrosis factor receptor (TNFR) super-family receptors; signaling domains used by NK and NKT cells, such as NKp30 (B7-H6), DAP12, NKG2D, NKp44, NKp46, DAP10, and CD3 zeta; and signaling domains from ITAM-containing human immunoglobulin receptors, such as Fc α RI, Fc γ RI, Fc γ RIIA, Fc γ RIIA, Fc γ RIIC, and FcRL5. Thus, in certain embodiments, the activation domain is from CD3 zeta (SEQ ID NO: 125), Fc epsilon receptor gamma (FCER1G), Fc gamma RIla, FcR beta (Fc epsilon Ib), CD3 gamma, CD3 delta, CD3epsilon, CD5, CD22, CD79a, CD79b, CD278 (also known as "ICOS"), Fc ϵ RI, or CD66d.

[0066] The c-MET CARs disclosed herein further comprises a transmembrane domain, a hinge (or spacer) domain, or both. The transmembrane domain facilitates insertion of the CAR into the cell membranes and can be inserted between the binding domain and a costimulatory domain. In some embodiments, the transmembrane domain can be inserted between the hinge region and the co-stimulatory domain. Any transmembrane domains and/or hinge domains commonly used in CAR constructs can be used here.

[0067] In some embodiments, the c-MET CAR further comprises a hinge between the extracellular binding domain (e.g., scFv) and the transmembrane domain. In this orientation, the hinge domain provides additional distance between the antigen binding domain and the cell membrane surface on which the CAR is expressed. In some examples, the hinge domain may be from CD28, CD8, or an IgG, such as IgG1 or IgG4. In some embodiments, the hinge domain comprises an amino sequence set forth in any one of SEQ ID NOs: 126-128. In some embodiments, the transmembrane domain is obtained from a suitable cell-surface receptor, such as the transmembrane domain of a cell surface receptor of the alpha, beta or zeta chain of the T cell receptor, CD28, CD3 epsilon, CD45, CD4, CD5, CD8, CD9, CD16, CD22, CD33, CD37, CD64, CD80, CD86, CD134, CD137, CD154, CD271, TNFRSF19, and killer cell immunoglobulin-like receptor (KIR). In certain embodiments, the transmembrane domain is from CD8 alpha, CD28, ICOS, or GITR, and optionally comprises an amino acid sequence set forth in SEQ ID NOs: 129-132, respectively.

[0068] In exemplary embodiments, the c-MET CAR comprises an amino acid sequence set forth in any one of SEQ ID NOs: 43-48, 50-54, 102-105, or 107-110.

[0069] In another aspect, the present disclosure provides a bispecific c-MET-CAR (or tandem c-MET-CAR (tanCAR)) comprising an additional extracellular antigen binding domain to provide improved specificity. In some embodiments, the present disclosure further contemplates a multi-specific c-MET-CAR targeting three or more antigens.

[0070] In one embodiment, the second extracellular binding domain in the bispecific CAR or tanCAR comprises an antibody fragment, such as an scFv. In some embodiments, within each antibody or antibody fragment (e.g., scFv) of a bispecific antibody molecule, the VH can be upstream or downstream of the VL. In some embodiments, the upstream antibody or antibody fragment (e.g., scFv) is arranged with its VH (VH $_1$) upstream of its VL (VL $_1$) and the downstream antibody or antibody fragment (e.g., scFv) is arranged with its VL (VL $_2$) upstream of its VH (VH $_2$), such that the overall bispecific antibody molecule has the arrangement VH $_1$ -VL $_1$ -VL $_2$ -VH $_2$. In other embodiments, the upstream antibody or antibody fragment (e.g., scFv) is arranged with its VL (VL $_1$) upstream of its VH (VH $_1$) and the downstream antibody or

antibody fragment (e.g., scFv) is arranged with its VH (VH₂) upstream of its VL (VL₂), such that the overall bispecific antibody molecule has the arrangement VL₁ VH₁-VH₂-VL₂. In some embodiments, a linker is disposed between the two antibodies or antibody fragments (e.g., scFvs), for example, between VL₁ and VL₂ if the construct is arranged as VH₁-VL₁-VL₂-VH₂, or between VH₁ and VH₂ if the construct is arranged as VL₁-VH₁-VH₂-VL₂. The linker may be a linker as described herein, e.g., a (Gly₃-Ser)_n or (Gly₄-Ser)_n linker, wherein n is 1, 2, 3, 4, 5, or 6. In general, the linker between the two scFvs should be long enough to avoid mispairing between the domains of the two scFvs. In some embodiments, a linker is disposed between the VL and VH of the first scFv. In some embodiments, a linker is disposed between the VL and VH of the second scFv. In constructs that have multiple linkers, any two or more of the linkers may be the same or different. Accordingly, in some embodiments, a bispecific CAR or tanCAR comprises VLs, VHs, and may further comprise one or more linkers in an arrangement as described herein.

[0071] In some embodiments, immune effector cells can be genetically modified one or more CARs recognizing target cells with combinatorial Boolean logic: one can engineer T cells with multi-receptor circuits that function as AND gates (requiring two antigens to be present), OR gates (requiring the presence of one of two possible antigens), and NOT gates (high expression of one antigen, low expression of another) to increase tumor selectivity by limiting cross-reactivity with healthy tissues that also express the CAR/TCR target antigen. Exemplary target antigen combinations for AND and AND-NOT logic gates are disclosed in Table 1 of WO 2022/036133 where an “AND” precedes or follows a target antigen present on the surface of a target cancer cell and a “NOT” precedes an antigen that is not present on the surface of a target cancer cell, but may be present on the surface of a non-cancerous cell.

[0072] Where a target antigen pair (or triple) provides for an AND logic gate, two (or three) antigens must be present on the surface of a target cancer cell in order for a genetically modified cytotoxic immune cell of the present disclosure to kill the target cancer cell, where in this case the genetically modified cytotoxic immune cell is genetically modified to express two or three antigen-triggered polypeptides, each recognizing one of the target antigens of the target antigen pair/triplet. For example, where a target antigen pair provides an AND gate logic, each of the target antigens of the target antigen pair must be present on the surface of a target cancer cell in order for a genetically modified cytotoxic immune cell of the present disclosure to kill the target cancer cell.

[0073] Where a target antigen pair/triple provides an AND-NOT logic gate (or, correspondingly, a NOT-AND logic gate), a genetically modified cytotoxic immune cell of the present disclosure: a) is activated to kill a target cancer cell that expresses the AND target cell surface antigen (e.g., the first target cell surface antigen), but not the NOT target cell surface antigen (e.g., the second and/or third target cell surface antigen), on its cell surface; and b) is inhibited from killing a non-cancerous cell if the non-cancerous cell expresses both the AND target cell surface antigen and the NOT target cell surface antigen(s) on its cell surface. In these cases, the genetically modified cytotoxic immune cell must express at least a first antigen-triggered polypeptide that specifically binds the AND target antigen of the target

antigen pair and a second triggered polypeptide that specifically binds the NOT antigen of the target antigen pair. For example, in some cases, binding of an antigen-triggered polypeptide to the NOT target cell surface antigen (expressed on a non-cancerous cell) inhibits T cell activation. In this manner, unintended/undesired killing of a non-cancerous cell is reduced, because the target cancer cell expressing the AND target antigen and not the NOT target antigen will be preferentially killed over the non-cancerous cell expressing both the AND target antigen and the NOT target antigen. Since the cancer cell does not express the NOT target cell surface antigen (expressed on a non-cancerous cell), binding of the first antigen-triggered polypeptide to the AND target antigen (present on the cancer cell surface) results in activation of the genetically modified cytotoxic T cell and killing of the cancer cell.

[0074] In some embodiments, the second extracellular binding is an scFv. In other embodiments, the second extracellular binding domain is a binding domain of a protein ligand. In some embodiments, the second extracellular binding domain targets a tumor-associated antigen (TAA), particularly one that is overexpressed in solid tumors and metastatic tumors of renal, lung, thyroid, and gastrointestinal tissues. Exemplary solid tumor antigen targets for the second binding domain may include but are not limited to target antigens selected from ICAM-1, EpCAM, CEACAM5, EGFRvIII, mesothelin, CS-1, GD2, Tn Ag, PSMA, TAG72, CD44v6, CEA, KIT, IL-13Ra2, GD3, CD171, IL-1Ra, PSCA, VEGFR2, Lewis Y, CD24, PDGFR-beta, SSEA-4, folate receptor alpha, ERBB2, Her2/neu, MUC1, EGFR, NCAM, Ephrin B2, CAIX, LMP2, sLe, HMWMAA, o-acetyl-GD2, folate receptor beta, TEM1/CD248, TEM7R, FAP, Legumam, HPV E6 or E7, CLDN6, TSHR, GPRC5D, ALK, Polysialic acid, PLAC1, globoH, NY-BR-1, UPK2, HAVCR1, ADRB3, PANX3, GPR20, Ly6k, OR51E2, TARP, and GFRa4.

[0075] In certain embodiments, the second extracellular antigen binding domain comprises an ICAM-1 binding domain. In one embodiment, the ICAM-1 binding domain comprises the I domain of the α L subunit (e.g., SEQ ID NO: 134) of human lymphocyte function-associated antigen-1 (LFA-1), which is known to target ICAM-1. A wild-type (WT) I domain encompasses amino acid residues 130-310 (SEQ ID NO: 134) of the 1145 amino acid long mature α L integrin subunit protein (SEQ ID NO: 133, which corresponds to amino acid residues 26-1170 of GenBank Accession No. NP_002200).

[0076] In some embodiments, the ICAM-1 binding domain comprises an I domain mutant disclosed in U.S. Pat. No. 10,428,136, which is incorporated herein by reference in its entirety. Such mutants differ in their affinity for ICAM-1. For example, I domain mutants having one mutation at F292A (Kd 20 μ M), F292S (Kd 1.24 μ M), L289G (Kd 196 nM), F265S (Kd 145 nM), and F292G (Kd 119 nM), or having two mutations at K287C/K294C (Kd 100 nM) in the wild-type I domain are suitable for the present invention. The above numbering of the amino acid residues is in reference to the amino acid sequence of the mature α L integrin of SEQ ID 133. Thus, in some embodiments, the I domain in the bispecific or multi-specific c-MET-CAR set forth in SEQ ID NO: 134 includes one mutation of F292A, F292S, L289G, F265S, or F292G, or with two mutations at K287C/K294C.

[0077] In a specific example, the CAR construct comprises, from the N-terminus to C-terminus, a c-Myc tag, an anti-c-MET scFv described herein, a CD8 hinge domain, a CD28 transmembrane domain, a CD28 co-stimulatory domain, and a CD3 ζ signaling domain. In some embodiments, the CAR construct further comprises a ribosome skipping element (e.g., P2A) following the CD3 ζ signaling domain and an SSTR2 coding region operably linked (i.e. fused in frame) to the ribosome skipping element. Exemplary c-MET-CARs suitable for use according to the present disclosure are set forth in SEQ ID NOs: 43-48 (encoded by SEQ ID NOs: 57-62, respectively), SEQ ID NOs: 50-55 (encoded by SEQ ID NOs: 64-69, respectively), SEQ ID NOs: 102-105 (encoded by SEQ ID NOs: 112-115, respectively), and SEQ ID NOs: 107-110 (encoded by SEQ ID NOs: 117-120, respectively). Additionally, the CAR typically comprises a signal peptide at the N-terminus of the CAR precursor protein, which is operably linked to the c-MET binding domain.

[0078] In some embodiments, the other CAR components in section (B) may be variants of their naturally occurring counterparts. In some embodiments, any of the other CAR components may contain an amino acid sequence at least 80% (e.g., at least 85%, 90%, 95%, 98%, 99% or above) identical to its natural counterpart. As used herein, the “percent identity” of two amino acid sequences is determined using the algorithm of Karlin and Altschul Proc. Natl. Acad. Sci. USA 87:2264-68, 1990, modified as in Karlin and Altschul Proc. Natl. Acad. Sci. USA 90:5873-77, 1993. Such an algorithm is incorporated into the NBLAST and XBLAST programs (version 2.0) of Altschul, et al. J. Mol. Biol. 215:403-10, 1990. BLAST protein searches can be performed with the XBLAST program, score=50, word-length=3 to obtain amino acid sequences homologous to the protein molecules of interest. Where gaps exist between two sequences, Gapped BLAST can be utilized as described in Altschul et al., Nucleic Acids Res. 25(17):3389-3402, 1997. When utilizing BLAST and Gapped BLAST programs, the default parameters of the respective programs (e.g., XBLAST and NBLAST) can be used.

[0079] In some examples, the costimulatory or activation domain contain up to 15 (e.g., up to 12, 10, 8, 6, 5, 4, 3, 2, or 1) amino acid residue substitutions relative to the wild-type counterpart. In some examples, the amino acid residue substitutions are conservative amino acid residue substitutions. As used herein, a “conservative amino acid substitution” refers to an amino acid substitution that does not alter the relative charge or size characteristics of the protein in which the amino acid substitution is made. Variants can be prepared according to routine methods for altering polypeptide sequences known to those skilled in the art. Conservative substitutions of amino acids may include substitutions within the following groups: ((a) A→G, S; (b) R→K, H; (c) N→Q, H; (d) D→E, N; (e) C→S, A; (f) Q→N; (g) E4→D, Q; (h) G4→A; (i) H→N, Q; (j) I→L, V; (k) L→I, V; (l) K→R, H; (m) M→L, I, Y; (n) F→Y, M, L; (o) P→A; (p) S→T; (q) T→S; (r) W→Y, F; (s) Y→W, F; and (t) V→I, L.

II. Genetically Modified Immune Cells

[0080] In another aspect, the present disclosure provides a population of immune cells comprising genetically modified immune cells (e.g., T cells) expressing one or more of the chimeric antigen receptor (CAR) constructs disclosed herein. Such modified immune cells express a CAR which

specifically binds c-MET, thereby eliminating the target disease cells via, e.g., the effector activity of the immune cells. In some embodiments, the population of immune cells comprises genetically modified cytotoxic effector cells (CAR-T cells) independently expressing a single CAR, a bispecific or multispecific CAR, or multiple CARs (e.g., dual CAR-T).

[0081] In some embodiments, the immune cells for gene transduction herein may be T cells, natural killer (NK) cells, tumor infiltrating lymphocytes, dendritic cells, monocytes, macrophages, B cells, neutrophils, eosinophils, basophils, mast cells, myeloid-derived suppressor cells, mesenchymal stem cells, precursors thereof, subtypes thereof, or combinations thereof.

[0082] T cells may be selected from the group consisting of cytotoxic T-lymphocytes (CD8+), including Tc1, Tc2, Tc9, Tc17, and Tc22 T cells; helper T-lymphocytes (CD4+), including Th1, Th2, Th17, Th9, and Tfh T cells; antigen-inexperienced naïve T cells (T_N), stem cell memory T cells (T_{scm} or T_{SCM}), central memory T cells (T_{cm} or T_{CM}), effector memory T cells (T_{em} or T_{EM}), effector T cells (T_{eff} , T_{EFF} or T_E), precursors to an exhausted T cell (T_{pex} or T_{PEX}), or exhausted T cells (T_{ex} or T_{EX}), central memory T cells, effector memory T cells, tissue resident memory T cells, virtual memory T cells, natural killer T cells (NKT cells), FOXP3+ T cells, FOXP3- T cells). T cells may be purified from peripheral blood lymphocytes by methods known to those skilled in the art. In some embodiments, T cells may be cultured, expanded, differentiated or de-differentiated to obtain particular T cell subsets prior to or following transduction, such as antigen-inexperienced naïve T cells (T_N), stem cell memory T cells (T_{scm} or T_{SCM}), and/or central memory T cells (T_{cm} or T_{CM}).

[0083] In some embodiments, the immune cells are stem cells or are derived from stem cells. The stem cells can be adult stem cells, non-human embryonic stem cells, more particularly non-human stem cells, mesenchymal stem cells, cord blood stem cells, progenitor cells, bone marrow stem cells, induced pluripotent stem cells, totipotent stem cells or hematopoietic stem cells. Representative human cells are CD34+ cells. In other embodiments, the immune cell is derived from the differentiation of a population of induced pluripotent cells (iPSCs).

[0084] In some embodiments, the immune cells are harvested directly from a subject, e.g., a human subject. The cells are genetically modified as described herein and the genetically engineered immune cells are infused back into the same subject, for example, in a CAR-T cell therapy. In this case, the genetically engineered immune cells are autologous to the subject receiving the CAR-T cell therapy. In another embodiment, the immune cells are harvested directly from a donor subject, modified, and the genetically engineered immune cells are infused into a recipient subject in need of therapy, e.g., a CAR-T cell therapy. In this case, the donor immune cells are HLA-matched to the recipient subject, i.e., the cells are allogeneic to the recipient subject. In some embodiments, the immune cells are harvested and isolated from the peripheral blood of the subject (e.g., peripheral blood lymphocytes) and expanded in vitro prior to the genetic modifications disclosed herein.

[0085] In some embodiments, the CAR-expressing immune cells, including CAR-T cells, are transduced to additionally express one or more gene products, including, but not limited to ICAM-1. In some embodiments, the

immune cells (e.g., T cells) are transduced to additionally express human somatostatin receptor 2 (SSTR2).

[0086] In some embodiments, the population of immune cells comprising the CAR-expressing cells further includes a second population of immune cells. In some embodiments, the second population of immune cells includes non-transduced immune cells, immune cells expressing another CAR, and/or immune cells expressing another gene product.

[0087] In some embodiments, the population of immune cells comprising the CAR disclosed herein may comprise at least about 10%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90%, or more of the total immune cell population, or a range between two of the foregoing amounts. In one embodiment, about 50-70% of the immune cells may express the CAR.

III. Methods of Preparing Modified Immune Cells

[0088] The c-MET CAR and optionally other exogenous nucleic acids can be introduced into suitable immune cells by routine methods and/or approaches. To generate modified immune cells expressing the c-MET-directed CARs described herein, coding sequences from one or more CARs, and optionally other gene products, may be cloned into a suitable expression vector (e.g., including but not limited to lentiviral vectors, retroviral vectors, adenoviral vectors, adeno-associated vectors, PiggyBac transposon vector and Sleeping Beauty transposon vector) and introduced into host immune cells using conventional recombinant technology known to those skilled in the art. As a result, modified immune cells of the present disclosure may comprise one or more exogenous nucleic acids encoding at least one CAR and optionally one or more other gene products described herein. In some instances, the coding sequences of such molecules are integrated into the genome of the cell for expression using viral expression vectors (e.g., lentivirus vectors) or by gene editing into suitable target sites. In other instances, the coding sequences of such molecules are not integrated into the genome of the cell.

[0089] An exogenous nucleic acid comprising a coding sequence of interest may further comprise a suitable promoter, which can be in operable linkage to the coding sequence. A promoter, as used herein, refers to a nucleotide sequence in a nucleic acid to which RNA polymerase can bind to initiate the transcription of a DNA coding region into mRNA, which will then be translated into the corresponding protein. A promoter is considered to be “operably linked” to a coding sequence when it is in a correct functional location and orientation relative to the coding sequence to control (“drive”) transcriptional initiation to produce the mRNA for translating the protein. In some instances, the promoter described herein can be constitutive, which initiates transcription independent of other regulatory factors. In some instances, the promoter described herein can be inducible, which is dependent on regulatory factors for transcription.

[0090] Additionally, the exogenous nucleic acids described herein may further contain, for example, some or all of the following: a selectable marker gene, such as the neomycin gene for selection of stable or transient transfectants in mammalian cells; enhancer/promoter sequences from the immediate early gene of human CMV for high levels of transcription; transcription termination and RNA processing signals from SV40 for mRNA stability; SV40 polyoma origins of replication and ColE1 for proper epi-

somal replication; versatile multiple cloning sites; and T7 and SP6 RNA promoters for in vitro transcription of sense and antisense RNA.

[0091] In some instances, one or more nucleic acids encoding the CAR(s) and/or other gene products disclosed herein can be inserted into a suitable expression cassette in a multi-cistronic manner such that the various molecules are expressed as separate polypeptides. In some examples, an internal ribosome entry site (IRES) can be inserted between two coding sequences to achieve this goal. Alternatively, a nucleotide sequence coding for a self-cleaving peptide (e.g., T2A or P2A) can be inserted between two coding sequences as described above.

[0092] For example, in some embodiments, T cells may be transduced with a nucleic acid encoding a second gene product, such as SSTR2, and a CAR disclosed herein. In certain embodiments, the nucleic acid encodes an SSTR2 polypeptide (aa 1-381; SEQ ID NO: 137) or a truncated SSTR2 polypeptide (aa 1-314; SEQ ID NO: 138). In a specific embodiment, the nucleic acid encodes a SSTR2-CAR fusion protein comprising an amino acid cleavage sequence between the CAR coding sequence and the SSTR2 coding sequence. The cleavage sequence may encode a self-cleaving 2A peptide from porcine teschovirus-1 (P2A; SEQ ID NO: 139), equine rhinitis A virus (E2A; SEQ ID NO: 140), *Thosea asigna* virus (T2A; SEQ ID NO: 141), or foot-and-mouth disease virus (F2A; SEQ ID NO: 142), or a combination thereof.

[0093] SSTR2 expressing cell compositions and methods for using SSTR2 as a reporter for CAR-T cell monitoring and use in cancer are disclosed in U.S. Pat. No. 10,577,408, which is expressly incorporated by reference herein.

IV. Therapeutic Applications

(A) Adoptive CAR-T Cell Therapy

[0094] In one aspect, immune cell populations comprising the modified immune cells as described herein may be used in an adoptive immune cell therapy (e.g., CAR-T) for treating a target disease, such as a solid tumor or tumor characterized by overexpression of c-MET. In an embodiment, a method for treating cancer comprises administering to a subject in need thereof, a population of immune cells comprising genetically engineered CAR-expressing cells (e.g., T cells) described herein in an amount suitable for treating the cancer.

[0095] To practice the therapeutic methods described herein, an effective amount of the immune cell population comprising the genetically modified immune cells described herein may be administered to a subject in need of cancer treatment via a suitable route of administration (e.g., intravenous infusion). One or more of the immune cell populations may be mixed with a pharmaceutically acceptable carrier to form a pharmaceutical composition prior to administration, which is also within the scope of the present disclosure. The immune cells may be autologous to the subject, e.g., obtained from the subject in need of the treatment, modified to express the c-MET CAR construct and optionally one or more additional exogenous gene products. The resultant modified immune cells can then be administered to the same subject. Administration of autologous cells to a subject may result in reduced rejection of the immune cells as compared to administration of non-autologous cells. Alternatively, the immune cells can be allogeneic

cells, e.g., the cells are obtained from a first subject, modified as described herein and administered to a second subject that is different from the first subject but of the same species. For example, allogeneic immune cells may be derived from a human donor and administered to a human recipient who is different from the donor.

[0096] The subject to be treated may be a mammal (e.g., human, mouse, pig, cow, rat, dog, guinea pig, rabbit, hamster, cat, goat, sheep, or monkey) suffering from cancer, particularly a human patient with a cancer characterized by overexpression of c-MET.

[0097] The term “an effective amount” as used herein refers to the amount of each active agent required to confer therapeutic effect on the subject, either alone or in combination with one or more active agents. Effective amounts may vary, as recognized by those skilled in the art, depending on the particular condition being treated, the severity of the condition, toxicity consideration, previous trial results, individual patient parameters including age, physical condition, size, gender and weight, the duration of treatment, route of administration, excipient usage, co-usage (if any) with other active agents and like factors within the knowledge and expertise of the health practitioner. The quantity to be administered depends on the subject to be treated, including, for example, the capacity of the individual's immune system to produce a cell-mediated immune response. Effective amounts of the genetically engineered CAR-T cells required to be administered depend on the judgment of the practitioner. However, suitable dosage ranges are readily determinable by one skilled in the art.

[0098] The term “treating” as used herein refers to the application or administration of a cell composition to a subject with cancer with the purpose to cure, heal, alleviate, relieve, alter, remedy, ameliorate, improve, affect progression of the cancer and its symptoms.

[0099] An effective amount of the immune cells may be administered to a human patient in need of the treatment via a suitable route, e.g., intravenous infusion. In some instances, about 1×10^6 to about 1×10^8 CAR+ T cells may be given to a human patient (e.g., a leukemia patient, a lymphoma patient, or a multiple myeloma patient). In some examples, a human patient may receive multiple doses of the immune cells. For example, the patient may receive two doses of the immune cells on two consecutive days. In some instances, the first dose is the same as the second dose. In other instances, the first dose is lower than the second dose, or vice versa.

[0100] In any of the treatment methods involving the use of the modified immune cells disclosed herein, the subject may be administered IL-2 concurrently with the cell therapy. More specifically, an effective amount of IL-2 may be given to the subject via a suitable route before, during, or after the cell therapy. In some embodiments, IL-2 is given to the subject after administration of the immune cells. In some embodiments, prior to the cell therapy, the subject receives a lymphodepleting treatment to condition the subject for the cell therapy.

[0101] In some embodiments, the cancer for treatment is a solid tumor. In some embodiments, the solid tumor is a carcinoma. In certain embodiments, the carcinoma is an adenocarcinoma or squamous cell carcinoma thereof. Examples of carcinomas include small cell lung carcinoma, non-small cell lung carcinoma, squamous cell lung carcinoma, large cell lung carcinoma, pancreatic carcinoma,

pancreatic ductal carcinoma, prostate carcinoma, esophageal carcinoma, breast carcinoma, ovarian carcinoma, prostate carcinoma, colorectal carcinoma, bladder carcinoma, cervical carcinoma, hepatocellular carcinoma, renal hepatocellular carcinoma, gastric carcinoma, papillary carcinoma, adrenocortical carcinoma, pituitary carcinoma, a head and neck carcinoma, an adenocarcinomas thereof, a squamous cell carcinomas thereof, and a metastatic cancer thereof. In some embodiments, the solid cancer is a glioblastoma or mesothelioma.

[0102] In some embodiments, the solid tumor expresses one or more antigen selected from the group consisting of: EpCAM, CEACAM5, EGFRvIII, mesothelin, CS-1, GD2, Tn Ag, PSMA, TAG72, CD44v6, CEA, KIT, IL-13Ra2, GD3, CD171, IL-1Ra, PSCA, VEGFR2, Lewis Y, CD24, PDGFR-beta, SSEA-4, folate receptor alpha, ERBB2, Her2/neu, MUC1, EGFR, NCAM, Ephrin B2, CAIX, LMP2, sLe, HMWMAA, o-acetyl-GD2, folate receptor beta, TEM1/CD248, TEM7R, FAP, Legumam, HPV E6 or E7, CLDN6, TSHR, GPRC5D, ALK, Polysialic acid, PLAC1, globoH, NY-BR-1, UPK2, HAVCR1, ADRB3, PANX3, GPR20, Ly6k, OR51E2, TARP, and GFRa4. In some embodiments, these antigens may serve as target antigens for the bispecific c-MET CARs or as secondary CARs to be used in conjunction with the c-MET CARs disclosed herein.

(B) Co-Administration of Tyrosine Kinase Inhibitors.

[0103] It is known that transient inhibition or modulation of TCR signaling and/or CAR signaling in human T cells can prevent or reverse T cell exhaustion and restore T cell function to CAR-T cells undergoing functional exhaustion. In particular, in vivo treatment of the CAR-T cells described herein with certain tyrosine kinase inhibitors inhibiting T cell receptor signaling (e.g., a Lck tyrosine kinase inhibitor (e.g., dasatinib)) can suppress exhaustion marker expression, augment memory formation, decrease expression of PD1 on CAR-T cells, and facilitate cell survival/proliferation as described in US 2020/101108 A1 and US 2021169880 A1, the disclosures of which are expressly incorporated by reference herein. Similar findings have been obtained using small molecules having a thiazole, imidazolepyridiazine or piperazinyl-methyl-aniline structure (US 2021/0393628 A1, incorporated by reference herein).

[0104] Thus, in some embodiments, a patient receiving CAR-T treatment in accordance with the present disclosure may be additionally administered a tyrosine kinase or small molecule having a thiazole, imidazolepyridiazine or piperazinyl-methyl-aniline structure to mitigate T cell exhaustion, augment memory T cell formation, and/or maintain and facilitate cell survival and proliferation. In other embodiments, these active agents may be used to improve ex vivo cell expansion and collection of CAR-T populations that are resistant and/or less prone to T cell exhaustion. Thus, in another aspect, the present disclosure further provides cell compositions comprising a population of CAR-T cells that are expanded in the presence of particular compounds described herein.

[0105] Exemplary tyrosine kinases for administration or cell treatment include dasatinib, ponatinib, saracatinib, bosutinib, nilotinib, and combinations thereof. Exemplary small molecules having a thiazole, imidazolepyridiazine or piperazinyl-methyl-aniline structure are disclosed in US 2021/0393628 A1, the disclosures of which are expressly incorporated by reference herein.

[0106] Such methods are not limited to particular manner of administration. In some embodiments, multiple cycles of treatment are administered to the subject. In some embodiments, the pharmaceutical composition is administered intermittently. In some embodiments, the pharmaceutical composition is administered for a period of time sufficient to restore at least partial T cell function then discontinued. In some embodiments, the pharmaceutical composition is administered orally.

[0107] In some embodiments, the pharmaceutical compositions are administered iteratively for purposes of facilitating periods of CAR-T cell inactivation (e.g., during pharmaceutical composition administration) and periods of CAR-T cell activation (e.g., during absence of pharmaceutical composition administration; following clearance of the pharmaceutical composition).

[0108] In some embodiments, patients undergoing CAR-T treatment are subjected to intermittent exposure to dasatinib (or other active agents above) to reduce exhaustion, and augment the engraftment, proliferation, and persistence of CAR-T cells in vivo., as well as antitumor function of the CAR-T cells.

[0109] The terms “intermittent administration” or “administered intermittently” in connection with the tyrosine kinase inhibitors described herein refer to the use of these tyrosine kinase inhibitors in an administration regime that causes intermittent changes between a state wherein the patient has tyrosine kinase inhibitor serum levels within the therapeutic window and a state wherein the patient has tyrosine kinase inhibitor serum levels below the therapeutic window. A therapeutic window of a given tyrosine kinase inhibitor can be determined by any methods known in the art.

[0110] Alternatively, the terms “intermittent administration” and “administered intermittently” in connection with a tyrosine kinase inhibitor as used herein refer to the use of a tyrosine kinase inhibitor in an administration regime causing: (1) intermittent changes between a state where the patient has tyrosine kinase inhibitor serum levels causing complete inhibition of the tyrosine kinase and a state where the patient has tyrosine kinase inhibitor serum levels causing partial inhibition of the tyrosine kinase; (2) intermittent changes between a state where the patient has tyrosine kinase inhibitor serum levels causing complete inhibition of the tyrosine kinase and a state where the patient has tyrosine kinase inhibitor serum levels causing no inhibition of the tyrosine kinase; or (3) intermittent changes between a state where the patient has tyrosine kinase inhibitor serum levels causing partial inhibition of the tyrosine kinase and a state where the patient has tyrosine kinase inhibitor serum levels causing no inhibition of the tyrosine kinase.

[0111] Such inhibition can be measured by any methods known in the art, e.g., by measuring the activity of the tyrosine kinase itself using appropriate enzyme assays, or by measuring cellular functions downstream of the kinase. In some embodiments, a partial inhibition refers to an inhibition of at least 25% to 75% compared to a situation in the absence of the inhibitor. As used herein, “no inhibition” refers to an inhibition of less than 25%, or less than 10% compared to a situation in the absence of the inhibitor.

[0112] In the case of CAR-T cells, inhibition of less than 25% or 10% can be an inhibition of the cytotoxic lysis, cytokine secretion, and/or proliferation of the T cells. Further, the inhibition of at least 25%, but no more than 75%,

can preferably be an inhibition of the cytotoxic lysis, cytokine secretion, and proliferation of the CAR-T cells.

[0113] In some embodiments, intermittent administration of dasatinib may cause intermittent changes between a state wherein the serum levels of dasatinib are above 50 nM and a state wherein the serum levels of dasatinib are at or below 50 nM. Intermittent administration may be achieved by using an administration interval longer than the terminal phase half-life of the tyrosine kinase inhibitor, longer than 2 times the terminal phase half-life of the tyrosine kinase inhibitor, or longer than 3 times, 4 times, or 5 times the terminal phase half-life of the tyrosine kinase inhibitor. For example, intermittent administration of dasatinib may be achieved using an administration interval of at least 6 hours for dasatinib or at least 12 hours for dasatinib. It will be understood by a person skilled in the art that for each administration regime, appropriate dosages of the respective tyrosine kinase inhibitors can be selected based on pharmacokinetic and pharmacodynamic experiments.

[0114] The terms “continuous administration” or “administered continuously” in connection with a tyrosine kinase inhibitor as used herein refer to the use of said tyrosine kinase inhibitor in an administration regime that causes a complete inhibition of the tyrosine kinase in a continuous manner. According to the invention, a complete inhibition refers to an inhibition of at least 75%, compared to a situation in the absence of the inhibitor. Such inhibition can be measured by any methods known in the art, e.g., by measuring the activity of the tyrosine kinase itself using appropriate enzyme assays, or by measuring cellular functions downstream of said kinase.

[0115] In the case of CAR-T cells, inhibition of at least 75% can be an inhibition of the cytotoxic lysis, cytokine secretion, and proliferation of T cells. Alternatively, the terms “continuous administration” and “administered continuously” in connection with a tyrosine kinase inhibitor described herein refer to use of the tyrosine kinase inhibitor in an administration regime that results in serum levels of the tyrosine kinase which are continuously within the therapeutic window. In some embodiments, continuous administration of dasatinib encompasses any administration wherein the serum levels of dasatinib are constantly maintained at or above 50 nM. In one embodiment, dasatinib is administered continuously, where the administration comprises oral administration of 50-200 mg dasatinib every 6-8 hours or 140 mg every 6 hours.

[0116] In some embodiments, the threshold serum level is within the range of 0.1 nM-1 μ M, 1 nM-500 nM, 5 nM-100 nM, 10 nM-75 nM, or 25 nM-50 nM.

(C) Monitoring CAR-T Cell Distribution in a Patient

[0117] In another aspect, the present disclosure provides a method for treating cancer and monitoring CAR-T cell distribution in a patient undergoing CAR-T cell therapy. The method involves the use of CAR-T cells co-expressing human somatostatin receptor 2 (SSTR2) as a cell surface marker for monitoring CAR-T cell distribution in a patient.

[0118] SSTR2 can be used in conjunction with FDA-approved positron emission tomography (PET) radiotracers currently used in clinics to probe for overexpressed SSTR2 in neuroendocrine tumors, specifically 68 Gallium conjugates of DOTATOC and DOTATATE. Single-photon emission computed tomography (SPECT)-based imaging is also available using 111 In-DTPAOC (Octreoscan) or 177 Lute-

tium. SSTR2 displays restricted basal expression in tissues and all major organs except in the kidneys and cerebrum making it ideal for detection of adoptively transferred CAR-T cells targeting a multitude of solid tumors.

[0119] It has previously been shown that SSTR2 facilitates rapid radiotracer uptake and this combined with swift renal clearance of unbound DOTATOC means that high quality, clinical-grade images can be obtained at one hour post DOTATOC injection. DOTATOC also has a short half-life of 68 min which, combined with its rapid clearance, delivers a low radiation dose to the patient. The fact that SSTR2 is of human origin also limits its immunogenicity which has plagued experiments using non-human genetic reporters.

[0120] SSTR2 compositions and methods for using SSTR2 as a reporter for CAR-T cell monitoring and use in cancer are disclosed in U.S. Pat. No. 10,577,408, which is expressly incorporated by reference herein.

[0121] In one embodiment, the method comprises: (a) incubating a population of CAR-T cells described herein with a radioactive label that binds to SSTR2; (b) intravenously infusing the labeled CAR-T cells into a patient in an amount of 10^4 - 10^8 cells/kg patient, and (c) detecting the labeled CAR-T cell distribution by positron emission tomography/computed tomography (PET/CT) imaging, wherein the labeled CAR-T cells are infiltrated into cancer cells to kill the cancer cells. In this method, SSTR2 is pre-labeled in vitro. In some embodiments, the labeled CAR-T cells are administered in an amount of 10^6 - 10^8 or 10^6 - 10^7 cells/kg of the patient.

[0122] In another embodiment, the method comprises: (a) intravenously infusing a population of CAR-T cells described herein into a patient; (b) injecting into the patient a radioactive label that binds to SSTR2 at least one hour prior to PET/CT imaging, and (c) detecting the labeled CAR-T cell distribution by PET/CT imaging, wherein the labeled CAR-T cells are infiltrated into cancer cells to kill the cancer cells. In this method, SSTR2 is labeled post-infusion in vivo. In certain embodiments, the CAR-T cells have been transduced to express at least 100,000 molecules of SSTR2 per T cell.

[0123] In some embodiments, the label is radioactively labeled DOTATOC or radioactively labeled DOTATATE, such as ^{68}Ga -DOTATOC or ^{68}Ga -DOTATATE. The methods for monitoring distribution of radiolabeled CAR-T cells may be used in connection with treating any of the cancers described herein, including but not limited to thyroid cancer, gastric cancer, pancreatic cancer, or breast cancer.

(D) Combination Therapies

[0124] The immune cell populations comprising e.g., the CAR-T cells described herein may be utilized in conjunction with other types of therapy or active agents for cancer, such as chemotherapy, surgery, radiation, gene therapy, and so forth. Such therapies can be administered simultaneously or sequentially (in any order) with the immunotherapy described herein. When co-administered with an additional therapeutic agent, suitable therapeutically effective dosages for each agent may be lowered due to additive or synergistic effects.

[0125] In some examples, the subject is treated with an anti-cancer therapy (e.g., those disclosed herein) to reduce tumor burden prior to the CAR-T therapy disclosed herein. For example, the subject (e.g., a human cancer patient) may

be treated with chemotherapy (e.g., comprising a single chemotherapeutic agent or a combination of two or more chemotherapeutic agents) at a dose that substantially reduces tumor burden. In some instances, the chemotherapy may reduce the total white blood cell count in the subject to lower than $10^8/\text{L}$, e.g., lower than $10^7/\text{L}$. Tumor burden of a patient after the initial anti-cancer therapy, and/or after the CAR-T cell therapy disclosed herein may be monitored via routine methods. If a patient showed a high growth rate of cancer cells after the initial anti-cancer therapy and/or after the CAR-T therapy, the patient may be subjected to a new round of chemotherapy to reduce tumor burden followed by any of the CAR-T therapies disclosed herein.

[0126] Non-limiting examples of other anti-cancer therapeutic agents for use in combination with the modified immune cells (e.g., CAR-T cells) described herein include, but are not limited to, immune checkpoint inhibitors (e.g., PD1, PD1, and CTLA4 inhibitors), anti-angiogenic agents (e.g., TNP-470, platelet factor 4, thrombospondin-1, tissue inhibitors of metalloproteases, prolactin, angiostatin, endostatin, bFGF soluble receptor, transforming growth factor beta, interferon alpha, interferon gamma, soluble KDR and FLT-1 receptors, and placental proliferin-related protein); VEGF antagonists (e.g., anti-VEGF antibodies, VEGF variants, soluble VEGF receptor fragments); chemotherapeutic compounds. In some embodiments, the anti-cancer therapeutic agents include pembrolizumab (KeytrudaTM) ipilimumab (YervoyTM), nivolumab (OpdivoTM), or atezolizumab (TecentriqTM).

[0127] Exemplary chemotherapeutic compounds include pyrimidine analogs (e.g., 5-fluorouracil, floxuridine, capecitabine, gemcitabine and cytarabine); purine analogs (e.g., fludarabine); folate antagonists (e.g., mercaptopurine and thioguanine); antiproliferative or antimitotic agents, for example, vinca alkaloids; microtubule disruptors such as taxane (e.g., paclitaxel, docetaxel), vincristin, vinblastin, nocodazole, epothilones and navelbine, and epidipodophyllotoxins; and DNA damaging agents (e.g., actinomycin, amsacrine, anthracyclines, bleomycin, busulfan, camptothecin, carboplatin, chlorambucil, cisplatin, cyclophosphamide, cytoxan, dactinomycin, daunorubicin, doxorubicin, epirubicin, hexamethylnelamineoxaliplatin, iphosphamide, melphalan, merchlorheptamine, mitomycin, mitoxantrone, nitrosourea, plicamycin, procarbazine, taxol, taxotere, teniposide, triethylenethiophosphoramide and etoposide).

[0128] In some embodiments, radiation or radiation and chemotherapy is used in combination with the cell populations comprising modified immune cells described herein. Additional useful agents and therapies can be found in Physician's Desk Reference, 59th edition, (2005), Thomson P D R, Montvale N.J.; Gennaro et al., Eds. Remington's The Science and Practice of Pharmacy 20th edition, (2000), Lippincott Williams and Wilkins, Baltimore Md.; Braunwald et al., Eds. Harrison's Principles of Internal Medicine, 15th edition, (2001), McGraw Hill, NY; Berkow et al., Eds. The Merck Manual of Diagnosis and Therapy, (1992), Merck Research Laboratories, Rahway N.J.

V. General Techniques

[0129] The practice of the present disclosure will employ, unless otherwise indicated, conventional techniques of molecular biology (including recombinant techniques), microbiology, cell biology, biochemistry, and immunology, which are within the skill of the art. Such techniques are

explained fully in the literature, such as *Molecular Cloning: A Laboratory Manual*, second edition (Sambrook, et al., 1989) Cold Spring Harbor Press; *Oligonucleotide Synthesis* (M. J. Gait, ed. 1984); *Methods in Molecular Biology*, Humana Press; *Cell Biology: A Laboratory Notebook* (J. E. Cellis, ed., 1989) Academic Press; *Animal Cell Culture* (R. I. Freshney, ed. 1987); *Introduction to Cell and Tissue Culture* (J. P. Mather and P. E. Roberts, 1998) Plenum Press; *Cell and Tissue Culture: Laboratory Procedures* (A. Doyle, J. B. Griffiths, and D. G. Newell, eds. 1993-8) J. Wiley and Sons; *Methods in Enzymology* (Academic Press, Inc.); *Handbook of Experimental Immunology* (D. M. Weir and C. C. Blackwell, eds.); *Gene Transfer Vectors for Mammalian Cells* (J. M. Miller and M. P. Calos, eds., 1987); *Current Protocols in Molecular Biology* (F. M. Ausubel, et al. eds. 1987); *PCR: The Polymerase Chain Reaction*, (Mullis, et al., eds. 1994); *Current Protocols in Immunology* (J. E. Coligan et al., eds., 1991); *Short Protocols in Molecular Biology* (Wiley and Sons, 1999); *Immunobiology* (C. A. Janeway and P. Travers, 1997); *Antibodies* (P. Finch, 1997); *Antibodies: a practice approach* (D. Catty., ed., IRL Press, 1988-1989); *Monoclonal antibodies: a practical approach* (P. Shepherd and C. Dean, eds., Oxford University Press, 2000); *Using antibodies: a laboratory manual* (E. Harlow and D. Lane, Cold Spring Harbor Laboratory Press, 1999); *The Antibodies* (M. Zanetti and J. D. Capra, eds. Harwood Academic Publishers, 1995); *DNA Cloning: A practical Approach*, Volumes I and II (D. N. Glover ed. 1985); *Nucleic Acid Hybridization* (B. D. Hames & S. J. Higgins eds. (1985; *Transcription and Translation* (B. D. Hames & S. J. Higgins, eds. (1984; *Animal Cell Culture* (R. I. Freshney, ed. (1986; *Immobilized Cells and Enzymes* (IRL Press, (1986); B. Perbal, *A practical Guide To Molecular Cloning* (1984); F. M. Ausubel et al. (eds.); *Chimeric Antigen Receptor (CAR) Immunotherapy* (D. W. Lee and N. N. Shah, eds., Elsevier, 2019, ISBN:9780323661812); *Basics of Chimeric Antigen Receptor (CAR) Immunotherapy* (M. Y. Balkhi, Academic Press, Elsevier Science, 2019, ISBN: 9780128197479); *Chimeric Antigen Receptor T Cells Development and Production* (V. Picanço-Castro, K. C. R. Malmegrim, K. Swiech, eds., Springer US, 2020, ISBN:

9781071601488); *Cell and Gene Therapies* (C. Bollard, S. A. Abutalib, M.-A. Perales eds., Springer International, 2018; ISBN:9783319543680) and *Developing Costimulatory Molecules for Immunotherapy of Diseases* (M. A. Mir, Elsevier Science, 2015, ISBN:9780128026755).
[0130] The present disclosure is not limited in its application to the details of construction and the arrangements of component set forth in the description herein or illustrated in the drawings. The present disclosure is capable of other embodiments and of being practice or of being carried out in various ways. Also, the phraseology and terminology used herein is for the purpose of description and should not be regarded as limiting. The use of “including,” “comprising,” or “having,” “containing,” “involving,” and variations thereof herein, is meant to encompass the items listed thereafter and equivalents thereof as well as additional items. As also used in this specification and the appended claims, the singular forms “a,” “an,” and “the” include plural references unless the context clearly dictates otherwise.
[0131] Without further elaboration, it is believed that one skilled in the art can, based on the above description, utilize the present invention to its fullest extent. The following specific embodiments are, therefore, to be construed as merely illustrative, and not limitative of the remainder of the disclosure in any way whatsoever. All publications cited herein are incorporated by reference for the purposes or subject matter referenced herein.

EXAMPLES

Example 1: Generation of Reduced Affinity c-MET CAR-T Cells

[0132] Site directed mutagenesis was used to create affinity tuned anti-c-MET variant single-chain variable fragments (scFvs) comprising variable heavy (V_H) regions with lower affinities than scFvs comprising parental versions of the V_H and variable light (V_L) chain regions. Edits to the CDR3 of the heavy chain were made using an NEB Q5 site-directed mutagenesis kit (NEB #E0554S) with individually designed primers outlined in Table 1 to create desired amino acid changes.

TABLE 1

Primers used for making antibodies							
Variant	Change	Primer #	Primer*	Primer SEQ ID NO	Tm	VH SEQ ID NOS	
Affy-Y6A	Y → A	TAC → gcg	AB_001	CTGTGCCACCG gcg GGCTCCTACGTGAGC	133	66	13, 19
			AB_002	TAGTACACGGCGCTGTTC	134		
Affy7-L9A	L → A	TTG → gcG	AB_015	CGTGAGCCCT gcg GATTATTGGGGC	135	67	14, 19
			AB_016	TAGGAGCCGTAGGTGGCA	136		
Affy7-Y6V	Y → V	TAC → gtC	AB_037	CTGTGCCACCG gt CGGCTCCTACG	137	66	15, 19
			AB_038	TAGTACACGGCGCTGTTC	138		
Affy7-Y6L	Y → L	TAC → ctC	AB_039	CTGTGCCACCG ct CGGCTCCTAC	139	65	16, 19
			AB_038	TAGTACACGGCGCTGTTC	138		
Affy7-Y6I	Y → I	TAC → atC	AB_040	CTGTGCCACCA at CGGCTCCTAC	140	65	17, 19
			AB_038	TAGTACACGGCGCTGTTC	138		
Affy7-Y6M	Y → M	TAC → atg	AB_042	CTGTGCCACCA atg GGCTCCTACGTAGCCC	141	65	18, 19
			AB_038	TAGTACACGGCGCTGTTC	138		

TABLE 1-continued

Primers used for making antibodies							
Variant	Change	Primer #	Primer*	Primer SEQ ID NO	Tm	VH SEQ ID NOS	
Affy6-I3A	I → A TAC → g cC	AB_086 AB_087	CCGCTCGGAG g cCACCACCGAG GCGCAGTAGTACACGGCC	142 143	68	80, 84	
Affy6-I3G	I → G ATC → g gC	AB_120 AB_117	CCGCTCGGAG g gCACCACCGAGTTC GCGCAGTAGTACACGGCC	147 145	70	83, 84	
Affy6-I3F	I → F ATC → t tC	AB_123 AB_117	CCGCTCGGAG t tCACCACCGAGT GCGCAGTAGTACACGGCC	148 146	71	82, 84	
Affy6-I3W	I → W ATC → t gg	AB_125 AB_117	CCGCTCGGAG t ggCACCACCGAGTTC GCGCAGTAGTACACGGCC	149 146	67	81, 84	

*mutant nucleotides are bolded in small case

[0133] Lentivirus constructs were prepared to facilitate characterization of the binding properties of the affinity tuned anti-c-MET scFvs in chimeric artificial receptors (CARs). C-MET CARs were prepared by cloning genetic sequences encoding parental (wild-type) c-MET Affy7 and Affy6 scFvs and the affinity-tuned c-MET Affy7 and Affy6 variant scFvs into a lentiviral vector between a c-myc tag and CAR element CD8 hinge. As shown in FIG. 1, the lentiviral vector construct comprises (5' to 3') the EF1a promoter, c-myc tag, c-MET scFv, CD8 hinge, CD28 transmembrane domain, CD28 co-stimulatory domain, CD3 zeta signaling domain, porcine teschovirus-1 2A (P2A) ribosome skipping element, and a human somatostatin receptor 2 (SSTR2) reporter gene for CAR-T cell imaging.

[0134] To obtain stably transformed CAR-expressing cells, lentivirus particles were produced by transiently transfecting HEK 293T cells using the TransIT®-Lenti transfection reagent, (Mirus Bio, MIR6604). Briefly, 8 µg of transfer gene and 12 µg of LV-MAX lentiviral packaging mix (ThermoFisher, A43237) were mixed with Opti-MEM and 50 µL of Trans-IT and incubated for 20 minutes at room temperature. The resulting solutions were added dropwise to 10 cm² cell culture dishes seeded with 6×10⁶ HEK 293T cells in 10 mL DMEM 24 hrs. previously. Transfection media was replaced 16-24 hrs. later. Media containing lentivirus was harvested at 48 hrs. post transfection, filtered through 0.45 µm filters and concentrated by adding Lenti-X concentrator (Takara, 631231) at 1/3 of final volume of supernatant. The solution was then incubated at 4° C. for at least 3 hrs. (as long as overnight) and then spun at 1,500 g for 1 hr at 4° C. The supernatant was removed, and the pellet was resuspended in 400 µL of sterile PBS and stored at -80° C.

Example 2: Evaluation of Binding Properties of Affinity-Tuned c-MET CARs

[0135] Binding of labeled c-MET to CAR-T cells expressing the affinity tuned c-MET scFvs was evaluated by transducing lentivirus anti-c-MET CAR particles into human T cells to form c-MET CAR-T cells incubated with AF647-labeled c-MET. Briefly, human T cells were transduced 24 hrs. post activation with ImmunoCult™ CD3/CD28 T cell activator (StemCell, 10971) in 10 mL of media containing IL7 (12.5 ng/mL), IL15 (12.5 ng/mL) and 10 µL of Lenti-

BOOST (Mayflower Bioscience, SBPLV10103) in G-Rex® 6M wells (Scale Ready, 80660M). Following a 48 hr. incubation at 37° C., the culture media was brought up to 100 mL per well and incubated at 37° C. for 10 days.

[0136] Binding curves were generated by serial diluting AF647 labeled c-met protein (AlexaFluor™ 647 antibody labeling kit, ThermoFisher, A20186) and staining each sample with c-myc antibody (Miltenyi, 130-116-485) at a 1:800 dilution. Cells were stained on ice for 30 mins and then washed with PBS. After washing, residually bound protein on CAR T cells were measured on a flow cytometer (Beckman, CytoFLEX®). The resulting binding data was fit using a non-linear regression analysis on GraphPad Prism to calculate half effective dose concentration (EC₅₀) values. The results of this analysis are shown in FIGS. 2-5 and Tables 2-5 below.

[0137] As compared to CAR-T cells expressing the parental anti-c-MET scFv binding domains (i.e., Affy7 and Affy6), CAR-T cells expressing the affinity-tuned anti-c-MET scFvs exhibiting reduced affinities for c-MET ranging between 20 to 7636-fold relative to the parental-expressing CARs (see FIGS. 2-5 and Tables 2-5).

TABLE 2

Binding Affinities of Parental- and Variant Affy7- Based MET CAR-T Cells for c-MET (Round 1)		
Parent/Variant	EC50 (µM)	Fold Change*
Affy7	0.05	N/A
Affy7-L9A	1.0	20
Affy7-Y6A	3.3	66

TABLE 3

Binding Affinities of Parental- and Variant Affy7- Based MET CAR-T Cells for c-MET (Round 2)		
Parent/Variant	EC50 (µM)	Fold Change*
Affy7	0.1	N/A
Affy7-Y6V	5.3	53
Affy7-Y6M	17.2	172
Affy7-Y6I	183.1	1831
Affy7-Y6L	619.2	6192

TABLE 4

Binding Affinities of Parental- and Variant Affy6-Based MET CAR-T Cells for c-MET (Round 1)		
Parent/Variant	EC50 (μM)	Fold Change*
Affy6	0.02	N/A
Affy6-I3A	20.9	956

TABLE 5

Binding Affinities of Parental- and Variant Affy6-Based MET CAR-T Cells for c-MET (Round 2)		
Parent/Variant	EC50 (μM)	Fold Change*
Affy6	0.02	N/A
Affy6-I3W	3.6	167
Affy6-I3F	20.5	937
Affy6-I3G	150.8	6899

Example 3: Cytotoxicity of Reduced Affinity c-MET CAR-T Cells in Cancer Cell Lines

[0138] One of the pitfalls of CAR-T cell therapies concerns toxicities and T cell exhaustion associated with recognition of antigen both on normal, non-target cells as well

as on cancer cells. However, when using reduced affinity c-MET CAR-T cells to reduce cell toxicities and T cell exhaustion, it is important to maintain effective killing of the targeted cancer cells.

[0139] To evaluate cytotoxicity of the reduced affinity c-MET CAR-T cells against human carcinoma cells, 5×10⁴ target cells (SNU638, Hs746T, and H1993) stably transduced to express GFP and firefly luciferase were co-cultured with c-met CAR-T cells at an effector to target ratio of 5:1. Co-cultures were carried out in T cell culture medium containing 150 μg/mL D-Luciferin (Fisher Scientific, NC1276267) with no cytokine supplementation. Luminescence was measured using a plate reader (BioTek®, Synergy Neo2) with readings every 24 hrs. for 3 days. In each case, readings were normalized to the no T cell control group and percent cytotoxicity was reported.

[0140] As shown in FIGS. 6-9 and Tables 6-9 below, all of the c-MET CAR-T cells of the present disclosure exhibited cytotoxicities similar to or moderately reduced when compared the high-affinity parental c-MET CAR-T cells in the 3 carcinoma cell lines tested. Although the cytotoxicities were significantly reduced in the 24 hour co-cultures compared to the high-affinity parental CAR-T cells, upon further incubation for 48 and 72 hours, the cytotoxicities typically approached the cytotoxicity levels obtained with the high-affinity parental CAR-T cells (see FIGS. 6-9 and Tables 6-9 below).

TABLE 6

In Vitro Cytotoxicity of Parental- and Variant Affy7-Based MET CAR-T Cells Against SNU-638 Human Stomach Carcinoma Cells								
SNU638	0 hrs	stdev	24 hrs	stdev	48 hrs	stdev	72 hrs	stdev
Affy7	0	0	78.6	7.1	95.9	1.9	99.5	0.3
Affy7-Y6A	0	0	79.2	5.8	93.5	3.3	98.7	0.4
Affy7-Y6I	0	0	67.0	3.9	83.6	2.4	93.2	0.7
Affy7-Y6L	0	0	66.9	2.2	84.5	2.8	93.5	0.3
Affy7-Y6V	0	0	68.1	8.2	81.9	5.4	93.2	2.1
Affy7-Y6M	0	0	61.2	3.0	77.5	1.8	90.6	0.4

TABLE 7

In Vitro Cytotoxicity of Parental- and Variant Affy7-Based MET CAR-T Cells Against Hs746T Human Stomach Carcinoma Cells								
Hs746T	0 hrs	stdev	24 hrs	stdev	48 hrs	stdev	72 hrs	stdev
Affy7	0	0	87.1	3.1	97.6	0.7	99.2	0.3
Affy7-Y6A	0	0	88.3	4.9	98.4	0.8	98.8	0.6
Affy7-Y6I	0	0	1.5	2.3	45.4	5.6	69.4	7.9
Affy7-Y6L	0	0	1.6	12.2	42.9	9.3	64.5	10.9
Affy7-Y6V	0	0	65.0	2.8	94.8	0.8	98.2	0.2
Affy7-Y6M	0	0	47.6	2	87	4.8	96.4	0.4

TABLE 8

In Vitro Cytotoxicity of Parental- and Variant Affy7-Based MET CAR-T Cells Against H1993 Non-Small Cell Lung Adenocarcinoma Cells								
H1993	0 hrs	stdev	24 hrs	stdev	48 hrs	stdev	72 hrs	stdev
Affy7	0	0	85.6	0.6	97.7	0.8	99.6	0.5
Affy7-Y6A	0	0	85.7	6.7	96.9	0.4	99.2	0.2
Affy7-Y6I	0	0	11.7	8.9	45.8	5.9	73.5	1.4

TABLE 8-continued

In Vitro Cytotoxicity of Parental- and Variant Affy7-Based MET CAR-T Cells Against H1993 Non-Small Cell Lung Adenocarcinoma Cells								
H1993	0 hrs	stdev	24 hrs	stdev	48 hrs	stdev	72 hrs	stdev
Affy7-Y6L	0	0	12.8	9.9	42.6	4.4	71.2	4.8
Affy7-Y6V	0	0	40.0	7.5	84.2	2.2	93.8	1.3
Affy7-Y6M	0	0	21.3	7	70.5	1.1	89.6	1.1

TABLE 9

In Vitro Cytotoxicity of Parental- and Variant Affy6-Based MET CAR-T Cells Against H1993 Non-Small Cell Lung Adenocarcinoma Cells								
H1993	0 hrs	stdev	24 hrs	stdev	48 hrs	stdev	72 hrs	stdev
Affy6	0	0						
Affy6-I3A	0	0	73.06	2.57	95.28	1.25	98.86	0.31
Affy6-I3F	0	0	64.67	2.71	91.32	1.73	96.99	0.21
Affy6-I3G	0	0	32.48	8.29	68.14	5.08	81.56	4.12
Affy6-I3W	0	0	74.88	5.65	93.5	0.88	98.20	0.24

Example 4. Cytotoxicity of Reduced Affinity c-MET CAR-T Cells in Primary Cells

[0141] As implicitly suggested above, an ideal CAR-T effector cell would be potent enough to kill c-MET overexpressing cancer cells while sparing normal c-MET expressing cells. Therefore, normal human cells were utilized as target cells for examining cytotoxicity in co-cultures with the reduced affinity c-MET CAR-T cells. More specifically, in vitro cytotoxicity assays were carried out by co-culturing the reduced affinity c-MET CAR-T cells with normal primary cells from human epithelial lung, liver, kidney, and colon cells.

[0142] Initially, the normal primary cells were evaluated for c-MET expression by flow cytometry. Briefly, 10 μ L of human HGFR/c-met PE-conjugated antibody (R&D Systems, FAB3582P) was added to 1×10^6 cells in a 100 μ L reaction volume containing 95 μ L staining buffer (BioLegend, 420201). The resulting mixture was then incubated on ice for 30 min., washed with staining buffer and run on the flow cytometer. As shown in FIGS. 10A-10D, the results of this analysis confirm expression of c-MET in normal human epithelial cells.

[0143] Next, cytotoxicity assays were carried out by co-culturing normal primary target cells (human epithelial lung, colon, kidney, and liver) with c-met CAR-T cells at an effector to target ratio of 2:1. The co-cultures were maintained in T cell culture medium with no cytokine supplementation over a period of 48 hours. Cytotoxicity was measured using label-free cellular impedance to continuously monitor and record cell killing using 96-well electronic microplates with the xCELLigence Real-Time Cell Analysis (RTCA) SP instrument (Agilent).

[0144] The results of this analysis are shown in Tables 10A-17B. The results show a correlation between the affinity of the binding between c-MET as shown in Tables 2-5 and cytotoxic activity. For example, among the Affy7 variants, Affy7-L9A and Affy7-Y6A CAR-T cells, which showed the

least comparative reduction in binding affinity to c-MET (relative to the parental Affy7 CAR-T control cells) exhibited the greatest cytotoxicity against normal human lung (Tables 10A-10B; FIGS. 11A-11B), liver (Tables 12A-12B; FIGS. 13A-13B), kidney (Tables 14A-14B; FIGS. 15A-15B), and colon cells (Tables 16A-16B; FIGS. 17A-17B).

[0145] On the other hand, Affy7-Y6M CAR-T cells having a greater reduction in binding affinity to c-MET exhibited greatly reduced cytotoxicity toward normal human lung (Table 10B, FIG. 11B) and colon cells (Table 16B, FIG. 17B). Likewise, Affy7-Y6L CAR-T cells exhibited greatly reduced cytotoxicity toward normal human liver cells (Table 12B, FIG. 13B), while Affy7-Y6V CAR-T cells exhibited greatly reduced cytotoxicity toward normal human lung (Table 10B, FIG. 11B) and kidney cells (Table 14B, FIG. 15B).

[0146] Among the Affy6 CAR-T cells variants, Affy6-I3F and I3G CAR-T cells were found to exhibit greatly reduced cytotoxicity toward normal human lung (Table 11B, FIG. 12B), liver (Table 13B, FIG. 14B), kidney (Table 15B, FIG. 16B), and colon cells (Table 17B, FIG. 18B).

TABLE 10A

Affy7 CAR-T Variants Co-cultured with Normal Human Lung Cells (FIG. 11A, Round 1)				
Time (hrs)	Affy7	Affy7_Y6A	Affy7_L9A	Negative
8	14.7	13.1	7.5	0
16	33.3	31.3	20.9	0
24	50.7	47.6	38.3	0
32	57.1	50.0	46.4	0
40	59.9	53.0	50.7	0
47	56.5	52.1	45.3	0

TABLE 10B

Affy7 CAR-T Variants Co-cultured with Normal Human Lung Cells (FIG. 11A, Round 2)							
Time (hrs)	Affy7	Affy7_Y6A	Affy7_Y6V	Affy7_Y6M	Affy7_Y6L	Affy7_Y6I	Negative
8	31.8	30.6	18.2	9.9	21.1	23.1	6.7
16	35.7	32.6	18.0	9.7	23.6	27.5	5.4
24	43.2	40.1	15.4	6.3	22.8	26.9	4.1
32	55.8	48.3	17.9	6.9	25.4	29.7	11.6
40	67.5	56.5	16.7	9.5	25.9	29.9	9.7
48	76.0	64.0	21.5	13.8	30.1	30.9	16.5

TABLE 11A

Affy6 CAR-T Variants Co-cultured with Normal Human Lung Cells (FIG. 12A Round 1)			
Time (hrs)	Affy6	Affy6-I3A	Negative
8	39.4	33.2	0
16	46.5	36.2	0
24	47.5	33.5	0
32	51.8	31.9	0
40	66.7	42.5	0
48	79.0	52.2	8.7

TABLE 13A

Affy6 CAR-T Variants Co-Cultured with Normal Human Liver Cells (FIG. 14A, Round 1)			
Time (hrs)	Affy6	I3A	Negative
8	36.0	23.9	11.8
16	47.2	28.2	10.3
24	60.8	32.7	16.2
32	69.8	35.7	20.0
40	79.3	38.4	18.0
48	86.0	35.5	24.3

TABLE 11B

Affy6 CAR-T Variants Co-cultured with Normal Human Lung Cells (FIG. 12B, Round 2)					
Time (hrs)	Affy6	I3W	I3F	I3G	Negative
8	38.4	14.3	17.1	14.8	4.3
16	47.1	20.9	16.4	17.3	2.6
24	54.7	30.6	19.1	19.5	8.9
32	63.1	34.0	18.0	18.4	8.8
40	76.4	41.2	29.4	28.0	4.3
48	85.2	42.7	34.8	33.1	4.2

TABLE 13B

Affy6 CAR-T Variants Co-Cultured with Normal Human Liver Cells (FIG. 14B, Round 2)					
Time (hrs)	Affy6	I3W	I3F	I3G	Negative
8	28.4	15.1	15.3	9.6	1.2
16	47.1	32.4	24.7	13.7	7.1
24	59.0	44.0	31.8	18.1	13.0
32	67.9	52.5	34.8	22.3	14.9
40	76.8	52.4	34.8	22.5	17.3
48	83.6	53.6	31.8	20.5	17.0

TABLE 12A

Affy7 CAR-T Variants Co-Cultured with Normal Human Liver Cells (FIG. 13A, Round 1)				
Time (hrs)	Affy7	Affy7_Y6A	Affy7_L9A	Negative
8	17.7	14.2	13.7	0
16	30.1	26.3	24.5	0
24	38.6	33.1	32.8	0
32	46.2	40.1	37.9	0
40	59.1	55.0	50.1	0
48	68.7	65.0	60.0	0

TABLE 14A

Affy7 CAR-T Variants Co-Cultured with Normal Human Kidney Cells (FIG. 15A, Round 1)				
Time (hrs)	Affy7	Affy7_Y6A	Affy7_L9A	Negative
8	12.5	15.8	12.9	0
16	27.2	27.1	23.9	0
24	32.4	29.4	30.7	0
32	36.9	31.1	38.8	0
40	47.4	42.6	46.9	0
48	59.0	54.7	54.7	0

TABLE 12B

Affy7 CAR-T Variants Co-Cultured with Normal Human Liver Cells (FIG. 13B, Round 2)							
Time (hrs)	Affy7	Affy7_Y6A	Affy7_Y6V	Affy7_Y6M	Affy7_Y6L	Affy7_Y6I	Negative
8	23.4	19.3	6.5	7.2	6.1	7.8	0
16	36.9	29.8	3.9	6.2	2.3	6.2	0
24	50.7	44.8	5.5	9.0	2.5	7.4	0
32	64.0	53.9	8.7	10.9	1.1	17.8	0
40	71.2	57.5	10.7	11.3	0	16.4	0
48	73.4	58.5	10.8	10.0	3.2	13.6	0

TABLE 14B

Affy7 CAR-T Variants Co-Cultured with Normal Human Kidney Cells (FIG. 15B, Round 2)							
Time (hrs)	Affy7	Affy7_Y6A	Affy7_Y6V	Affy7_Y6M	Affy7_Y6L	Affy7_Y6I	Negative
8	32.7	29.2	9.3	8.8	11.7	20.5	4.6
16	43.5	39.3	9.7	10.0	12.9	20.9	2.1
24	53.8	48.9	10.6	12.0	14.8	25.8	3.7
32	61.2	53.9	10.3	13.3	16.2	27.6	4.2
40	70.2	60.5	11.0	15.9	17.1	29.7	4.4
48	76.5	65.9	14.8	19.7	21.4	33.9	4.9

TABLE 15A

Affy6 CAR-T Variants Co-Cultured with Normal Human Kidney Cells (FIG. 16A, Round 1)			
Time (hrs)	Affy6	I3A	Negative
8	35.5	28.0	9.7
16	45.2	35.0	14.5
24	56.9	41.0	15.4
32	68.9	46.5	11.0
40	78.5	41.0	6.5
48	85.5	40.2	10.0

TABLE 16A-continued

Affy7 CAR-T Variants Co-Cultured with Normal Human Colon Cells (FIG. 17A, Round 1)				
Time (hrs)	Affy7	Affy7_Y6A	Affy7_L9A	Negative
32	42.8	35.8	38.2	0
40	54.1	45.4	48.5	0
48	60.1	45.6	49.6	0

TABLE 16B

Affy7 CAR-T Variants Co-Cultured with Normal Human Colon Cells (FIG. 17B, Round 2)							
Time (hrs)	Affy7	Affy7_Y6A	Affy7_Y6V	Affy7_Y6M	Affy7_Y6L	Affy7_Y6I	Negative
8	28.5	26.5	9.4	1.4	12.1	15.1	0
16	33.9	36.3	13.5	1.7	16.4	18.8	0
24	43.2	42.5	14.0	1.4	13.6	20.8	0
32	57.0	51.5	18.2	7.1	19.0	28.9	0
40	69.7	62.7	20.3	1.8	20.0	32.5	5.4
48	79.4	70.7	22.2	3.6	17.2	36.4	4.3

TABLE 15B

Affy6 CAR-T Variants Co-Cultured with Normal Human Kidney Cells (FIG. 16B, Round 2)					
Time (hrs)	Affy6	I3W	I3F	I3G	Negative
8	25.7	23.6	19.2	10.1	12.2
16	45.4	39.4	28.0	17.9	18.8
24	57.0	46.0	30.4	17.2	17.3
32	65.5	48.6	27.8	15.8	17.1
40	73.0	43.6	18.8	10.2	13.4
48	78.8	39.5	16.5	14.0	5.9

TABLE 17A

Affy6 CAR-T Variants Co-Cultured with Normal Human Colon Cells (FIG. 18A, Round 1)			
Time (hrs)	Affy6	I3A	Negative
8	31.9	19.4	2.7
16	39.2	22.5	4.4
24	50.3	30.9	7.8
32	57.8	31.8	6.2
40	65.7	31.1	5.2
48	76.8	30.0	2.9

TABLE 16A

Affy7 CAR-T Variants Co-Cultured with Normal Human Colon Cells (FIG. 17A, Round 1)				
Time (hrs)	Affy7	Affy7_Y6A	Affy7_L9A	Negative
8	13.8	13.1	12.1	0
16	29.3	25.0	25.0	0
24	34.9	25.6	27.2	0

TABLE 17B

Affy6 CAR-T Variants Co-Cultured with Normal Human Colon Cells (FIG. 18B, Round 2)					
Time (hrs)	Affy6	I3W	I3F	I3G	Negative
8	24.6	18.8	13.2	8.2	8.9
16	37.7	22.7	19.2	14.4	13.7

TABLE 17B-continued

Affy6 CAR-T Variants Co-Cultured with Normal Human Colon Cells (FIG. 18B, Round 2)					
Time (hrs)	Affy6	I3W	I3F	I3G	Negative
24	46.4	27.0	22.4	16.4	16.6
32	57.3	29.9	13.3	11.4	13.6

TABLE 17B-continued

Affy6 CAR-T Variants Co-Cultured with Normal Human Colon Cells (FIG. 18B, Round 2)					
Time (hrs)	Affy6	I3W	I3F	I3G	Negative
40	72.3	30.9	1.6	2.0	10.0
48	83.8	38.6	8.3	10.7	14.1

LIST OF SEQUENCES

[0147] Exemplary amino acid sequences described herein are provided in Table 18 below:

TABLE 18

Amino Acid and Nucleotide Sequences for Exemplary Polypeptides Described Herein		
SEQ ID NO:	Name	Sequence
1	Affy7 HCDR1	SGYTFTSY
2	Affy7 HCDR2	LEWIGMIDPSNSDTRFN
3	Affy7-Y6A CDR3	AGSYVSPLDY
4	Affy7-L9A HCDR3	YGSYVSPADY
5	Affy7-Y6V HCDR3	VGSYVSPLDY
6	Affy7-Y6L HCDR3	LGSYVSPLDY
7	Affy7-Y6I HCDR3	IGSYVSPLDY
8	Affy7-Y6M HCDR3	MGSYVSPLDY
9	Affy7 LCDR1	QSLLYTSSQKN
10	Affy7 LCDR2	PKLLIYWASTRE
11	Affy7 LCDR3	YYAYP
12	Affy7 VH (wt)	QVQLQQSGPELVRPGASVKMSCRASGYTFTSYWLHW VKQRPQGQLEWIGMIDPSNSDTRFNPFDKATLNVD RSSNTAYMLLSSLTSADSAVYYCATYGSYVSPLDYWG QGTSVTVSS
13	Affy7-Y6A VH*	QVQLQQSGPELVRPGASVKMSCRASGYTFTSYWLHW VKQRPQGQLEWIGMIDPSNSDTRFNPFDKATLNVD RSSNTAYMLLSSLTSADSAVYYCATAGSYVSPLDYWG QGTSVTVSS
14	Affy7-L9A VH	QVQLQQSGPELVRPGASVKMSCRASGYTFTSYWLHW VKQRPQGQLEWIGMIDPSNSDTRFNPFDKATLNVD RSSNTAYMLLSSLTSADSAVYYCATYGSYVSPADYWG QGTSVTVSS
15	Affy7-Y6V VH	QVQLQQSGPELVRPGASVKMSCRASGYTFTSYWLHW VKQRPQGQLEWIGMIDPSNSDTRFNPFDKATLNVD RSSNTAYMLLSSLTSADSAVYYCATVGSYVSPLDYWG QGTSVTVSS
16	Affy7-Y6L VH	QVQLQQSGPELVRPGASVKMSCRASGYTFTSYWLHW VKQRPQGQLEWIGMIDPSNSDTRFNPFDKATLNVD RSSNTAYMLLSSLTSADSAVYYCATLGSYVSPLDYWG QGTSVTVSS
17	Affy7-Y6I VH	QVQLQQSGPELVRPGASVKMSCRASGYTFTSYWLHW VKQRPQGQLEWIGMIDPSNSDTRFNPFDKATLNVD RSSNTAYMLLSSLTSADSAVYYCATIGSYVSPLDYWGQ GTSVTVSS
18	Affy7-Y6M VH	QVQLQQSGPELVRPGASVKMSCRASGYTFTSYWLHW VKQRPQGQLEWIGMIDPSNSDTRFNPFDKATLNVD RSSNTAYMLLSSLTSADSAVYYCATMGSYVSPLDYWG QGTSVTVSS

TABLE 18-continued

Amino Acid and Nucleotide Sequences for Exemplary Polypeptides Described Herein		
SEQ ID NO:	Name	Sequence
19	Affy7 VL (wt)	DIMMSQSPSSLTVSVGEKVTVSCCKSS <u>QSLLYTSSQKNYL</u> AWYQQKPGQS <u>PKLLIYWASTRES</u> GVDPDRFTGSGSGTDF TLTITSVKADDLAVYYCQ <u>QYYAYFW</u> TFGGGKLEIKR
20	Affy7 VH (wt)	CAGGTGCAGCTCCAGCAGAGCGGCCCTGAGCTGGT GCGCCCCGGCGCTTCCGTGAAGATGTCATGCCGAG CTTCTGGCTACAGCTTACCAGCTATTGGCTTACTG GGTCAAGCAGAGGCCCGGACAGGGTCTGGAGTGA <u>TCGGTATGATTGACCCGTCAAATAGTGACACCCGCTTC</u> <u>AACCCCAACTTCAAGGACAAGGCCACTCTCAACGTG</u> GACCGCTCAGCAACACGGCCTACATGCTGCTGTCC TCTCTGACCTCGGCGGACAGCGCGTGTACTACTGT GCCACC <u>TACGGCTCCTACGTGAGCCCTTGGATTATTG</u> GGGCCAGGGCACCTCCGTACCGTGTCTCTCC
21	Affy7-Y6A VH	CAGGTGCAGCTCCAGCAGAGCGGCCCTGAGCTGGT GCGCCCCGGCGCTTCCGTGAAGATGTCATGCCGAG CTTCTGGCTACAGCTTACCAGCTATTGGCTTACTG GGTCAAGCAGAGGCCCGGACAGGGTCTGGAGTGA <u>TCGGTATGATTGACCCGTCAAATAGTGACACCCGCTTC</u> <u>AACCCCAACTTCAAGGACAAGGCCACTCTCAACGTG</u> GACCGCTCAGCAACACGGCCTACATGCTGCTGTCC TCTCTGACCTCGGCGGACAGCGCGTGTACTACTGT GCCACC <u>GCGGGCTCCTACGTGAGCCCTTGGATTATT</u> GGGGCCAGGGCACCTCCGTACCGTGTCTCTCC
22	Affy7-L9A VH	CAGGTGCAGCTCCAGCAGAGCGGCCCTGAGCTGGT GCGCCCCGGCGCTTCCGTGAAGATGTCATGCCGAG CTTCTGGCTACAGCTTACCAGCTATTGGCTTACTG GGTCAAGCAGAGGCCCGGACAGGGTCTGGAGTGA <u>TCGGTATGATTGACCCGTCAAATAGTGACACCCGCTTC</u> <u>AACCCCAACTTCAAGGACAAGGCCACTCTCAACGTG</u> GACCGCTCAGCAACACGGCCTACATGCTGCTGTCC TCTCTGACCTCGGCGGACAGCGCGTGTACTACTGT GCCACC <u>TACGGCTCCTACGTGAGCCCTGCGGATTATT</u> GGGGCCAGGGCACCTCCGTACCGTGTCTCTCC
23	Affy7-Y6V VH	CAGGTGCAGCTCCAGCAGAGCGGCCCTGAGCTGGT GCGCCCCGGCGCTTCCGTGAAGATGTCATGCCGAG CTTCTGGCTACAGCTTACCAGCTATTGGCTTACTG GGTCAAGCAGAGGCCCGGACAGGGTCTGGAGTGA <u>TCGGTATGATTGACCCGTCAAATAGTGACACCCGCTTC</u> <u>AACCCCAACTTCAAGGACAAGGCCACTCTCAACGTG</u> GACCGCTCAGCAACACGGCCTACATGCTGCTGTCC TCTCTGACCTCGGCGGACAGCGCGTGTACTACTGT GCCACC <u>GTCGGCTCCTACGTGAGCCCTTGGATTATT</u> GGGGCCAGGGCACCTCCGTACCGTGTCTCTCC
24	Affy7-Y6L VH	CAGGTGCAGCTCCAGCAGAGCGGCCCTGAGCTGGT GCGCCCCGGCGCTTCCGTGAAGATGTCATGCCGAG CTTCTGGCTACAGCTTACCAGCTATTGGCTTACTG GGTCAAGCAGAGGCCCGGACAGGGTCTGGAGTGA <u>TCGGTATGATTGACCCGTCAAATAGTGACACCCGCTTC</u> <u>AACCCCAACTTCAAGGACAAGGCCACTCTCAACGTG</u> GACCGCTCAGCAACACGGCCTACATGCTGCTGTCC TCTCTGACCTCGGCGGACAGCGCGTGTACTACTGT GCCACC <u>CTCGGCTCCTACGTGAGCCCTTGGATTATTG</u> GGGCCAGGGCACCTCCGTACCGTGTCTCTCC
25	Affy7-Y6I VH	CAGGTGCAGCTCCAGCAGAGCGGCCCTGAGCTGGT GCGCCCCGGCGCTTCCGTGAAGATGTCATGCCGAG CTTCTGGCTACAGCTTACCAGCTATTGGCTTACTG GGTCAAGCAGAGGCCCGGACAGGGTCTGGAGTGA <u>TCGGTATGATTGACCCGTCAAATAGTGACACCCGCTTC</u> <u>AACCCCAACTTCAAGGACAAGGCCACTCTCAACGTG</u> GACCGCTCAGCAACACGGCCTACATGCTGCTGTCC TCTCTGACCTCGGCGGACAGCGCGTGTACTACTGT GCCACC <u>ATCGGCTCCTACGTGAGCCCTTGGATTATTG</u> GGGCCAGGGCACCTCCGTACCGTGTCTCTCC

TABLE 18-continued

Amino Acid and Nucleotide Sequences for Exemplary Polypeptides Described Herein		
SEQ ID NO:	Name	Sequence
26	Affy7-Y6M VH	CAGGTGCAGCTCCAGCAGAGCGGCCCTGAGCTGGT GCGCCCCGGCGCTTCCGTGAAGATGTCATGCCGAG CTCTGGCTACACGTTACACGCTATTGGCTACACTG GGTCAGCAGAGGCCCGGACAGGGTCTGGAGTGGGA TCGGTATGATTGACCCGTCAAATAGTGACACCCGCTTC AACCCCAACTTCAAGGACAAGGCCACTCTCAACGTG GACCGCTCGAGCAACACGGCCTACATGCTGCTGTCC TCTCTGACCTCGGCGGACAGCGCGTGTACTACTGT GCCACCATGGGCTCCTACGTGAGCCCTTTGGATTATT GGGGCCAGGGCACCTCCGTCACCGTGTCTCTC
27	Affy7 VL	GACATCATGATGTCTCAGAGTCCCTCTTCTCTGACC GTGTCCGTGGGCGAGAAGGTGACCGTGTCAATGCAA GTCTTACAGAGCCTTCTGTACACAGCTCCCGAAG AACTACCTGGCGTGGTACCAGCAGAAGCCCGGACA GAGC CCAAGCTGCTGATCTATTGGGCTTCCACCCGC GAGAGCGGCGTCCCGACCGCTTACCCGCTCCGGC TCCGGGACCGACTTCACCTGACCATCACCTCCGTG AAGGCCGATGACCTGGCCGTGTACTACTGTCAACA GTATTACGCCTACCCGTGGACCTTCGGAGGCGGCAC CAAGCTGGAGATCAAGCGC
28	Affy7 scFv (control)	QVQLQQSGPELVRPGASVKMSCRASGYTFTSYWLHW VKQRPQGQLEWIGMIDPSNSDTRFNPFDKATLNVD RSSNTAYMLLSLTSADSAVYYCATYGSYVSPLDYWG QGTSVTVSSGGGSGGGSGGGSDIMMSQSPSSLTV SVGEKVTVCKSSQSLLYTSSQKNYLAWYQQKPGQSP KLLIYWASTRESGVPDRFTGSGSGTDFTLTITSVKADD LAVYYCQQYYAYPWTFGGGTKLEIKR
29	Affy7-Y6A scFv	QVQLQQSGPELVRPGASVKMSCRASGYTFTSYWLHW VKQRPQGQLEWIGMIDPSNSDTRFNPFDKATLNVD RSSNTAYMLLSLTSADSAVYYCATAGSYVSPLDYWG QGTSVTVSSGGGSGGGSGGGSDIMMSQSPSSLTV SVGEKVTVCKSSQSLLYTSSQKNYLAWYQQKPGQSP KLLIYWASTRESGVPDRFTGSGSGTDFTLTITSVKADD LAVYYCQQYYAYPWTFGGGTKLEIKR
30	Affy7-L9A scFv	QVQLQQSGPELVRPGASVKMSCRASGYTFTSYWLHW VKQRPQGQLEWIGMIDPSNSDTRFNPFDKATLNVD RSSNTAYMLLSLTSADSAVYYCATYGSYVSPADYWG QGTSVTVSSGGGSGGGSGGGSDIMMSQSPSSLTV SVGEKVTVCKSSQSLLYTSSQKNYLAWYQQKPGQSP KLLIYWASTRESGVPDRFTGSGSGTDFTLTITSVKADD LAVYYCQQYYAYPWTFGGGTKLEIKR
31	Affy7-Y6V scFv	QVQLQQSGPELVRPGASVKMSCRASGYTFTSYWLHW VKQRPQGQLEWIGMIDPSNSDTRFNPFDKATLNVD RSSNTAYMLLSLTSADSAVYYCATVGSYVSPLDYWG QGTSVTVSSGGGSGGGSGGGSDIMMSQSPSSLTV SVGEKVTVCKSSQSLLYTSSQKNYLAWYQQKPGQSP KLLIYWASTRESGVPDRFTGSGSGTDFTLTITSVKADD LAVYYCQQYYAYPWTFGGGTKLEIKR
32	Affy7-Y6L scFv	QVQLQQSGPELVRPGASVKMSCRASGYTFTSYWLHW VKQRPQGQLEWIGMIDPSNSDTRFNPFDKATLNVD RSSNTAYMLLSLTSADSAVYYCATLGSYVSPLDYWG QGTSVTVSSGGGSGGGSGGGSDIMMSQSPSSLTV SVGEKVTVCKSSQSLLYTSSQKNYLAWYQQKPGQSP KLLIYWASTRESGVPDRFTGSGSGTDFTLTITSVKADD LAVYYCQQYYAYPWTFGGGTKLEIKR
33	Affy7-Y6I scFv	QVQLQQSGPELVRPGASVKMSCRASGYTFTSYWLHW VKQRPQGQLEWIGMIDPSNSDTRFNPFDKATLNVD RSSNTAYMLLSLTSADSAVYYCATIGSYVSPLDYWGQ GTSVTVSSGGGSGGGSGGGSDIMMSQSPSSLTVS VGEKVTVCKSSQSLLYTSSQKNYLAWYQQKPGQSP KLLIYWASTRESGVPDRFTGSGSGTDFTLTITSVKADD LAVYYCQQYYAYPWTFGGGTKLEIKR

TABLE 18-continued

Amino Acid and Nucleotide Sequences for Exemplary Polypeptides Described Herein		
SEQ ID NO:	Name	Sequence
34	Affy7-Y6M scFv	QVQLQQSGPELVRPGASVKMSCRASGYTFTSYWLHW VKQRPQGLEWIGMIDPSNSDTREPNFKDKATLNVD RSSNTAYMLLSLTSADSAVYYCATMGSYVSPLDYWG QGTSVTVSSGGGSGGGGSGGGSDIMMSQSPSSLTV SVGEKVTVSCKSSQSLLYTSSQKNYLAWYQQKPGQSP KLLIYWASTRESGVPRFTGSGSGTDFTLTITSVKADD LAVYYCQQYYAYPWTFFGGGKLEIKR
35	Affy7 scFv (control)	CAGGTGCAGCTCCAGCAGAGCGGCCCTGAGCTGGT GCGCCCCGGCGCTTCCGTGAAGATGTCATGCCGAG CTTCCTGGCTACACGTTACACAGCTATTGGCTACACT GGGTCAAGCAGAGGCCCGGACAGGGTCTGGAGTGG ATCGGTATGATTGACCCGTCAAATAGTGACACCCGC TTCAACCCCAACTTCAAGGACAAGGCCACTCTCAAC GTGGACCGCTCGAGCAACACGGCCTACATGCTGCT GTCCCTCTGACCTCGGCGGACAGCGCCGTGTACTA CTGTGCCACC <u>TACGGCTCCTACGTGAGCCCTTTGGATT</u> <u>ATTGGGGCCAGGGCACCTCCGTACCGTGTCTCCG</u> <u>GTGGAGGCGGTT</u> CAGGCGGAGGTGGCTCTGGCGGT GGCGGATCGACATCATGATGTCTCAGAGTCCCTCT TTCTTGACCGTGTCCGTGGGCGAGAAGGTGACCGT GTCATGCAAGTCTCAGAGAGCCTTCTGTACACCAG CTCCAGAGAAGTAACCTGGCGTGGTACCAGCAGA AGCCCGGACAGAGCCCCAAGCTGTCTGATCTATTGG GCTTCCACCCGCGAGAGCGGCGTCCCGACCGCTTC ACCGGCTCCGGCTCCGGGACCGACTTACCCCTGACC ATCACCTCCGTGAAGGCCGATGACCTGGCCGTGTAC TACTGTCAACAGTATTACGCCTACCCGTGGACCTTC GGAGGCGGCACCAAGCTGGAGATCAAGCGC
36	Affy7-Y6A scFv	CAGGTGCAGCTCCAGCAGAGCGGCCCTGAGCTGGT GCGCCCCGGCGCTTCCGTGAAGATGTCATGCCGAG CTTCCTGGCTACACGTTACACAGCTATTGGCTACACT GGGTCAAGCAGAGGCCCGGACAGGGTCTGGAGTGG ATCGGTATGATTGACCCGTCAAATAGTGACACCCGC TTCAACCCCAACTTCAAGGACAAGGCCACTCTCAAC GTGGACCGCTCGAGCAACACGGCCTACATGCTGCT GTCCCTCTGACCTCGGCGGACAGCGCCGTGTACTA CTGTGCCACC <u>GCGGGCTCCTACGTGAGCCCTTTGGAT</u> <u>TATTGGGGCCAGGGCACCTCCGTACCGTGTCTCC</u> <u>GGTGGAGGCGGTT</u> CAGGCGGAGGTGGCTCTGGCGG TGGCGGATCGACATCATGATGTCTCAGAGTCCCTC TTCTTGACCGTGTCCGTGGGCGAGAAGGTGACCGT GTCATGCAAGTCTCAGAGAGCCTTCTGTACACCAG CTCCAGAGAAGTAACCTGGCGTGGTACCAGCAGA AGCCCGGACAGAGCCCCAAGCTGTCTGATCTATTGG GCTTCCACCCGCGAGAGCGGCGTCCCGACCGCTTC ACCGGCTCCGGCTCCGGGACCGACTTACCCCTGACC ATCACCTCCGTGAAGGCCGATGACCTGGCCGTGTAC TACTGTCAACAGTATTACGCCTACCCGTGGACCTTC GGAGGCGGCACCAAGCTGGAGATCAAGCGC
37	Affy7-L9A scFv	CAGGTGCAGCTCCAGCAGAGCGGCCCTGAGCTGGT GCGCCCCGGCGCTTCCGTGAAGATGTCATGCCGAG CTTCCTGGCTACACGTTACACAGCTATTGGCTACACT GGGTCAAGCAGAGGCCCGGACAGGGTCTGGAGTGG ATCGGTATGATTGACCCGTCAAATAGTGACACCCGC TTCAACCCCAACTTCAAGGACAAGGCCACTCTCAAC GTGGACCGCTCGAGCAACACGGCCTACATGCTGCT GTCCCTCTGACCTCGGCGGACAGCGCCGTGTACTA CTGTGCCACC <u>TACGGCTCCTACGTGAGCCCTGCGGAT</u> <u>TATTGGGGCCAGGGCACCTCCGTACCGTGTCTCC</u> <u>GGTGGAGGCGGTT</u> CAGGCGGAGGTGGCTCTGGCGG TGGCGGATCGACATCATGATGTCTCAGAGTCCCTC TTCTTGACCGTGTCCGTGGGCGAGAAGGTGACCGT GTCATGCAAGTCTCAGAGAGCCTTCTGTACACCAG CTCCAGAGAAGTAACCTGGCGTGGTACCAGCAGA AGCCCGGACAGAGCCCCAAGCTGTCTGATCTATTGG GCTTCCACCCGCGAGAGCGGCGTCCCGACCGCTTC ACCGGCTCCGGCTCCGGGACCGACTTACCCCTGACC ATCACCTCCGTGAAGGCCGATGACCTGGCCGTGTAC TACTGTCAACAGTATTACGCCTACCCGTGGACCTTC GGAGGCGGCACCAAGCTGGAGATCAAGCGC

TABLE 18-continued

Amino Acid and Nucleotide Sequences for Exemplary Polypeptides Described Herein		
SEQ ID NO:	Name	Sequence
		GTCATGCAAGTCTCAGAGCCTTCTGTACACCAG CTCCCAGAAGAACTACCTGGCGTGGTACCAGCAGA AGCCCGGACAGAGCCCCAAGCTGCTGATCTATTGG GCTTCCACCCGCGAGAGCGGCGTCCCGACCGCTTC ACCGGCTCCGGCTCCGGGACCGACTTCACCCGTGACC ATCACCTCCGTGAAGGCCGATGACCTGGCCGTGTAC TACTGTCAACAGTATTACGCCACCCGTGGACCTTC GGAGGCGGCACCAAGCTGGAGATCAAGCGC
38	Affy7-Y6V scFv	CAGGTGCAGCTCCAGCAGAGCGGCCCTGAGCTGGT GCGCCCCGCGCTTCCGTGAAGATGTCATGCCGAG CTTCTGGCTACACGTTACACAGCTATTGGCTACACT GGGTCAAGCAGAGGCCCGGACAGGGTCTGGAGTGG ATCGGTATGATTGACCCGTCAAATAGTGACACCCGC TTCAACCCCAACTTCAAGGACAAGGCCACTCTCAAC GTGGACCGCTCGAGCAACACGGCCTACATGCTGCT GTCCCTCTGACCTCGGCGGACAGCGCCGTGACTA CTGTGCCACC <u>CTCGGCTCCTACGTGAGCCCTTGGAT</u> <u>TATTGGGGCCAGGGCACCTCCGTACCCGTGTCCTCC</u> GGTGGAGGCGGTTACGGCGGAGGTGGCTCTGGCGG TGGCGGATCGACATCATGATGTCTCAGAGTCCCTC TTCTCTGACCGTGTCCGTGGGCGAGAAGGTGACCGT GTCATGCAAGTCTCAGAGCCTTCTGTACACCAG CTCCCAGAAGAACTACCTGGCGTGGTACCAGCAGA AGCCCGGACAGAGCCCCAAGCTGCTGATCTATTGG GCTTCCACCCGCGAGAGCGGCGTCCCGACCGCTTC ACCGGCTCCGGCTCCGGGACCGACTTCACCCGTGACC ATCACCTCCGTGAAGGCCGATGACCTGGCCGTGTAC TACTGTCAACAGTATTACGCCACCCGTGGACCTTC GGAGGCGGCACCAAGCTGGAGATCAAGCGC
39	Affy7-Y6L scFv	CAGGTGCAGCTCCAGCAGAGCGGCCCTGAGCTGGT GCGCCCCGCGCTTCCGTGAAGATGTCATGCCGAG CTTCTGGCTACACGTTACACAGCTATTGGCTACACT GGGTCAAGCAGAGGCCCGGACAGGGTCTGGAGTGG ATCGGTATGATTGACCCGTCAAATAGTGACACCCGC TTCAACCCCAACTTCAAGGACAAGGCCACTCTCAAC GTGGACCGCTCGAGCAACACGGCCTACATGCTGCT GTCCCTCTGACCTCGGCGGACAGCGCCGTGACTA CTGTGCCACC <u>CTCGGCTCCTACGTGAGCCCTTGGAT</u> <u>TATTGGGGCCAGGGCACCTCCGTACCCGTGTCCTCC</u> GGTGGAGGCGGTTACGGCGGAGGTGGCTCTGGCGG TGGCGGATCGACATCATGATGTCTCAGAGTCCCTC TTCTCTGACCGTGTCCGTGGGCGAGAAGGTGACCGT GTCATGCAAGTCTCAGAGCCTTCTGTACACCAG CTCCCAGAAGAACTACCTGGCGTGGTACCAGCAGA AGCCCGGACAGAGCCCCAAGCTGCTGATCTATTGG GCTTCCACCCGCGAGAGCGGCGTCCCGACCGCTTC ACCGGCTCCGGCTCCGGGACCGACTTCACCCGTGACC ATCACCTCCGTGAAGGCCGATGACCTGGCCGTGTAC TACTGTCAACAGTATTACGCCACCCGTGGACCTTC GGAGGCGGCACCAAGCTGGAGATCAAGCGC
40	Affy7-Y6I scFv	CAGGTGCAGCTCCAGCAGAGCGGCCCTGAGCTGGT GCGCCCCGCGCTTCCGTGAAGATGTCATGCCGAG CTTCTGGCTACACGTTACACAGCTATTGGCTACACT GGGTCAAGCAGAGGCCCGGACAGGGTCTGGAGTGG ATCGGTATGATTGACCCGTCAAATAGTGACACCCGC TTCAACCCCAACTTCAAGGACAAGGCCACTCTCAAC GTGGACCGCTCGAGCAACACGGCCTACATGCTGCT GTCCCTCTGACCTCGGCGGACAGCGCCGTGACTA CTGTGCCACC <u>ATCGGCTCCTACGTGAGCCCTTGGAT</u> <u>TATTGGGGCCAGGGCACCTCCGTACCCGTGTCCTCC</u> GGTGGAGGCGGTTACGGCGGAGGTGGCTCTGGCGG TGGCGGATCGACATCATGATGTCTCAGAGTCCCTC TTCTCTGACCGTGTCCGTGGGCGAGAAGGTGACCGT GTCATGCAAGTCTCAGAGCCTTCTGTACACCAG CTCCCAGAAGAACTACCTGGCGTGGTACCAGCAGA AGCCCGGACAGAGCCCCAAGCTGCTGATCTATTGG GCTTCCACCCGCGAGAGCGGCGTCCCGACCGCTTC ACCGGCTCCGGCTCCGGGACCGACTTCACCCGTGACC ATCACCTCCGTGAAGGCCGATGACCTGGCCGTGTAC TACTGTCAACAGTATTACGCCACCCGTGGACCTTC GGAGGCGGCACCAAGCTGGAGATCAAGCGC

TABLE 18-continued

Amino Acid and Nucleotide Sequences for Exemplary Polypeptides Described Herein		
SEQ ID NO:	Name	Sequence
		CTCCCAGAAGAACTACCTGGCGTGGTACCAGCAGA AGCCCGGACAGAGCCCCAGCTGCTGATCTATTGG GCTTCCACCCGCGAGAGCGCGTCCCCGACCGCTTC ACCGGCTCCGGCTCCGGGACCGACTTCACCTGACC ATCACCTCCGTGAAGCGGATGACCTGGCCGTGTAC TACTGTCAACAGTATTACGCCCTACCCGTGGACCTTC GGAGCGGCACCAAGCTGGAGATCAAGCGC
41	Affy7-Y6M scFv	CAGGTGCAGCTCCAGCAGAGCGGCCCTGAGCTGGT GCGCCCCGGCGCTTCCGTGAAGATGTCATGCCGAG CTTCTGGCTACACGTTCACCAGCTATTGGCTACACT GGGTCAAGCAGAGGCCCGGACAGGGTCTGGAGTGG ATCGGTATGATTGACCCGTCAAATAGTGACACCCGC TTCAACCCCAACTTCAAGGACAAGGCCACTCTCAAC GTGGACCGCTCGAGCAACACGGCCTACATGTGCT GTCTCTCTGACCTCGGCGGACAGCGCCGTGTACTA CTGTGCCACC ATGGGCTCCTACGTGAGCCCTTGGAT TATTGGGGCCAGGGCACCTCCGTACCCGTGTCCTCC GGTGAGGCGGTTCAAGCGGAGGTGGCTCTGGCGG TGGCGGATCGGACATCATGATGTCTCAGAGTCCCTC TTCTCTGACCGTGTCCGTGGGCGAGAAGGTGACCGT GTCATGCAAGTCTCACAGAGCCTTCTGTACACCAG CTCCCAGAAGAACTACCTGGCGTGGTACCAGCAGA AGCCCGGACAGAGCCCCAAGCTGCTGATCTATTGG GCTTCCACCCGCGAGAGCGCGTCCCCGACCGCTTC ACCGGCTCCGGCTCCGGGACCGACTTCACCTGACC ATCACCTCCGTGAAGCGGATGACCTGGCCGTGTAC TACTGTCAACAGTATTACGCCCTACCCGTGGACCTTC GGAGCGGCACCAAGCTGGAGATCAAGCGC
42	Affy7 CAR (control)	QVQLQQSGPELVRPGASVKMSCRASGYTFTSYWLHW VKQRPQGLEWIGMIDPSNSDTRFNPFDKATLNV RSSNTAYMLLSLTSADSAVYYCATYGSYVSPLDYWG QGTSVTVSSGGGSGGGGSGGGSDIMMSQSPSSLTV SVGEKVTVCCKSSQSLLYTSSQKNYLAWYQQKPGQSP KLLIYWASTRESGVDPDRFTGSGSGTDFTLTITSVKADD LAVYYCQYYAYPWTFGGKLEIKRASTTTTPAPRPP TPAPTIASQPLSLRPEACRPAAGGAVHTRGLDFACDF WVLVVVGGVLACYSLLVTVAFII FWVRSKRSLHSD YMNMTPRRPGPTRKHYQPYAPPRDFAAYRSLRVKFS RSADAPAYQQGQNQLYNELNLGRREEYDVLDKRRGR DPEMGGKPRRKNPQEGLYNELQKDKMAEAYSEIGMK GERRRGKGHDGLYQGLSTATKDTYDALHMQLPPR
43	Affy7-Y6A CAR	QVQLQQSGPELVRPGASVKMSCRASGYTFTSYWLHW VKQRPQGLEWIGMIDPSNSDTRFNPFDKATLNV RSSNTAYMLLSLTSADSAVYYCAT AGSYVSPLDYWG QGTSVTVSSGGGSGGGGSGGGSDIMMSQSPSSLTV SVGEKVTVCCKSSQSLLYTSSQKNYLAWYQQKPGQSP KLLIYWASTRESGVDPDRFTGSGSGTDFTLTITSVKADD LAVYYCQYYAYPWTFGGKLEIKRASTTTTPAPRPP TPAPTIASQPLSLRPEACRPAAGGAVHTRGLDFACDF WVLVVVGGVLACYSLLVTVAFII FWVRSKRSLHSD YMNMTPRRPGPTRKHYQPYAPPRDFAAYRSLRVKFS RSADAPAYQQGQNQLYNELNLGRREEYDVLDKRRGR DPEMGGKPRRKNPQEGLYNELQKDKMAEAYSEIGMK GERRRGKGHDGLYQGLSTATKDTYDALHMQLPPR
44	Affy7-L9A CAR	QVQLQQSGPELVRPGASVKMSCRASGYTFTSYWLHW VKQRPQGLEWIGMIDPSNSDTRFNPFDKATLNV RSSNTAYMLLSLTSADSAVYYCATYGSYVSPADYWG QGTSVTVSSGGGSGGGGSGGGSDIMMSQSPSSLTV SVGEKVTVCCKSSQSLLYTSSQKNYLAWYQQKPGQSP KLLIYWASTRESGVDPDRFTGSGSGTDFTLTITSVKADD LAVYYCQYYAYPWTFGGKLEIKRASTTTTPAPRPP TPAPTIASQPLSLRPEACRPAAGGAVHTRGLDFACDF WVLVVVGGVLACYSLLVTVAFII FWVRSKRSLHSD YMNMTPRRPGPTRKHYQPYAPPRDFAAYRSLRVKFS RSADAPAYQQGQNQLYNELNLGRREEYDVLDKRRGR DPEMGGKPRRKNPQEGLYNELQKDKMAEAYSEIGMK GERRRGKGHDGLYQGLSTATKDTYDALHMQLPPR

TABLE 18-continued

Amino Acid and Nucleotide Sequences for Exemplary Polypeptides Described Herein		
SEQ ID NO:	Name	Sequence
45	Affy7-Y6V CAR	QVQLQQSGPELVRPGASVKMSCRASGYTFTSYWLHW VKQRPQGLEWIGMIDPSNSDTRFNPFDKATLNVD RSSNTAYMLLSSLTADSAVYYCATVGSYVSPLDYWG QGTSVTVSSGGGSGGGGSGGGSDIMMSQSPSSLTV SVGEKVTVSCKSSQSLLYTSSQKNYLAWYQQKPGQSP KLLIYWASTRESGVDPDRFTGSGSGTDFTLTITSVKADD LAVYYCQQYYAYPWTFGGGTKLEIKRASTTTPAPRPP TPAPTIASQPLSLRPEACRPAAGGAVHTRGLDFACDF WVLVVGVLACYSLLVTVAFIIFWVRSKRSRLHSD YMNMTPRRPGPTRKHYQPYAPPRDFAAYRSLRVKFS RSADAPAYQQGQNQLYNELNLGRREEYDVLDKRRGR DPENGGKPRRKNPQEGLYNELQKDKMAEAYSEIGMK GERRRGKGGHDGLYQGLSTATKDTYDALHMQLALPPR
46	Affy7-Y6L CAR	QVQLQQSGPELVRPGASVKMSCRASGYTFTSYWLHW VKQRPQGLEWIGMIDPSNSDTRFNPFDKATLNVD RSSNTAYMLLSSLTADSAVYYCATLGSYVSPLDYWG QGTSVTVSSGGGSGGGGSGGGSDIMMSQSPSSLTV SVGEKVTVSCKSSQSLLYTSSQKNYLAWYQQKPGQSP KLLIYWASTRESGVDPDRFTGSGSGTDFTLTITSVKADD LAVYYCQQYYAYPWTFGGGTKLEIKRASTTTPAPRPP TPAPTIASQPLSLRPEACRPAAGGAVHTRGLDFACDF WVLVVGVLACYSLLVTVAFIIFWVRSKRSRLHSD YMNMTPRRPGPTRKHYQPYAPPRDFAAYRSLRVKFS RSADAPAYQQGQNQLYNELNLGRREEYDVLDKRRGR DPENGGKPRRKNPQEGLYNELQKDKMAEAYSEIGMK GERRRGKGGHDGLYQGLSTATKDTYDALHMQLALPPR
47	Affy7-Y6I CAR	QVQLQQSGPELVRPGASVKMSCRASGYTFTSYWLHW VKQRPQGLEWIGMIDPSNSDTRFNPFDKATLNVD RSSNTAYMLLSSLTADSAVYYCATIGSYVSPLDYWGQ GTSVTVSSGGGSGGGGSGGGSDIMMSQSPSSLTVS VGEKVTVSCKSSQSLLYTSSQKNYLAWYQQKPGQSP KLLIYWASTRESGVDPDRFTGSGSGTDFTLTITSVKADD LAVYYCQQYYAYPWTFGGGTKLEIKRASTTTPAPRPP TPAPTIASQPLSLRPEACRPAAGGAVHTRGLDFACDF WVLVVGVLACYSLLVTVAFIIFWVRSKRSRLHSD YMNMTPRRPGPTRKHYQPYAPPRDFAAYRSLRVKFS RSADAPAYQQGQNQLYNELNLGRREEYDVLDKRRGR DPENGGKPRRKNPQEGLYNELQKDKMAEAYSEIGMK GERRRGKGGHDGLYQGLSTATKDTYDALHMQLALPPR
48	Affy7-Y6M CAR	QVQLQQSGPELVRPGASVKMSCRASGYTFTSYWLHW VKQRPQGLEWIGMIDPSNSDTRFNPFDKATLNVD RSSNTAYMLLSSLTADSAVYYCATMGSYVSPLDYWG QGTSVTVSSGGGSGGGGSGGGSDIMMSQSPSSLTV SVGEKVTVSCKSSQSLLYTSSQKNYLAWYQQKPGQSP KLLIYWASTRESGVDPDRFTGSGSGTDFTLTITSVKADD LAVYYCQQYYAYPWTFGGGTKLEIKRASTTTPAPRPP TPAPTIASQPLSLRPEACRPAAGGAVHTRGLDFACDF WVLVVGVLACYSLLVTVAFIIFWVRSKRSRLHSD YMNMTPRRPGPTRKHYQPYAPPRDFAAYRSLRVKFS RSADAPAYQQGQNQLYNELNLGRREEYDVLDKRRGR DPENGGKPRRKNPQEGLYNELQKDKMAEAYSEIGMK GERRRGKGGHDGLYQGLSTATKDTYDALHMQLALPPR
49	MycT-Affy7 CAR- SSTR2 (control)	GSEQKLISEEDLGSQVQLQQSGPELVRPGASVKMSCR ASGYTFTSYWLHWVKQRPQGLEWIGMIDPSNSDTR FNPFDKATLNVDRSSNTAYMLLSSLTADSAVYYC ATYGSYVSPLDYWGQGTSTVTVSSGGGSGGGGSGGGG SDIMMSQSPSSLTVSVGEKVTVSCKSSQSLLYTSSQKN YLAWYQQKPGQSPKLLIYWASTRESGVDPDRFTGSGSG TDFTLTITSVKADDLAVYYCQQYYAYPWTFGGGTKL EIKRASTTTPAPRPTTPAPTIASQPLSLRPEACRPAAGG AVHTRGLDFACDFWVLVVGVLACYSLLVTVAFIIF WVRSKRSRLHSDYMNMTPRRPGPTRKHYQPYAPPR DFAAYRSLRVKFSRSADAPAYQQGQNQLYNELNLGR REEYDVLDKRRGRDPENGGKPRRKNPQEGLYNELQK DKMAEAYSEIGMKGERRRGKGGHDGLYQGLSTATKDT YDALHMQLALPPRTSGSGATNFSLLKQAGDVEENPGFS

TABLE 18-continued

Amino Acid and Nucleotide Sequences for Exemplary Polypeptides Described Herein		
SEQ ID NO:	Name	Sequence
		RMDMADEPLNGSHTWLSIPFDLNGSVVSTNTSNQTEP YYDLTSNAVLTFIYFVVCIIGLCGNTLVIYVILRYAKM KTITNIYIILNLAIADELFMLGLPFLAMQVALVHWPF GK AICRVVMTVDGINQFTSIFCLTVMSIDRYLAVVHPIKS AKWRRPRTAKMITMAVWGVSLVLIPIMIYAGLR SNQ WGRSSCTINWPGESGAWYTGFIITYFILGFLVPLTIICL CYLFII IKVKSSGIRVGSSKRKKSEKKVTRMVSIVVAVF IFCWLPFYIFNVSSVSMASPTPALKGMDFVVLTYA NSCANPILYAF LSDNFKKS FQNVLC LVK VSGTDDGER SDSKQDKSRLNETTETQRTLLNGDLQTSI
50	MycT-Affy7-Y6A CAR-SSTR2	GSEQKLISEEDLGSQVQLQQSGPELV R PGASVKMSCR ASGYTFTSYWLHWVKQRPGGLEWIGMIDPSNSDTR FNP NFKDKATLNVDRSSNTAYMLLSLTSADSAVYYC ATAGSYVSPLDYWGQGTSVTVSSGGGGSGGGSGGGG SDIMMSQSPSSLTVSVGEKVTVSCSSQSLLYTSSQKN YLAWYQQKPGQSPKLLIYWASTRESGVDRFTGSGSG TDFTLTITSVKADDLAVYYCQYYAYPWTFGGGTKL EIKRASTTTPAPRPPTPAPTIASQPLSLRPEACRPAAGG AVHTRGLDFACDFWVLVVVGGVLACYSLLVTVAFIIF WVRSKRSRLLHSDYMNMTPRRPGPTRKHYQPYAPPR DFAAYRSLRVKFSRSADAPAYQQGQNQLYNELNLGR REEYDVLDKRRGRDP EMGGKPRRKNPQEGLYNELQK DKMAEAYS EIGMKGERRRKGHDGLYQGLSTATKDT YDALHMQUALPPRTSGSGATNFSLLKQAGDVEENPGPS RMDMADEPLNGSHTWLSIPFDLNGSVVSTNTSNQTEP YYDLTSNAVLTFIYFVVCIIGLCGNTLVIYVILRYAKM KTITNIYIILNLAIADELFMLGLPFLAMQVALVHWPF GK AICRVVMTVDGINQFTSIFCLTVMSIDRYLAVVHPIKS AKWRRPRTAKMITMAVWGVSLVLIPIMIYAGLR SNQ WGRSSCTINWPGESGAWYTGFIITYFILGFLVPLTIICL CYLFII IKVKSSGIRVGSSKRKKSEKKVTRMVSIVVAVF IFCWLPFYIFNVSSVSMASPTPALKGMDFVVLTYA NSCANPILYAF LSDNFKKS FQNVLC LVK VSGTDDGER SDSKQDKSRLNETTETQRTLLNGDLQTSI
51	MycT-Affy7-L9A CAR-SSTR2	GSEQKLISEEDLGSQVQLQQSGPELV R PGASVKMSCR ASGYTFTSYWLHWVKQRPGGLEWIGMIDPSNSDTR FNP NFKDKATLNVDRSSNTAYMLLSLTSADSAVYYC ATYGSYVSPADYWGQGTSVTVSSGGGGSGGGSGGGG SDIMMSQSPSSLTVSVGEKVTVSCSSQSLLYTSSQKN YLAWYQQKPGQSPKLLIYWASTRESGVDRFTGSGSG TDFTLTITSVKADDLAVYYCQYYAYPWTFGGGTKL EIKRASTTTPAPRPPTPAPTIASQPLSLRPEACRPAAGG AVHTRGLDFACDFWVLVVVGGVLACYSLLVTVAFIIF WVRSKRSRLLHSDYMNMTPRRPGPTRKHYQPYAPPR DFAAYRSLRVKFSRSADAPAYQQGQNQLYNELNLGR REEYDVLDKRRGRDP EMGGKPRRKNPQEGLYNELQK DKMAEAYS EIGMKGERRRKGHDGLYQGLSTATKDT YDALHMQUALPPRTSGSGATNFSLLKQAGDVEENPGPS RMDMADEPLNGSHTWLSIPFDLNGSVVSTNTSNQTEP YYDLTSNAVLTFIYFVVCIIGLCGNTLVIYVILRYAKM KTITNIYIILNLAIADELFMLGLPFLAMQVALVHWPF GK AICRVVMTVDGINQFTSIFCLTVMSIDRYLAVVHPIKS AKWRRPRTAKMITMAVWGVSLVLIPIMIYAGLR SNQ WGRSSCTINWPGESGAWYTGFIITYFILGFLVPLTIICL CYLFII IKVKSSGIRVGSSKRKKSEKKVTRMVSIVVAVF IFCWLPFYIFNVSSVSMASPTPALKGMDFVVLTYA NSCANPILYAF LSDNFKKS FQNVLC LVK VSGTDDGER SDSKQDKSRLNETTETQRTLLNGDLQTSI
52	MycT-Affy7-Y6V CAR-SSTR2	GSEQKLISEEDLGSQVQLQQSGPELV R PGASVKMSCR ASGYTFTSYWLHWVKQRPGGLEWIGMIDPSNSDTR FNP NFKDKATLNVDRSSNTAYMLLSLTSADSAVYYC ATVGSYVSPLDYWGQGTSVTVSSGGGGSGGGSGGGG SDIMMSQSPSSLTVSVGEKVTVSCSSQSLLYTSSQKN YLAWYQQKPGQSPKLLIYWASTRESGVDRFTGSGSG TDFTLTITSVKADDLAVYYCQYYAYPWTFGGGTKL EIKRASTTTPAPRPPTPAPTIASQPLSLRPEACRPAAGG AVHTRGLDFACDFWVLVVVGGVLACYSLLVTVAFIIF WVRSKRSRLLHSDYMNMTPRRPGPTRKHYQPYAPPR

TABLE 18-continued

Amino Acid and Nucleotide Sequences for Exemplary Polypeptides Described Herein		
SEQ ID NO:	Name	Sequence
		DFAAYRSLRVKFSRSADAPAYQQGQNQLYNELNLGR REEYDVLDKRRGRDPEMGGKPRRKNPQEGLYNELQK DKMAEAYSEIGMKGERRRGKGHDGLYQGLSTATKDT YDALHMQUALPPRTSGSGATNFSLLKQAGDVEENPGPS RMDMADEPLNGSHTWLSIPFDLNGSVVSTNTSNQTEP YYDLTSNAVLTFIYFVVCIIIGLCGNTLVIYVILRYAKM KTITNIYIILNLAIADELFMLGLPFLAMQVALVHWPFGK AICRVVMTVDGINQFTSIFCLTVMSIDRYLAVVHPIKS AKWRRPRTAKMITMAVWGVSLVILPIMIYAGLRSNQ WGRSSCTINWPGESGAWYTGFIITYFILGFLVPLTIICL CYLFII IKVKSSGIRVGSSKRKKSEKKVTRMVSIVVAVF IFCWLFPFIYFNVSSVSMASPTPALKGMFDFVVVLTYA NSCANPILYAF LSDNFKKS FQNVLC LVK VSGTDDGER SDSKQDKSRLNETTETQRTLNLNGDLQTSI
53	MycT-Affy7-Y6L CAR-SSTR2	GSEQKLISEEDLGSQVQLQQSGPELVRPGASVKMSCR ASGYTFTSYWLHWVKQRPQGGLWIGMIDPSNSDTR FNPENFKDKATLNVDRSSNTAYMLLSLTSADSAVYYC <u>ATLGSYVSPLDYWGQGT</u> SVTVSSGGGGSGGGSGGGG SDIMMSQSPSSLTVSVGEKVTVSCSSQSLLYTSSQKN YLAWYQQKPGQSPKLLIYWASTRESGVDPDRFTGSGSG TDFTLTITSVKADDLAVYYCQYYAYPWTFGGGTKL EIKRASTTTTPAPRPPTPAPTIASQPLSLRPEACRPAAGG AVHTRGLDFACDFWVLVVVGGVLACYSLLVTVAFIIF WVRSKRSLRLHSDYMNMTPRRPGPTRKHYQPYAPPR DFAAYRSLRVKFSRSADAPAYQQGQNQLYNELNLGR REEYDVLDKRRGRDPEMGGKPRRKNPQEGLYNELQK DKMAEAYSEIGMKGERRRGKGHDGLYQGLSTATKDT YDALHMQUALPPRTSGSGATNFSLLKQAGDVEENPGPS RMDMADEPLNGSHTWLSIPFDLNGSVVSTNTSNQTEP YYDLTSNAVLTFIYFVVCIIIGLCGNTLVIYVILRYAKM KTITNIYIILNLAIADELFMLGLPFLAMQVALVHWPFGK AICRVVMTVDGINQFTSIFCLTVMSIDRYLAVVHPIKS AKWRRPRTAKMITMAVWGVSLVILPIMIYAGLRSNQ WGRSSCTINWPGESGAWYTGFIITYFILGFLVPLTIICL CYLFII IKVKSSGIRVGSSKRKKSEKKVTRMVSIVVAVF IFCWLFPFIYFNVSSVSMASPTPALKGMFDFVVVLTYA NSCANPILYAF LSDNFKKS FQNVLC LVK VSGTDDGER SDSKQDKSRLNETTETQRTLNLNGDLQTSI
54	MycT-Affy7-Y6I CAR-SSTR2	GSEQKLISEEDLGSQVQLQQSGPELVRPGASVKMSCR ASGYTFTSYWLHWVKQRPQGGLWIGMIDPSNSDTR FNPENFKDKATLNVDRSSNTAYMLLSLTSADSAVYYC <u>ATIGSYVSPLDYWGQGT</u> SVTVSSGGGGSGGGSGGGG SDIMMSQSPSSLTVSVGEKVTVSCSSQSLLYTSSQKN YLAWYQQKPGQSPKLLIYWASTRESGVDPDRFTGSGSG TDFTLTITSVKADDLAVYYCQYYAYPWTFGGGTKL EIKRASTTTTPAPRPPTPAPTIASQPLSLRPEACRPAAGG AVHTRGLDFACDFWVLVVVGGVLACYSLLVTVAFIIF WVRSKRSLRLHSDYMNMTPRRPGPTRKHYQPYAPPR DFAAYRSLRVKFSRSADAPAYQQGQNQLYNELNLGR REEYDVLDKRRGRDPEMGGKPRRKNPQEGLYNELQK DKMAEAYSEIGMKGERRRGKGHDGLYQGLSTATKDT YDALHMQUALPPRTSGSGATNFSLLKQAGDVEENPGPS RMDMADEPLNGSHTWLSIPFDLNGSVVSTNTSNQTEP YYDLTSNAVLTFIYFVVCIIIGLCGNTLVIYVILRYAKM KTITNIYIILNLAIADELFMLGLPFLAMQVALVHWPFGK AICRVVMTVDGINQFTSIFCLTVMSIDRYLAVVHPIKS AKWRRPRTAKMITMAVWGVSLVILPIMIYAGLRSNQ WGRSSCTINWPGESGAWYTGFIITYFILGFLVPLTIICL CYLFII IKVKSSGIRVGSSKRKKSEKKVTRMVSIVVAVF IFCWLFPFIYFNVSSVSMASPTPALKGMFDFVVVLTYA NSCANPILYAF LSDNFKKS FQNVLC LVK VSGTDDGER SDSKQDKSRLNETTETQRTLNLNGDLQTSI
55	MycT-Affy7-Y6M CAR-SSTR2	GSEQKLISEEDLGSQVQLQQSGPELVRPGASVKMSCR ASGYTFTSYWLHWVKQRPQGGLWIGMIDPSNSDTR FNPENFKDKATLNVDRSSNTAYMLLSLTSADSAVYYC <u>ATMGSYVSPLDYWGQGT</u> SVTVSSGGGGSGGGSGGGG GSDIMMSQSPSSLTVSVGEKVTVSCSSQSLLYTSSQK NYLAWYQQKPGQSPKLLIYWASTRESGVDPDRFTGSGS

TABLE 18-continued

Amino Acid and Nucleotide Sequences for Exemplary Polypeptides Described Herein		
SEQ ID NO:	Name	Sequence
		GTDFTLTITSVKADDLAVYYCQQYYAYPWTFGGGTK LEIKRASTTTPAPRPPTPAPTIASQPLSLRPEACRPAAG GAVHTRGLDFACDFWVLVVVGGVLACYSLLVTVAFI I FWVRSKRSRLHSDYMNMTPRRPGPTRKHYQPYAPP RDFAAYRSLRVKFSRSADAPAYQQQNQLYNELNLG RREEYDVLDKRRGRDPEMGGKPRRKNPQEGLYNELQ KDKMAEAYSEIGMKGERRRGKGDGLYQGLSTATK DTYDALHMQALPRTSGSGATNFSLLKQAGDVEENP GPSRMDMADEPLNGSHTWLSIPFDLNGSVVSTNTSNQ TEPYYDLTSSNAVLTFIYFVVCII GLCGNTLVIYVILRYA KMKTIITNIYILNLAIADLFMLGLPFLAMQVALVHWP FGKAICRVMTVDGINQFTSIFCLTVMSIDRYLAVVHP IKSAKWRRPRTAKMITMAVWGVSLVILPIMIYAGLR SNQWGRSSCTINWPGESGAWYTGFI IYTFILGFLVPLTI ICLCYLFII I KVKSSGIRVGSSKRKKSEKKVTRMVSIVV AVFI FCWLPFYIFNVSSVSMASPTPALKGMFDFVVVL TYANSCANPILYAFSLDNFKSFQNVLCVVKVSGTDD GERSDSKQDKSRLNETTETQRTLNLGDLQTSI
56	Affy7 CAR (control)	CAGGTGCAGCTCCAGCAGAGCGGCCCTGAGCTGGT GCGCCCCGCGCTTCCGTGAAGATGTCATGCCGAG CTTCGGCTACACGTTACACAGCTATTGGCTACACT GGGTCAAGCAGAGGCCCGACAGGGTCTGGAGTGG ATCGGTATGATTGACCCGTCAAATAGTGACACCCGC TTCAACCCCACTTCAAGGACAAGGCCACTCTCAAC GTGGACCGCTCGAGCAACACGGCTACATGCTGCT GTCCTCTCTGACCTCGGCGGACAGCGCCGTGTA CTGTGCCACC <u>TACGGCTCCTACGTGAGCCCTTGAT</u> <u>ATTGGGGCCAGGGCACCTCCGTACCGTGCTCCGg</u> TggaggcggttcaggcgagggtggctctggcggtggcggtcgGACATCAT GATGTCTCAGAGTCCCTCTTCTCTGACCGTGTCCT GGGCGAGAAGGTGACCGTGTCATGCAAGTCCTCAC AGAGCCTTCTGTACACAGCTCCAGAAGAACTAC CTGGCGTGGTACCAGCAGAAGCCCGGACAGAGCC CAAGCTGCTGATCTATTGGGCTTCCACCCGCGAGAG CGGCGTCCCGACCGCTTACCGGCTCCGGCTCCGG GACCGACTTACCCCTGACCATCACCTCCGTGAAGC CGATGACCTGGCCGTGTACTACTGTCAACAGTATTA CGCCTACCCGTGGACCTTCGGAGGCGGCACCAAGC TGGAGATCAAGCGCgctagcagcagccactccggcgccgccccaccg actccggcccccaactatcgcgagccagccccctgtcgctgaggccggaagcatgcccg ccctcgccgcccggagggtgctgtgcatacccggggattggacttcgcatgcgacTTT TGGGTGCTGGTGGTGGTGGTGGAGTCTGGCTTGC TATAGCTTGCTAGTAACAGTGGCCTTATTATTTTCT GGGTGAGGAGTAAGAGGAGCAGGCTCTGCACAGT GACTACATGAACATGACTCCCCGCCGCCCGGGCC CACCCGCAAGCATTACAGCCCTATGCCCCACCACG CGACTTCGAGCCTATCGCTCCctgagagtgaagttcagcagga gcgcagacgcccccgctaccagcagggccagaaccagctctataacgagctcaa tctaggacgaagagaggagtacgatgtttggacaagagacgtggccgggaccctg agatggggggaaagccgagaaggaagaaccctcaggaaggcctgtacaatgaact gcagaaagat aagatggcgaggccctacagtgcagatgggatgaaaggcgagcgc cggaggggcaaggggcacgatggcctttaccagggtctcagtacagccaccaagg acacctacgagcccttcacatgcaggccctgccccctcgc
57	Affy7-Y6A CAR	CAGGTGCAGCTCCAGCAGAGCGGCCCTGAGCTGGT GCGCCCCGCGCTTCCGTGAAGATGTCATGCCGAG CTTCGGCTACACGTTACACAGCTATTGGCTACACT GGGTCAAGCAGAGGCCCGACAGGGTCTGGAGTGG ATCGGTATGATTGACCCGTCAAATAGTGACACCCGC TTCAACCCCACTTCAAGGACAAGGCCACTCTCAAC GTGGACCGCTCGAGCAACACGGCTACATGCTGCT GTCCTCTCTGACCTCGGCGGACAGCGCCGTGTA CTGTGCCACC <u>GCGGGCTCCTACGTGAGCCCTTGAT</u> <u>TATTGGGGCCAGGGCACCTCCGTACCGTGCTCC</u> ggtggaggcggttcaggcgagggtggctctggcggtggcggtcgGACATC ATGATGTCTCAGAGTCCCTCTTCTCTGACCGTGTC GTGGCGAGAAGGTGACCGTGTCATGCAAGTCCTC ACAGAGCCTTCTGTACACAGCTCCAGAAGAACT ACCTGGCGTGGTACCAGCAGAAGCCCGGACAGAGC CCCAAGCTGCTGATCTATTGGGCTTCCACCCGCGAG

TABLE 18-continued

Amino Acid and Nucleotide Sequences for Exemplary Polypeptides Described Herein		
SEQ ID NO:	Name	Sequence
		AGCGGCGTCCCGACCGCTTCACCGGCTCCGGCTCC GGGACCGACTTCACCTGACCATCACCTCCGTGAAG GCCGATGACCTGGCCGTGTACTACTGTCAACAGTAT TACGCCCTACCGTGGACCTTCGGAGGCGGCACCAA GCTGGAGATCAAGCGCGctagcagcaccactccggcgccgcgcca ccgactccggccccaaactatcgcgagccagccccctgtcgctgaggccggaagcatg ccgcctctgcgcgaggtgctgtgcataccggggattggacttcgcatgcgacT TTTGGGTGCTGGTGGTGGTGGTGGAGTCTGGCTT GCTATAGCTTGCTAGTAACAGTGGCCTTTATTATTT TCTGGTGAGGAGTAAGAGGAGCAGGCTCCTGCAC AGTGACTACATGAACATGACTCCCGCCGCCCGG GCCCCACCGCAAGCATTACCAGCCCTATGCCCCACC ACGCGACTTCGCGACCTATCGCTCCctgagagtgaagttcagc aggagcgcagacgcccccgctaccagcagggccagaaccagctctataacgag ctcaatctaggacgaagagaggagtacgatgtttggacaagagacgtggccggga ccctgagatgggggaaagccgagaaggaagaaccctcaggaaggcctgtacaat gaactgcagaaagataagatggcgaggcctacagtgagattgggatgaaaggcg agcgccggaggggcaaggggcacgatggcctttaccagggtctcagtacagccac caaggacacctacgacgccttcacatgcaggccctgccccctcgc
58	Affy7-L9A CAR	CAGGTGCAGCTCCAGCAGAGCGGCCCTGAGCTGGT GCGCCCCGCGCTTCCGTGAAGATGTCATGCCGAG CTTCTGGCTACACGTTCACAGCTATTGGCTACACT GGGTCAAGCAGAGGCCCGGACAGGGTCTGGAGTGG ATCGGTATGATTGACCCGTCAAATAGTGACACCCGC TTCAACCCCAACTTCAAGGACAAGGCCACTCTCAAC GTGGACCGCTCGAGCAACACGGCCTACATGCTGCT GTCTCTCTGACCTCGGCGGACAGCGCCGTGTAATA CTGTGCCACC <u>TACGGCTCCTACGTGAGCCCTTGCGAT</u> <u>TATTGGGGCCAGGGCACCTCCGTCACCGTGTCTCTCC</u> ggtagaggcggttcaggcggaagtggtctggcggtggcgatcgGACATC ATGATGTCTCAGAGTCCCTCTTCTCTGACCGTGTCC GTGGCGGAGAAGGTGACCGTGTATGCAAGTCCTC ACAGAGCCTTCTGTACACAGCTCCAGAAGAAT ACCTGGCGTGTACACGAGAAGCCCGGACAGAGC CCCAAGCTGCTGATCTATTGGGCTTCCACCCGCGAG AGCGCGCTCCCGACCGCTTCACCGGCTCCGGCTCC GGGACCGACTTCACCTGACCATCACCTCCGTGAAG GCCGATGACCTGGCCGTGTACTACTGTCAACAGTAT TACGCCCTACCGTGGACCTTCGGAGGCGGCACCAA GCTGGAGATCAAGCGCGctagcagcaccactccggcgccgcgcca ccgactccggccccaaactatcgcgagccagccccctgtcgctgaggccggaagcatg ccgcctctgcgcgaggtgctgtgcataccggggattggacttcgcatgcgacT TTTGGGTGCTGGTGGTGGTGGTGGAGTCTGGCTT GCTATAGCTTGCTAGTAACAGTGGCCTTTATTATTT TCTGGTGAGGAGTAAGAGGAGCAGGCTCCTGCAC AGTGACTACATGAACATGACTCCCGCCGCCCGG GCCCCACCGCAAGCATTACCAGCCCTATGCCCCACC ACGCGACTTCGCGACCTATCGCTCCctgagagtgaagttcagc aggagcgcagacgcccccgctaccagcagggccagaaccagctctataacgag ctcaatctaggacgaagagaggagtacgatgtttggacaagagacgtggccggga ccctgagatgggggaaagccgagaaggaagaaccctcaggaaggcctgtacaat gaactgcagaaagataagatggcgaggcctacagtgagattgggatgaaaggcg agcgccggaggggcaaggggcacgatggcctttaccagggtctcagtacagccac caaggacacctacgacgccttcacatgcaggccctgccccctcgc
59	Affy7-Y6V CAR	CAGGTGCAGCTCCAGCAGAGCGGCCCTGAGCTGGT GCGCCCCGCGCTTCCGTGAAGATGTCATGCCGAG CTTCTGGCTACACGTTCACAGCTATTGGCTACACT GGGTCAAGCAGAGGCCCGGACAGGGTCTGGAGTGG ATCGGTATGATTGACCCGTCAAATAGTGACACCCGC TTCAACCCCAACTTCAAGGACAAGGCCACTCTCAAC GTGGACCGCTCGAGCAACACGGCCTACATGCTGCT GTCTCTCTGACCTCGGCGGACAGCGCCGTGTAATA CTGTGCCACC <u>GTGGCTCCTACGTGAGCCCTTTGGAT</u> <u>TATTGGGGCCAGGGCACCTCCGTCACCGTGTCTCTCC</u> ggtagaggcggttcaggcggaagtggtctggcggtggcgatcgGACATC ATGATGTCTCAGAGTCCCTCTTCTCTGACCGTGTCC GTGGCGGAGAAGGTGACCGTGTATGCAAGTCCTC ACAGAGCCTTCTGTACACAGCTCCAGAAGAAT ACCTGGCGTGTACACGAGAAGCCCGGACAGAGC

TABLE 18-continued

Amino Acid and Nucleotide Sequences for Exemplary Polypeptides Described Herein		
SEQ ID NO:	Name	Sequence
		CCCAAGCTGCTGATCTATTGGGCTTCCACCCGCGAG AGCGGCGTCCCGACCGCTTCACCGGCTCCGGCTCC GGGACCGACTTCACCTGACCATCACCTCCGTGAAG GCCGATGACCTGGCCGTGTACTACTGTCAACAGTAT TACGCCCTACCGTGGACCTTCGGAGGCGGCACCAA GCTGGAGATCAAGCGCgctagcacgaccactccggcgccgcgccc ccgactccggccccaactatcgcgagccagcccctgctgctgagccggaagcatg ccgcctgcccgcggaggtgctgtgcataccggggattggacttcgcatgcgacT TTTGGGTGCTGGTGGTGGTGGTGGAGTCCCTGGCTT GCTATAGCTTGCTAGTAACAGTGGCCTTATTATT TCTGGGTGAGGAGTAAGAGGAGCAGGCTCCTGCAC AGTGACTACATGAACATGACTCCCGCCGCCCGG GCCACCCGCAAGCATTACCAGCCCTATGCCCAACC ACGGACTTCGCAGCTATCGTCCctgagagtgaagttcagc aggagcgcagacgcccccgctaccagcagggccagaaccagctctataacgag ctcaatctaggacgaagagaggagtacgatgtttggacaagagacgtggccggga ccctgagatgggggaaagccgagaaggaaccctcaggaaggcctgtacaat gaactgcagaaagataagatggcggaggcctacagtgagattgggatgaaaggcg agcgccggaggggcaaggggcacgatggccttaccagggtctcagtacagccac caaggacacctacgacgccttcacatgcaggccctgccccctcgc
60	Affy7-Y6L CAR	CAGGTGCAGCTCCAGCAGAGCGGCCCTGAGCTGGT GCGCCCCGGCGCTTCCGTGAAGATGTCATGCCGAG CTTCTGGCTACACGTTTACCAGCTATTGGCTACACT GGGTCAAGCAGAGGCCCGGACAGGGTCTGGAGTGG ATCGGTATGATTGACCCGTCAAATAGTGACACCCGC TTCAACCCCAACTTCAAGGACAAGGCCACTCTCAAC GTGGACCGCTCGAGCAACACGGCCTACATGCTGCT GTCCCTCTGACCTCGGCGGACAGCCCGTGTACTA CTGTGCCACCCTCGGCTCCTACGTGAGCCCTTTGGAT <u>TATTGGGGCCAGGGCACCTCCGTACCCGTGCCTCC</u> ggTggaggcggttcaggcgagggtggctctggcggtggcgatcgGACATC ATGATGTCTCAGAGTCCCTCTTCTCTGACCGTGTCC GTGGGCGAGAAGGTGACCGTGTATGCAAGTCCTC ACAGAGCCTTCTGTACACCAGCTCCAGAGAAGT ACCTGGCGTGGTACCAGCAGAAGCCCGACAGAGC CCCAAGCTGCTGATCTATTGGGCTTCCACCCGCGAG AGCGGCGTCCCGACCGCTTCACCGGCTCCGGCTCC GGGACCGACTTCACCTGACCATCACCTCCGTGAAG GCCGATGACCTGGCCGTGTACTACTGTCAACAGTAT TACGCCCTACCGTGGACCTTCGGAGGCGGCACCAA GCTGGAGATCAAGCGCgctagcacgaccactccggcgccgcgccc ccgactccggccccaactatcgcgagccagcccctgctgctgagccggaagcatg ccgcctgcccgcggaggtgctgtgcataccggggattggacttcgcatgcgacT TTTGGGTGCTGGTGGTGGTGGTGGAGTCCCTGGCTT GCTATAGCTTGCTAGTAACAGTGGCCTTATTATT TCTGGGTGAGGAGTAAGAGGAGCAGGCTCCTGCAC AGTGACTACATGAACATGACTCCCGCCGCCCGG GCCACCCGCAAGCATTACCAGCCCTATGCCCAACC ACGGACTTCGCAGCTATCGTCCctgagagtgaagttcagc aggagcgcagacgcccccgctaccagcagggccagaaccagctctataacgag ctcaatctaggacgaagagaggagtacgatgtttggacaagagacgtggccggga ccctgagatgggggaaagccgagaaggaaccctcaggaaggcctgtacaat gaactgcagaaagataagatggcggaggcctacagtgagattgggatgaaaggcg agcgccggaggggcaaggggcacgatggccttaccagggtctcagtacagccac caaggacacctacgacgccttcacatgcaggccctgccccctcgc
61	Affy7-Y6I CAR	CAGGTGCAGCTCCAGCAGAGCGGCCCTGAGCTGGT GCGCCCCGGCGCTTCCGTGAAGATGTCATGCCGAG CTTCTGGCTACACGTTTACCAGCTATTGGCTACACT GGGTCAAGCAGAGGCCCGGACAGGGTCTGGAGTGG ATCGGTATGATTGACCCGTCAAATAGTGACACCCGC TTCAACCCCAACTTCAAGGACAAGGCCACTCTCAAC GTGGACCGCTCGAGCAACACGGCCTACATGCTGCT GTCCCTCTGACCTCGGCGGACAGCCCGTGTACTA CTGTGCCACC <u>ATCGGCTCCTACGTGAGCCCTTTGGAT</u> <u>TATTGGGGCCAGGGCACCTCCGTACCCGTGCCTCC</u> ggTggaggcggttcaggcgagggtggctctggcggtggcgatcgGACATC ATGATGTCTCAGAGTCCCTCTTCTCTGACCGTGTCC GTGGGCGAGAAGGTGACCGTGTATGCAAGTCCTC ACAGAGCCTTCTGTACACCAGCTCCAGAGAAGT ACCTGGCGTGGTACCAGCAGAAGCCCGACAGAGC CCCAAGCTGCTGATCTATTGGGCTTCCACCCGCGAG AGCGGCGTCCCGACCGCTTCACCGGCTCCGGCTCC GGGACCGACTTCACCTGACCATCACCTCCGTGAAG GCCGATGACCTGGCCGTGTACTACTGTCAACAGTAT TACGCCCTACCGTGGACCTTCGGAGGCGGCACCAA GCTGGAGATCAAGCGCgctagcacgaccactccggcgccgcgccc ccgactccggccccaactatcgcgagccagcccctgctgctgagccggaagcatg ccgcctgcccgcggaggtgctgtgcataccggggattggacttcgcatgcgacT TTTGGGTGCTGGTGGTGGTGGTGGAGTCCCTGGCTT GCTATAGCTTGCTAGTAACAGTGGCCTTATTATT TCTGGGTGAGGAGTAAGAGGAGCAGGCTCCTGCAC AGTGACTACATGAACATGACTCCCGCCGCCCGG GCCACCCGCAAGCATTACCAGCCCTATGCCCAACC ACGGACTTCGCAGCTATCGTCCctgagagtgaagttcagc aggagcgcagacgcccccgctaccagcagggccagaaccagctctataacgag ctcaatctaggacgaagagaggagtacgatgtttggacaagagacgtggccggga ccctgagatgggggaaagccgagaaggaaccctcaggaaggcctgtacaat gaactgcagaaagataagatggcggaggcctacagtgagattgggatgaaaggcg agcgccggaggggcaaggggcacgatggccttaccagggtctcagtacagccac caaggacacctacgacgccttcacatgcaggccctgccccctcgc

TABLE 18-continued

Amino Acid and Nucleotide Sequences for Exemplary Polypeptides Described Herein		
SEQ ID NO:	Name	Sequence
		ACCTGGCGTGGTACCAGCAGAAGCCCGACAGAGC CCCAAGCTGCTGATCTATTGGGCTTCCACCCGCGAG AGCGGCGTCCCGACCGCTTCACCGGCTCCGGCTCC GGGACCGACTTCACCTGACCATCACCTCCGTGAAG GCCGATGACCTGGCCGTGTACTACTGTCAACAGTAT TACGCCTACCCGTGGACCTTCGGAGGCGGCACCAA GCTGGAGATCAAGCGCgctagcagcaccactccggcgccgcgccc ccgactccggccccaactatcgcgagccagcccctgtcgctgaggccggaagcatg ccgcctcgcccgagggtgctgtgcataccgggggattggacttcgatgcgacT TTTGGGTGCTGGTGGTGGTGGTGGAGTCTGGCTT GCTATAGCTTGCTAGTAACAGTGGCCTTATTATT TCTGGGTGAGGAGTAAGAGGAGCAGGCTCCTGCAC AGTGACTACATGAACATGACTCCCCGCCGCCCGG GCCCCCGCAAGCATTACAGCCCTATGCCCCACC ACGCGACTTCGCAGCCTATCGTCCctgagagtgaagttcagc aggagcgagagcgcgcgcgcgtaccagcaggccagaaccagctctataacgag ctcaatctaggacgaagagaggagtacgatgtttggacaagagacgtggccggga ccctgagatgggggaaagccgagaaggaacccctcaggaaggcctgtacaat gaactgcagaaagataagatggcggaggccctacagtgagattgggatgaaaggcg agcgccggaggggcaaggggcacgatggcctttaccagggtctcagtagagccac caaggacacctacgacgccttcacatgcaggccctgccccctcgc
62	Affy7-Y6M CAR	CAGGTGCAGCTCCAGCAGAGCGGCCCTGAGCTGGT GCGCCCCGCGCTTCCGTGAAGATGTCATGCCGAG CTTCTGGCTACACGTTACACAGCTATTGGCTACACT GGGTCAAGCAGAGGCCCGACAGGGTCTGGAGTGG ATCGGTATGATTGACCCGTCAAATAGTGACACCCGC TTCAACCCCACTTCAAGGACAAGGCCACTCTCAAC GTGGACCGCTCGAGCAACACGGCTACATGCTGCT GTCTCTCTGACCTCGGCGGACAGCGCCGTGTACTA CTGTGCCACC <u>ATGGGCTCCTACGTGAGCCCTTGGAT</u> <u>TATTGGGGCCAGGGCACCTCCGTCACCGTGTCTCTCC</u> gggtggaggcggttcaggcgagggtggctctggcggtggcggtcgGACATC ATGATGTCTCAGAGTCCCTCTCTCTGACCGTGTCC GTGGGCGAGAAGGTGACCGTGTATGCAAGTCCTC ACAGAGCCTTCTGTACACAGCTCCCAGAAGAACT ACCTGGCGTGGTACCAGCAGAAGCCCGACAGAGC CCCAAGCTGCTGATCTATTGGGCTTCCACCCGCGAG AGCGGCGTCCCGACCGCTTCACCGGCTCCGGCTCC GGGACCGACTTCACCTGACCATCACCTCCGTGAAG GCCGATGACCTGGCCGTGTACTACTGTCAACAGTAT TACGCCTACCCGTGGACCTTCGGAGGCGGCACCAA GCTGGAGATCAAGCGCgctagcagcaccactccggcgccgcgccc ccgactccggccccaactatcgcgagccagcccctgtcgctgaggccggaagcatg ccgcctcgcccgagggtgctgtgcataccgggggattggacttcgatgcgacT TTTGGGTGCTGGTGGTGGTGGTGGAGTCTGGCTT GCTATAGCTTGCTAGTAACAGTGGCCTTATTATT TCTGGGTGAGGAGTAAGAGGAGCAGGCTCCTGCAC AGTGACTACATGAACATGACTCCCCGCCGCCCGG GCCCCCGCAAGCATTACAGCCCTATGCCCCACC ACGCGACTTCGCAGCCTATCGTCCctgagagtgaagttcagc aggagcgagagcgcgcgcgcgtaccagcaggccagaaccagctctataacgag ctcaatctaggacgaagagaggagtacgatgtttggacaagagacgtggccggga ccctgagatgggggaaagccgagaaggaacccctcaggaaggcctgtacaat gaactgcagaaagataagatggcggaggccctacagtgagattgggatgaaaggcg agcgccggaggggcaaggggcacgatggcctttaccagggtctcagtagagccac caaggacacctacgacgccttcacatgcaggccctgccccctcgc
63	EF1a-MycT-Affy7 CAR-SSTR2 (control)	ggctccggtgcccgtcagtgggcgagcgccacatcgcccacagtcgccgagaagtt ggggggagggggtcggaatgaaacgggtgcctagagaagggtggcgccgggtaa ctgggaaagtgatgtcggtactggctccgccttttcccgagggtgggggagaacc gtataaagtgcagtagtcgcccgtgaacgttcttttcgcaacgggtttgcccgcagaa cacaggtaagtgcggtgtgtggttcccccgggcctggcctctttacgggttatggccc ttgcgtgcctgaaattacttcacctggctgcagtagctgattcttgatcccgagcttcgg gttggaagtgggtgggagagttcgaggcccttcgcttaaggagcccccttcgctcgt gcttgagttgaggcctggcctggcgctggggcgccgcgctgcgaatctggtggca ccttcgcgctgtctcgctgctttcgataagctctagccatttaaaatttttgatgacctg ctgcgacgcttttttctggcaagatagttctgtaaatgcgggccaagatctgcacactg gtatttcggtttttggggcgcgggggcgacggggcccgctgcgtcccagcgacat gttcggcgaggcggggctgcgagcgggccaccgagaatcggaagggggtagt ctcaagctggccggcctgctctggtgctggtctcgcgcccgctgtatcgccccgc

TABLE 18-continued

Amino Acid and Nucleotide Sequences for Exemplary Polypeptides Described Herein		
SEQ ID NO:	Name	Sequence
		cctgggcggaaggctggcccggtcggcaccagttgctgagcggaagatggcc gcttcccgccctgctgcaggagctcaaatggaggacggcgctcgaggagag cgggcggtgagtcacccacacaaaggaaaaggcccttccgtcctcagccgtcgc ttcatgtgactccacggagtacgggcccgtccaggcacctcgattagttctcgagc ttttggagtacgtcgtctttaggtggggggaggggtttatgcatggagtttcccac actgagtggtggagactgaagttagggcagcttggcactgatgtaattctccttgga atttgcccttttgagtttgatcttggttcattctcaagcctcagacagtggttcaaagtt ttttcttccatttcaggtgtcgtgacaagttgtacaaaaagcaggctgccaccatggc cttaccagtgacggccttgcctcgtccgctggccttgcctgctccacggccgacggccg ggatccgaacaaaaactcatctcagaagaggatctgggatcgCAGGTGCAG CTCCAGCAGAGCGGCCCTGAGCTGGTGCGCCCGG CGCTTCCGTGAAGATGTCATGCCGAGCTTCTGGCTA CACGTTACACAGCTATTGGCTACACTGGGTCAAGCA GAGGCCCGACAGGGTCTGGAGTGGATCGGTATGA TTGACCCGTCAAATAGTGACACCCGCTTCAACCCCA ACTTCAAGGACAAGGCCACTCTCAACGTGGACCGC TCGAGCAACACGGCTACATGCTGTCTCTCTG ACCTCGGCGGACAGCGCGTGTACTACTGTGCCACC TACGGCTCTACGTGAGCCCTTGGATTATTGGGGCCA GGGCACCTCCGTACCGTGTCTCCggtggaggcggttcagg cggaggtggctctggcggtggcggtcgGACATCATGATGTCTCA GAGTCCCTCTTCTGACCGTGTCCGTGGGCGAGAA GGTGACCGTGTATGCAAGTCTCACAGAGCCTTCT GTACACAGCTCCCAGAAGAACTACCTGGCGTGGT ACCAGCAGAAGCCCGGACAGAGCCCAAGCTGCTG ATCTATTGGGCTTCCACCCGCGAGAGCGCGTCCCC GACCGCTTACCCGGCTCCGGCTCCGGGACCGACTTC ACCTGACCATCACTCCGTGAAGGCCGATGACCTG GCCGTGTACTACTGTCAACAGTATTACGCTACCCG TGGACCTTCGGAGGCGGCACCAAGCTGGAGATCAA GCGCgtagcagcagcactccggcgccgcgccacagcactccggcccaacta tcgagagccagccctgtcgtgagggcggaagcatgcgcctcggccggagg tgctgtgcataccggggattggaattcgatgcgacTTTGGGTGCTGG TGGTGGTTGGTGGAGTCTGGCTTGCTATAGCTTGC TAGTAACAGTGGCTTTATTATTTCTGGGTGAGGA GTAAGAGGAGCAGGCTCTGCACAGTGACTACATG AACATGACTCCCCCGCCCCGGGCCACCCGCAA GCATTACAGCCCTATGCCCCACCGGACTTCGC AGCCTATCGCTCCctgagagtgagttcagcaggagcgcagacgcccc cgcgtaaccagcagggccagaaccagctctataacgagctcaatctaggacgaagag aggagtagcatgttttgacaagagacgtggccgggacccctgagatggggggaaa gccgagaaggaagaacctcaggaaggcctgtacaatgaactgcagaagataag atggcggaaggcctacagtgagattgggatgaaaggcgagcgccggaggggcaag gggcacgatggccttaccagggtctcagtacagccaccaaggacacctacgacgc ccttcacatgcaggccctgccccctcgcactagtggaggcgagctactaacttcag cctgctgaagcaggctggagacgtggaggagaacctggacctctagaatggaca tggcggtgagccactcaatggaagccacacatggctatccatttgcatttgacctcaat ggctctgtggtgtcaaccaacacctcaaaccagacagagccgactatgacctgaca agcaatgcagctcctcacatctcatctatttgggtctgcatattgggtgtgtggcaaca cacttgtcatctatgtcatcctccgctatgcccaagatgaagacctcaccaacattacat cctcaacctggccatcgagatgagctcttcatgctgggtctgcctttcttggctatgca gggtgctctgggtccactggcccttggcaaggccattggcggtgggtcatgactgtg gatggcatcaatcagttcaccagcatctctgcctgacagtcagagcatcgaccgata cctggctgtggtccaccccatcaagtggcccaagtggaggagaccccgagcgcc aagatgacacatggctgtgtggggagttctctgctgggtcatcttggccatcatgata tatgtcgggtcctcgaggaacacagtgggggagaagcagctgcacatcaactggcc aggtgaatctggggcttggtacacaggggttcatcatctacactttcattctggggttcct ggtaacctcaccatcatctgtcttctgctacgtgttcatctacacaaagtggaagtccctg gaatccgagtggtcctctaagaggaagaagtcgtgagaagaaggtcaccggaatg gtgtccatcgtgggtggtgtcttcatctctgctgggtccctctacatatccaactttc ttccgtctccatggccatcagccccacccagccctaaaggcatgttggactttgtgggtg gtcctcacctatgctaacagctgtgccaaacctatcctatagccttctgtctgacaact tcaagaagagcttccagaatgtcctctgcttggtcaaggtgagcggcacagatgatgg ggagcggagtacagtaagcaggacaaatccggctgaatgagaccacggagac ccagaggacctcctcaatggagacctccaaccagtatctaa
64	EF1a-MycT-Affy7-Y6A CAR-SSTR2	ggctccgggtgccctcagtgggcagagcgacatcgcccacagtcgccgagaagtt ggggggagggtcggaatgaaccgggtgcctagagaaggtggcgcggtgaaa ctgggaaagtgtgctgtactggctccgctttttcccgaggtgggggagaacc gtatataagtgcagtagtcgctgaaagttcttttcgcaacgggtttgcgcgagaa cacaggtaaagtgccgtgtgtgttcccggggcctggcctctttacgggttatggccc

TABLE 18-continued

Amino Acid and Nucleotide Sequences for Exemplary Polypeptides Described Herein		
SEQ ID NO:	Name	Sequence
		ttgcgtgccttgaattacttcacctggctgcagtagctgattcttgatcccgagcttcgg gttggaagtgggtgggagagttcgaggccttgcgcttaaggagccccttcgcctcgt gcttgagttgaggcctggcctggcgctggggcccgcgctgcgaatctggtggca ccttcgcgcctgtctcgctgctttcgataagttcttagccatttaaaattttgatgacctg ctgcgacgctttttctggcaagatagttctgtaaatgcgggccaagatctgcacactg gtatttcggtttttggggcgcgggcggcgacggggcccgctgcgtcccagcgacat gttcggcgaggcggggcctgcgagcgcggccaccgagaatcggacgggggtagt ctcaagctggcgggctgctctggtgctggtctcgcgccgcgctgatatcgcccgc cctgggcggcaaggctggcccggctcgccaccagttgctgagcggaaagatggcc gcttcccggcctgctgcaggagctcaaaatggaggacgcggcctcgggagag cgggcgggtgagtcacccacacaaaggaaaaggcccttccgtcctcagccgtcgc ttcattgtactccacggagtagcgggcgcgctccaggcacctcgattagttctcgagc ttttgagtagctgctctttaggtggggggagggtttatgcatggagtttcccac actgagtggtggagactgaagttagggcagcttgccacttgatgtaattctccttgga atttgcccttttgagtttgatcttggttcattctcaagcctcagacagtggttcaaggt ttttcttccatttcaggtgctgtagaagtttgtaaaaaaagcaggctgccaccatggc cttaccagtgaccgccttgctcctgccgtggccttgctgctccacgcgcgcaggccg ggatccgaacaaaaactcatctcagaaggagatctgggtagcCAGGTGCAG CTCCAGCAGAGCGGCCCTGAGCTGGTGCGCCCGG CGCTTCCGTGAAGATGTCATGCCGAGCTTCTGGCTA CACGTTACACGACTATTGGCTACACTGGGTCAAGCA GAGGCCCGACAGGGTCTGGAGTGGATCGGTATGA TTGACCCGTCAAATAGTGACACCCGCTTCAACCCCA ACTTCAAGGACAAGGCCACTCTCAACGTGGACCGC TCGAGCAACACGGCTTACATGCTGTCTCTCTG ACCTCGGCGGACAGCGCGTGTACTACTGTGCCACC <u>GCGGGCTCCTACGTGAGCCCTTGGATTATTGGGGCC</u> AGGGCACCTCCGTACCGTGTCTCCggtggaggcggttca ggccgaggtggctctggcggtggcggtcgGACATCATGATGTCTC AGAGTCCCTCTTCTGACCGTGTCCGTGGGCGAGA AGGTGACCGTGTATGCAAGTCTCACAGAGCCTTC TGTACACGAGCTCCAGAGAAGTACCTGGCGTGG TACCAGCAGAAGCCCGGACAGAGCCCAAGCTGCT GATCTATTGGGCTTCCACCCGCGAGAGCGGCTGCC CGACCGCTTACCCGCTCCGGCTCCGGGACCGACTT CACCTGACCATCACCTCCGTGAAGCCGATGACCT GGCCGTGTACTACTGTCAACAGTATTACGCTTACCC GTGGACCTTCGGAGGCGGCACCAAGCTGGAGATCA AGCGCgttagcacgaccactccggcgccgcgcccaccgactccggccccaac tatcgcgagccagcccctgtcgctgaggccggaagcatgccgccctgcccggga ggtgctgtgcataccggggatgtgacttcgcatgcgacTTTGGGTGCTG GTGGTGGTGGTGGAGTCTGGCTTGCTATAGCTTG CTAGTAACAGTGGCCTTTATTATTTCTGGGTGAGG AGTAAGAGGAGCAGGCTCTGCACAGTGACTACAT GAACATGACTCCCCGCGCCCCGGGCCACCCGCA AGCATTACCAGCCCTATGCCCCACCACGCACTTCG CAGCCTATCGCTCCctgagagtgaaagttcagcaggagcgcagacgccc ccgctaccagcaggccagaaccagctctataacgagctcaatctaggacgaaga gaggagtagcatgttttgacaagagacgtggccgggaccctgagatggggggaa agccgagaagggaagaccctcaggaaggcctgtacaatgaactgcagaagataa gatggcggaggccctacagtgagattgggatgaaaggcgagcgcggaggggcaa ggggcacgatggcctttaccagggtctcagtagcaccacgaaggacacctacgacg cccttcacatgaggccctgccccctcgcaactagtggaaagcggagctactaacttca gcctgctgaagcaggctggagacgtggaggagaacctggacctctagaatggac atggcggtgagccactcaatggaagccacacatggctatccattccatttgacctca atggctctgtggtgtcaaccaacacctcaaaccagacagagccgtactatgacctgac aagcaatgcagtcctcacattcatctattttggtgtcgcatcatgggttggtggcaac acacttgctatttatgtcatcctcgctatgccaaagatgaagaccatcaccaacatttac atcctcaacctggccatcgagatgagctcttcattgctgggtctgccttcttggtatg cagggtggctctggtccactggccctttggcaaggccatttgccgggtggtcatgactg tggatggcatcaatcagttcaccagcatctctgctgacagtcagagcatcgaccga taactggctgtggtccaccccatcaagtcggccaagtggaggagacccggagcggc caagatgataccatggctgtgtggggagctctctgctggtcatcttgcccatcatgat atatgctgggctccggagcaaccagtgggggagaagcagctgcaccatcaactggc

TABLE 18-continued

Amino Acid and Nucleotide Sequences for Exemplary Polypeptides Described Herein		
SEQ ID NO:	Name	Sequence
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65	EF1 α -MycT-Affy7- L9A CAR-SSTR2	ggctccgggtgcccgctcagtgggcagagcgcacatcgcccacagtcctccgagaagtt ggggggagggtcggaatgaaccgggtgcctagagaaggtggcgcggggtaaa ctgggaaagtgatgtcggtactggctccgctttttcccgagggtgggggagaacc gtataaagtgcagtagtcgcgtgaacgttcttttgcgaacgggtttgcccgcagaa cacaggtaaagtgcgtgtgtggttcccccgggcctggcctctttacgggttatggccc ttgcgtgcctgaattacttccacctggctgcagtagctgattcttgatcccgagcttcgg gttgaagtggtgggagagtgcaggccttgctgttaaggagcccttcgctcgt gcttgagttgaggcctggcctggcgctggggcccgcgctgcgaatctggtggca ccttcgcgctgtctcgctgctttcgataagctctagccatttaaaatttttgatgacctg ctgcgacgcttttttctggcaagatagttctgtaatgcgggccaagatctgcacactg gtatttcggtttttggggccgcgggcgcgagcggggcccgctgcgtcccagcgcacat gttcggcgaggcggggcctgcgagcggggccaccgagaatcgagcggggtagt ctcaagctggccggcctgctctggtgcttggtctcgcgcccgctgtatcgcccgc cctggggcggaaggctggcccggctggcaccagttgctgagcggaaagatggcc gcttcccgccctgctgcaggagctcaaaatggaggacgcgcgctcgggagag cgggggggtgagtcacccacacaaaaggaaaaggcctttccgtcctcagcgcgtgc ttcatgtgactccacggagtaccgggcccgtccaggcactcgattagttctcgagc ttttggagtacgtcgtctttaggttggggggagggttttatgcatggagtttccccac actgagtggttgagactgaagttaggccagcttggaacttgatgtaattctccttgga atttgccctttttgagtttggtcttggttcattctcaagcctcagacagtggttcaaggt ttttcttccatttcaggtgtcgtgacaagtttgtaaaaaaagcaggctgccacctggc cttaccagtgaccgccttgctcctgcccgtggccttgctgctccacgcccgcaggccg ggatccgaacaaaaactcatctcagaagaggatctgggatcgCAGGTGCAG CTCCAGCAGAGCGGCCCTGAGCTGGTGCGCCCGG CGCTTCCGTGAAGATGTCATGCCGAGCTTCTGGCTA CACGTTACACGCTATTGGCTACACTGGGTCAAGCA GAGGCCCGACAGGGTCTGGAGTGGATCGGTATGA TTGACCCGTCAAAATAGTGACACCCGCTTCAACCCCA ACTTCAAGGACAAGGCCACTCTCAACGTGGACCGC TCGAGCAACACGGCTACATGCTGCTGTCTCTCTG ACCTCGGCGGACAGCGCGGTACTACTGTGCCACC TACGGCTCCTACGTGAGCCCTGCGGATTATTGGGGCC AGGGCACTCCGTACCGTGTCTCCggtggaggcggttca ggcggaggtggtctggcggtggcggatcgGACATCATGATGTCTC AGAGTCCCCTCTCTGACCGGTGTCGGTGGGCGAGA AGGTGACCGTGCATGCAAGTCTTACAGAGCCTTC TGTAACCAAGTCCCAGAAGAACTACCTGGCGTGG TACCAGCAGAAGCCCGGACAGAGCCCCAAGCTGCT GATCTATTGGGCTTCCACCCGCGAGAGCGGCTCCC CGACCGCTTACCGGCTCCGGCTCCGGGACCGACTT CACCTGACCATCACCTCCGTGAAGGCCGATGACCT GGCCGTGTACTACTGTCAACAGTATTACGCTACCC GTGGACCTTCGGAGGCGGCACCAAGCTGGAGATCA AGCGCgctagcacgaccactccggcgccgcccaccgactccggccccaac tatcgagccagccctgtcgtgagccggaagcatgccgcccgtgcccgga ggtgctgtgcataccggggataggactcgcacgagctTTTGGGTGCTG GTGGTGGTGGTGGAGTCTGGCTTGCTATAGCTTG CTAGTAACAGTGGCCTTATTATTTCTGGGTGAGG AGTAAGAGGAGCAGGCTCTGCACAGTGACTACAT GAACATGACTCCCCGCCGCCCGGGCCACCCGCA AGCATTACAGCCCTATGCCACCAACGCGACTTCG CAGCTTATCGCTCCctgagagtgaaagttcagcaggagcgcagacgccc ccgctaccagcaggggccagaaccagctctataacgagctcaatctaggacgaaga gaggagtacgatgttttgacaagagacgtggccgggacccctgagatgggggaa agccgagaaggaaccctcaggaaggcctgtacaatgaaatgcagaagataa gatggcgaggcctacagtgagattgggatgaaaggcgagcgcggaggggcaa ggggacagatggcctttaccagggtctcagtagcaccaccaaggacacctacgag cccttcacatcaggccctgcccctcgcaactagtgaagcggagctactaaactca gcctgctgaagcaggctggagacgtggaggagaacctggaccttctagaatggac atggcggtgagccactcaatggaagccacacatggctatccattccatttgacctca atggctctgtggtgtcaaccaacacctcaaccagacagagccgtaactatgacctgac

TABLE 18-continued

Amino Acid and Nucleotide Sequences for Exemplary Polypeptides Described Herein		
SEQ ID NO:	Name	Sequence
		aagcaatgcagtcctcacattcatctatTTTTgtggtctgcatcattgggttggtggcaac acacttgatcattatgtcatcctccgctatgccaatgaagaccatcaccaacatttac atcctcaacctggccatgcagatgagctcttcatgctgggtctgcctttcttggtatg caggtggctctgtgctcactggcctttggcaaggccatttgccgggtggtcatgactg tggatggcatcaatcagttcaccagcatcttctgcctgacagtcatgagcatcgaccga tacctggctgtggtccaccccatcaagtgcgccaagtggaggagaccccgacggc caagatgatcaccatggctgtgtggggagtctctctgctggtcatcttgcccatcatgat atatgctgggtccggagcaaccagtgggggagaagcagctgcaccatcaactggc caggtgaatctggggttggtacacagggttcacatctacactttcatcttggggttcc tggtaccctcaccatcatctgtcttctgctacctgtcattatcatcaaggtgaagtcctct ggaatccgagtggtgctcctctaaaggagaagctctgagaagaaggtcacccgaat ggtgtccatcgtggtggctgtcttcatcttctgctgggttcccttctacatattcaactgtt cttcggtctccatggccatcagccccacccagccttaaaggcatgtttgactttgtggt ggtcctcacctatgctaacagctgtgccaacctatcctatagcctctctgtctgacaac ttcaagaagagcttcagaatgtctctgcttggtcaaggtgagcggcacagatgatg gggagcggagtgcagtaagcaggacaaatcccgctgaatgagaccacggaga ccagaggacctcctcaatggagacctccaaccagtatctaa
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TABLE 18-continued

Amino Acid and Nucleotide Sequences for Exemplary Polypeptides Described Herein		
SEQ ID NO:	Name	Sequence
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TABLE 18-continued

Amino Acid and Nucleotide Sequences for Exemplary Polypeptides Described Herein		
SEQ ID NO:	Name	Sequence
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68	EF1α-MycT-Affy7- Y6I CAR-SSTR2	ggctccggtgccctcagtgggcagagcgcacatcgcccacagtcgccgagaagtt ggggggagggggtcggaatgaaccgggtgcctagagaaggtggcgcggggtaaa ctgggaaagtgatgtcgtgactggctccgctttttcccgaggggtgggggagaacc gtataaagtgcagtagtcgcggtgaacgttcttttcgcaacgggtttgcccagaa cacaggtaaagtccgtgtgtggttcccgcggtcctggcctctttacgggttatggccc ttgcgtgccttgaattacttcacactggctgcagtagctgattcttgatcccgagcttcgg gttgaagtgggtgggagagttcgaggccttgcgcttaaggagcccttcgcctcgt gcttgagttgaggcctggcctggcgctggggcccgcgctgcgaatctgggtggca ccttcgcgcctgtctcgtgctttcgataagttctagccatttaaaatttttgatgacctg ctgcgacgcttttttctggcaagatagttctgtaaatgcgggccaagatctgcacactg gtatttcggtttttggggcccgggcgcgacggggccgctgcgtccagcgcacat gttcggcgaggcggggcctgcgagcgcggccaccgagaatcggaagggggtagt ctcaagctggcgggcctgctctgggtgctggctctcgcccgccgctgtatcgccccgc cctggcgcgcaaggctggccccgttcggcaccagttgctgagcggaaagatggcc gcttcccgccctgctgcaggagctcaaaatggaggacgcggcgtcgggagag cgggggggtgagtcacccacacaaaggaaaaggcccttccgtcctcagccgtgcg ttcatgtgactccacggagtacggggcgcgtccaggcacctcgattagttctcgagc ttttggagtacgtcgtctttaggtggggggagggttttatgcgatggagtttccccac actgagtggtgggagactgaagttaggccagcttggcacttgatgaattctccttgga atttgccctttttgagtttggtcttggttcattctcaagcctcagacagtggttcaaaagttt ttttctccatttcagggtgctgacaaagtttgtaaaaaagcaggctgccaccatggc cttaccagtgacccgcttgcctcctgcgctggccttgcgtgctccacgcgcggcggcg ggatccgaacaaaaactcatctcagaaggagatctgggatcgCAGGTGCAG CTCCAGCAGAGCGGCCCTGAGCTGGTGCGCCCCGG CGCTTCCGTGAAGATGTATGCCGAGCTTCTGGCTA CACGTTACACGACTATTGGCTACACTGGGTCAAGCA GAGGCCCGACAGGGTCTGGAGTGGATCGGTATGA TTGACCCGTCAAATAGTGACACCGCTTCAACCCCA ACTTCAAGGACAAGGCCACTCTCAACGTGGACCGC

TABLE 18-continued

Amino Acid and Nucleotide Sequences for Exemplary Polypeptides Described Herein		
SEQ ID NO:	Name	Sequence
		TCGAGCAACACGGCCTACATGCTGTCTCTCTCTG ACCTCGGCGGACAGCGCCGTGTACTACTGTGCCACC ATCGGCTCCTACGTGAGCCCTTTGGATTATTGGGGCC AGGGCACCTCCGTACCGTGTCTCCggtggaggcggttca ggcggaggtggctctggcggtggcggtcgGACATCATGATGTCTC AGAGTCCCTCTTCTGTACCGTGTCCGTGGGCGAGA AGGTGACCGTGTATGCAAGTCTCACAGAGCCTTC TGTACACAGCTCCCAGAAGAACTACCTGGCGTGG TACCAGCAGAAGCCCGACAGAGCCCAAGCTGCT GATCTATTGGGCTTCCACCCGAGAGCGGCGTCCC CGACCGCTTACCGGCTCCGGTCCGGGACCGACTT CACCTGACCATCACCTCCGTGAAGGCCGATGACCT GGCCGTGTACTACTGTCAACAGTATTACGCCTACCC GTGGACCTTCGGAGGCGGCACCAAGCTGGAGATCA AGCGCgtctagcagcaccactccggcgccgcccaccgactccggccccaac tatecgagccagccctgtcgctgaggccggaagcatgcccctgcccggga ggtgctgtgcataccggggatttgacttcgcatcgacTTTTGGGTGCTG GTGGTGGTGGTGGAGTCTGGCTTGCTATAGCTTG CTAGTAACAGTGGCCTTTATTATTTCTGGGTGAGG AGTAAGAGGAGCAGGCTCTGCACAGTACTACAT GAACATGACTCCCCGCGCCCGGGCCACCCGCA AGCATTACCAGCCCTATGCCCCACCACGCACTTCG CAGCCTATCGCTCCctgagagtgaagttcagcaggagcgagacgccc ccgctaccagcaggggcagaaccagctctataacgagctcaatctaggacgaaga gaggagtacgatgttttgacaagagacgtggccgggacctgagatggggggaa agccgagaaggaagaaccctcaggaaggcctgtacaatgaactgcagaaagataa gatggcggaggcctacagtgagattgggatgaaaggcgagcgccggaggggcaa ggggcacgatggcctttaccagggctcagtagcaccacaaaggacacctacgacg cccttcacatgcaggccctgccccctcgcaactagtgaagcggagctactaaactca gcctgctgaagcaggctggagacgtggaggagaaccctggacctctagaatggac atggcggtgagccactcaatggaagccacacatggctatccattccatttgacctca atggctctgtggtgtcaaccaacacctcaaacagacagagccgtagtagctgac aagcaatgcagtcctcaccattcatctattttgtggtctgcatcatgggtgtgtggcaac acacttgtcatttatgtcatcctccgctatgccaatgaagaccatcaccacatttac atcctcaacctggccatcgagatgagctcttcagtgctgggtctgccttcttggtatg cagggtgctctggtccactggcctttggcaaggccatttgccgggtggtcatgactg tggatggcatcaatcagttcaccagcatctctgcctgacagtcagagcatcgaccga tacctggctgtggtccaccccatcaagtggcgaagtggaggagaccccgagcgc caagatgatcaccatggctgtgtgggagctctctgctggtcatcttgcccatcatgat atagctgggtccggagcaaccagtgggggagaagcagctgcaccatcaactggc caggtgaatctggggcttggtacacagggttcacatctacacttcatctgggggtcc tggtagccctcaccatcatctgtcttgctacctgttcattatcatcaagtggaagtcctct ggaatccgagtggtgctctctaaaggaagaagctcgagaagaaggtcaccgaat gggtccatcggtgggtgctgtctcatctctgctggcttccctctacattcaacgttcc ttccgtctccatggccatcagccccacccagcccttaaaggcatgtttgactttgtggt ggtcctcaccatgctaaccagctgtgccaacctatcctatagcctctctgtctgacaac ttcaagaagagcttcagaatgtctctgcttggtcaaggtgagcggcacagatgatg gggagcggagtgacagtaagcaggacaaatccggctgaatgagaccacggaga cccagaggacctcctcaatggagacctccaaccagtaactaa
69	EF1α-MycT-Affy7-Y6M CAR-SSTR2	ggctccggtgcccgtcagtgggcagagcgacatcgcccacagtcgccgagaagtt ggggggagggtcggaatgaaccgggtgcctagagaaggtggcgcggggtaaa ctgggaaagtgatgtcggtactggctccgctttttcccgagggtgggggagaacc gtatataagtgagtagtcgccgtgaacgttcttttcgcaacgggtttgcgcgcagaa cacaggtaaagtgcgtgtgtggttcccgcgggctggcctctttacgggttatggccc ttgctgcttgaattacttcacctggctgcagtagctgattcttgatccgagcttcgg gttgaagtgggtgggagagtgcaggccttgctgcttaaggagcccttcgctcgt gcttgagttagggcctggcctggcgctggggccgcgcgtgcgaatctggtggca ccttcgcgcctgtctcgctgctttcgataagctctagccatttaaaattttgatgacctg ctgcgacgcttttttctggcaagatagcttctgtaaatgcgggccaagatctgcacactg gtatttcggtttttggggccgcggcgggcgacggggcccgctgcgtcccagcgacat gttcggcgaggcggggcctgcgagcgcgccaccgagaatcggacgggggtagt ctcaagctggcggcctgctctggtgctggtctcgcgcccgctgtatcgccccgc cctggggcggaaggtggccgggtcgccaccagttgctgagcgggaagatggcc gcttcccggcctgctgcaggagctcaaaatggaggacgcggcgtcgggagag

TABLE 18-continued

Amino Acid and Nucleotide Sequences for Exemplary Polypeptides Described Herein		
SEQ ID NO:	Name	Sequence
		cgggcggtgagtcacccacacaaaggaaaaggcccttccgtcctcagccgtcgc ttcatgtgactccacggagtagccggcgccgtccaggcacctcgattagttctcgagc ttttggagtagctcgtctttaggttggggggagggtttatgcgatggagtttccccac actgagtggttgagactgaagttaggccagcttggcacttgatgaattctccttgga atttgcccttttgagtttgatcttggttcattctcaagcctcagacagtgggtcaaaagttt ttttcttccatttcagggtgctgacaaagttgtacaaaaaagcaggctgccaccatggc cttaccagtgacgccttgctcctgcccgtggccttgcgtctccacgcgcgcaggccg ggatccgaacaaaaactcatctcagaaggagatctgggatcgCAGGTGCAG CTCCAGCAGAGCGGCCCTGAGCTGGTGCCGCCCGG CGCTTCCGTGAAGATGTCATGCCGAGCTTCTGGCTA CACGTTACCAGCTATTGGCTACACTGGGTCAAGCA GAGGCCCGACAGGGTCTGGAGTGGATCGGTATGA TTGACCCGTCAAATAGTGACACCCGCTTCAACCCCA ACTTCAAGGACAAGGCCACTCTCAACGTGGACCGC TCGAGCAACACGGCTACATGCTGCTGCTCTCTG ACCTCGGCGGACAGCGCGTGTACTACTGTGCCACC <u>ATGGGCTCTCTAGTGAGCCCTTTGGATTATTGGGGCC</u> AGGGCACCTCCGTCACCGTGTCTCCggtggaggcggttca ggcgaggtggctctggcggtggcgatcgGACATCATGATGTCTC AGAGTCCCTCTTCTGTACCGTGTCCGTGGCGAGA AGGTGACCGTGTCTGCAAGTCTCACAGAGCCTTC TGTAACCCAGCTCCCAGAAGAACTACCTGGCGTGG TACCAGCAGAAGCCCGGACAGAGCCCCAAGCTGCT GATCTATTGGGCTTCCACCCGCGAGAGCGGCGTCCC CGACCGCTTACCCGGCTCCGGCTCCGGGACCGACTT CACCTGACCATCACTCCGTGAAGGCCGATGACCT GGCCGTGTACTACTGTCAACAGTATTACGCCTACCC GTGGACCTTCGGAGGCGGCACCAAGCTGGAGATCA AGCGCgtagcacgaccactccggcgccgcgcccaccgactccggccccaac tatcgcgagccagccccgtcgctgagggcggaagcatgcgcctcgcccgga ggtgctgtgcataccggggttggaactcgcatgcgactTTTGGGTGCTG GTGGTGGTTGGTGGAGTCTGGCTTGCTATAGCTTG CTAGTAACAGTGGCCTTTATTATTTCTGGGTGAGG AGTAAGAGGAGCAGGCTCTGCACAGTGACTACAT GAACATGACTCCCCGCCGCCCGGGCCCAACCGCA AGCATTACCAGCCCTATGCCCCACCACGCACTTCG CAGCCTATCGCTCCctgagagtgaagttcagcaggagcgacgacccc ccgctaccagcagggccagaaccagctctataacgagctcaatctaggacgaaga gaggagtacgatgtttggacaagagacgtggccgggaccctgagatggggggaa agccgagaaggaagaaccctcaggaaggcctgtacaatgaactgcagaaagataa gatggcgaggcctacagtgagattgggatgaaaggcgagcgccggaggggcaa ggggcacgatggccttaccagggctcagtagacgccaccaaggacacctacgacg cccttcacatgcaggccctgccccctcgcaactagtgggaagcgagctactaactca gcctgctgaagcaggtggagacgtggaggagaaccctggaacctctagaatggac atggcggtatgagccactcaatggaagccacacatggctatccattccatttgacctca atggctctgtggtgtcaaccaacacctcaaacagacagagccgactatgacctgac aagcaatgcagtcctcacattcatctatttgtggtctgcatcatggggtgtgtggcaac acacttgtcatttatgtcatctccgctatgccaaagtgaagaccatcaccaacattac atcctcaacctggccatcgcatgagctctcatgctgggtctgccttcttggtctag caggtggctctgggtccactggccctttggcaaggccatttgccgggtgggtcatgactg tggatggcatcaatcagttcaccagcatctctgctgacagtcagtagcatcgaccga tacctggctgtgggtccaccccatcaagtggccaaagtggaggagaccccggaacgc caagatgatcaccatggctgtgtggggagtctctgctggtcatcttgcccatcatgat atatgctgggtccggagcaaccagtgggggagaagcagctgcaccatcaactggc caggtgaatctggggttggtacacagggttcatcatctacacttcatctggggttcc tggtaacctcaccatcatctgcttctgctacctgttcattatcatcaaggtgaagtctct ggaatccgagtggtcctctaaaggaagaagctgagaagaaggtcacccgaat gggtccatcggtgggtgtctcatctctgctgggttcccttctacatattcaaggttcc ttccgtctccatggccatcagccccacccagccctaaaggcatgtttgacttttggt ggtcctcacctatgctaacagctgtgccaacctatcctatatgccttctgtctgacaac ttcaagaagagcttcagaatgtcctctgcttggtcaaggtgagcggcacagatgatg gggagcggagtacagtaagcaggacaaatcccgctgaatgagaccacggaga cccagaggacctcctcaatggagacctccaaccagatctata
70	Affy6 HCDR1	GYIFTAY
71	Affy6 HCDR2	LEWMGWIKPNNGLANLY
72	Affy6-I3A HCDR3	SEATTEFDY
73	Affy6-I3W HCDR3	SEWTTEFDY

TABLE 18-continued

Amino Acid and Nucleotide Sequences for Exemplary Polypeptides Described Herein		
SEQ ID NO:	Name	Sequence
74	Affy6-I3F HCDR3	SEFTTEFDY
75	Affy6-I3G HCDR3	SEGTTEFDY
76	Affy6 LDR1	ESVDSY
77	Affy6 LDR2	PKLLIYRASTRE
78	Affy6 LDR3	SKEDPL
79	Affy6 VH (wt)	QVQLVQSGAEVKKPGASVKVSCKASGYIFTAYTMHW VRQAPGQGLEWMGWIKPNNGLANYAQKFQGRVTMTR DTSISTAYMELSRRLRSDDTAVYYCAR <u>SEITTEFDY</u> WGQ GTLVTVSS
80	Affy6-I3A VH	QVQLVQSGAEVKKPGASVKVSCKASGYIFTAYTMHW VRQAPGQGLEWMGWIKPNNGLANYAQKFQGRVTMTR DTSISTAYMELSRRLRSDDTAVYYCAR <u>SEATTEFDY</u> WG QGLTVTVSS
81	Affy6-I3W VH	QVQLVQSGAEVKKPGASVKVSCKASGYIFTAYTMHW VRQAPGQGLEWMGWIKPNNGLANYAQKFQGRVTMTR DTSISTAYMELSRRLRSDDTAVYYCAR <u>SEWTTEFDY</u> WG QGLTVTVSS
82	Affy6-I3F VH	QVQLVQSGAEVKKPGASVKVSCKASGYIFTAYTMHW VRQAPGQGLEWMGWIKPNNGLANYAQKFQGRVTMTR DTSISTAYMELSRRLRSDDTAVYYCAR <u>SEFTTEFDY</u> WG QGLTVTVSS
83	Affy6-I3G VH	QVQLVQSGAEVKKPGASVKVSCKASGYIFTAYTMHW VRQAPGQGLEWMGWIKPNNGLANYAQKFQGRVTMTR DTSISTAYMELSRRLRSDDTAVYYCAR <u>SEGTTEFDY</u> WG QGLTVTVSS
84	Affy6 VL (wt)	DIVLTQSPDLSAVSLGERATINCKSS <u>ESVDSY</u> ANSFMH WYQQKPGQPPKLLIYRASTRSGVDPDRFSGSGSRDFTL TISSLQAEADVAVYYCQ <u>SKEDPL</u> TFGGGTKVEIK
85	Affy6 VH (wt)	CAGGTGCAGCTCGTGCAGAGCGGCGCCGAGGTGAA GAAGCCCGGCGCTTCCGTGAAGGTGTCCTGTAAGG CCTCTGGCTACATCTTCACCGCTACACCATGCACTG GGTGCGCCAGGCCCGGGACAGGGGCTGGAGTGGA <u>TGGGCTGGATCAAGCCTAACAACGGTCTGGCCA</u> ACTAC <u>GCGCAGAAGTTCCAGGGCCGCGTGACCATGACTCG</u> GGACACCAGCATCTCTACCGCTACATGGAGCTGTC CCGCCTGCGCTCCGATGACACGGCCGTGTACTACTG CGCCCGCTCGGAGATCACACCGAGTTCGACTATTGG GGCCAGGGCACCCCTGGTGACCGTGTCGTCC
86	Affy6-I3A VH	CAGGTGCAGCTCGTGCAGAGCGGCGCCGAGGTGAA GAAGCCCGGCGCTTCCGTGAAGGTGTCCTGTAAGG CCTCTGGCTACATCTTCACCGCTACACCATGCACTG GGTGCGCCAGGCCCGGGACAGGGGCTGGAGTGGA <u>TGGGCTGGATCAAGCCTAACAACGGTCTGGCCA</u> ACTAC <u>GCGCAGAAGTTCCAGGGCCGCGTGACCATGACTCG</u> GGACACCAGCATCTCTACCGCTACATGGAGCTGTC CCGCCTGCGCTCCGATGACACGGCCGTGTACTACTG CGCCCGCTCGGAGGCCACACCGAGTTCGACTATTGG GGCCAGGGCACCCCTGGTGACCGTGTCGTCC

TABLE 18-continued

Amino Acid and Nucleotide Sequences for Exemplary Polypeptides Described Herein		
SEQ ID NO:	Name	Sequence
87	Affy6-I3W VH	CAGGTGCAGCTCGTGCAGAGCGGCGCCGAGGTGAA GAAGCCCGGCGCTTCCGTGAAGGTGTCTGTAAGG CCTCTGGCTACATCTTCACCGCTACACCATGCACTG GGTGCGCCAGGCCCGGGACAGGGGCTGGAGTGGG <u>TGGGCTGGATCAAGCCTAACAACGGTCTGGCCAACTAC</u> <u>GCGCAGAAGTTCCAGGGCCGCGTGACCATGACTCG</u> GGACACCAGCATCTCTACCGCTACATGGAGCTGTC CCGCCTGCGCTCCGATGACACGGCCGTGTACTACTG CGCCCGCTCGGAGTGGACCACCGAGTTCGACTATTGG GGCCAGGGCACCCTGGTGACCGTGTCTGTC
88	Affy6-I3F VH	CAGGTGCAGCTCGTGCAGAGCGGCGCCGAGGTGAA GAAGCCCGGCGCTTCCGTGAAGGTGTCTGTAAGG CCTCTGGCTACATCTTCACCGCTACACCATGCACTG GGTGCGCCAGGCCCGGGACAGGGGCTGGAGTGGG <u>TGGGCTGGATCAAGCCTAACAACGGTCTGGCCAACTAC</u> <u>GCGCAGAAGTTCCAGGGCCGCGTGACCATGACTCG</u> GGACACCAGCATCTCTACCGCTACATGGAGCTGTC CCGCCTGCGCTCCGATGACACGGCCGTGTACTACTG CGCCCGCTCGGAGTTCACACCGAGTTCGACTATTGG GGCCAGGGCACCCTGGTGACCGTGTCTGTC
89	Affy6-I3G VH	CAGGTGCAGCTCGTGCAGAGCGGCGCCGAGGTGAA GAAGCCCGGCGCTTCCGTGAAGGTGTCTGTAAGG CCTCTGGCTACATCTTCACCGCTACACCATGCACTG GGTGCGCCAGGCCCGGGACAGGGGCTGGAGTGGG <u>TGGGCTGGATCAAGCCTAACAACGGTCTGGCCAACTAC</u> <u>GCGCAGAAGTTCCAGGGCCGCGTGACCATGACTCG</u> GGACACCAGCATCTCTACCGCTACATGGAGCTGTC CCGCCTGCGCTCCGATGACACGGCCGTGTACTACTG CGCCCGCTCGGAGGGCACCACCGAGTTCGACTATTGG GGCCAGGGCACCCTGGTGACCGTGTCTGTC
90	Affy6 VL (wt)	GACATCGTGCTGACCCAGAGCCCTGACAGCCTGGC CGTGTCCTGGGCGAGCGCGCCACCATTAACTGCA AGAGCTCCGAGAGTGTCTAGCTACGCCAACTCCTT CATGCACTGGTACCAGCAGAAGCCCGGCCAGCCAC <u>CCAAGCTGCTGATCTACAGGGCTTCACCCGCGAGAG</u> CGGCGTCCCCGACAGGTTTTCAGGTTCTGGCTCCCG CACCGACTTCACCTGACCATCTCTCTCTGCAGGC CGAGGATGTGGCGGTGTACTACTGTCAACAGTCCAA <u>GGAGGACCCGCTTACCTTCGGCGGCGGCACCAAGT</u> GGAGATCAAG
91	Affy6 scFv (control)	QVQLVQSGAEVKKPGASVKVSKASGYIFTAYTMHW VRQAPGQGLEWMGWI KPNNGLANYAQKFGQGRVTMT RDTISITAYMELSR LRSDDTAVYYCARSEITTEFDYW QGT LVT VSSGGGSGGGSGGGSDIVLTQSPDSLAV SLGERATINCKSSESVD SYAN SFMHYQQKPGQPPKL LIYRASTRESGV PDRFSGSGSRDFTLTISSLQAEDVAV YYCQQSKEDPLTFGGGT KVEIK
92	Affy6-I3A scFv	QVQLVQSGAEVKKPGASVKVSKASGYIFTAYTMHW VRQAPGQGLEWMGWI KPNNGLANYAQKFGQGRVTMT RDTISITAYMELSR LRSDDTAVYYCARSEATTEFDYW GQGT LVT VSSGGGSGGGSGGGSDIVLTQSPDSLAV VSLGERATINCKSSESVD SYAN SFMHYQQKPGQPPKL LLIYRASTRESGV PDRFSGSGSRDFTLTISSLQAEDVA VYYCQQSKEDPLTFGGGT KVEIK
93	Affy6-I3W scFv	QVQLVQSGAEVKKPGASVKVSKASGYIFTAYTMHW VRQAPGQGLEWMGWI KPNNGLANYAQKFGQGRVTMT RDTISITAYMELSR LRSDDTAVYYCARSEWTFEFDYW GQGT LVT VSSGGGSGGGSGGGSDIVLTQSPDSLAV

TABLE 18-continued

Amino Acid and Nucleotide Sequences for Exemplary Polypeptides Described Herein		
SEQ ID NO:	Name	Sequence
		VSLGERATINCKSSESVDSYANSFMHWYQQKPGQPPK LLIYRASTRESGVPDRFSGSGSRDFTLTISSLQAEDVA VYYCQQSKEDPLTFGGGTKVEIK
94	Affy6-I3F scFv	QVQLVQSGAEVKKPGASVKVSCASGYIFTAYTMHW VRQAPGQGLEWMGWIKPNNGLANIYAQKPFQGRVTMT RDTISITAYMELSRRLSDDTAVYYCARSEFTTEFDYW GQGT LVT VSSGGGSGGGSGGGSDIVLTQSPDSL VSLGERATINCKSSESVDSYANSFMHWYQQKPGQPPK LLIYRASTRESGVPDRFSGSGSRDFTLTISSLQAEDVA VYYCQQSKEDPLTFGGGTKVEIK
95	Affy6-I3G scFv	QVQLVQSGAEVKKPGASVKVSCASGYIFTAYTMHW VRQAPGQGLEWMGWIKPNNGLANIYAQKPFQGRVTMT RDTISITAYMELSRRLSDDTAVYYCARSEGTTEFDYW GQGT LVT VSSGGGSGGGSGGGSDIVLTQSPDSL VSLGERATINCKSSESVDSYANSFMHWYQQKPGQPPK LLIYRASTRESGVPDRFSGSGSRDFTLTISSLQAEDVA VYYCQQSKEDPLTFGGGTKVEIK
96	Affy6 scFv (control)	CAGGTGCAGCTCGTGCAGAGCGGCGCCGAGGTGAA GAAGCCCGCGCTTCCGTGAAGGTGTCTGTAAGG CCTCTGGCTACATCTTACCGCTACACCATGCACT GGGTGCGCCAGGCCCCGGGACAGGGCTGGAGTGG ATGGGCTGGATCAAGCCTAACACGGTCTGGCCAA CTACGCGCAGAAGTTCAGGGCCGCGTGACCATGA CTCGGGACACCAGCATCTCTACCGCGTACATGGAG CTGTCCCGCTGCGCTCCGATGACACGGCCGTGTAC TACTGCGCCCGCTCGGAGATCACACCGAGTTCGACT ATTGGGGCCAGGGCACCCCTGGTGACCGTGTCTCGTCCG GTGGAGGCGGTTT CAGGCGGAGGTGGCTCTGGCGGT GGCGGATCGGACATCGTGCTGACCCAGAGCCCTGA CAGCCTGGCCGTGTCCCTGGGCGAGCGGCCACCA TTAACTGCAAGAGCTCCGAGAGTGTCTGATAGCTAC GCCAACTCCTTCATGCACTGGTACCAGCAGAAGCCC GGCCAGCCACCAAGCTGTGTATCTACAGGGCTTCC ACCCGCGAGAGCGGCGTCCCCGACAGGTTTTCAGG TTCTGGCTCCCGCACCGACTTACCCGTGACCATCTC TTCTCTGAGGCGGAGGATGTGGCGGTGTACTACTG TCAACAGTCCAAGGAGGACCCGCTTACCTTCGGCG GCGGCACCAAGGTGGAGATCAAG
97	Affy6-I3A scFv	CAGGTGCAGCTCGTGCAGAGCGGCGCCGAGGTGAA GAAGCCCGCGCTTCCGTGAAGGTGTCTGTAAGG CCTCTGGCTACATCTTACCGCTACACCATGCACT GGGTGCGCCAGGCCCCGGGACAGGGCTGGAGTGG ATGGGCTGGATCAAGCCTAACACGGTCTGGCCAA CTACGCGCAGAAGTTCAGGGCCGCGTGACCATGA CTCGGGACACCAGCATCTCTACCGCGTACATGGAG CTGTCCCGCTGCGCTCCGATGACACGGCCGTGTAC TACTGCGCCCGCTCGGAGGCCACACCGAGTTCGACT ATTGGGGCCAGGGCACCCCTGGTGACCGTGTCTCGTCCG GTGGAGGCGGTTT CAGGCGGAGGTGGCTCTGGCGGT GGCGGATCGGACATCGTGCTGACCCAGAGCCCTGA CAGCCTGGCCGTGTCCCTGGGCGAGCGGCCACCA TTAACTGCAAGAGCTCCGAGAGTGTCTGATAGCTAC GCCAACTCCTTCATGCACTGGTACCAGCAGAAGCCC GGCCAGCCACCAAGCTGTGTATCTACAGGGCTTCC ACCCGCGAGAGCGGCGTCCCCGACAGGTTTTCAGG TTCTGGCTCCCGCACCGACTTACCCGTGACCATCTC TTCTCTGAGGCGGAGGATGTGGCGGTGTACTACTG TCAACAGTCCAAGGAGGACCCGCTTACCTTCGGCG GCGGCACCAAGGTGGAGATCAAG
98	Affy6-I3W scFv	CAGGTGCAGCTCGTGCAGAGCGGCGCCGAGGTGAA GAAGCCCGCGCTTCCGTGAAGGTGTCTGTAAGG CCTCTGGCTACATCTTACCGCTACACCATGCACT GGGTGCGCCAGGCCCCGGGACAGGGCTGGAGTGG ATGGGCTGGATCAAGCCTAACACGGTCTGGCCAA CTACGCGCAGAAGTTCAGGGCCGCGTGACCATGA CTCGGGACACCAGCATCTCTACCGCGTACATGGAG CTGTCCCGCTGCGCTCCGATGACACGGCCGTGTAC TACTGCGCCCGCTCGGAGGCCACACCGAGTTCGACT ATTGGGGCCAGGGCACCCCTGGTGACCGTGTCTCGTCCG GTGGAGGCGGTTT CAGGCGGAGGTGGCTCTGGCGGT GGCGGATCGGACATCGTGCTGACCCAGAGCCCTGA CAGCCTGGCCGTGTCCCTGGGCGAGCGGCCACCA TTAACTGCAAGAGCTCCGAGAGTGTCTGATAGCTAC GCCAACTCCTTCATGCACTGGTACCAGCAGAAGCCC GGCCAGCCACCAAGCTGTGTATCTACAGGGCTTCC ACCCGCGAGAGCGGCGTCCCCGACAGGTTTTCAGG TTCTGGCTCCCGCACCGACTTACCCGTGACCATCTC TTCTCTGAGGCGGAGGATGTGGCGGTGTACTACTG TCAACAGTCCAAGGAGGACCCGCTTACCTTCGGCG GCGGCACCAAGGTGGAGATCAAG

TABLE 18-continued

Amino Acid and Nucleotide Sequences for Exemplary Polypeptides Described Herein		
SEQ ID NO:	Name	Sequence
		CTGTCCCGCCTGCGCTCCGATGACACGGCCGTGTAC TACTGCGCCCGC <u>TCGGAGTGG</u> ACCACCGAGTTCGACT <u>ATTGGGGCCAGGGCACCC</u> TGGTGACCGTGTCTGTCG GTGGAGGCGGTTT CAGGCGGAGGTGGCTCTGGCGGT GGCGGATCGGACATCGTGCTGACCCAGAGCCCTGA CAGCCTGGCCGTGTCCCTGGGCGAGCGCGCCACCA TTAAGTGAAGAGCTCCGAGAGTGTCTGATAGCTAC GCCAACTCCTTCATGCACTGGTACCAGCAGAAGCCC GGCCAGCCACCAAGCTGCTGATCTACAGGGCTTCC ACCCGCGAGAGCGGCGTCCCCGACAGGTTTTCAGG TTCTGGCTCCCGACCGACTTCACCTGACCATCTC TTCTCTGACGGCCGAGGATGTGGCGGTGTACTACTG TCAACAGTCCAAGGAGGACCCGCTTACCTTCGGCG GCGGCACCAAGGTGGAGATCAAG
99	Affy6-I3F scFv	CAGGTGCAGCTCGTGCAGAGCGGCGCGAGGTGAA GAAGCCCGGCGCTTCCGTGAAGGTGTCTGTAAAG CCTCTGGCTACATCTTACCGCCTACACCATGCACT GGGTGCGCCAGGCCCCGGGACAGGGGTGGAGTGG ATGGGCTGGATCAAGCCTAACAAACGGTCTGGCCAA CTACGCGCAGAAGTTCAGGGCCGCGTGACCATGA CTCGGGACACCGCATCTCTACCGCGTACATGGAG CTGTCCCGCCTGCGCTCCGATGACACGGCCGTGTAC TACTGCGCCCGC <u>TCGGAGTTC</u> ACCACCGAGTTCGACT <u>ATTGGGGCCAGGGCACCC</u> TGGTGACCGTGTCTGTCG GTGGAGGCGGTTT CAGGCGGAGGTGGCTCTGGCGGT GGCGGATCGGACATCGTGCTGACCCAGAGCCCTGA CAGCCTGGCCGTGTCCCTGGGCGAGCGCGCCACCA TTAAGTGAAGAGCTCCGAGAGTGTCTGATAGCTAC GCCAACTCCTTCATGCACTGGTACCAGCAGAAGCCC GGCCAGCCACCAAGCTGCTGATCTACAGGGCTTCC ACCCGCGAGAGCGGCGTCCCCGACAGGTTTTCAGG TTCTGGCTCCCGACCGACTTCACCTGACCATCTC TTCTCTGACGGCCGAGGATGTGGCGGTGTACTACTG TCAACAGTCCAAGGAGGACCCGCTTACCTTCGGCG GCGGCACCAAGGTGGAGATCAAG
100	Affy6-I3G scFv	CAGGTGCAGCTCGTGCAGAGCGGCGCGAGGTGAA GAAGCCCGGCGCTTCCGTGAAGGTGTCTGTAAAG CCTCTGGCTACATCTTACCGCCTACACCATGCACT GGGTGCGCCAGGCCCCGGGACAGGGGTGGAGTGG ATGGGCTGGATCAAGCCTAACAAACGGTCTGGCCAA CTACGCGCAGAAGTTCAGGGCCGCGTGACCATGA CTCGGGACACCGCATCTCTACCGCGTACATGGAG CTGTCCCGCCTGCGCTCCGATGACACGGCCGTGTAC TACTGCGCCCGC <u>TCGGAGGG</u> CACCACCGAGTTCGACT <u>ATTGGGGCCAGGGCACCC</u> TGGTGACCGTGTCTGTCG GTGGAGGCGGTTT CAGGCGGAGGTGGCTCTGGCGGT GGCGGATCGGACATCGTGCTGACCCAGAGCCCTGA CAGCCTGGCCGTGTCCCTGGGCGAGCGCGCCACCA TTAAGTGAAGAGCTCCGAGAGTGTCTGATAGCTAC GCCAACTCCTTCATGCACTGGTACCAGCAGAAGCCC GGCCAGCCACCAAGCTGCTGATCTACAGGGCTTCC ACCCGCGAGAGCGGCGTCCCCGACAGGTTTTCAGG TTCTGGCTCCCGACCGACTTCACCTGACCATCTC TTCTCTGACGGCCGAGGATGTGGCGGTGTACTACTG TCAACAGTCCAAGGAGGACCCGCTTACCTTCGGCG GCGGCACCAAGGTGGAGATCAAG

TABLE 18-continued

Amino Acid and Nucleotide Sequences for Exemplary Polypeptides Described Herein		
SEQ ID NO:	Name	Sequence
101	Affy6 CAR (control)	QVQLVQSGAEVKKPGASVKVSCKASGYIFTAYTMHW VRQAPGQGLEWMGWI KPNNGLANYAQKFGQGRVTMT RDTISISTAYMELSRLRSDDTAVYYCARSEITTEFDYW GGTLVTVSSGGGGSGGGSGGGSDIVLTQSPDSLAV SLGERATINCKSSESVDSYANSFMHWYQKPGQPPK LIYRASTRESGVPDRFSGSGSRDTFTLTISLQAEDVAV YYCQQSKEDPLTFGGGTKEIKASTTTPAPRPPTPAPT ASQPLSLRPEACRPAAGGAVHTRGLDFACDFWVLV VGGVLACYSLLVTVAFIIFWVRSKRSLRLHSDYMNMT PRRPGPTRKHYPYAPPRDFAAYRSLRVKFSRSADAP APAYQQGQNQLYNELNLRREEYDVLDRRGRDPEMG GKPRRKNPQEGLYNELQDKMAEAYSEIGMKGERRR RRGKHDGLYQGLSTATKDTYDALHMQALPPR
102	Affy6-I3A CAR	QVQLVQSGAEVKKPGASVKVSCKASGYIFTAYTMHW VRQAPGQGLEWMGWI KPNNGLANYAQKFGQGRVTMT RDTISISTAYMELSRLRSDDTAVYYCARSEATTEFDYW GGTLVTVSSGGGGSGGGSGGGSDIVLTQSPDSLAV VSLGERATINCKSSESVDSYANSFMHWYQKPGQPPK LLIYRASTRESGVPDRFSGSGSRDTFTLTISLQAEDVA VYYCQQSKEDPLTFGGGTKEIKASTTTPAPRPPTPAP TIASQPLSLRPEACRPAAGGAVHTRGLDFACDFWVLV VGGVLACYSLLVTVAFIIFWVRSKRSLRLHSDYMN MTPRRPGPTRKHYPYAPPRDFAAYRSLRVKFSRSAD APAYQQGQNQLYNELNLRREEYDVLDRRGRDPE MGGKPRRKNPQEGLYNELQDKMAEAYSEIGMKGE RRRGKHDGLYQGLSTATKDTYDALHMQALPPR
103	Affy6-I3W CAR	QVQLVQSGAEVKKPGASVKVSCKASGYIFTAYTMHW VRQAPGQGLEWMGWI KPNNGLANYAQKFGQGRVTMT RDTISISTAYMELSRLRSDDTAVYYCARSEWTFEFDYW GGTLVTVSSGGGGSGGGSGGGSDIVLTQSPDSLAV VSLGERATINCKSSESVDSYANSFMHWYQKPGQPPK LLIYRASTRESGVPDRFSGSGSRDTFTLTISLQAEDVA VYYCQQSKEDPLTFGGGTKEIKASTTTPAPRPPTPAP TIASQPLSLRPEACRPAAGGAVHTRGLDFACDFWVLV VGGVLACYSLLVTVAFIIFWVRSKRSLRLHSDYMN MTPRRPGPTRKHYPYAPPRDFAAYRSLRVKFSRSAD APAYQQGQNQLYNELNLRREEYDVLDRRGRDPE MGGKPRRKNPQEGLYNELQDKMAEAYSEIGMKGE RRRGKHDGLYQGLSTATKDTYDALHMQALPPR
104	Affy6-I3F CAR	QVQLVQSGAEVKKPGASVKVSCKASGYIFTAYTMHW VRQAPGQGLEWMGWI KPNNGLANYAQKFGQGRVTMT RDTISISTAYMELSRLRSDDTAVYYCARSEFTTEFDYW GGTLVTVSSGGGGSGGGSGGGSDIVLTQSPDSLAV VSLGERATINCKSSESVDSYANSFMHWYQKPGQPPK LLIYRASTRESGVPDRFSGSGSRDTFTLTISLQAEDVA VYYCQQSKEDPLTFGGGTKEIKASTTTPAPRPPTPAP TIASQPLSLRPEACRPAAGGAVHTRGLDFACDFWVLV VGGVLACYSLLVTVAFIIFWVRSKRSLRLHSDYMN MTPRRPGPTRKHYPYAPPRDFAAYRSLRVKFSRSAD APAYQQGQNQLYNELNLRREEYDVLDRRGRDPE MGGKPRRKNPQEGLYNELQDKMAEAYSEIGMKGE RRRGKHDGLYQGLSTATKDTYDALHMQALPPR
105	Affy6-I3G CAR	QVQLVQSGAEVKKPGASVKVSCKASGYIFTAYTMHW VRQAPGQGLEWMGWI KPNNGLANYAQKFGQGRVTMT RDTISISTAYMELSRLRSDDTAVYYCARSEGTTEFDYW GGTLVTVSSGGGGSGGGSGGGSDIVLTQSPDSLAV VSLGERATINCKSSESVDSYANSFMHWYQKPGQPPK LLIYRASTRESGVPDRFSGSGSRDTFTLTISLQAEDVA VYYCQQSKEDPLTFGGGTKEIKASTTTPAPRPPTPAP TIASQPLSLRPEACRPAAGGAVHTRGLDFACDFWVLV VGGVLACYSLLVTVAFIIFWVRSKRSLRLHSDYMN MTPRRPGPTRKHYPYAPPRDFAAYRSLRVKFSRSAD APAYQQGQNQLYNELNLRREEYDVLDRRGRDPE MGGKPRRKNPQEGLYNELQDKMAEAYSEIGMKGE RRRGKHDGLYQGLSTATKDTYDALHMQALPPR

TABLE 18-continued

Amino Acid and Nucleotide Sequences for Exemplary Polypeptides Described Herein		
SEQ ID NO:	Name	Sequence
106	MycT-Affy6 CAR-SSTR2 (control)	GSEQKLISEEDLGSQVQLVQSGAEVKKPGASVKVSCK ASGYIFTAYTMHWVRQAPQGGLWGMWIKPNNGLA NYAQKFQGRVTMTDRDTSISTAYMELSRRLSDDTAVY YCARSEITTEFDYWGQGLVTVSSGGGSGGGGSGGG GSDIVLTQSPDLSAVSLGERATINCKSSESVDSYANSF MHWYQQKPGQPPKLLIYRASTRESGVDRFSGSGSR DFTLTISSLQAEDVAVYYCQSKEDPLTFGGGKVEIK ASTTTPAPRPPTPAPTIASQPLSLRPEACRPAAGGAVHT RGLDFACDFWVLVVVGGVLACYSLLVTVAFIIFWVRS KRSRLHSDYMNMTPRRPGPTRKHYQPYAPPRDFAA YRSLRVKFSRSADAPAYQQGQNQLYNELNLGRREY DVLDKRRGRDPEMGGKPRRKNPQEGLYNELQKDKM AEAYSEIGMKGERRRRGKHDGLYQGLSTATKDTYDA LHMQUALPPRTSGSGATNFSLLKQAGDVEENPGPSRMD MADEPLNGSHTWLSIPFDLNGSVVSTNTSNQTEPYD LTSNAVLTFIYFVVCIIIGLCGNTLVIYVILRYAKMKTIT NIYILNLAIADELFMLGLPFLAMQVALVHWPFGKAIC RVVMTVDGINQFTSIFCLTVMSIDRYLAVVHPKSAK WRRPRTAKMITMAVWGVSLVLIPIMYAGLRSNQW GRSSCTINWPGESGAWYTGFIITYFILGFLVPLTIICLCY LFIIIKVKSSGIRVGSSKRKKSEKKVTRMVSIVVAVFIF CWLFPYIFNVSSVSMASPTPALKGMDFVVLTYAN SCANPILYAFSLDNFKKSFQNVLCVVKVSGTDDGERS DSKQDKSRLNETTETQRTLLNGDLQTSI
107	MycTaffy6-I3A CAR-SSTR2	GSEQKLISEEDLGSQVQLVQSGAEVKKPGASVKVSCK ASGYIFTAYTMHWVRQAPQGGLWGMWIKPNNGLA NYAQKFQGRVTMTDRDTSISTAYMELSRRLSDDTAVY YCARSEATTEFDYWGQGLVTVSSGGGSGGGGSGGG GGSDIVLTQSPDLSAVSLGERATINCKSSESVDSYANS FMHWYQQKPGQPPKLLIYRASTRESGVDRFSGSGSR TDFTLTISSLQAEDVAVYYCQSKEDPLTFGGGKVEI KASTTTPAPRPPTPAPTIASQPLSLRPEACRPAAGGAV HTRGLDFACDFWVLVVVGGVLACYSLLVTVAFIIFW VRSKRSRLHSDYMNMTPRRPGPTRKHYQPYAPPRDF AAYRSLRVKFSRSADAPAYQQGQNQLYNELNLGRRE EYDVLDKRRGRDPEMGGKPRRKNPQEGLYNELQKD KMAEAYSEIGMKGERRRRGKHDGLYQGLSTATKDTY DALHQUALPPRTSGSGATNFSLLKQAGDVEENPGPSR MDMADEPLNGSHTWLSIPFDLNGSVVSTNTSNQTEPY YDLTNAVLTFIYFVVCIIIGLCGNTLVIYVILRYAKMK TITNIYILNLAIADELFMLGLPFLAMQVALVHWPFGKA ICRVVMTVDGINQFTSIFCLTVMSIDRYLAVVHPKSA KWRPRTAKMITMAVWGVSLVLIPIMYAGLRSNQ WGRSSCTINWPGESGAWYTGFIITYFILGFLVPLTIICL CYLFIIKVKSSGIRVGSSKRKKSEKKVTRMVSIVVAVF IFCWLFPYIFNVSSVSMASPTPALKGMDFVVLTYA NSCANPILYAFSLDNFKKSFQNVLCVVKVSGTDDGER SDSKQDKSRLNETTETQRTLLNGDLQTSI
108	MycT-Affy6-I3W CAR-SSTR2	GSEQKLISEEDLGSQVQLVQSGAEVKKPGASVKVSCK ASGYIFTAYTMHWVRQAPQGGLWGMWIKPNNGLA NYAQKFQGRVTMTDRDTSISTAYMELSRRLSDDTAVY YCARSEWTEFDYWGQGLVTVSSGGGSGGGGSGGG GGSDIVLTQSPDLSAVSLGERATINCKSSESVDSYANS FMHWYQQKPGQPPKLLIYRASTRESGVDRFSGSGSR TDFTLTISSLQAEDVAVYYCQSKEDPLTFGGGKVEI KASTTTPAPRPPTPAPTIASQPLSLRPEACRPAAGGAV HTRGLDFACDFWVLVVVGGVLACYSLLVTVAFIIFW VRSKRSRLHSDYMNMTPRRPGPTRKHYQPYAPPRDF AAYRSLRVKFSRSADAPAYQQGQNQLYNELNLGRRE EYDVLDKRRGRDPEMGGKPRRKNPQEGLYNELQKD KMAEAYSEIGMKGERRRRGKHDGLYQGLSTATKDTY DALHQUALPPRTSGSGATNFSLLKQAGDVEENPGPSR MDMADEPLNGSHTWLSIPFDLNGSVVSTNTSNQTEPY YDLTNAVLTFIYFVVCIIIGLCGNTLVIYVILRYAKMK TITNIYILNLAIADELFMLGLPFLAMQVALVHWPFGKA ICRVVMTVDGINQFTSIFCLTVMSIDRYLAVVHPKSA KWRPRTAKMITMAVWGVSLVLIPIMYAGLRSNQ WGRSSCTINWPGESGAWYTGFIITYFILGFLVPLTIICL CYLFIIKVKSSGIRVGSSKRKKSEKKVTRMVSIVVAVF

TABLE 18-continued

Amino Acid and Nucleotide Sequences for Exemplary Polypeptides Described Herein		
SEQ ID NO:	Name	Sequence
109	MycT-Affy6-I3F CAR-SSTR2	IFCWLPPYIFNVSSVSMASPTPALKGMFDFVVVLTYA NSCANPILYAF LSDNFKKSFQNVLC LVK VSGTDDGER SDSKQDKSRLNETTETQRTLLNGDLQTSI
		GSEQKLISEEDLGSQVQLVQSGAEVKKPGASVKVSCK ASGYIFTAYTMHWVRQAPGQGLEWMGWIKPNNGLA NYAQKFQGRVTMTDRDTSISTAYMELSR LRSDDTAVY YCARSEFTTEFDYWGQGLTVTVSSGGGSGGGGSGG GGSDIVLTQSPDSLAVSLGERATINCKSSESVD SYANS FMHWYQQKPGQPPKLLIYRASTRESGVDRFSGSGSR TDFTLTISSLQAEDVAVYYCQSKEDPLTFGGGTKEI KASTTTPAPRPPTPAPTIASQPLSLRPEACRPAAGGAV HTRGLDFACDFWLVVVGGVLACYSLLVTVAFIIFW VRSKSRLLHSDYMNMTPRRPGPTRKHYPYAPPRDF AAYRSLRVKFSRSADAPAYQQGNQLYNELNLRRE EYDVLDKRRGRDPEMGGKPRRKNPQEGLYNELQKD KMAEAYSEIGMKGERRRKGHDGLYQGLSTATKDTY DALHMQALPPRTSGSGATNFSLLKQAGDVEENPGPSR MDMADEPLNGSHTWLSIPFDLNGSVVSTNTSNQTEPY YDLTSSNAVLTFIYFVVCIIIGLCGNTLVIIYVILRYAKMK TITNIYILNLAIADLFMLGLPFLAMQVALVHWPFGKA ICRVMTVDGINQFTSIFCLTVMSIDRYLAVVHPKSA KWRRPRTAKMITMAVWGVSLLVILPIMIYAGLR SNQ WGRSSCTINWPGESGAWYTGFIITYFILGFLVPLTIICL CYLFIIIVKSSGIRVGSSKRKKSEKKVTRMVSIVVAVF IFCWLPPYIFNVSSVSMASPTPALKGMFDFVVVLTYA NSCANPILYAF LSDNFKKSFQNVLC LVK VSGTDDGER SDSKQDKSRLNETTETQRTLLNGDLQTSI
110	MycT-Affy6-I3G CAR-SSTR2	GSEQKLISEEDLGSQVQLVQSGAEVKKPGASVKVSCK ASGYIFTAYTMHWVRQAPGQGLEWMGWIKPNNGLA NYAQKFQGRVTMTDRDTSISTAYMELSR LRSDDTAVY YCARSEGTTEFDYWGQGLTVTVSSGGGSGGGGSGG GGSDIVLTQSPDSLAVSLGERATINCKSSESVD SYANS FMHWYQQKPGQPPKLLIYRASTRESGVDRFSGSGSR TDFTLTISSLQAEDVAVYYCQSKEDPLTFGGGTKEI KASTTTPAPRPPTPAPTIASQPLSLRPEACRPAAGGAV HTRGLDFACDFWLVVVGGVLACYSLLVTVAFIIFW VRSKSRLLHSDYMNMTPRRPGPTRKHYPYAPPRDF AAYRSLRVKFSRSADAPAYQQGNQLYNELNLRRE EYDVLDKRRGRDPEMGGKPRRKNPQEGLYNELQKD KMAEAYSEIGMKGERRRKGHDGLYQGLSTATKDTY DALHMQALPPRTSGSGATNFSLLKQAGDVEENPGPSR MDMADEPLNGSHTWLSIPFDLNGSVVSTNTSNQTEPY YDLTSSNAVLTFIYFVVCIIIGLCGNTLVIIYVILRYAKMK TITNIYILNLAIADLFMLGLPFLAMQVALVHWPFGKA ICRVMTVDGINQFTSIFCLTVMSIDRYLAVVHPKSA KWRRPRTAKMITMAVWGVSLLVILPIMIYAGLR SNQ WGRSSCTINWPGESGAWYTGFIITYFILGFLVPLTIICL CYLFIIIVKSSGIRVGSSKRKKSEKKVTRMVSIVVAVF IFCWLPPYIFNVSSVSMASPTPALKGMFDFVVVLTYA NSCANPILYAF LSDNFKKSFQNVLC LVK VSGTDDGER SDSKQDKSRLNETTETQRTLLNGDLQTSI
		CAGGTGCAGCTCGTGCAGAGCGGCGCCGAGGTGAA GAAGCCCGCGCTTCCGTGAAGGTGTCCTGTAAGG CCTCTGGCTACATCTTACCGCCTACACCATGCACT GGGTGCGCCAGGCCCCGGGACAGGGCTGGAGTGG ATGGGCTGGATCAAGCCTAACACGGTCTGGCCAA CTACGCGCAGAAGTCCAGGGCCGCTGACCATGA CTCGGACACCGCATCTCTACCGGTACATGGAG CTGTCCCGCTGCGCTCCGATGACACGGCCGTGTAC TACTGCGCCCGCTCGGAGATCACCAAGGTTCTGACT ATGGGGCCAGGGCACCTGGTGACCGTGTCTGTCG gtggaggcggttcaggcgaggtggctctggcggtggcgatcgGACATCG TGCTGACCAGAGCCCTGACAGCCTGGCCGTGTCCC TGGCGAGCGCGCCACCATTAACGCAAGAGCTCC GAGAGTGTGATAGTACGCCAATCTCTCATGCAC TGGTACCAGCAGAAGCCCGCCAGCCACCAAGCT GCTGATCTACAGGCTTCCACCGCGAGAGCGGCG TCCCCGACAGGTTTTTCAGGTTCTGGCTCCCGCACCG
111	Affy6 CAR (control)	

TABLE 18-continued

Amino Acid and Nucleotide Sequences for Exemplary Polypeptides Described Herein		
SEQ ID NO:	Name	Sequence
		ACTTCACCCTGACCATCTCTTCTCTGCAGGCCGAGG ATGTGGCGGTGTAAGTCTCAACAGTCCAAAGGAG GACCCGCTTACCTTCGGCGGCGGCACCAAGGTGGA GATCAAGGctagcacgaccactccggcgccgccccaccgactccggcccc aactatcgcgagccagccccctgtcgctgaggccggaagcatgccgccctgccgcc ggaggtgctgtgcataccggggatggacttcgcatgcgacTTTGGGTG CTGGTGGTGGTTGGTGGAGTCTGGCTTGCTATAGC TTGCTAGTAACAGTGGCCTTTATTATTTCTGGGTG AGGAGTAAGAGGAGCAGGCTCCTGCACAGTGACTA CATGAACATGACTCCCCGCGCCCCGGGCCACCC GCAAGCATTACAGCCCTATGCCCCACACGCGACT TCGCAGCCTATCGCTCCctgagagtgaagtccagcaggagcgcaga cgcccccgctaccagcagggccagaaccagctctataacgagctcaatctaggac gaagagaggagtacgatgtttggacaagagacgtggccgggaccctgagatggg gggaaagccgagaaggaagaacctcaggaaaggcctgtacaatgaactgcagaaa gataagatggcgaggcctacagtgaattgggatgaaaggcgagcgccggagg ggcaaggggcacgatggcctttaccagggtctcagtagccaccaaggacaccta cgacgccccctcacatgcaggccctgccccctcgc
112	Affy6-I3A CAR	CAGGTGCAGCTCGTGACAGCGGCGCCGAGGTGAA GAAGCCCGCGCTTCCGTGAAGGTGTCCTGTAAGG CCTCTGGCTACATCTTCACCGCTACACCATGCACT GGGTGCGCCAGGCCCGGGACAGGGCTGGAGTGG ATGGGCTGGATCAAGCCTAACACGGTCTGGCCAA CTACGCGCAGAAGTTCCAGGGCCGCGTGACCATGA CTCGGGACACACGATCTCTACCGCTACATGGAG CTGTCCCGCTGCGCTCCGATGACACGGCCGTGTAC TACTGCGCCCGCTCGGAGGCCACCCAGTTCGACT <u>ATTGGGGCCAGGGCACCCCTGGTGACCGTGTCTGCTCCg</u> gtggaggcggttcaggcgagggtggctctggcggtggcggatcgGACATCG TGCTGACCCAGAGCCCTGACAGCCTGGCCGTGTCCC TGGGCGAGCGCGCCACCATTAAGTCAAGAGCTCC GAGAGTGTGATAGCTACGCCAACTCCTTCATGCAC TGGTACCAGCAGAAGCCCGGCCAGCCACCCAAGCT GCTGATCTACAGGGCTTCCACCCGCGAGAGCGGCG TCCCCGACAGGTTTTTCAGTTCCTGGCTCCCGCACCG ACTTCACCCTGACCATCTCTTCTCTGCAGGCCGAGG ATGTGGCGGTGTAAGTCTCAACAGTCCAAAGGAG GACCCGCTTACCTTCGGCGGCGGCACCAAGGTGGA GATCAAGGctagcacgaccactccggcgccgccccaccgactccggcccc aactatcgcgagccagccccctgtcgctgaggccggaagcatgccgccctgccgcc ggaggtgctgtgcataccggggatggacttcgcatgcgacTTTGGGTG CTGGTGGTGGTTGGTGGAGTCTGGCTTGCTATAGC TTGCTAGTAACAGTGGCCTTTATTATTTCTGGGTG AGGAGTAAGAGGAGCAGGCTCCTGCACAGTGACTA CATGAACATGACTCCCCGCGCCCCGGGCCACCC GCAAGCATTACAGCCCTATGCCCCACACGCGACT TCGCAGCCTATCGCTCCctgagagtgaagtccagcaggagcgcaga cgcccccgctaccagcagggccagaaccagctctataacgagctcaatctaggac gaagagaggagtacgatgtttggacaagagacgtggccgggaccctgagatggg gggaaagccgagaaggaagaacctcaggaaaggcctgtacaatgaactgcagaaa gataagatggcgaggcctacagtgaattgggatgaaaggcgagcgccggagg ggcaaggggcacgatggcctttaccagggtctcagtagccaccaaggacaccta cgacgccccctcacatgcaggccctgccccctcgc
113	Affy6-I3W CAR	CAGGTGCAGCTCGTGACAGCGGCGCCGAGGTGAA GAAGCCCGCGCTTCCGTGAAGGTGTCCTGTAAGG CCTCTGGCTACATCTTCACCGCTACACCATGCACT GGGTGCGCCAGGCCCGGGACAGGGCTGGAGTGG ATGGGCTGGATCAAGCCTAACACGGTCTGGCCAA CTACGCGCAGAAGTTCCAGGGCCGCGTGACCATGA CTCGGGACACACGATCTCTACCGCTACATGGAG CTGTCCCGCTGCGCTCCGATGACACGGCCGTGTAC TACTGCGCCCGCTCGGAGTGGACCCAGTTCGACT <u>ATTGGGGCCAGGGCACCCCTGGTGACCGTGTCTGCTCCg</u> gtggaggcggttcaggcgagggtggctctggcggtggcggatcgGACATCG TGCTGACCCAGAGCCCTGACAGCCTGGCCGTGTCCC TGGGCGAGCGCGCCACCATTAAGTCAAGAGCTCC GAGAGTGTGATAGCTACGCCAACTCCTTCATGCAC TGGTACCAGCAGAAGCCCGGCCAGCCACCCAAGCT GCTGATCTACAGGGCTTCCACCCGCGAGAGCGGCG

TABLE 18-continued

Amino Acid and Nucleotide Sequences for Exemplary Polypeptides Described Herein		
SEQ ID NO:	Name	Sequence
		TCCCCGACAGGTTTTCAGGTTCTGGCTCCCGCACCG ACTTCACCCTGACCATCTCTTCTCTGCAGGCCGAGG ATGTGGCGGTGTACTACTGTCAACAGTCCAAGGAG GACCCGCTTACCTTCGGCGGCGGCACCAAGGTGGA GATCAAGGctagcacgaccactccggcgccgcccaccgactccggcccc aactatcgcgagccagccctgtcgctgaggccggaagcatgccgccctgccgcc ggaggtgctgtgcataccggggatggacttcgcatgcgacTTTGGGTG CTGGTGGTGGTGGTGGAGTCTGGCTTGCTATAGC TTGCTAGTAACAGTGGCCTTTATTATTTCTGGGTG AGGAGTAAGAGGAGCAGGCTCCTGCACAGTGACTA CATGAACATGACTCCCCGCCGCCCGGGCCCCACCC GCAAGCATTACCAGCCCTATGCCCCACACGCGACT TCGCAGCCTATCGCTCCctgagagtgaagttcagcaggagcgaga cgcccccggtaccagcagggccagaaccagctctataacgagctcaatctaggac gaagagagagtagtacyatgttttggaacaagagacgtggccgggacccctgagatggg gggaagccgagaaggaagaaccctcaggaaggcctgtacaatgaactgcagaaa gataagatggcgaggcctacagttagattgggatgaaaggcgagcgccggagg ggcaaggggacagatggcctttaccagggtctcagtagcaccacaaggacaccta cgacgcccttcacatgcaggccctgccccctgcg
114	Affy6-I3F CAR	CAGGTGCAGCTCGTGCAGAGCGGCGCCGAGGTGAA GAAGCCCGCGCTTCCGTGAAGGTGTCTGTAAAGG CCTCTGGCTACATCTTACCGCTTACCATGCACT GGGTGCGCCAGGCCCCGGGACAGGGGCTGGAGTGG ATGGGCTGGATCAAGCCTAACACGGTCTGGCCAA CTACGCGCAGAAGTTCCAGGGCCGCTGACCATGA CTCGGGACACAGCATCTCTACCGCGTACATGGAG CTGTCCCGCTGCGCTCCGATGACACGGCCGTGTAC TACTGCGCCCGCTCGGAGTTCACACCGAGTTCGACT <u>ATTGGGGCCAGGGCACCCCTGGTGACCGTGTCTGTCG</u> gtggaggcggttcaggcgagggtggctctggcggtggcgatcgGACATCG TGCTGACCCAGAGCCCTGACAGCCTGGCCGTGTCC TGGGCGAGCGCGCCACCATTAACTGCAAGAGCTCC GAGAGTGTGATAGCTACGCCAACTCCTTCATGCAC TGGTACCAGCAGAAGCCCGGCCAGCCACCCAAGCT GCTGATCTACAGGGCTTCCACCCGCGAGAGCGGCG TCCCCGACAGGTTTTTACAGTTCTGGCTCCCGCACCG ACTTCACCCTGACCATCTCTTCTCTGCAGGCCGAGG ATGTGGCGGTGTACTACTGTCAACAGTCCAAGGAG GACCCGCTTACCTTCGGCGGCGGCACCAAGGTGGA GATCAAGGctagcacgaccactccggcgccgcccaccgactccggcccc aactatcgcgagccagccctgtcgctgaggccggaagcatgccgccctgccgcc ggaggtgctgtgcataccggggatggacttcgcatgcgacTTTGGGTG CTGGTGGTGGTGGTGGAGTCTGGCTTGCTATAGC TTGCTAGTAACAGTGGCCTTTATTATTTCTGGGTG AGGAGTAAGAGGAGCAGGCTCCTGCACAGTGACTA CATGAACATGACTCCCCGCCGCCCGGGCCCCACCC GCAAGCATTACCAGCCCTATGCCCCACACGCGACT TCGCAGCCTATCGCTCCctgagagtgaagttcagcaggagcgaga cgcccccggtaccagcagggccagaaccagctctataacgagctcaatctaggac gaagagagagtagtacyatgttttggaacaagagacgtggccgggacccctgagatggg gggaagccgagaaggaagaaccctcaggaaggcctgtacaatgaactgcagaaa gataagatggcgaggcctacagttagattgggatgaaaggcgagcgccggagg ggcaaggggacagatggcctttaccagggtctcagtagcaccacaaggacaccta cgacgcccttcacatgcaggccctgccccctgcg
115	Affy6-I3G CAR	CAGGTGCAGCTCGTGCAGAGCGGCGCCGAGGTGAA GAAGCCCGCGCTTCCGTGAAGGTGTCTGTAAAGG CCTCTGGCTACATCTTACCGCTTACCATGCACT GGGTGCGCCAGGCCCCGGGACAGGGGCTGGAGTGG ATGGGCTGGATCAAGCCTAACACGGTCTGGCCAA CTACGCGCAGAAGTTCCAGGGCCGCTGACCATGA CTCGGGACACAGCATCTCTACCGCGTACATGGAG CTGTCCCGCTGCGCTCCGATGACACGGCCGTGTAC TACTGCGCCCGCTCGGAGGGCACCAACCGAGTTCGACT <u>ATTGGGGCCAGGGCACCCCTGGTGACCGTGTCTGTCG</u> gtggaggcggttcaggcgagggtggctctggcggtggcgatcgGACATCG TGCTGACCCAGAGCCCTGACAGCCTGGCCGTGTCC TGGGCGAGCGCGCCACCATTAACTGCAAGAGCTCC GAGAGTGTGATAGCTACGCCAACTCCTTCATGCAC TGGTACCAGCAGAAGCCCGGCCAGCCACCCAAGCT GCTGATCTACAGGGCTTCCACCCGCGAGAGCGGCG TCCCCGACAGGTTTTTACAGTTCTGGCTCCCGCACCG ACTTCACCCTGACCATCTCTTCTCTGCAGGCCGAGG ATGTGGCGGTGTACTACTGTCAACAGTCCAAGGAG GACCCGCTTACCTTCGGCGGCGGCACCAAGGTGGA GATCAAGGctagcacgaccactccggcgccgcccaccgactccggcccc aactatcgcgagccagccctgtcgctgaggccggaagcatgccgccctgccgcc ggaggtgctgtgcataccggggatggacttcgcatgcgacTTTGGGTG CTGGTGGTGGTGGTGGAGTCTGGCTTGCTATAGC TTGCTAGTAACAGTGGCCTTTATTATTTCTGGGTG AGGAGTAAGAGGAGCAGGCTCCTGCACAGTGACTA CATGAACATGACTCCCCGCCGCCCGGGCCCCACCC GCAAGCATTACCAGCCCTATGCCCCACACGCGACT TCGCAGCCTATCGCTCCctgagagtgaagttcagcaggagcgaga cgcccccggtaccagcagggccagaaccagctctataacgagctcaatctaggac gaagagagagtagtacyatgttttggaacaagagacgtggccgggacccctgagatggg gggaagccgagaaggaagaaccctcaggaaggcctgtacaatgaactgcagaaa gataagatggcgaggcctacagttagattgggatgaaaggcgagcgccggagg ggcaaggggacagatggcctttaccagggtctcagtagcaccacaaggacaccta cgacgcccttcacatgcaggccctgccccctgcg

TABLE 18-continued

Amino Acid and Nucleotide Sequences for Exemplary Polypeptides Described Herein		
SEQ ID NO:	Name	Sequence
		GCTGATCTACAGGGCTTCCACCCGCGAGAGCGGCG TCCCCGACAGGTTTTTCAGGTTCTGGCTCCCCGACCG ACTTCACCTGACCATCTCTCTCTGACGCGCGAGG ATGTGGCGGTGTAATACTGTCAACAGTCCAAGGAG GACCCGCTTACCTTCGGCGGCGGCACCAAGGTGGA GATCAAGGctagcacgaccactccggcgccgcgcccaccgactccggcccc aactatcgcgagccagccctgtcgctgaggcggaagcatgccgccctgccgc ggaggtgctgtgcataccggggattggacttcgcatgcgacTTTGGGTG CTGGTGGTGGTGGTGGAGTCTGGCTTGCTATAGC TTGCTAGTAACAGTGGCCTTTATTTCTGGGTG AGGAGTAAGAGGAGCAGGCTCCTGCACAGTGACTA CATGAACATGACTCCCCGCGCGCCGGGCCACCC GCAAGCATTACCAGCCCTATGCCCCACCACGCGACT TCGCAGCCTATCGCTCCctgagagtgaagttcagcaggagcgaga cgcccccggtaccagcagggccagaaccagctctataacgagctcaatctaggac gaagagaggagtagcatgttttgacaagagacgtggccgggacccctgagatggg gggaaagccgagaaggaagaacctcaggaaggcctgtacaatgaactgcagaaa gataagatggcgaggccctacagtgagattgggatgaaaggcgagcgccggagg ggcaaggggcacagtgccctttaccagggtctcagtagccaccaaggacaccta cgacgcccttcacatgcaggccctgccccctcgc
116	EF1 α -MycT-Affy6 CAR-SSTR2 (control)	ggctccgggtgcccgctcagtgggcagagcgacacatcgcccacagtccccgagaagtt gggggagggggtcggaatgaaccgggtgcctagagaaggtggcgcggggtaaa ctgggaaagtgatgtcggtactggctccgccttttcccgagggtgggggagaacc gtataaagtgcagtagtcgctggaacgttcttttcgcaacgggtttgccgccagaa cacaggtaagtgcggtgtgtggttcccgcgggcctggcctctttacgggttatggccc ttgctgtgcctgaattacttcacctggtgcagtagtgattcttgatcccgagcttcgg gttggaagtggggggagagttcgaggccttgcgcttaaggagcccttcgcctcgt gcttgagttgaggcctggcctggcgctggggccgcgcgtgcgaatctggtggca ccttcgcgcctgtctcgctgctttcgataagttcttagccatttaaaattttgatgacctg ctgcgacgtctttttctggcaagatagttctgtaaatggggccaagatctgcacactg gtatttcggtttttggggccgcgggcgagcgaggggcccgctgcgtcccagcgacacat gttccggcgaggcgggcctgcgagcgcgggccaccgagaatcggaagggggtagt ctcaagctggccggcctgctctggtgcctggtctcgcgccgcctgtatcgccccgc cctggcgcgcaaggctggcccggtcggcaccagttgctgagcggaagatggcc gcttcccgccctgtctcagggagctcaaaatggaggacgcggcgctcgggagag cgggcggtgagtcacccacacaaagaaaaggcccttccgtcctcagccgtcgc ttcattgtgactccacggagtaccgggcccgtccaggcacctcgattagttctcgagc ttttgagtagctcgtcttttaggttggggggagggttttatgcatggagtttccccac actgagtggtggagactgaagttaggccagcttgccacttgatgtaattctccttgga atttgccctttttgagtttgatcttggttcattctcaagcctcagacagtggttcaaagttt tttctctccatttcaggtgtcgtgacaagtttgtaaaaaagcaggctgccaccatggc cttaccagtgaccgccttgcctcgcgctggccttgcgtcctccacgcccgcaggccg ggatccgaacaaaaactcatctcagaagagatctgggatcgCAGGTGCAG CTCGTGCAGAGCGCGCCGAGGTGAAGAAGCCCGG CGCTTCCGTGAAGGTGTCTGTAAAGCCTCTGGCTA CATCTTACCCGCTACACCATGCATGGGTGCGCCA GGCCCCGGGACAGGGGCTGGAGTGGATGGGCTGGA TCAAGCCTAACACGGTCTGGCCAACTACGCGCAG AAGTTCAGGGCCGCGTGACCATGACTCGGGACAC CAGCATCTCTACCGGTACATGGAGCTGTCCGCGCT GCGCTCCGATGACACGGCCGTGTACTACTGCGCCCG <u>CTCGGAGATCACACCGAGTTCGACTATTGGGCCAG</u> GGCACTTGGTGACCGTGTCTGTCggtggaggcggttcaggc ggaggtggctctggcggtggcggtcgGACATCGTGTGACCCAG AGCCCTGACAGCCTGGCCGTGTCCCTGGGCGAGCG CGCCACCATTAAGTGAAGAGTCCGAGAGTGTCG ATAGCTACGCCAACTCCTTCATGCACTGGTACCAGC AGAAGCCCGGCCAGCCACCAAGCTGCTGATCTAC AGGGCTTCACCCGCGAGAGCGGCGTCCCCGACAG

TABLE 18-continued

Amino Acid and Nucleotide Sequences for Exemplary Polypeptides Described Herein		
SEQ ID NO:	Name	Sequence
		GTTTTCAGGTTCTGGCTCCCGACCGACTTCACCCT GACCATCTCTTCTCTGAGGCCGAGGATGTGGCGGT GTACTACTGTCAACAGTCCAAGGAGGCCCGCTTA CCTTCGGCGCGGCCACCAAGGTGGAGATCAAGgctag cagcaccactccggcgccgcgcccaccgactccggccccaactatcgcgagccag cccctgtcgctgagggccgaagcatgccgcctgccgcggaggtgctgtgcatac ccggggattggacttcgcatcgacTTTGGGTGCTGGTGGTGGT TGGTGGAGTCTGGCTTGTATAGCTTGTCTAGTAAC AGTGGCCTTATTATTTCTGGGTGAGGAGTAAGAG GAGCAGGCTCCTGCACAGTGAATACATGAACATGA CTCCCGCGCGCCCGGGCCACCCGCAAGCATTACC AGCCCTATGCCCCACACGCGACTTCGCAGCCTATC GCTCCctgagagtgaagttcagcaggagcgagacgcccccgctaccagca ggggcagaaacagctctataacgagctcaatctaggacgaagagaggagtagcatg ttttggacaagagacgtggccgggacccctgagatggggggaagccgagaaggaa gaaccttcaggaaggcctgtacaatgaactgcagaaagataagatggcgaggcct acagtgagat tgggatgaaaggcgagcgccggaggggcaaggggcacgatggcc tttaccagggtctcagtacagccaccaaggacacctacgagcccttcacatgcagg cccctgccccctcgactagtggaggcgagctactaacttcagcctgctgaagcagg ctggagacgtggaggagaacctggaccttctagaatggacatggcgatgagcca ctcaatggaagccacacatggctatccattccatttgacctcaatggctctgtggtgtca accaacacctcaaacagacagagcgctactatgacctgacaagcaatgcagtcctc acattcatctattttgtggtctgcatcattgggttgtgtggcaacacacttgtcatttatgtc atcctccgctatgccaaagatgaagaccatcaccaacattacatcctcaacctggccat cgcagatgagctcttcagctgggtctgcctttcttggctatgcaggtggctctgggtcca ctggccctttggcaaggccatttgccgggggtcatgactgtggtatggcatcaatcagt tcaccagcatcttctgctgacagtcagagcatcgaccgatacctggctgtggtccac cccatcaagtccggccaagtggaggagaccccgacggccaagatgatcaccatgg ctgtgtggggagtcctctgctgggtcatcttgcctcatgat atatgctgggtcccgga gcaaccagtggggggagaagcagctgcaccatcaactggccagggtgaatctggggc ttggtagacagggttcacatctacaactttcattctgggttcctggtaacctcaccatc atctgtcttttgctacctgttcattatcatcaaggtgaagtcctctggaatccagagtggtg cctctaagaggaagaagtcctgagaagaaggtcacccgaatgggtgtccatcgtggtgg ctgtcttcattctgctgggttcctctacatattcaacgtttcttcgctctccatggcca tcagccccacccagccctaaaggcatgttgactttgtggtggtcctcactatgcta acagctgtgccaacctatcctatatgctctctgtctgacaacttcaagaagagcttcc agaatgtcctctgcttggtcaaggtgagcgccacagatgatggggagcggagtgac agtaagcaggacaatccccgctgaatgagaccacggagaccacaggaccctcc tcaatggagacctccaaacagtatctaa
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TABLE 18-continued

Amino Acid and Nucleotide Sequences for Exemplary Polypeptides Described Herein		
SEQ ID NO:	Name	Sequence
		CTCGGAGGCCACCACCGAGTTCGACTATGGGGCCAG GGCACCTGGTGACCGTGTCTCCggtggaggcggttcaggc ggaggtggctctggcggtggcggtatcgGACATCGTGTGACCCAG AGCCCTGACAGCCTGGCCGTGTCCCTGGGCGAGCG CGCCACCATTAACTGCAAGAGTCCGAGAGTGTCG ATAGCTACGCCAACTCCTTCATGCACTGGTACCAGC AGAAGCCCGGCCAGCCACCAAGCTGCTGATCTAC AGGGCTTCACCCGCGAGAGCGGCGTCCCCGACAG GTTTTCAAGTTCTGGCTCCCGCACCGACTTCACCCT GACCATCTCTTCTGTCAGGCGGAGGATGTGGCGGT GTACTACTGTCAACAGTCCAAGGAGACCCGCTTA CCTTCGGCGGCGGCCAACAGGTGGAGATCAAGgctag cacgaccactccggcgccgcgcccaccgactccggcccaactatcgcgagccag ccoctgtcgctgaggccggaagcatgccgcctcgcccgagggtgctgtgcatac ccggggattggacttcgcatgcgacTTTGGGTGCTGGTGGTGGT TGGTGGAGTCTGGCTTGCTATAGCTTGCTAGTAAC AGTGGCCTTATTATTTCTGGGTGAGGAGTAAGAG GAGCAGGCTCTGCACAGTGACTACATGAACATGA CTCCCCGCGCCCCGGGCCACCCGCAAGCATTACC AGCCCTATGCCCCACCACGCGACTTCGACGCTATC GCTCCctgagagtgaagttcagcaggagcgagacgcccccggtaccagca gggcccagaaccagctctataacgagctcaatctaggacgaagagagggtacgatg tttggacaagagacgtggccgggaccctgagatgggggaaagccgagaaggaa gaaccctcaggaagcctgtacaatgaactgcagaaagataagatggcgaggcct acagtggatggaagcgagcgccggaggggcaaggggcacgatggcc ttaccagggtctcagtagcagccaccaaggacacctacgacgcccttcacatgcagg ccctgccccctcgactagtgaagcgagctactaacttcagcctgctgaagcagg ctggagacgtggaggagaaccctggaccttctagaatggacatggcgatgagcca ctcaatggaagccacacatggctatccattccatttgacctcaatggctctgtggtgtca accaacacctcaaacagacagagccgtactatgacctgacaagcaatgcagtcctc acattcatctatttgggtctgcacatctgggttggtggcaacacacttgctattatgtc atcctccgctatgccaaagatgaagaccatcaccaacatttacatcctcaacctggccat cgcagatgagctcttcagctgggtctgaccttcttggtatgacaggtgctctgggtcca ctggccctttggcaaggccatttgccgggtggtcatgactgtggatggcatcaatcagt tcaccagcatcttctgcctgacagtcagtagcatcgaccgatacctggctgtggtccac cccatcaagtccggcaagtggaggagacccggacggccaagatgatcaccatgg ctgtgtgggagtcctctgctgggtcatcttggccatcatgatatactggggtccggga gcaaccagtgggggagaagcagctgcacatcaactggccaggtgaatctggggc ttggtacacaggggtcatcatctacactttcattctggggttcctgggtacccctcaccatc atctgtcttctgctacctgttcattatcatcaagtggaagtcctctggaatccagtggtgct cctctaagaggaagaagctctgagaagaaggtcacccgaatggtgtccatcgtggtgg ctgtcttcatcttctgctgggttcctctcatatcaacgtttcttcggtctccatggcca tcagccccccccagcccttaaggcattgttgactttggtgggtggtcctcactatgcta acagctgtgccaacctatcctatatgcctcttctgtctgacaacttcaagaagagcttcc agaatgtcctctgcttggtcaaggtgagcggcacagatgatggggagcggagtgac agtaagcaggacaaatccggctgaatgagaccaggagaccagaggacctcc tcaatggagacctccaaaccagtatctaa
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TABLE 18-continued

Amino Acid and Nucleotide Sequences for Exemplary Polypeptides Described Herein		
SEQ ID NO:	Name	Sequence
		CTCGTGCAGAGCGGCGCCGAGGTGAAGAAGCCCGG CGCTTCCGTGAAGGTGTCTGTAAAGCCTCTGGCTA CATCTTACCCGCTACACCATGCACCTGGGTGCGCCA GGCCCCGGACAGGGGCTGGAGTGGATGGGCTGGA TCAAGCCTAACACGGTCTGGCCAACTACGCGCAG AAGTTCCAGGGCCGCTGACCATGACTCGGGACAC CAGCATCTCTACCGGTACATGGAGCTGTCCGCCT GCGCTCCGATGACACGGCCGTGTACTACTGCGCCCG CTCGGAGTGGACCACCGAGTTCGACTATTGGGGCCAG GGCACCTTGGTGACCGTGTCTGTCGgtggaggcggttcaggc ggaggtggctctggcggtggcgatcgGACATCGTGCTGACCCAG AGCCCTGACAGCCTGGCCGTGTCTTGGGCGAGCG CGCCACCATTAACTGCAAGAGCTCCGAGAGTGTCTG ATAGCTACGCCAACTCCTTCATGCACTGGTACCAGC AGAAGCCCGGCCAGCCACCAAGCTGTGTATCTAC AGGGCTTCCACCCGCGAGAGCGGCGTCCCGACAG GTTTTCAGGTTCTGGCTCCCGCACCGACTTCACCT GACCATCTCTCTCTGCAAGGCCGAGGATGTGGCGGT GTACTACTGTCAACAGTCCAAGGAGGACCCGCTTA CCTTCGGCGGCGCACCAAGGTGGAGATCAAGgctag cagcaccactccggcgccgccccaccgactccggcccaactatcgcgagccag cccctgtcgtgagggcggaagcatgcccgcctgcccggaggtgctgtgcatatc ccggggattggacttcgcatgcgactTTTGGGTGTGTGTGTGGT TGGTGAGATCCTGGCTTGTCTATAGCTTGTCTAGTAAC AGTGGCCTTATTATTTCTGGGTGAGGAGTAAGAG GAGCAGGCTCCTGCACAGTGACTACATGAACATGA CTCCCCGCGCCCCGGGCCACCCGCAAGCATTACC AGCCCTATGCCCCACACGCGACTTCGCAACCTATC GTTCCctgagagtgaagttcagcaggagcgagacgccccgcgtaccagca gggccagaaccagctctataacgagctcaatctaggacgaagagaggagtacgatg ttttggacaagagacgtggccgggaccctgagatgggggaaagccgagaagga gaaccctcaggaaggcctgtacaatgaactgcagaaagataagatggcgaggcct acagtgagattgggatgaaaggcgagcgccggaggggcaaggggcacgatggcc ttaccagggtctcagtagcagccaccaaggacacctacgacgcccctcacatgacgg ccctgccccctcgactagtgaagcggagctactaacttcagcctgctgaagcagg ctgggagcgtggaggagaaccctggacctcttagaatggacatggcggtgagcca ctcaatggaagccacacatggctatccattccatttgacctcaatggctctgtggtgca accaacacctcaaacagacagagccgtactatgacctgacaagcaatgcagtcctc acattcatctattttgggtctgcatcatgggttggttggaacacacttgctattatgtc atcctccgctatgccaaagtgaagaccatcaccaacattacatcctcaacctggccat cgcagatgagctctcatgctgggtctgccttcttggtatgacaggtggctctgggtcca ctggccctttggcaaggccatttgcggggtggctcatgactgtggtggcatcaatcagt tcaccagcatctctgcctgacagtcagtagcatcgaccgataacctggctgtggtccac cccatcaagtccggcaagtgaggagaccccggaaggccaagatgatcaccatgg ctgtgtggggagttctctgctgggtcatcttgcccatcatgatatgctgggtcccgga gcaaccagtgggggagaagcagctgcaccatcaactggccaggtgaatctggggc ttggtacacagggttcacatctacactttcattctgggggttccctggtaaccctcaccatc atctgtctttgctacctgttcattatcatcaagggtgaagtcctctggaatccgagtggtgct ccttaaggaagaagtcctgagaagaaggtcaccgcaatggtgtccatcgtgggtgg ctgtcttcatctctgctggcttccctctacatattcaacgtttctctcgtctccatggcca tcagccccaccacagccctaaaggcatgtttgactttgtgggtggtctccatcatgcta acagctgtgccaaacctatcctatatgccttctgtctgacaactcaagaagagcttcc agaatgtcctctgcttgggtcaaggtgagcgccacagatgatggggagcggagtgac agtaagcaggacaatcccggtgtaatgagaccacggagaccagaggacctcc tcaatggagagctccaaaccagtatctaa
119	EF1a-MycT-Affy6- I3F CAR-SSTR2	ggctccgggtgcccgtcagtgggcgagcgcacatcgcccacagtcgcccgagaagtt ggggggaggggtcggaatgaaccgggtgcctagagaaggtggcgcggggtaaa ctgggaaagtgtgctgctgactggctccgcctttttcccgagggtgggggagaacc gtataaagtgcagtagtcgcgctgaacgttcttttgcgaacgggtttgcccgcagaa cacaggtaagtccgctgtgtggttcccgcgccctggcctctttacgggttatggccc ttgctgtgcctgaattacttccacctggctgcagtagctgattcttgatcccgagcttcgg gttggaagtgggtgggagagttcgaggccttgccgttaaggagcccttcgctcgt gcttgagttagggcctggcctggcgctggggccgcgcgtgcgaatctgggtgga ccttcgcgccctgtctcgtgctttcgataaagtctctagccatttaaaattttgatgacctg ctgagacgcttttttctggcaagatagttctgtaaatggggccaagatctgcacactg gtattcgtttttggggccgcgggcgagcggggcccgctgcgtcccagcgacat gttcggcgaggcgggcctgcgagcgcgccaccgagaatcgagcggggtagt ctcaagctggccggcctgctctggtgctggtctcgcgcccgctgtatcgccccgc cctggggcggaaggctggcccggtcggaaccagttgctgagcgggaaagatggcc gcttcccgccctgctgcaggagctcaaaatggaggacgcggcgctcgggagag

TABLE 18-continued

Amino Acid and Nucleotide Sequences for Exemplary Polypeptides Described Herein		
SEQ ID NO:	Name	Sequence
120	EF1 α -MycT-Affy6- I3G CAR-SSTR2	cgggcggtgagtcacccacacaaaggaaaaggcctttccgtcctcagccgtcgc ttcatgtgactccacggagtagcgggcccgtccaggcacctcgattagttctcgagc ttttggagtagctcgtctttaggttgggggaggggtttatgcgtaggagtttccccac actgagtggtggagactgaagttaggccagcttggcacttgatgaattctccttgga atttgccctttttaggttggatcttggttcattctcaagcctcagacagtgggtcaaaagttt ttttcttccatttcagggtgctgacaaagtttgtacaaaaagcaggctgccaccatggc cttaccagtgacgccttgctcctgccgtggccttgctgctccacgcgcggcaggccg ggatccgaacaaaaactcatctcagaaggagatctgggatcgCAGGTGCAG CTCGTGACAGAGCGGCGCGAGGTGAAGAAGCCCGG CGCTTCCGTGAAGGTGTCTGTAGGCCCTCTGGCTA CATCTTACC GCCTACACCATGCACTGGGTGCGCCA GGCCCCGGACAGGGGCTGGAGTGGATGGGTGGA TCAAGCCTAACACGGTCTGGCCAACTACGCGCAG AAGTTCAGGGCGCGGTGACCATGACTCGGGACAC CAGCATCTCTACCGGTACATGGAGCTGTCCGCCT GCGCTCCGATGACACGGCGGTGACTACTGCGCCG CTCGGAGTTCACACCGAGTTCGACTATTGGGGCCAG GGCACCTGGTGACCGTGTCTCGTCCggtggagggcggttcaggc ggaggtggctctggcggtggcggtcgGACATCGTGTGACCCAG AGCCCTGACAGCCTGGCCGTGTCTGGGCGAGCG CGCCACCATTAAGTCAAGAGCTCCGAGAGTGTCTG ATAGCTACGCCAACTCCTTATGCACTGGTACCAGC AGAAGCCCGGCGAGCCACCAAGTGTCTGATCTAC AGGGCTTCACCCGCGAGAGCGGCGTCCCGACAG GTTTTAGGTTCTGGCTCCCGACCGACTTCACCT GACCATCTCTTCTCTGAGGCGAGGATGTGGCGGT GTACTACTGTCAACAGTCCAAGGAGGACCCGCTTA CCTTCGGCGCGGCACCAAGGTGGAGATCAAGgctag cacgaccactccggcgccgcgcccacccgactccgcccccaatcgcgcgagccag ccccgtcgctgagggccgaagcatgcgcgcctgcgcgcggaggtgctgtgcatac ccggggattggacttcgcatgcgacTTTGGGTGCTGGTGGTGGT TGGTGGAGTCTGGCTTGCTATAGCTTGCTAGTAAC AGTGCCCTTTATTATTTCTGGGTGAGGAGTAAGAG GAGCAGGCTCCTGCACAGTGACTACATGAACATGA CTCCCCGCGCCCCGGGCCACCCGCAAGCATTACC AGCCCTATGCCCCACACGCGACTTCGCGAGCCTATC GTCCTctgagagtgaagttcagcaggagcgcagacgcccccgctaccagca ggggcagaaacagctctataacgagctcaatctaggacgaagagaggagtagcatg ttttggacaagagacgtggccgggacccctgagatggggggaagccgagaaggaa gaacctcaggaaggcctgtacaatgaactgcagaaagataagatggcggaggcct acagtgagatgggatgaaaggcgcgcgcggagggggaaggggcacgatggcc tttaccagggtctcagtacagccaccaaggacacctacgacgcccctcacatgcagg ccctgccccctcgactagtggagcggagctactaactcagcctgctgaagcagg ctggagacgtggaggagaacctggacctctagaatggacatggcggatgagcca ctcaatggaagccacacatggctatccattccatttgacctcaatggctctgtggtgtca accaacacctcaaacagacagagccgactatgacctgacaagcaatgcagctctc acattcatctattttgtggtctgcatcattgggttgtgtggcaacacacttgtcattatgtc atcctcgcctatgccaaagtgaagaccatcaccaacattacatcctcaacctggccat cgcagatgagctctcatgctgggtctgccttcttggctatgcagggtggctctgggtcca ctggccctttggcaaggccatttgcgggtggctcatgactgtggatggcatcaatcagt tcaccagcatctctgctgacagtcagagcatcgaccgatacctggctgtggtccac cccatcaagtgcggcaagtggaggagacccggacggccaagatgatcaccatgg ctgtgtgggagttctctgctgggtcatcttgcccatcatgatcatgctgggtccgga gcaaccagtgggggagaagcagctgcaccatcaactggccagggtgaatctggggc ttggtagacagggttcacatctacacttctctgggttctcgggtaccctcaccatc atctgtctttgtacctgttcatatcatcaagggtgaagtcctctggaatccagagtggtg cctctaagaggaagaagtcgtgagaagaaggtcacccgaatgggtgcatcgtggtgg ctgtcttcatctctgctgggttccctctacatattcaacggttcttccgctcctatggcca tcagccccacccagccctaaaggcatgttgactttgtggtggtcctcaccatgcta acagctgtgccaacctatcctatatgcttcttgtctgacaacttcaagaagagcttcc agaatgctctgcttgggtcaaggtagagcgccacagatgatggggagcggagtgac agtaagcaggacaaatcccggtgaatgagaccacggagaccagaggacctcc tcaatggagacctccaaaccagtatctaa
		ggctccggtgcccgctcagtgggcagagcgcacatcgcccacagtcctccgagaagtt ggggggagggtcggaatgaacccggtgcctagagaaggtggcgcggggtaaa ctgggaagtgatgctggtactggctccgcttttcccgagggtgggggagaaac gtataaagtcagtagtcgctgaaagcttcttttcgcaacgggtttgcgcgcagaa cacaggtgaagtgcggtgtggttcccgcggtcctggtctttacgggttatggccc ttgctgcttgaattacttccacgtgctgcagtagctgattctgacccgagcttcgg gttggaggtgggggagagttcgaggccttgccgttaaggagcccttcgctcgt

TABLE 18-continued

Amino Acid and Nucleotide Sequences for Exemplary Polypeptides Described Herein		
SEQ ID NO:	Name	Sequence
		gcttgagttgaggcctggcctggcgctggggccgcgcgtgcgaatctggtggca ccttcgcgcctgtctcgctgcttctcgataagctctcagccatttaaaatttttgatgacctg ctggcgacgcttttttctggcaagatagctctgtaaaatgcggggccaagatctgcacactg gtatttcggtttttggggcgcggcgcgacggggcccgctgcgtccagcgacat gttcggcgaggcggggcctgcgagcgcggccaccgagaatcggacgggggtagt ctcaagctggcggcctgctctggtgacctggtctcgccgcgcgtgtatcgcccgc cctggcgaggcaaggctggcccggtcggcaccagttgcgtgagcggaaagatggcc gcttcccgccctgctgcaggagctcaaaatggaggacgcggcgtcgggagag cggggcggtgagtcacccacacaaaggaaggcccttccgtcctcagccgtcgc ttcatgtgactccacggagtacggcgccgtccaggcacctcgattagttctcgagc ttttggagtacgtcgtcttaggttgggggaggggttttatgcgatggagtttccccac actgagtggtggagactgaagttagggcagcttggcacttgatgtaattctccttgga atttgcctttttaggttggtatcttggttcattctcaagcctcagacagtggttcaagttt ttttcttccatttcaggtgctgacaaagtttgtacaaaaagcaggctgccaccatggc cttaccagtgcgccttgcctcgcgcgtggccttgcgtgctccacgcgcgcaggccg ggatccgaacaaaaactcatctcagaagagatctgggatcgCAGGTGCAG CTCGTGACAGCGCGCCGAGGTGAAGAGCCCGG CGCTTCCGTGAAGGTGCTCTGAAGGCTCTGGCTA CATCTTACCGCTACACCATGCATGGGTGCGCA GGCCCCGGACAGGGCTGGAGTGGATGGGTGGA TCAAGCCTAACACGGTCTGGCCAACACGCGCAG AAGTTCAGGGCCGCTGACCATGACTCGGGACAC CAGCATCTTACCGGTACATGGAGCTGTCCGCT GCGCTCCGATGACACGGCGTGTACTACTGCGCCCG CTCGGAGGACACCGAGTTCGACTATTGGGGCCAG GGCACCCTGGTGACGTGTCTGTCGGtgaggcggttcaggc ggaggtggctctggcggtggcggtatcgGACATCGTGTCGACCCAG AGCCCTGACAGCCTGGCCGTGTCTTGGCGAGCG CGCCACCATTAACTGCAAGAGCTCCGAGAGTGTCTG ATAGCTACGCCAACTCCTTATGCACTGGTACCAGC AGAAGCCCCGCCAGCCACCAAGCTGCTGATCTAC AGGGCTTCCACCGCGAGAGCGGCGTCCCCGACAG GTTTTCAGTTCTGGCTCCCGACCGACTTACCCT GACCATCTTCTCTGCAAGCGAGGATGTGGCGGT GTACTACTGTCAACAGTCCAAGGAGACCCGCTTA CCTTCGGCGGCGCACCAAGGTGGAGATCAAGgctag cacgaccactccggcgccgcgcacccagactccggcccaactatcgcgagccag cccctgtcgtgaggccggaagcatgcccgcctgcgcgcggaggtgctgtgcatac cgggggattggacttcgcatacgacTTTGGGTGCTGGTGGTGGT TGGTGGAGTCTGGCTTGCTATAGCTTGCTAGTAAC AGTGGCTTTATTATTTCTGGGTGAGGAGTAAGAG GAGCAGGCTCTGCACAGTGACTACATGAACATGA CTCCCCGCGCCCGGGCCACCCGCAAGCATTACC AGCCCTATGCCCCACCACGCACTTCGACGCTATC GCTCCctgagagtgaagttcagcaggagcgcagacgcccccgctaccagca ggggcagaaccagctctataacgagctcaatctaggacgaagagaggatgacgatg ttttggacaagagacgtggccgggacccctgagatgggggaaagccgagaaggaa gaacctcaggaagccctgtacaatgaactgcagaaagataagatggcgaggccct acagtgagatgggatgaaaggcgagcgcggaggggcaaggggcacgatggcc tttaccagggtctcagtagacagcccaaggacacctacgacgccccttcacatgcagg ccctgccccctcgactagtggagcggagctactaacttcagcctgctgaagcagg ctggagacgtggaggagaacctggaccttctagaatggacatggcgatgagcca ctcaatggaagccacacatggctatccattccatttgacctcaatggctctgtggtgtca accaacacctcaaacagacagagccgactatgacctgacaagcaatgcagtcctc acattcatctattttggtctgcatcattgggttggtggcaacacacttgctattatgtc atcctccgctatgccaaagatgaagaccatcaccaacattacatcctcaacctggccat cgcagatgagctcttcagctgggtctgccttctctggctatgcagggtgctctgggtcca ctggcctcttggaagggccatttgccgggtggtcatgactgtggatggcatcaatcagt tcaccagcatctctgcctgacagtcatgagcatcgaccgatacctggctgtggtccac cccatcaagtcgcccaagtggaggagaccccgacggccaagatgatcaccatgg ctgtgtggggagtctctgctgggtcatctgcccacatgatataatgctgggtccggga gcaaccagtgggggagaagcagctgcaccatcaactggccaggtgaatctggggc ttggtacacaggggtcatcatctacactttcattctgggggttctgggtacccctcaccatc atctgtcttgctacctgttcattatcatcaagggtgaagtctctgggaatccgagtggtc cctctaagaggaagaagtctgagaagaaggtcacccgaatggtgtccatcgtggtgg ctgtctcatctctgtggttcccttctacatatacaagcttctctcgtctccatggcca tcagccccccccagccctaaaggcatgttgactttggtggtggtcctcactatgcta acagctgtgccaaacctatcctatatgcctctctgtctgacaacttcaagaagagcttcc agaatgtcctctgcttggtcaaggtgagcggcacagatgatggggagcggagtgac agtaagcaggacaaatcccggtgaatgagaccagagaccagaggacctcc tcaatggagacctccaaccagtatctaa

TABLE 18-continued

Amino Acid and Nucleotide Sequences for Exemplary Polypeptides Described Herein		
SEQ ID NO:	Name	Sequence
121	(GGGGS)x3 linker	GGGSGGGSGGGGS
122	4-1BB costimulatory domain	KRGRKLLYIFKQPFMRPVQTTQEEDGCSCRFPEEEE GGCEL
123	CD28 costimulatory domain	RSKRSRLHSDYMNMTPRRPGPTRKHYPYAPPRDFA AYRS
124	OX-40 costimulatory domain	ALYLLRRDQRLPPDAHKKPPGGSFRTPIQEEQADAHS TLAKI
125	CD3 zeta signaling domain	RVKFSRSADAPAYKQGQNLYNELNLGRREEYDVLD KRRGRDPENGGKPRKNPQEGLYNELQDKMAEAY SEIGMKGERRRGKHDGLYQGLSTATKDTYDALHMQ ALPPR
126	CD8 alpha hinge	TTTPAPRPPTPAPTIASQPLSLRPEACRPAAGGAVHTR GLDFACD
127	CD28 hinge	RAAAIEVMYPPPYLDNEKSNGTIIHVKGKHLCPSPFP GPSKPKDPK
128	IgG4 hinge	SKYGPPCPSCP
129	CD8 alpha transmembrane domain	IYIWAPLAGTCGVLLLSLVITLYC
130	CD28 transmembrane domain	FWVLVVGGVLACYSLLVTVAFII
131	ICOS transmembrane domain	FWLPIGCAAFVVVCILGCILICWL
132	GITR transmembrane domain	LGWLTVVLLAVAACVLLLTSAQLGL
133	α L subunit of LFA-1	YNLDVRGARSFSPPRAGRHFYRVLQVGNVIVGAP GEGNSTGSLYQCQSGTGHCPLVTLRGSNYTSKYLGM TLATDPTDGSILACDPGLSRTCDQNTYLSGLCYLFRQN LQGPMLQGRPGFQECIKGNVDLVFLFDGMSLQPDF QKILDFMKDVMKKLSNTSYQFAAVQFSTSYPKTEPDFS DYVKWKDPDALLKHVKHMLLLTNTFGAINYVATEVF REELGARPDATKVLIIITDGEATDSGNIDAAKDIIIR YIIG IGKHFQTKESQETLHKFASKPASEFVKILDTFEKLKDL FTELQKKIYVIEGTSKQDLTSFNMELSSSGISADLSRGH AVVGAVGAKDWAGGFLLDLKADLQDDTFIGNEPLTPE VRAGYLGTVTWLPSRQKTSLLASGAPRYQHMGRLV LFQEPQGGHWSQVQTIHGTQIGSYFGGELCGVDVD QDGETELLLIGAPLFYGEQGRGVFIYQRRQLGFEEVS ELQGDPGYPLGRFGEAITALTDINGDGLVDVAVGAPL EEQGAVYIFNGRHGGLSPQPSQRIEGTQVLSGIQWFGR SIHGVDLEGDGLADVAVGAESQMIVLSRPPVDMV TLMSFSPAEPVHEVECSYSTSNMKKEGVNITICFQIKS LYPQFQGRLVANLTYTLQLDGHRTRRRGLFPGGRHEL RRNIAVTTSMSCDTSFHFPCVQDLISPINVSLNFSW EEEGTPRDQRAQKDIPIILRPSLHSETWEIPFEKNCGE DKKCEANLRVSFSPARSRALRLTAPASLSVELSLNLE

TABLE 18-continued

Amino Acid and Nucleotide Sequences for Exemplary Polypeptides Described Herein		
SEQ ID NO:	Name	Sequence
		EDAYWVQLDLHFPPGLSFRKVEMLKPHSQIPVSCCEL PEESRLLSRALSCNVSSPIFKAGHSVALQMMENTLVNS SWGDSVELHANVTCNNEDSDLEDNSATTIIPILYPINI LIQDQEDSTLYVSFTPKGPKIHQVKHMYQVRIQPSIHD HNIPTLEAVVGVPQPPSEGPITHQWSVQMEPPVPCHYE DLERLPDAEPCPLGALFRCPVVFRRQEIILVQVIGTLEL VGEIEASSMFLCSSLISFNSSKHFHLYGSNASLAQVV MKVDVVYEKQMLYLYVLSGIGGLLLLLIFIVLYKVG FFKRNLKEKMEAGRGVPNGIPAEDSEQLASGQEAGDP GCLKPLHEKDESSEGGGKD
134	I domain of α L subunit of LFA-1	VDLVFLFDGMSLQPDFQKILDFMKDVMKKLSNTSY QFAAVQFSTSYKTEPDFSDYVKWDPDALLKHVKHM LLLTNTFGAINVATEVFREELGARPDATKVLIIITDGE ATDSGNIDAADKIIRYIIIGIGKHFQTKESQETLHKFASK PASEFVKILDTFEKLKDLFTELQKKIYVIE
135	SSTR2 (aa1-381)	MDMADEPLNGSHTWLSIPFDLNGSVVSTNTSNQTEPY YDLTSNAVLTFIYFVVCIIIGLCNTLVIYVILRYAKMK TITNIYILNLAIADELFMLGLPFLAMQVALVHWPPGKA ICRVVMTVDGINQFTSIFCLTVMSIDRYLAVVHPKSA KWRPRPTAKMITMAVWGVSLLVILPIMIIYAGLRSNQ WGRSSCTINWPGESGAWYTGFIITYFILGFLVPLTIICL CYLFIIIVKVKSSGIRVGSSKRKKSEKKVTRMVSIVVAVF IFCWLPFFYIFNVSSVSMAISPTPAKGMFDFVVVLTYA NSCANPILYAFSLDNFKKSFQNVCLVKVSGTDDGER SDSKQDKSRLNETTETQRTLLNGDLQTSI
136	SSTR2 (aa 1-314)	MDMADEPLNGSHTWLSIPFDLNGSVVSTNTSNQTEPY YDLTSNAVLTFIYFVVCIIIGLCNTLVIYVILRYAKMK TITNIYILNLAIADELFMLGLPFLAMQVALVHWPPGKA ICRVVMTVDGINQFTSIFCLTVMSIDRYLAVVHPKSA KWRPRPTAKMITMAVWGVSLLVILPIMIIYAGLRSNQ WGRSSCTINWPGESGAWYTGFIITYFILGFLVPLTIICL CYLFIIIVKVKSSGIRVGSSKRKKSEKKVTRMVSIVVAVF IFCWLPFFYIFNVSSVSMAISPTPAKGMFDFVVVLTYA NSCANPILYAF
137	SSTR2 (aa 1-381) nucleotide sequence	ATGGACATGGCGGATGAGCCACTCAATGGAAGCCA CACATGGCTATCCATTCCATTGACCTCAATGGCTC TGTGGTGTCAACCAACACCTCAAAACAGACAGAGC CGTACTATGACCTGACAAGCAATGCAGTCCTCACAT TCATCTATTTGTGGTCTGCATCATTTGGTGTGTGG CAACACACTTGTCAATTTATGTATCTCCGCTATGC CAAGATGAAGACCATCACCAACATTTACATCCTCA ACCTGGCCATCGCAGATGAGCTTTCATGCTGGGTC TGCCCTTCTTGCGTATGCAGGTGGCTCTGGTCCACT GGCCCTTTGGCAAGGCCATTTGCCGGGTGGTCATGA CTGTGGATGGCATCAATCAGTTCACCAGCATCTTCT GCCTGACAGTCATGAGCATCGACCGATACCTGGCT GTGGTCCACCCCATCAAGTCGGCCAAGTGAGGAG ACCCCGGACGGCCAAGATGATCACCATGGCTGTGT GGGAGTCTCTCTGCTGGTCATCTTGCCCATCATGA TATATGCTGGGCTCCGGAGCAACAGTGGGGGAGA AGCAGCTGACCATCAACTGGCAGGTGAATCTGG GGCTTGGTACACAGGTTTCATCATCTACACTTTCAT TCTGGGTTCTCTGGTACCCCTCACCATCATCTGTCTT TGCTACCTGTTTATTATCATCAAGGTGAAGTCTCT GGAATCCGAGTGGGCTCTCTAAGAGGAAGAAGTC TGAGAAGAAGGTACCCGAATGGTGTCATCGTGG TGGCTGTCTTCATCTTCTGCTGGCTTCCCTTCTACAT ATTCAACGTTTCTTCCGTCTCCATGGCCATCAGCCC CACCCAGCCCTTAAAGGCATGTTTGACTTTGTGGT GGTCTCACCTATGCTAACAGCTGTGCCAACCTAT CCTATATGCCTTCTTGTCTGACAACTTCAAGAAGAG CTTCCAGAATGCTCTGCTTGGTCAAGGTGAGCGG CACAGATGATGGGGAGCGGAGTGACAGTAAGCAGG ACAAATCCCGGCTGAATGAGACCACGGAGACCAG AGGACCCTCTCAATGGAGACCTCCAAACAGTAT CTAA

TABLE 18-continued

Amino Acid and Nucleotide Sequences for Exemplary Polypeptides Described Herein		
SEQ ID NO:	Name	Sequence
138	SSTR2 (aa1-314) nucleotide sequence	ATGGACATGGCGGATGAGCCACTCAATGGAAGCCA CACATGGCTATCCATTCCATTTGACCTCAATGGCTC TGTGGTGTCAACCAACACCTCAAACCAGACAGAGC CGTACTATGACCTGACAAGCAATGCAGTCCTCACAT TCATCTATTTTGTGGTCTGCATCATTTGGGTTGTGTGG CAACACACTTGTCAATTTATGTCATCCTCCGCTATGC CAAGATGAAGACCATCACCAACATTACATCCTCA ACCTGGCCATCGCAGATGAGCTCTTCATGCTGGGTC TGCCTTTCTTGGCTATGCAGGTGGCTCTGGTCCACT GGCCCTTTGGCAAGGCCATTTGCCGGGTGGTCATGA CTGTGGATGGCATCAATCAGTTCACCAGCATCTTCT GCCTGACAGTCATGAGCATCGACCGATACTGGCT GTGGTCCACCCCATCAAGTCGGCCAAGTGGAGGAG ACCCGGACGGCCAAGATGATCACCATGGCTGTGT GGGGAGTCTCTCTGCTGGTCATCTTGCCCATCATGA TATATGCTGGGCTCCGAGCAACAGTGGGGGAGA AGCAGCTGCACCATCAACTGGCCAGGTGAATCTGG GGCTTGGTACACAGGGTTCATCATCTACACTTTCAT TCTGGGTTCTCTGGTACCCCTCACCATCATCTGTCTT TGCTACCTGTTTATTATCATCAAGGTGAAGTCTCT GGAATCCGAGTGGGCTCCTCTAAGAGGAAGAAGTC TGAGAAGAAGGTCAACCGAATGGTGTCCATCGTGG TGGCTGTCTTCTCTCTGCTGGCTCCCTTCTACAT ATTCACGTTTCTTCCGTCTCCATGGCCATCAGCCC CACCCAGCCCTTAAAGGCATGTTTGACTTTGTGGT GGTCTCACCTATGCTAACAGCTGTGCCAACCTAT CGTATAT
139	P2A peptide	GSGATNFSLLKQAGDVEENPGP
140	E2A peptide	GSGQCTNYALLKLAGDVESNPGP
141	T2A peptide	GSGEGRGSLLTCGDVEENPGP
142	F2A peptide	GSGVKQTLNFDLLKLAGDVESNPGP
143	Linker	(GGGS) n
144	AB_001 primer	CTGTGCCACCgcgGGCTCCTACGTGAGC
145	AB_002 primer	TAGTACACGGCGCTGTCC
146	AB_015 primer	CGTGAGCCCTgcgGATTATTGGGGC
147	AB_016 primer	TAGGAGCCGTAGGTGGCA
148	AB_037 primer	CTGTGCCACCgtCGGCTCCTACG
149	AB_038 primer	TAGTACACGGCGCTGTCC
150	AB_039 primer	CTGTGCCACCctCGGCTCCTAC
151	AB_040 primer	CTGTGCCACCatCGGCTCCTAC
152	AB_042 primer	CTGTGCCACCatgGGCTCCTACGTGAGCCC
153	AB_086 primer	CCGCTCGGAGgcCACCACCGAG
154	AB_087 primer	GCGCAGTAGTACACGCC
155	AB_088 primer	CTCGGAGATCgCCACCGAGTTC
156	AB_089 primer	CGGGCGCAGTAGTACACG
157	AB_117 primer	GCGCAGTAGTACACGCC
158	AB_120 primer	CCGCTCGGAGggcACCACCGAGTTC
159	AB_123 primer	CCGCTCGGAGttcACCACCGAGT
160	AB_125 primer	CCGCTCGGAGtgACCACCGAGTTC

TABLE 18-continued

Amino Acid and Nucleotide Sequences for Exemplary Polypeptides Described Herein		
SEQ ID NO:	Name	Sequence
161	EF1a promoter	GGCTCCGGTGCCCGTCAGTGGGCAGAGCGCACATC GCCCACAGTCCCGGAGAAGTTGGGGGAGGGGTCG GCAATTGAACCGGTGCTTAGAGAAGGTGGCGCGGG GTAAACTGGGAAAGTGATGTCGTGACTGGCTCCG CCTTTTCCCGAGGGTGGGGGAGAACCCTATATAA GTGCAGTAGTCGCCGTGAACGTTCTTTTCGCAACG GGTTTGCCGCCAGAACACAGGTAAGTGCCTGTGT GGTCCCGCGGGCCTGGCCTCTTTACGGGTTATGGC CCTTGCGTGCCTTGAATTACTTCCACCTGGCTGCAG TACGTGATTCTTGATCCCGAGCTTCGGTTGGAAGT GGGTGGGAGAGTTCGAGGCCTTGCGCTTAAGGAGC CCCTTCGCCTCGTGCTTGAGTTGAGGCCTGGCCTGG GCGCTGGGGCCGCCGCTGCGAATCTGGTGGCACC TTCGCGCCTGTCTCGCTGCTTTTCGATAAGTCTCTAG CCATTTAAATTTTGGATGACCTGCTGCGACGCTTT TTTTCTGGCAAGATAGTCTTGTAATGCGGGCCAAG ATCTGCACACTGGTATTTTCGGTTTTTGGGGCCGCGG GCGGCGACGGGCGCGTGCCTCCAGCGCACATGT TCGGCGAGGCGGGCCTGCGAGCGCGGCCACCGAG AATCGGACGGGGTAGTCTCAAGCTGGCCGGCCTG CTCTGGTGCTGGTCTCGCGCCGCGGTATCGCCC CGCCCTGGGCGGCAAGGCTGGCCCGGTGCGCACC GTTGCGTGAGCGGAAAGATGGCCGCTTCCCGGCC TGCTGCAGGGAGCTCAAATGGAGGACGCGCGCT CGGAGAGCGGGCGGTGAGTCAACACACAAGG AAAAGGGCCTTCCGTCCTCAGCCGTCGCTTCATGT GACTCCACGAGTACCGGCGCGCTCCAGGCACCT CGATTAGTTCTCGAGCTTTTGAGTACGTCGCTTT AGGTTGGGGGAGGGGTTTTATGCGATGGAGTTTC CCCACTGAGTGGGTGGAGACTGAAGTTAGGCCA GCTTGGCACTTGATGTAATTCCTTGGAAATTGCC CTTTTGAAGTTGGATCTTGGTTCAATCTAAGCCTC AGACAGTGGTTCAAAGTTTTTTCTTCCATTTCAAG TGTCGTGA
162	CD8 alpha chain signal peptide	MALPVTALLPLALLHAARP
163	CD8 alpha chain signal peptide	ATGGCCTTACCAGTGACCGCCTTGCTCCTGCCGCTG GCCTTGCTGCTCCACGCCGCCAGGCGG

*CDR regions underlined and italicized. CDR1, CD2, and CD3 regions are underlined and italicized in the VH and VL sequences. Only HCDR3 regions are underlined and italicized in the scFv and CAR sequences. Mutated amino acids of nucleotides in the HCDR3 regions are bolded.

Other Embodiments

[0148] All of the features disclosed in this specification may be combined in any combination. Each feature disclosed in this specification may be replaced by an alternative feature serving the same, equivalent, or similar purpose. Thus, unless expressly stated otherwise, each feature disclosed is only an example of a generic series of equivalent or similar features.

[0149] From the above description, one skilled in the art can easily ascertain the essential characteristics of the present invention, and without departing from the spirit and scope thereof, can make various changes and modifications of the invention to adapt it to various usages and conditions. Thus, other embodiments are also within the claims.

EQUIVALENTS

[0150] While several inventive embodiments have been described and illustrated herein, those of ordinary skill in the art will readily envision a variety of other means and/or structures for performing the function and/or obtaining the

results and/or one or more of the advantages described herein, and each of such variations and/or modifications is deemed to be within the scope of the inventive embodiments described herein. More generally, those skilled in the art will readily appreciate that all parameters, dimensions, materials, and configurations described herein are meant to be exemplary and that the actual parameters, dimensions, materials, and/or configurations will depend upon the specific application or applications for which the inventive teachings is/are used. Those skilled in the art will recognize, or be able to ascertain using no more than routine experimentation, many equivalents to the specific inventive embodiments described herein. It is, therefore, to be understood that the foregoing embodiments are presented by way of example only and that, within the scope of the appended claims and equivalents thereto, inventive embodiments may be practiced otherwise than as specifically described and claimed. Inventive embodiments of the present disclosure are directed to each individual feature, system, article, material, kit, and/or method described herein. In addition, any combination of two or more such features, systems, articles, mate-

rials, kits, and/or methods, if such features, systems, articles, materials, kits, and/or methods are not mutually inconsistent, is included within the inventive scope of the present disclosure.

[0151] All definitions, as defined and used herein, should be understood to control over dictionary definitions, definitions in documents incorporated by reference, and/or ordinary meanings of the defined terms.

[0152] All references, patents and patent applications disclosed herein are incorporated by reference with respect to the subject matter for which each is cited, which in some cases may encompass the entirety of the document.

[0153] The indefinite articles “a” and “an,” as used herein in the specification and in the claims, unless clearly indicated to the contrary, should be understood to mean “at least one.”

[0154] The phrase “and/or,” as used herein in the specification and in the claims, should be understood to mean “either or both” of the elements so conjoined, e.g., elements that are conjunctively present in some cases and disjunctively present in other cases. Multiple elements listed with “and/or” should be construed in the same fashion, e.g., “one or more” of the elements so conjoined. Other elements may optionally be present other than the elements specifically identified by the “and/or” clause, whether related or unrelated to those elements specifically identified. Thus, as a non-limiting example, a reference to “A and/or B”, when used in conjunction with open-ended language such as “comprising” can refer, in one embodiment, to A only (optionally including elements other than B); in another embodiment, to B only (optionally including elements other than A); in yet another embodiment, to both A and B (optionally including other elements); etc.

[0155] As used herein in the specification and in the claims, “or” should be understood to have the same meaning as “and/or” as defined above. For example, when separating items in a list, “or” or “and/or” shall be interpreted as being inclusive, e.g., the inclusion of at least one, but also including more than one of a number or list of elements, and,

optionally, additional unlisted items. Only terms clearly indicated to the contrary, such as “only one of” or “exactly one of,” or, when used in the claims, “consisting of,” will refer to the inclusion of exactly one element of a number or list of elements. In general, the term “or” as used herein shall only be interpreted as indicating exclusive alternatives (e.g., “one or the other but not both”) when preceded by terms of exclusivity, such as “either,” “one of,” “only one of,” or “exactly one of.” “Consisting essentially of,” when used in the claims, shall have its ordinary meaning as used in the field of patent law.

[0156] As used herein in the specification and in the claims, the phrase “at least one,” in reference to a list of one or more elements, should be understood to mean at least one element selected from any one or more of the elements in the list of elements, but not necessarily including at least one of each and every element specifically listed within the list of elements and not excluding any combinations of elements in the list of elements. This definition also allows that elements may optionally be present other than the elements specifically identified within the list of elements to which the phrase “at least one” refers, whether related or unrelated to those elements specifically identified. Thus, as a non-limiting example, “at least one of A and B” (or, equivalently, “at least one of A or B,” or, equivalently “at least one of A and/or B”) can refer, in one embodiment, to at least one, optionally including more than one, A, with no B present (and optionally including elements other than B); in another embodiment, to at least one, optionally including more than one, B, with no A present (and optionally including elements other than A); in yet another embodiment, to at least one, optionally including more than one, A, and at least one, optionally including more than one, B (and optionally including other elements); etc.

[0157] It should also be understood that, unless clearly indicated to the contrary, in any methods claimed herein that include more than one step or act, the order of the steps or acts of the method is not necessarily limited to the order in which the steps or acts of the method are recited.

SEQUENCE LISTING

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Sequence total quantity: 163
SEQ ID NO: 1          moltype = AA  length = 8
FEATURE              Location/Qualifiers
source                1..8
                     mol_type = protein
                     organism = synthetic construct

SEQUENCE: 1
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SEQ ID NO: 2          moltype = AA  length = 17
FEATURE              Location/Qualifiers
source                1..17
                     mol_type = protein
                     organism = synthetic construct

SEQUENCE: 2
LEWIGMIDPS NSDTFRN                             17

SEQ ID NO: 3          moltype = AA  length = 10
FEATURE              Location/Qualifiers
source                1..10
                     mol_type = protein
                     organism = synthetic construct

SEQUENCE: 3
AGSYVSPLDY                                     10
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SEQ ID NO: 4	moltype = AA length = 10	
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	mol_type = protein	
	organism = synthetic construct	
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SEQ ID NO: 5	moltype = AA length = 10	
FEATURE	Location/Qualifiers	
source	1..10	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 5		
VGSYVSPLDY		10
SEQ ID NO: 6	moltype = AA length = 10	
FEATURE	Location/Qualifiers	
source	1..10	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 6		
LGSYVSPLDY		10
SEQ ID NO: 7	moltype = AA length = 10	
FEATURE	Location/Qualifiers	
source	1..10	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 7		
IGSYVSPLDY		10
SEQ ID NO: 8	moltype = AA length = 10	
FEATURE	Location/Qualifiers	
source	1..10	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 8		
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SEQ ID NO: 9	moltype = AA length = 11	
FEATURE	Location/Qualifiers	
source	1..11	
	mol_type = protein	
	organism = synthetic construct	
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QSLLYTSSQK N		11
SEQ ID NO: 10	moltype = AA length = 12	
FEATURE	Location/Qualifiers	
source	1..12	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 10		
PKLLIYWAST RE		12
SEQ ID NO: 11	moltype = AA length = 5	
FEATURE	Location/Qualifiers	
source	1..5	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 11		
YYAYP		5
SEQ ID NO: 12	moltype = AA length = 119	
FEATURE	Location/Qualifiers	
source	1..119	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 12		
QVQLQQSGPE LVRPGASVKM SCRASGYTFT SYWLHWVKQR PGQGLEWIGM IDPSNSDTRF		60
NPNFKDKATL NVDRSSNTAY MLLSSLTSAD SAVYYCATYG SYVSPLDYWG QGTSVTVSS		119
SEQ ID NO: 13	moltype = AA length = 119	
FEATURE	Location/Qualifiers	
source	1..119	

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mol_type = protein
organism = synthetic construct

SEQUENCE: 13
QVQLQQSGPE LVRPGASVKM SCRASGYTFT SYWLHWVKQR PGQGLEWIGM IDPSNSDTRF 60
NPNFKDKATL NVDRSSNTAY MLLSSLTSAD SAVYYCATAG SYVSPLDYWG QGTSVTVSS 119

SEQ ID NO: 14      moltype = AA length = 119
FEATURE           Location/Qualifiers
source            1..119
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 14
QVQLQQSGPE LVRPGASVKM SCRASGYTFT SYWLHWVKQR PGQGLEWIGM IDPSNSDTRF 60
NPNFKDKATL NVDRSSNTAY MLLSSLTSAD SAVYYCATYG SYVSPADYWG QGTSVTVSS 119

SEQ ID NO: 15      moltype = AA length = 119
FEATURE           Location/Qualifiers
source            1..119
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 15
QVQLQQSGPE LVRPGASVKM SCRASGYTFT SYWLHWVKQR PGQGLEWIGM IDPSNSDTRF 60
NPNFKDKATL NVDRSSNTAY MLLSSLTSAD SAVYYCATVG SYVSPLDYWG QGTSVTVSS 119

SEQ ID NO: 16      moltype = AA length = 119
FEATURE           Location/Qualifiers
source            1..119
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 16
QVQLQQSGPE LVRPGASVKM SCRASGYTFT SYWLHWVKQR PGQGLEWIGM IDPSNSDTRF 60
NPNFKDKATL NVDRSSNTAY MLLSSLTSAD SAVYYCATLG SYVSPLDYWG QGTSVTVSS 119

SEQ ID NO: 17      moltype = AA length = 119
FEATURE           Location/Qualifiers
source            1..119
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 17
QVQLQQSGPE LVRPGASVKM SCRASGYTFT SYWLHWVKQR PGQGLEWIGM IDPSNSDTRF 60
NPNFKDKATL NVDRSSNTAY MLLSSLTSAD SAVYYCATIG SYVSPLDYWG QGTSVTVSS 119

SEQ ID NO: 18      moltype = AA length = 119
FEATURE           Location/Qualifiers
source            1..119
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 18
QVQLQQSGPE LVRPGASVKM SCRASGYTFT SYWLHWVKQR PGQGLEWIGM IDPSNSDTRF 60
NPNFKDKATL NVDRSSNTAY MLLSSLTSAD SAVYYCATMG SYVSPLDYWG QGTSVTVSS 119

SEQ ID NO: 19      moltype = AA length = 114
FEATURE           Location/Qualifiers
source            1..114
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 19
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ESGVPDRFTG SGSGTDFTLT ITSVKADDLA VYYCQQYYAY PWTFGGGTKL EIKR 114

SEQ ID NO: 20      moltype = DNA length = 357
FEATURE           Location/Qualifiers
source            1..357
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 20
cagggtgcagc tccagcagag cggccctgag ctggtgcgcc ccggcgcttc cgtgaagatg 60
tcatgccgag cttctggcta caggttcacc agctattggc tacactgggt caagcagagg 120
cccgacaggg gtctggagtg gatcggtatg attgaccctg caaatagtga caccgccttc 180
aaccccaact tcaaggacaa ggccactctc aacgtggacc gctcgagcaa caccggcctac 240
atgctgtgtg cctctctgac ctctggcgga agcgccgtgt actactgtgc cactacgggc 300
tctacgtga gccctttgga ttattggggc cagggcacct ccgtcacgtg gtectccc 357

SEQ ID NO: 21      moltype = DNA length = 357
FEATURE           Location/Qualifiers
source            1..357

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mol_type = other DNA
organism = synthetic construct

SEQUENCE: 21

cagggtgcagc	tccagcagag	cggccctgag	ctggtgcgcc	ccggcgcttc	cgtgaagatg	60
tcatgccgag	cttctggcta	cacgttcacc	agctattggc	tacactgggt	caagcagagg	120
cccgacagg	gtctggagt	gatcggtatg	attgacccgt	caaatagtga	caccgccttc	180
aaccccaact	tcaaggacaa	ggccactctc	aacgtggacc	gctcgagcaa	cacggcctac	240
atgctgctgt	cctctctgac	ctcggcggac	agcgccgtgt	actactgtgc	caccgcgggc	300
tcctacgtga	gccctttgga	ttattggggc	cagggcacct	ccgtcacctg	gtcctcc	357

SEQ ID NO: 22

FEATURE

source

moltype = DNA length = 357
Location/Qualifiers
1..357
mol_type = other DNA
organism = synthetic construct

SEQUENCE: 22

cagggtgcagc	tccagcagag	cggccctgag	ctggtgcgcc	ccggcgcttc	cgtgaagatg	60
tcatgccgag	cttctggcta	cacgttcacc	agctattggc	tacactgggt	caagcagagg	120
cccgacagg	gtctggagt	gatcggtatg	attgacccgt	caaatagtga	caccgccttc	180
aaccccaact	tcaaggacaa	ggccactctc	aacgtggacc	gctcgagcaa	cacggcctac	240
atgctgctgt	cctctctgac	ctcggcggac	agcgccgtgt	actactgtgc	caccgcgggc	300
tcctacgtga	gccctttgga	ttattggggc	cagggcacct	ccgtcacctg	gtcctcc	357

SEQ ID NO: 23

FEATURE

source

moltype = DNA length = 357
Location/Qualifiers
1..357
mol_type = other DNA
organism = synthetic construct

SEQUENCE: 23

cagggtgcagc	tccagcagag	cggccctgag	ctggtgcgcc	ccggcgcttc	cgtgaagatg	60
tcatgccgag	cttctggcta	cacgttcacc	agctattggc	tacactgggt	caagcagagg	120
cccgacagg	gtctggagt	gatcggtatg	attgacccgt	caaatagtga	caccgccttc	180
aaccccaact	tcaaggacaa	ggccactctc	aacgtggacc	gctcgagcaa	cacggcctac	240
atgctgctgt	cctctctgac	ctcggcggac	agcgccgtgt	actactgtgc	caccgcgggc	300
tcctacgtga	gccctttgga	ttattggggc	cagggcacct	ccgtcacctg	gtcctcc	357

SEQ ID NO: 24

FEATURE

source

moltype = DNA length = 357
Location/Qualifiers
1..357
mol_type = other DNA
organism = synthetic construct

SEQUENCE: 24

cagggtgcagc	tccagcagag	cggccctgag	ctggtgcgcc	ccggcgcttc	cgtgaagatg	60
tcatgccgag	cttctggcta	cacgttcacc	agctattggc	tacactgggt	caagcagagg	120
cccgacagg	gtctggagt	gatcggtatg	attgacccgt	caaatagtga	caccgccttc	180
aaccccaact	tcaaggacaa	ggccactctc	aacgtggacc	gctcgagcaa	cacggcctac	240
atgctgctgt	cctctctgac	ctcggcggac	agcgccgtgt	actactgtgc	caccgcgggc	300
tcctacgtga	gccctttgga	ttattggggc	cagggcacct	ccgtcacctg	gtcctcc	357

SEQ ID NO: 25

FEATURE

source

moltype = DNA length = 357
Location/Qualifiers
1..357
mol_type = other DNA
organism = synthetic construct

SEQUENCE: 25

cagggtgcagc	tccagcagag	cggccctgag	ctggtgcgcc	ccggcgcttc	cgtgaagatg	60
tcatgccgag	cttctggcta	cacgttcacc	agctattggc	tacactgggt	caagcagagg	120
cccgacagg	gtctggagt	gatcggtatg	attgacccgt	caaatagtga	caccgccttc	180
aaccccaact	tcaaggacaa	ggccactctc	aacgtggacc	gctcgagcaa	cacggcctac	240
atgctgctgt	cctctctgac	ctcggcggac	agcgccgtgt	actactgtgc	caccgcgggc	300
tcctacgtga	gccctttgga	ttattggggc	cagggcacct	ccgtcacctg	gtcctcc	357

SEQ ID NO: 26

FEATURE

source

moltype = DNA length = 357
Location/Qualifiers
1..357
mol_type = other DNA
organism = synthetic construct

SEQUENCE: 26

cagggtgcagc	tccagcagag	cggccctgag	ctggtgcgcc	ccggcgcttc	cgtgaagatg	60
tcatgccgag	cttctggcta	cacgttcacc	agctattggc	tacactgggt	caagcagagg	120
cccgacagg	gtctggagt	gatcggtatg	attgacccgt	caaatagtga	caccgccttc	180
aaccccaact	tcaaggacaa	ggccactctc	aacgtggacc	gctcgagcaa	cacggcctac	240
atgctgctgt	cctctctgac	ctcggcggac	agcgccgtgt	actactgtgc	caccgcgggc	300
tcctacgtga	gccctttgga	ttattggggc	cagggcacct	ccgtcacctg	gtcctcc	357

SEQ ID NO: 27

moltype = DNA length = 342

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FEATURE
source Location/Qualifiers
1..342
mol_type = other DNA
organism = synthetic construct

SEQUENCE: 27
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gtgtcatgca agtcctcaca gaccccttctg tacaccagct cccagaagaa ctacctggcg 120
tggtagcagc agaagcccg acagagcccc aagctgctga tctattgggc tccacccgc 180
gagagcgcg tccccgaccg cttcaccggc tccggtcccg ggaccgactt caccctgacc 240
atcacctccg tgaaggccga tgacctggcc gtgtactact gtcaacagta ttacgcctac 300
ccgtggacct tcggaggcgg caccaagctg gagatcaagc gc 342

SEQ ID NO: 28 moltype = AA length = 248
FEATURE Location/Qualifiers
source 1..248
mol_type = protein
organism = synthetic construct

SEQUENCE: 28
QVQLQQSGPE LVRPGASVKM SCRASGYTFT SYWLHWVKQR PGQGLEWIGM IDPSNSDTRF 60
NPNFKDKATL NVDRSSNTAY MLLSSLTSAD SAVYYCATYG SYVSPLDYWG QGTSVTVSSG 120
GGSGGGGSG GGGSDIMMSQ SPSSLTVSVG EKVTVCCKSS QSLLYTSSQK NYLAWYQQKP 180
GQSPKLLIYW ASTRESGVPD RFTGSGSGTD FTLTITSVKA DDLAVVYCQQ YYAYPWTFGG 240
GTKLEIKR 248

SEQ ID NO: 29 moltype = AA length = 248
FEATURE Location/Qualifiers
source 1..248
mol_type = protein
organism = synthetic construct

SEQUENCE: 29
QVQLQQSGPE LVRPGASVKM SCRASGYTFT SYWLHWVKQR PGQGLEWIGM IDPSNSDTRF 60
NPNFKDKATL NVDRSSNTAY MLLSSLTSAD SAVYYCATAG SYVSPLDYWG QGTSVTVSSG 120
GGSGGGGSG GGGSDIMMSQ SPSSLTVSVG EKVTVCCKSS QSLLYTSSQK NYLAWYQQKP 180
GQSPKLLIYW ASTRESGVPD RFTGSGSGTD FTLTITSVKA DDLAVVYCQQ YYAYPWTFGG 240
GTKLEIKR 248

SEQ ID NO: 30 moltype = AA length = 248
FEATURE Location/Qualifiers
source 1..248
mol_type = protein
organism = synthetic construct

SEQUENCE: 30
QVQLQQSGPE LVRPGASVKM SCRASGYTFT SYWLHWVKQR PGQGLEWIGM IDPSNSDTRF 60
NPNFKDKATL NVDRSSNTAY MLLSSLTSAD SAVYYCATYG SYVSPADYWG QGTSVTVSSG 120
GGSGGGGSG GGGSDIMMSQ SPSSLTVSVG EKVTVCCKSS QSLLYTSSQK NYLAWYQQKP 180
GQSPKLLIYW ASTRESGVPD RFTGSGSGTD FTLTITSVKA DDLAVVYCQQ YYAYPWTFGG 240
GTKLEIKR 248

SEQ ID NO: 31 moltype = AA length = 248
FEATURE Location/Qualifiers
source 1..248
mol_type = protein
organism = synthetic construct

SEQUENCE: 31
QVQLQQSGPE LVRPGASVKM SCRASGYTFT SYWLHWVKQR PGQGLEWIGM IDPSNSDTRF 60
NPNFKDKATL NVDRSSNTAY MLLSSLTSAD SAVYYCATVG SYVSPLDYWG QGTSVTVSSG 120
GGSGGGGSG GGGSDIMMSQ SPSSLTVSVG EKVTVCCKSS QSLLYTSSQK NYLAWYQQKP 180
GQSPKLLIYW ASTRESGVPD RFTGSGSGTD FTLTITSVKA DDLAVVYCQQ YYAYPWTFGG 240
GTKLEIKR 248

SEQ ID NO: 32 moltype = AA length = 248
FEATURE Location/Qualifiers
source 1..248
mol_type = protein
organism = synthetic construct

SEQUENCE: 32
QVQLQQSGPE LVRPGASVKM SCRASGYTFT SYWLHWVKQR PGQGLEWIGM IDPSNSDTRF 60
NPNFKDKATL NVDRSSNTAY MLLSSLTSAD SAVYYCATLG SYVSPLDYWG QGTSVTVSSG 120
GGSGGGGSG GGGSDIMMSQ SPSSLTVSVG EKVTVCCKSS QSLLYTSSQK NYLAWYQQKP 180
GQSPKLLIYW ASTRESGVPD RFTGSGSGTD FTLTITSVKA DDLAVVYCQQ YYAYPWTFGG 240
GTKLEIKR 248

SEQ ID NO: 33 moltype = AA length = 248
FEATURE Location/Qualifiers
source 1..248
mol_type = protein

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                                organism = synthetic construct
SEQUENCE: 33
QVQLQQSGPE LVRPGASVKM SCRASGYTFT SYWLHWVKQR PGQGLEWIGM IDPSNSDTRF 60
NPNFKDKATL NVDRSSNTAY MLLSSLTSAD SAVVYCATIG SYVSPLDYWG QGTSVTVSSG 120
GGSGGGGGSG GGGSDIMMSQ SPSSLTVSVG EKVTVSCKSS QSLLYTSSQK NYLAWYQQKP 180
GQSPKLLIYW ASTRESGVPD RFTGSGSGTD FTLTITSVKA DDLAVYYCQQ YYAYPWTFGG 240
GTKLEIKR 248

SEQ ID NO: 34      moltype = AA length = 248
FEATURE            Location/Qualifiers
source              1..248
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 34
QVQLQQSGPE LVRPGASVKM SCRASGYTFT SYWLHWVKQR PGQGLEWIGM IDPSNSDTRF 60
NPNFKDKATL NVDRSSNTAY MLLSSLTSAD SAVVYCATMG SYVSPLDYWG QGTSVTVSSG 120
GGSGGGGGSG GGGSDIMMSQ SPSSLTVSVG EKVTVSCKSS QSLLYTSSQK NYLAWYQQKP 180
GQSPKLLIYW ASTRESGVPD RFTGSGSGTD FTLTITSVKA DDLAVYYCQQ YYAYPWTFGG 240
GTKLEIKR 248

SEQ ID NO: 35      moltype = DNA length = 744
FEATURE            Location/Qualifiers
source              1..744
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 35
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tcatgccagc cttctggcta caggttcacc agctattggc tacactgggt caagcagagg 120
cccgacagag gtctggagtg gatcggtatg attgacccgt caaatagtga caccgccttc 180
aaccccaact tcaaggacaa ggccactctc aacgtggacc gctcgagcaa caggcctac 240
atgctgctgt cctctctgac ctcgccggac agcgccgtgt actactgtgc cactacggc 300
tcctacgtga gccctttgga ttattggggc cagggcacct ccgtcacgtg gtcctccggt 360
ggaggcggtt caggcggagg tggctctggc ggtggcggat cggacatcat gatgtctcag 420
agtcctcttt ccttgaccgt gtcctggggc gagaaggtga ccgtgtcatg caagtctca 480
cagagccttc tgtacaccag ctcccagaag aactacctgg cgtggtacca gcagaagccc 540
ggacagagcc ccaagctgct gatctattgg gcttccaccc gcgagagcgg cgtccccgac 600
cgcttcaccg gctcggctc cgggaccgac ttcacctga ccatcacctc cgtgaaggcc 660
gatgacctgg ccgtgtacta ctgtcaacag tattacgcct acccgtggac cttcggaggc 720
ggcaccacgc tggagatcaa gcgc 744

SEQ ID NO: 36      moltype = DNA length = 744
FEATURE            Location/Qualifiers
source              1..744
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 36
cagggtgcagc tccagcagag cggccctgag ctggtgcgcc ccggcgcttc cgtgaagatg 60
tcatgccagc cttctggcta caggttcacc agctattggc tacactgggt caagcagagg 120
cccgacagag gtctggagtg gatcggtatg attgacccgt caaatagtga caccgccttc 180
aaccccaact tcaaggacaa ggccactctc aacgtggacc gctcgagcaa caggcctac 240
atgctgctgt cctctctgac ctcgccggac agcgccgtgt actactgtgc caccgcggc 300
tcctacgtga gccctttgga ttattggggc cagggcacct ccgtcacgtg gtcctccggt 360
ggaggcggtt caggcggagg tggctctggc ggtggcggat cggacatcat gatgtctcag 420
agtcctcttt ccttgaccgt gtcctggggc gagaaggtga ccgtgtcatg caagtctca 480
cagagccttc tgtacaccag ctcccagaag aactacctgg cgtggtacca gcagaagccc 540
ggacagagcc ccaagctgct gatctattgg gcttccaccc gcgagagcgg cgtccccgac 600
cgcttcaccg gctcggctc cgggaccgac ttcacctga ccatcacctc cgtgaaggcc 660
gatgacctgg ccgtgtacta ctgtcaacag tattacgcct acccgtggac cttcggaggc 720
ggcaccacgc tggagatcaa gcgc 744

SEQ ID NO: 37      moltype = DNA length = 744
FEATURE            Location/Qualifiers
source              1..744
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 37
cagggtgcagc tccagcagag cggccctgag ctggtgcgcc ccggcgcttc cgtgaagatg 60
tcatgccagc cttctggcta caggttcacc agctattggc tacactgggt caagcagagg 120
cccgacagag gtctggagtg gatcggtatg attgacccgt caaatagtga caccgccttc 180
aaccccaact tcaaggacaa ggccactctc aacgtggacc gctcgagcaa caggcctac 240
atgctgctgt cctctctgac ctcgccggac agcgccgtgt actactgtgc cactacggc 300
tcctacgtga gcccttcgga ttattggggc cagggcacct ccgtcacgtg gtcctccggt 360
ggaggcggtt caggcggagg tggctctggc ggtggcggat cggacatcat gatgtctcag 420
agtcctcttt ccttgaccgt gtcctggggc gagaaggtga ccgtgtcatg caagtctca 480
cagagccttc tgtacaccag ctcccagaag aactacctgg cgtggtacca gcagaagccc 540
ggacagagcc ccaagctgct gatctattgg gcttccaccc gcgagagcgg cgtccccgac 600

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cgcttcaccg gctccggctc cgggaccgac ttcaccctga ccatcacctc cgtgaaggcc 660
gatgacctgg ccgtgtacta ctgtcaacag tattacgcct acccgtggac cttcggaggc 720
ggcaccaagc tggagatcaa gcgc 744

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SEQ ID NO: 38      moltype = DNA length = 744
FEATURE           Location/Qualifiers
source            1..744
                  mol_type = other DNA
                  organism = synthetic construct

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SEQUENCE: 38
cagggtgcagc tccagcagag cggccctgag ctggtgcgcc ccggcgcttc cgtgaagatg 60
tcatgccgag cttctggcta caggttcacc agctattggc tacactgggt caagcagagg 120
cccgacaggg gtctggagtg gatcggtatg attgaccctg caaatagtga caccgccttc 180
aaccccaact tcaaggacaa ggccactctc aacgtggacc gctcgagcaa cacggcctac 240
atgctgctgt cctctctgac ctccggcgac agcgccgtgt actactgtgc caccctcggc 300
tcctacgtga gccctttgga ttattggggc cagggcacct ccgtcacctg gtcctccggt 360
ggaggcggtt caggcggagg tggctctggc ggtggcggat cggacatcat gatgtctcag 420
agtcctcttt ctctgaccgt gtcctggggc gagaagggtg ccgtgtcatg caagtctcctca 480
cagagccttc tgtacaccag ctcccagaag aactacctgg cgtgggtacca gcagaagccc 540
ggacagagcc ccaagctgct gatctattgg gcttccaccc gcgagagcgg cgtccccgac 600
cgcttcaccg gctccggctc cgggaccgac ttcaccctga ccatcacctc cgtgaaggcc 660
gatgacctgg ccgtgtacta ctgtcaacag tattacgcct acccgtggac cttcggaggc 720
ggcaccaagc tggagatcaa gcgc 744

```

```

SEQ ID NO: 39      moltype = DNA length = 744
FEATURE           Location/Qualifiers
source            1..744
                  mol_type = other DNA
                  organism = synthetic construct

```

```

SEQUENCE: 39
cagggtgcagc tccagcagag cggccctgag ctggtgcgcc ccggcgcttc cgtgaagatg 60
tcatgccgag cttctggcta caggttcacc agctattggc tacactgggt caagcagagg 120
cccgacaggg gtctggagtg gatcggtatg attgaccctg caaatagtga caccgccttc 180
aaccccaact tcaaggacaa ggccactctc aacgtggacc gctcgagcaa cacggcctac 240
atgctgctgt cctctctgac ctccggcgac agcgccgtgt actactgtgc caccctcggc 300
tcctacgtga gccctttgga ttattggggc cagggcacct ccgtcacctg gtcctccggt 360
ggaggcggtt caggcggagg tggctctggc ggtggcggat cggacatcat gatgtctcag 420
agtcctcttt ctctgaccgt gtcctggggc gagaagggtg ccgtgtcatg caagtctcctca 480
cagagccttc tgtacaccag ctcccagaag aactacctgg cgtgggtacca gcagaagccc 540
ggacagagcc ccaagctgct gatctattgg gcttccaccc gcgagagcgg cgtccccgac 600
cgcttcaccg gctccggctc cgggaccgac ttcaccctga ccatcacctc cgtgaaggcc 660
gatgacctgg ccgtgtacta ctgtcaacag tattacgcct acccgtggac cttcggaggc 720
ggcaccaagc tggagatcaa gcgc 744

```

```

SEQ ID NO: 40      moltype = DNA length = 744
FEATURE           Location/Qualifiers
source            1..744
                  mol_type = other DNA
                  organism = synthetic construct

```

```

SEQUENCE: 40
cagggtgcagc tccagcagag cggccctgag ctggtgcgcc ccggcgcttc cgtgaagatg 60
tcatgccgag cttctggcta caggttcacc agctattggc tacactgggt caagcagagg 120
cccgacaggg gtctggagtg gatcggtatg attgaccctg caaatagtga caccgccttc 180
aaccccaact tcaaggacaa ggccactctc aacgtggacc gctcgagcaa cacggcctac 240
atgctgctgt cctctctgac ctccggcgac agcgccgtgt actactgtgc caccctcggc 300
tcctacgtga gccctttgga ttattggggc cagggcacct ccgtcacctg gtcctccggt 360
ggaggcggtt caggcggagg tggctctggc ggtggcggat cggacatcat gatgtctcag 420
agtcctcttt ctctgaccgt gtcctggggc gagaagggtg ccgtgtcatg caagtctcctca 480
cagagccttc tgtacaccag ctcccagaag aactacctgg cgtgggtacca gcagaagccc 540
ggacagagcc ccaagctgct gatctattgg gcttccaccc gcgagagcgg cgtccccgac 600
cgcttcaccg gctccggctc cgggaccgac ttcaccctga ccatcacctc cgtgaaggcc 660
gatgacctgg ccgtgtacta ctgtcaacag tattacgcct acccgtggac cttcggaggc 720
ggcaccaagc tggagatcaa gcgc 744

```

```

SEQ ID NO: 41      moltype = DNA length = 744
FEATURE           Location/Qualifiers
source            1..744
                  mol_type = other DNA
                  organism = synthetic construct

```

```

SEQUENCE: 41
cagggtgcagc tccagcagag cggccctgag ctggtgcgcc ccggcgcttc cgtgaagatg 60
tcatgccgag cttctggcta caggttcacc agctattggc tacactgggt caagcagagg 120
cccgacaggg gtctggagtg gatcggtatg attgaccctg caaatagtga caccgccttc 180
aaccccaact tcaaggacaa ggccactctc aacgtggacc gctcgagcaa cacggcctac 240
atgctgctgt cctctctgac ctccggcgac agcgccgtgt actactgtgc caccctgggc 300
tcctacgtga gccctttgga ttattggggc cagggcacct ccgtcacctg gtcctccggt 360

```

-continued

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ggaggcgggt caggcggagg tggtctgtgc ggtggcggat cggacatcat gatgtctcag 420
agtcctcttt ctctgaccgt gtcctgtggg gagaaaggta ccgtgtcatg caagtctcca 480
cagagccttc tgtacaccag ctcccagaag aactacctgg cgtggtacca gcagaagccc 540
ggacagagcc ccaagctgct gatctattgg gcttccaccc gcgagagcgg cgtccccgac 600
cgcttcaccg gctccggctc cgggaccgac ttcacctga ccatacctc cgtgaaggcc 660
gatgacctgg ccgtgtacta ctgtcaacag tattacgcct acccgtggac cttcggaggc 720
ggaccaaacg tggagatcaa gcgc 744

```

```

SEQ ID NO: 42      moltype = AA length = 476
FEATURE           Location/Qualifiers
source            1..476
                  mol_type = protein
                  organism = synthetic construct

```

```

SEQUENCE: 42
QVQLQQSGPE LVRPGASVKM SCRASGYTFT SYWLHWVKQR PGQGLEWIGM IDPSNSDTRF 60
NPNFKDKATL NVDRSSNTAY MLLSSLTSAD SAVYYCATYG SYVSPLDYWG QGTSVTVSSG 120
GGSGGGGGSG GGGSDIMMSQ SPSSLTVSVG EKVTVSCKSS QSLLYTSSQK NYLAWYQQKP 180
GQSPKLLIYW ASTRESGVPD RFTGSGSGTD FTLTITSVKA DDLAVYYCQQ YYAYPWTFGG 240
GKLEIKRAS TTTAPRPPT PAPTIASQPL SLRPEACRPA AGGAVHTRGL DFACDFWVLV 300
VVGGLVACYS LLVTVAFIIF WVRSKRSRL HSDYMMNTPR RPPGTRKHYQ PYAPPRDFAA 360
YRSLRVKFSR SADAPAYQQG QNQLYNELNL GRREYDVLD KRRGRDPEMG GKPRRKNPQE 420
GLYNELQDK MAEAYSEIGM KGERRRGKGH DGLYQGLSTA TKD TYDALHM QALPPR 476

```

```

SEQ ID NO: 43      moltype = AA length = 476
FEATURE           Location/Qualifiers
source            1..476
                  mol_type = protein
                  organism = synthetic construct

```

```

SEQUENCE: 43
QVQLQQSGPE LVRPGASVKM SCRASGYTFT SYWLHWVKQR PGQGLEWIGM IDPSNSDTRF 60
NPNFKDKATL NVDRSSNTAY MLLSSLTSAD SAVYYCATAG SYVSPLDYWG QGTSVTVSSG 120
GGSGGGGGSG GGGSDIMMSQ SPSSLTVSVG EKVTVSCKSS QSLLYTSSQK NYLAWYQQKP 180
GQSPKLLIYW ASTRESGVPD RFTGSGSGTD FTLTITSVKA DDLAVYYCQQ YYAYPWTFGG 240
GKLEIKRAS TTTAPRPPT PAPTIASQPL SLRPEACRPA AGGAVHTRGL DFACDFWVLV 300
VVGGLVACYS LLVTVAFIIF WVRSKRSRL HSDYMMNTPR RPPGTRKHYQ PYAPPRDFAA 360
YRSLRVKFSR SADAPAYQQG QNQLYNELNL GRREYDVLD KRRGRDPEMG GKPRRKNPQE 420
GLYNELQDK MAEAYSEIGM KGERRRGKGH DGLYQGLSTA TKD TYDALHM QALPPR 476

```

```

SEQ ID NO: 44      moltype = AA length = 476
FEATURE           Location/Qualifiers
source            1..476
                  mol_type = protein
                  organism = synthetic construct

```

```

SEQUENCE: 44
QVQLQQSGPE LVRPGASVKM SCRASGYTFT SYWLHWVKQR PGQGLEWIGM IDPSNSDTRF 60
NPNFKDKATL NVDRSSNTAY MLLSSLTSAD SAVYYCATYG SYVSPADYWG QGTSVTVSSG 120
GGSGGGGGSG GGGSDIMMSQ SPSSLTVSVG EKVTVSCKSS QSLLYTSSQK NYLAWYQQKP 180
GQSPKLLIYW ASTRESGVPD RFTGSGSGTD FTLTITSVKA DDLAVYYCQQ YYAYPWTFGG 240
GKLEIKRAS TTTAPRPPT PAPTIASQPL SLRPEACRPA AGGAVHTRGL DFACDFWVLV 300
VVGGLVACYS LLVTVAFIIF WVRSKRSRL HSDYMMNTPR RPPGTRKHYQ PYAPPRDFAA 360
YRSLRVKFSR SADAPAYQQG QNQLYNELNL GRREYDVLD KRRGRDPEMG GKPRRKNPQE 420
GLYNELQDK MAEAYSEIGM KGERRRGKGH DGLYQGLSTA TKD TYDALHM QALPPR 476

```

```

SEQ ID NO: 45      moltype = AA length = 476
FEATURE           Location/Qualifiers
source            1..476
                  mol_type = protein
                  organism = synthetic construct

```

```

SEQUENCE: 45
QVQLQQSGPE LVRPGASVKM SCRASGYTFT SYWLHWVKQR PGQGLEWIGM IDPSNSDTRF 60
NPNFKDKATL NVDRSSNTAY MLLSSLTSAD SAVYYCATYG SYVSPLDYWG QGTSVTVSSG 120
GGSGGGGGSG GGGSDIMMSQ SPSSLTVSVG EKVTVSCKSS QSLLYTSSQK NYLAWYQQKP 180
GQSPKLLIYW ASTRESGVPD RFTGSGSGTD FTLTITSVKA DDLAVYYCQQ YYAYPWTFGG 240
GKLEIKRAS TTTAPRPPT PAPTIASQPL SLRPEACRPA AGGAVHTRGL DFACDFWVLV 300
VVGGLVACYS LLVTVAFIIF WVRSKRSRL HSDYMMNTPR RPPGTRKHYQ PYAPPRDFAA 360
YRSLRVKFSR SADAPAYQQG QNQLYNELNL GRREYDVLD KRRGRDPEMG GKPRRKNPQE 420
GLYNELQDK MAEAYSEIGM KGERRRGKGH DGLYQGLSTA TKD TYDALHM QALPPR 476

```

```

SEQ ID NO: 46      moltype = AA length = 476
FEATURE           Location/Qualifiers
source            1..476
                  mol_type = protein
                  organism = synthetic construct

```

```

SEQUENCE: 46
QVQLQQSGPE LVRPGASVKM SCRASGYTFT SYWLHWVKQR PGQGLEWIGM IDPSNSDTRF 60
NPNFKDKATL NVDRSSNTAY MLLSSLTSAD SAVYYCATLG SYVSPLDYWG QGTSVTVSSG 120

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GGSGGGGGSG	GGGSDIMMSQ	SPSSLTVSVG	EKVTVSCCKSS	QSLLYTSSQK	NYLAWYQQKP	180
GQSPKLLIYW	ASTRESGVPD	RFTGSGSGTD	FTLTITSVKA	DDLAVYYCQQ	YYAYPWTFGG	240
GTKLEIKRAS	TTTPAPRPPT	PAPTIASQPL	SLRPEACRPA	AGGAVHTRGL	DFACDFWVLV	300
VVGGVLACYS	LLVTVAFIIF	WVRSKRSRL	HSDYMMNTPR	RPGPTRKHYQ	PYAPPRDFAA	360
YRSLRVKFSR	SADAPAYQQG	QNQLYNELNL	GRREEYDVL	KRRGRDPEMG	GKPRRKNPQE	420
GLYNELQDKK	MAEAYSEIGM	KGERRRGKGH	DGLYQGLSTA	TKDTYDALHM	QALPPR	476

SEQ ID NO: 47 moltype = AA length = 476
 FEATURE Location/Qualifiers
 source 1..476
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 47

QVQLQQSGPE	LVRPGASVKM	SCRASGYTFT	SYWLHWVKQR	PGQGLEWIGM	IDPSNSDTRF	60
NNPFKDKATL	NVDGSSNTAY	MLLSSSLTSAD	SAVYYCATIG	SYVSPLDYWG	QGTSTVTVSSG	120
GGSGGGGGSG	GGGSDIMMSQ	SPSSLTVSVG	EKVTVSCCKSS	QSLLYTSSQK	NYLAWYQQKP	180
GQSPKLLIYW	ASTRESGVPD	RFTGSGSGTD	FTLTITSVKA	DDLAVYYCQQ	YYAYPWTFGG	240
GTKLEIKRAS	TTTPAPRPPT	PAPTIASQPL	SLRPEACRPA	AGGAVHTRGL	DFACDFWVLV	300
VVGGVLACYS	LLVTVAFIIF	WVRSKRSRL	HSDYMMNTPR	RPGPTRKHYQ	PYAPPRDFAA	360
YRSLRVKFSR	SADAPAYQQG	QNQLYNELNL	GRREEYDVL	KRRGRDPEMG	GKPRRKNPQE	420
GLYNELQDKK	MAEAYSEIGM	KGERRRGKGH	DGLYQGLSTA	TKDTYDALHM	QALPPR	476

SEQ ID NO: 48 moltype = AA length = 476
 FEATURE Location/Qualifiers
 source 1..476
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 48

QVQLQQSGPE	LVRPGASVKM	SCRASGYTFT	SYWLHWVKQR	PGQGLEWIGM	IDPSNSDTRF	60
NNPFKDKATL	NVDGSSNTAY	MLLSSSLTSAD	SAVYYCATMG	SYVSPLDYWG	QGTSTVTVSSG	120
GGSGGGGGSG	GGGSDIMMSQ	SPSSLTVSVG	EKVTVSCCKSS	QSLLYTSSQK	NYLAWYQQKP	180
GQSPKLLIYW	ASTRESGVPD	RFTGSGSGTD	FTLTITSVKA	DDLAVYYCQQ	YYAYPWTFGG	240
GTKLEIKRAS	TTTPAPRPPT	PAPTIASQPL	SLRPEACRPA	AGGAVHTRGL	DFACDFWVLV	300
VVGGVLACYS	LLVTVAFIIF	WVRSKRSRL	HSDYMMNTPR	RPGPTRKHYQ	PYAPPRDFAA	360
YRSLRVKFSR	SADAPAYQQG	QNQLYNELNL	GRREEYDVL	KRRGRDPEMG	GKPRRKNPQE	420
GLYNELQDKK	MAEAYSEIGM	KGERRRGKGH	DGLYQGLSTA	TKDTYDALHM	QALPPR	476

SEQ ID NO: 49 moltype = AA length = 885
 FEATURE Location/Qualifiers
 source 1..885
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 49

GSEQKLISEE	DLGSQVQLQQ	SGPELVRPGA	SVKMSCRASG	YTFTSYWLHW	VKQRPQGGL	60
WIGMIDPSNS	DTRFNPFPKD	KATLNVDRSS	NTAYMLLSSL	TSADSAVYYC	ATYGSYVSPL	120
DYWGQGTSTV	VSSGGGGSGG	GGSGGGGSDI	MMSQSPSSLT	VSVGEKVTVS	CKSSQSLLYT	180
SSQKNYLAWY	QQKPGQSPKL	LIYWASTRES	GVPDRFTGSG	SGTDFTLTIT	SVKADDLAVY	240
YQQYYAYPW	TFGGGTLEI	KRASTTPAP	RPPTPAPTIA	SQPLSLRPEA	CRPAAGGAVH	300
TRGLDFACDF	WVLVVVGV	ACYSLLVTVA	FIIFWVRSKR	SRLLHSDYMN	MTPRRPGPTR	360
KHYQPYAPPR	DFAAYRSLRV	KFSRSADAPA	YQQGQNQLYN	ELNLGRREEY	DVLDKRRGRD	420
PEMGKPRRK	NPQEGLYNEL	QKDKMAEAYS	EIGMKGERRR	GKGHDGLYQG	LSTATKDTYD	480
ALHMQALPPR	TSGSGATNFS	LLKQAGDVEE	NPGPSRMDMA	DEPLNGSHTW	LSIPFDLNGS	540
VVSTNTSNQT	EPYYDLTSNA	VLTFIYFVVC	IIGLCGNTLV	IYVILRYAKM	KTITNIYILN	600
LAIADLFML	GLPFLAMQVA	LVHWPFGKAI	CRVVMVTDGI	NQFTSIFCLT	VMSIDRYLAV	660
VHPIKSAKWR	RPRTAKMITM	AVWGVSLLVI	LPIMIYAGLR	SNQWGRSCT	INWPGESGAW	720
YTGFIIYTFI	LGFLVPLTII	CLCYLFIIIK	VKSSGIRVGS	SKRKKSEKKV	TRMVSIVVAV	780
FIFCWLPFYI	FNVSSVSMAI	SPTPALKGMF	DFVVVLTIAN	SCANPILYAF	LSNPFKKSFO	840
NVLCVLKVS	G	TDDGERSDSK	QDKSRLNETT	ETQRTLNGD	LQTSI	885

SEQ ID NO: 50 moltype = AA length = 885
 FEATURE Location/Qualifiers
 source 1..885
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 50

GSEQKLISEE	DLGSQVQLQQ	SGPELVRPGA	SVKMSCRASG	YTFTSYWLHW	VKQRPQGGL	60
WIGMIDPSNS	DTRFNPFPKD	KATLNVDRSS	NTAYMLLSSL	TSADSAVYYC	ATAGSYVSPL	120
DYWGQGTSTV	VSSGGGGSGG	GGSGGGGSDI	MMSQSPSSLT	VSVGEKVTVS	CKSSQSLLYT	180
SSQKNYLAWY	QQKPGQSPKL	LIYWASTRES	GVPDRFTGSG	SGTDFTLTIT	SVKADDLAVY	240
YQQYYAYPW	TFGGGTLEI	KRASTTPAP	RPPTPAPTIA	SQPLSLRPEA	CRPAAGGAVH	300
TRGLDFACDF	WVLVVVGV	ACYSLLVTVA	FIIFWVRSKR	SRLLHSDYMN	MTPRRPGPTR	360
KHYQPYAPPR	DFAAYRSLRV	KFSRSADAPA	YQQGQNQLYN	ELNLGRREEY	DVLDKRRGRD	420
PEMGKPRRK	NPQEGLYNEL	QKDKMAEAYS	EIGMKGERRR	GKGHDGLYQG	LSTATKDTYD	480
ALHMQALPPR	TSGSGATNFS	LLKQAGDVEE	NPGPSRMDMA	DEPLNGSHTW	LSIPFDLNGS	540
VVSTNTSNQT	EPYYDLTSNA	VLTFIYFVVC	IIGLCGNTLV	IYVILRYAKM	KTITNIYILN	600
LAIADLFML	GLPFLAMQVA	LVHWPFGKAI	CRVVMVTDGI	NQFTSIFCLT	VMSIDRYLAV	660

```
SEQ ID NO: 51      moltype = AA  length = 885
FEATURE            Location/Qualifiers
source             1..885
                  mol_type = protein
                  organism = synthetic construct
```

```
SEQ ID NO: 52      moltype = AA  length = 885
FEATURE            Location/Qualifiers
source             1..885
                  mol_type = protein
                  organism = synthetic construct
```

```
SEQ ID NO: 53      moltype = AA  length = 885
FEATURE            Location/Qualifiers
source             1..885
                  mol_type = protein
                  organism = synthetic construct
```

```
SEQ ID NO: 54      moltype = AA  length = 885
FEATURE           Location/Qualifiers
source            1..885
                  mol_type = protein
                  organism = synthetic construct
```

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SEQUENCE: 54

GSEQKLISEE	DLGSQVQLQQ	SGPELVRPGA	SVKMSCRASG	YTFTSYWLHW	VKQRPQGGL	60
WIGMIDPSNS	DTRFNPFPKD	KATLNVDRSS	NTAYMLLSSL	TSADSAVYYC	ATIGSYVSPL	120
DYWGQGTSTV	VSSGGGSGG	GGSGGGSDI	MMSQSPSSLT	VSVGEKTVS	CKSSQSLLYT	180
SSQKNYLAWY	QQKPGQSPKL	LIYWASTRES	GVPDRFTGSG	SGTDFTLTIT	SVKADDLAVY	240
YQQYYAYPW	TFGGGTGLEI	KRASTTPAP	RPPTPAPTIA	SQPLSLRPEA	CRPAAGGAVH	300
TRGLDFACDF	WVLVVVGVL	ACYSLLVTV	FIIFWVRSKR	SRLHSDYMN	MTPRRPGPTR	360
KHYQPYAPPR	DFAAYRSLRV	KFSRSADAPA	YQQGQNQLYN	ELNLGRREEY	DVLDKRRGRD	420
PEMGGKPRRK	NPQEGLYNEL	QKDKMAEAYS	EIGMKGERRR	GKGHDGLYQG	LSTATKDTYD	480
ALHMQALPPR	TSGSGATNFS	LLKQAGDVEE	NPGPSRMDMA	DEPLNGSHTW	LSIPFDLNGS	540
VVSTNTSNQT	EPYYDLTSNA	VLTFFIYFVVC	IIGLCGNTLV	IYVILRYAKM	KTITNIYILN	600
LAIADLFML	GLPFLAMQVA	LVHWPFKAI	CRVVMVDGI	NQFTSIFCLT	VMSIDRYLAV	660
VHPIKSAKWR	RPRTAKMITM	AVWGVSLTV	LPIMIYAGLR	SNQWGRSSCT	INWPGESGAW	720
YTGFIIYTFI	LGFLVPLTII	CLCYLFIIIK	VKSSGIRVGS	SKRKKSEKVV	TRMVSIVVAV	780
FIFCWLPFYI	FNVSVMMAI	SPTPALKGMF	DFVVVLTYAN	SCANPILYAF	LSDNFKKSFQ	840
NVLCVLKVS	G	TDDGERSDSK	QDKSRLNETT	ETQRTLLNGD	LQTSI	885

SEQ ID NO: 55

moltype = AA length = 885

FEATURE

Location/Qualifiers

source

1..885

mol_type = protein

organism = synthetic construct

SEQUENCE: 55

GSEQKLISEE	DLGSQVQLQQ	SGPELVRPGA	SVKMSCRASG	YTFTSYWLHW	VKQRPQGGL	60
WIGMIDPSNS	DTRFNPFPKD	KATLNVDRSS	NTAYMLLSSL	TSADSAVYYC	ATMGSYVSPL	120
DYWGQGTSTV	VSSGGGSGG	GGSGGGSDI	MMSQSPSSLT	VSVGEKTVS	CKSSQSLLYT	180
SSQKNYLAWY	QQKPGQSPKL	LIYWASTRES	GVPDRFTGSG	SGTDFTLTIT	SVKADDLAVY	240
YQQYYAYPW	TFGGGTGLEI	KRASTTPAP	RPPTPAPTIA	SQPLSLRPEA	CRPAAGGAVH	300
TRGLDFACDF	WVLVVVGVL	ACYSLLVTV	FIIFWVRSKR	SRLHSDYMN	MTPRRPGPTR	360
KHYQPYAPPR	DFAAYRSLRV	KFSRSADAPA	YQQGQNQLYN	ELNLGRREEY	DVLDKRRGRD	420
PEMGGKPRRK	NPQEGLYNEL	QKDKMAEAYS	EIGMKGERRR	GKGHDGLYQG	LSTATKDTYD	480
ALHMQALPPR	TSGSGATNFS	LLKQAGDVEE	NPGPSRMDMA	DEPLNGSHTW	LSIPFDLNGS	540
VVSTNTSNQT	EPYYDLTSNA	VLTFFIYFVVC	IIGLCGNTLV	IYVILRYAKM	KTITNIYILN	600
LAIADLFML	GLPFLAMQVA	LVHWPFKAI	CRVVMVDGI	NQFTSIFCLT	VMSIDRYLAV	660
VHPIKSAKWR	RPRTAKMITM	AVWGVSLTV	LPIMIYAGLR	SNQWGRSSCT	INWPGESGAW	720
YTGFIIYTFI	LGFLVPLTII	CLCYLFIIIK	VKSSGIRVGS	SKRKKSEKVV	TRMVSIVVAV	780
FIFCWLPFYI	FNVSVMMAI	SPTPALKGMF	DFVVVLTYAN	SCANPILYAF	LSDNFKKSFQ	840
NVLCVLKVS	G	TDDGERSDSK	QDKSRLNETT	ETQRTLLNGD	LQTSI	885

SEQ ID NO: 56

moltype = DNA length = 1428

FEATURE

Location/Qualifiers

source

1..1428

mol_type = other DNA

organism = synthetic construct

SEQUENCE: 56

cagggtgcagc	tccagcagag	cggccctgag	ctgggtgcgcc	ccggcgcttc	cgtgaagatg	60
tcatgccagc	cttctggcta	cacgttcacc	agctattggc	tacactgggt	caagcagagg	120
cccgacacgg	gtctggagtg	gatcggtatg	attgaccctg	caaatagtga	caccgccttc	180
aaacccaact	tcaaggacaa	ggccactctc	aacgtggacc	gctcgagcaa	cacggcctac	240
atgctgctgt	cctctctgac	ctcgccggac	agcgccgtgt	actactgtgc	cacctacggc	300
tcctacgtga	gccctttgga	ttattggggc	cagggcacct	ccgtcacctg	gtcctccggt	360
ggaggcggtg	caggcggagg	tggtctgggc	gggtggcgat	cggacatcat	gatgtctcag	420
agtcctcttt	ctctgaccgt	gtccgtgggc	gagaagggtg	ccgtgtcatg	caagtctcca	480
cagagccttc	tgtacaccag	ctcccagaag	aactacctgg	cgtggtacca	gcagaagccc	540
ggacagagcc	caaagctgtg	gatctattgg	gcttccaccc	gcgagagcgg	cgcccccgac	600
cgcttcacgg	gctccggctc	ggggaccgac	ttcacctcta	ccatcacctc	cgtgaaggcc	660
gatgacctgg	ccgtgtacta	ctgtcaacag	tattacgcct	accctgggac	cttcggaggc	720
ggcaccacgc	tgagatcaaa	cgcgcctagc	acgaccactc	cggcgccgag	cccaccgact	780
ccggcccaaa	ctatcgcgag	ccagccctct	tcgctgaggg	cgggaagcat	ccgccctgcc	840
gccggaggtg	ctgtgcatac	ccggggattg	gacttcgcac	gcgacttttg	gggtgctggtg	900
gtgggttggtg	gagtcctggc	ttgctatagc	ttgctagtaa	cagtggccct	tattattttc	960
tggttgaggga	gtaagaggag	caggctcctg	cacagtgtac	acatgaacat	gactccccgc	1020
cgccccgggc	ccacccgcaa	gcattaccag	ccctatgccc	caccacgcga	cttcgagacc	1080
tatcgctccc	tgagagtga	gttcagcagg	agcgcagacg	cccccgcgta	ccagcagggc	1140
cagaaccagc	tctataacga	gtcacaatcta	ggacgaagag	aggagtacga	tggtttggac	1200
aagagacgtg	gccgggaccc	tgagatgggg	ggaaagccga	gaaggaaaga	ccctcaggaa	1260
ggcctgtaca	atgaactgca	gaaagataag	atggcggagg	cctacagtga	gattggggatg	1320
aaaggcgagc	gcgggagggg	caaggggcac	gatggccttt	accagggtct	cagtacagcc	1380
accaaggaca	cctacgacgc	ccttcacatg	caggccctgc	ccccctgc		1428

SEQ ID NO: 57

moltype = DNA length = 1428

FEATURE

Location/Qualifiers

source

1..1428

mol_type = other DNA

organism = synthetic construct

SEQUENCE: 57

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cagggtgcagc tccagcagag cggccctgag ctggtgcgcc ccggcgcttc cgtgaagatg 60
tcatgccagc cttctggcta cacgttcacc agctattggc tacactgggt caagcagagg 120
cccgacaggg gtctggagtg gatcggtatg attgacccgt caaatagtga caccgcgttc 180
aaccccaact tcaaggacaa ggccactctc aacgtggacc gctcgagcaa cacggcctac 240
atgctgctgt cctctctgac ctcgccggac agcgccgtgt actactgtgc caccgcgggc 300
tctacgtga gccctttgga ttattggggc cagggcacct ccgtcacctg gtccctcggg 360
ggaggcgggt caggcggagg tggtctggc ggtggcggat cggacatcat gatgtctcag 420
agtcctctct cctcgaccgt gtccgtgggc gagaagggtga ccgtgtcatg caagtctca 480
cagagccttc tgtacaccag ctcccagaag aactacctgg cgtggtacca gcagaagccc 540
ggacagagcc ccaagctgct gatctattgg gcttccacc gcgagagcgg cgtccccgac 600
cgcttcaccg gctccggctc cgggaccgac ttcacctga ccatcacctc cgtgaaggcc 660
gatgacctgg ccgtgtacta ctgtcaacag tattacgcct acccgtggac cttcggaggc 720
ggcaccaaagc tggagatcaa gcgcgctagc acgaccactc cggcgccggc cccaccgact 780
ccggcccaaa ctatcgcgag ccagccctg tcgctgaggc cgggaagcatg ccgcctgcc 840
gcccggagtg ctgtgcatac ccggggattg gacttcgcat gcgacttttg ggtgctgggtg 900
gtggttggtg gactcctggc ttgctatagc ttgctagtaa cagtggcctt tattattttc 960
tgggtgagga gtaagaggag caggctcctg cacagtgaat acatgaacat gactccccgc 1020
cgccccgggc ccacccgcaa gcattaccag ccctatgccc caccacgcga cttcgcagcc 1080
tatcgctccc tgagagtga gttcagcagg agcgcagacg ccccgcgta ccagcagggc 1140
cagaaccagc tctataacga gctcaatcta ggacgaagag aggagtagca tgttttggac 1200
aagagacgtg gccgggaccc tgagatgggg ggaagccga gaaggaagaa cctcaggaa 1260
ggcctgtaca atgaactgca gaaagataag atggcggagg cctacagtga gattgggatg 1320
aaaggcgagc gccggagggg caaggggcac gatggcctt accagggtct cagtacagcc 1380
accaaggaca cctacgacgc ccttcacatg caggccctgc cccctcgc 1428

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SEQ ID NO: 58      moltype = DNA length = 1428
FEATURE           Location/Qualifiers
source            1..1428
                  mol_type = other DNA
                  organism = synthetic construct

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SEQUENCE: 58
cagggtgcagc tccagcagag cggccctgag ctggtgcgcc ccggcgcttc cgtgaagatg 60
tcatgccagc cttctggcta cacgttcacc agctattggc tacactgggt caagcagagg 120
cccgacaggg gtctggagtg gatcggtatg attgacccgt caaatagtga caccgcgttc 180
aaccccaact tcaaggacaa ggccactctc aacgtggacc gctcgagcaa cacggcctac 240
atgctgctgt cctctctgac ctcgccggac agcgccgtgt actactgtgc caccacgggc 300
tctacgtga gccctttgga ttattggggc cagggcacct ccgtcacctg gtccctcggg 360
ggaggcgggt caggcggagg tggtctggc ggtggcggat cggacatcat gatgtctcag 420
agtcctctct cctcgaccgt gtccgtgggc gagaagggtga ccgtgtcatg caagtctca 480
cagagccttc tgtacaccag ctcccagaag aactacctgg cgtggtacca gcagaagccc 540
ggacagagcc ccaagctgct gatctattgg gcttccacc gcgagagcgg cgtccccgac 600
cgcttcaccg gctccggctc cgggaccgac ttcacctga ccatcacctc cgtgaaggcc 660
gatgacctgg ccgtgtacta ctgtcaacag tattacgcct acccgtggac cttcggaggc 720
ggcaccaaagc tggagatcaa gcgcgctagc acgaccactc cggcgccggc cccaccgact 780
ccggcccaaa ctatcgcgag ccagccctg tcgctgaggc cgggaagcatg ccgcctgcc 840
gcccggagtg ctgtgcatac ccggggattg gacttcgcat gcgacttttg ggtgctgggtg 900
gtggttggtg gactcctggc ttgctatagc ttgctagtaa cagtggcctt tattattttc 960
tgggtgagga gtaagaggag caggctcctg cacagtgaat acatgaacat gactccccgc 1020
cgccccgggc ccacccgcaa gcattaccag ccctatgccc caccacgcga cttcgcagcc 1080
tatcgctccc tgagagtga gttcagcagg agcgcagacg ccccgcgta ccagcagggc 1140
cagaaccagc tctataacga gctcaatcta ggacgaagag aggagtagca tgttttggac 1200
aagagacgtg gccgggaccc tgagatgggg ggaagccga gaaggaagaa cctcaggaa 1260
ggcctgtaca atgaactgca gaaagataag atggcggagg cctacagtga gattgggatg 1320
aaaggcgagc gccggagggg caaggggcac gatggcctt accagggtct cagtacagcc 1380
accaaggaca cctacgacgc ccttcacatg caggccctgc cccctcgc 1428

```

```

SEQ ID NO: 59      moltype = DNA length = 1428
FEATURE           Location/Qualifiers
source            1..1428
                  mol_type = other DNA
                  organism = synthetic construct

```

```

SEQUENCE: 59
cagggtgcagc tccagcagag cggccctgag ctggtgcgcc ccggcgcttc cgtgaagatg 60
tcatgccagc cttctggcta cacgttcacc agctattggc tacactgggt caagcagagg 120
cccgacaggg gtctggagtg gatcggtatg attgacccgt caaatagtga caccgcgttc 180
aaccccaact tcaaggacaa ggccactctc aacgtggacc gctcgagcaa cacggcctac 240
atgctgctgt cctctctgac ctcgccggac agcgccgtgt actactgtgc caccgcgggc 300
tctacgtga gccctttgga ttattggggc cagggcacct ccgtcacctg gtccctcggg 360
ggaggcgggt caggcggagg tggtctggc ggtggcggat cggacatcat gatgtctcag 420
agtcctctct cctcgaccgt gtccgtgggc gagaagggtga ccgtgtcatg caagtctca 480
cagagccttc tgtacaccag ctcccagaag aactacctgg cgtggtacca gcagaagccc 540
ggacagagcc ccaagctgct gatctattgg gcttccacc gcgagagcgg cgtccccgac 600
cgcttcaccg gctccggctc cgggaccgac ttcacctga ccatcacctc cgtgaaggcc 660
gatgacctgg ccgtgtacta ctgtcaacag tattacgcct acccgtggac cttcggaggc 720
ggcaccaaagc tggagatcaa gcgcgctagc acgaccactc cggcgccggc cccaccgact 780
ccggcccaaa ctatcgcgag ccagccctg tcgctgaggc cgggaagcatg ccgcctgcc 840

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gccggagggtg ctgtgcatac ccgggggattg gacttcgcat gcgacttttg ggtgctgggtg 900
gtgggttggtg gagtcctggc ttgctatagc ttgctagtaa cagtggcctt tattattttc 960
tgggtgaggga gtaagaggag caggctcctg cacagtgact acatgaacat gactccccgc 1020
cgccccgggc ccacccgcaa gcattaccag ccctatgccc caccacgcga cttcgcagcc 1080
tatcgctccc tgagagtga gttcagcagg agcgcagacg cccccgcgt a ccagcagggc 1140
cagaaccagc tctataacga gctcaatcta ggacgaagag aggagtacga tgttttggac 1200
aagagacgtg gccgggaccc tgagatgggg ggaaagccga gaaggaagaa cctcaggaa 1260
ggcctgtaca atgaactgca gaaagataag atggcggagg cctacagtga gattgggatg 1320
aaaggcgagc gccggagggg caaggggcac gatggccttt accagggtct cagtacagcc 1380
accaaggaca cctacgacgc ccttcacatg caggccctgc cccctcgc 1428

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```

SEQ ID NO: 60      moltype = DNA length = 1428
FEATURE            Location/Qualifiers
source              1..1428
                    mol_type = other DNA
                    organism = synthetic construct

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SEQUENCE: 60
cagggtgcagc tccagcagag cggccctgag ctggtgcgcc ccggcgcttc cgtgaagatg 60
tcatgccagc cttctggcta cacgttcacc agctattggc tacactgggt caagcagagg 120
cccgacagag gtctggagtg gatcggtatg attgaccctg caaatagtga caccgccttc 180
aaccccaact tcaaggacaa ggccactctc aacgtggacc gctcgagcaa caccggcctac 240
atgctgctgt cctctctgac ctcgccggac agcgccgtgt actactgtgc caccctcggc 300
tcctacgtga gccctttgga ttattggggc cagggcacct ccgtcacgtg gtccctcggg 360
ggaggcggtt caggcggagg ttgctctggc ggtggcggat cggacatcat gatgtctcag 420
agtccctctt ctctgaccgt gtcctggggc gagaaggtga ccgtgtcatg caagtccctca 480
cagagccttc tgtacaccag ctcccagaag aactacctgg cgtggtacca gcagaagccc 540
ggacagagcc ccaagctgct gatctattgg gcttccaccc gcgagagcgg cgtccccgac 600
cgcttcaccg gctccggctc cgggaccgac ttcaccctga ccacacctc cgtgaaggcc 660
gatgacctgg ccgtgtacta ctgtcaacag tattacgcct acccgtggac cttcggaggc 720
ggcaccgaag tggagatcaa gcgcgctagc acgaccactc cggcgccggc cccaccgact 780
ccggcccca ctatcgcgag ccagccctctg tcgctgaggc cgggaagcatg ccgccctgcc 840
gccggagggtg ctgtgcatac ccggggattg gacttcgcat gcgacttttg ggtgctgggtg 900
gtgggttggtg gagtcctggc ttgctatagc ttgctagtaa cagtggcctt tattattttc 960
tgggtgaggga gtaagaggag caggctcctg cacagtgact acatgaacat gactccccgc 1020
cgccccgggc ccacccgcaa gcattaccag ccctatgccc caccacgcga cttcgcagcc 1080
tatcgctccc tgagagtga gttcagcagg agcgcagacg cccccgcgt a ccagcagggc 1140
cagaaccagc tctataacga gctcaatcta ggacgaagag aggagtacga tgttttggac 1200
aagagacgtg gccgggaccc tgagatgggg ggaaagccga gaaggaagaa cctcaggaa 1260
ggcctgtaca atgaactgca gaaagataag atggcggagg cctacagtga gattgggatg 1320
aaaggcgagc gccggagggg caaggggcac gatggccttt accagggtct cagtacagcc 1380
accaaggaca cctacgacgc ccttcacatg caggccctgc cccctcgc 1428

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SEQ ID NO: 61      moltype = DNA length = 1428
FEATURE            Location/Qualifiers
source              1..1428
                    mol_type = other DNA
                    organism = synthetic construct

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SEQUENCE: 61
cagggtgcagc tccagcagag cggccctgag ctggtgcgcc ccggcgcttc cgtgaagatg 60
tcatgccagc cttctggcta cacgttcacc agctattggc tacactgggt caagcagagg 120
cccgacagag gtctggagtg gatcggtatg attgaccctg caaatagtga caccgccttc 180
aaccccaact tcaaggacaa ggccactctc aacgtggacc gctcgagcaa caccggcctac 240
atgctgctgt cctctctgac ctcgccggac agcgccgtgt actactgtgc caccctcggc 300
tcctacgtga gccctttgga ttattggggc cagggcacct ccgtcacgtg gtccctcggg 360
ggaggcggtt caggcggagg ttgctctggc ggtggcggat cggacatcat gatgtctcag 420
agtccctctt ctctgaccgt gtcctggggc gagaaggtga ccgtgtcatg caagtccctca 480
cagagccttc tgtacaccag ctcccagaag aactacctgg cgtggtacca gcagaagccc 540
ggacagagcc ccaagctgct gatctattgg gcttccaccc gcgagagcgg cgtccccgac 600
cgcttcaccg gctccggctc cgggaccgac ttcaccctga ccacacctc cgtgaaggcc 660
gatgacctgg ccgtgtacta ctgtcaacag tattacgcct acccgtggac cttcggaggc 720
ggcaccgaag tggagatcaa gcgcgctagc acgaccactc cggcgccggc cccaccgact 780
ccggcccca ctatcgcgag ccagccctctg tcgctgaggc cgggaagcatg ccgccctgcc 840
gccggagggtg ctgtgcatac ccggggattg gacttcgcat gcgacttttg ggtgctgggtg 900
gtgggttggtg gagtcctggc ttgctatagc ttgctagtaa cagtggcctt tattattttc 960
tgggtgaggga gtaagaggag caggctcctg cacagtgact acatgaacat gactccccgc 1020
cgccccgggc ccacccgcaa gcattaccag ccctatgccc caccacgcga cttcgcagcc 1080
tatcgctccc tgagagtga gttcagcagg agcgcagacg cccccgcgt a ccagcagggc 1140
cagaaccagc tctataacga gctcaatcta ggacgaagag aggagtacga tgttttggac 1200
aagagacgtg gccgggaccc tgagatgggg ggaaagccga gaaggaagaa cctcaggaa 1260
ggcctgtaca atgaactgca gaaagataag atggcggagg cctacagtga gattgggatg 1320
aaaggcgagc gccggagggg caaggggcac gatggccttt accagggtct cagtacagcc 1380
accaaggaca cctacgacgc ccttcacatg caggccctgc cccctcgc 1428

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SEQ ID NO: 62      moltype = DNA length = 1428
FEATURE            Location/Qualifiers
source              1..1428

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mol_type = other DNA
organism = synthetic construct

SEQUENCE: 62
cagggtgcagc tccagcagag cgccctgag ctggtgcgcc ccggcgcttc cgtgaagatg 60
tcatgcccag cttctggcta cacgttcacc agctattggc tacactgggt caagcagagg 120
cccgacacag gtctggagtg gatcggtatg attgacccgt caaatagtga caccgcgttc 180
aaccccaact tcaaggacaa ggccactctc aacgtggacc gctcgagcaa cacggcctac 240
atgctgctgt cctctctgac ctccgggac agcgccgtgt actactgtgc caccatgggc 300
tctacgtga gccctttgga ttattggggc cagggcacct ccgtcacctg gtcctccggt 360
ggaggcgggt caggcggagg ttgctctggc ggtggcggat cggacatcat gatgtctcag 420
agtcctctct ctctgacagt gtccgtgggc gagaaggatg ccgtgtcatg caagtctcca 480
cagagccttc tgtacaccag ctcccagaag aactacctgg cgtggtacca gcagaagccc 540
ggacagagcc ccaagctgct gatctattgg gcttccacc ccgagagcgg cgtccccgac 600
cgcttcaccg gctccggctc cgggacgcac ttcacctga ccatcacctc cgtgaaggcc 660
gatgacctgg ccgtgtacta ctgtcaacag tattacgcct acccgtggac cttcggaggc 720
ggcaccagc tggagatcaa gcgcgctagc acgaccactc cggcgccgcg cccaccgact 780
ccggccccaa ctatccgag ccagccctg tccgtgaggg cggaaagcat ccgccctgcc 840
gcccggagtg ctgtgcatac cgggggattg gacttcgcat gcgacttttg ggtgctgggtg 900
gtggttggtg gactcctggc ttgctatagc ttgctagtaa cagtggcctt tattattttc 960
tgggtgagga gtaagaggag caggctcctg cacagtgaat acatgaacat gactccccgc 1020
gcgcccgggc ccacccgcaa gcattaccag ccctatgccc caccacgcga cttcgcagcc 1080
tatcgtctcc ccagagttaa gttcagcagg agcgcagacg ccccgcgta ccagcagggc 1140
cagaaccagc tctataacga gctcaatcta ggacgaagag aggagtagca tgttttggac 1200
aagagacgtg gccgggaccc tgagatgggg ggaagccga gaaggaagaa ccctcaggaa 1260
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aaaggcgcgc gccggagggg caaggggcac gatggcctt accagggtct cagtacagcc 1380
accaaggaca cctacgacgc ccttcacatg caggccctgc cccctcgc 1428

SEQ ID NO: 63      moltype = DNA length = 3930
FEATURE            Location/Qualifiers
source              1..3930
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 63
ggctccggtg cccgtcagtg ggcagagcgc acatcgccca cagtccccga gaagtggggg 60
ggaggggtcg gcaattgaac cgggtgcctag agaagggtggc gcggggtaaa ctgggaaagt 120
gatgtcgtgt actggtcccg cctttttccc gagggtgggg gagaaccgta tataagtga 180
gtagtcgccc tgaacgtttc ttttcgaac ggggttgccc ccagaacaca ggtaagtgcc 240
gtgtgtggtt cccgcggggc tggcctcttt acgggttatg gcccttgctg gcttgaatt 300
acttccacct ggtgcagta cgtgattctt gatcccgagc ttcgggttgg aagtgggtgg 360
gagagttcga ggccttgccc ttaaggagcc ccttcgctc gtgcttgagt tagggcctgg 420
cctgggctgt ggggcgcgct cgtgcgaatc tgggtggacc ttcgcgctg tctcgtgct 480
ttcgataagt cttagccatt taaaatttt tgatgacctg ctgcgacctg tttttctgg 540
caagatagtc ttgtaaatgc gggccaagat ctgcacactg gtatttcggt ttttggggcc 600
gcgggcggcg acggggcccg tgcgtcccag cgcacatggt cggcgaggcg gggcctgcga 660
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ggtctcgcgc cgcctgtgat cgcgccgccc tggggcgcaa ggctggcccg gtcggcacca 780
gttcggtgag cggaaagatg gccgcttccc ggccctgctg cagggagctc aaaatggagg 840
acgcgggcgt cgggagagcg ggcgggtgag tcacccacac aaaggaaaag ggcctttccg 900
tctcagcccg tgcgttcagt tgactccacg gagtaccggg cgcgctccag gcacctcgat 960
tagttctcga gcttttgagg tacgtcgtct ttaggttggg gggaggggtt ttatgcgatg 1020
gagtttcccc aactgagtg ggtggagact gaagttaggc cagcttggca cttgatgtaa 1080
ttctccttgg aatttgccct ttttgagttt ggatcttggg tcattctcaa gcctcagaca 1140
gtggttcaaa gtttttttct tcatttcag gtgctgtgac aagtttgtac aaaaaagcag 1200
gctgccacca tggccttacc agtgaccgccc ttgctcctgc cgtggcctt gctgctccac 1260
gccgccagcg cggggtccga acaaaaaact atctcagaag aggatctggg atcgcagggtg 1320
cagctccagc agagcggccc tgagctgggt cgcgccggcg cttccgtgaa gatgtcatgc 1380
cgagctcttg gctacacgtt caccagctat tggctacact gggtcaagca gaggcccgga 1440
caggggtctg agtggatcgg tatgattgac ccgtcaaata gtgacacccg cttcaacccc 1500
aacttcaagg acaaggccac tctcaacgtg gaccgctcga gcaacacggc ctacatgctg 1560
ctgtcctctc tgaacctcgg ggaacgcgcc gtgtactact gtgccacctc cggctcctac 1620
gtgagccctt tggattattg gggccagggc acctccgtca ccgtgtcctc cgggtggaggc 1680
ggttcaggcg gaggtggctc tggcggtggt ggatcggaca tcatgatgct tcagagtccc 1740
tcttctctga ccgtgtccgt gggcgagaag gtgaccgtgt catgcaagtc ctcacagagc 1800
cttctgtaca ccagctccca gaagaactac ctggcggtgt accagcagaa gcccgacag 1860
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accggctccg gctccgggac cgacttcacc ctgaccatca cctccgtgaa ggccgatgac 1980
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ccaactatcg cgagccagcc cctgtcgtg aggcgggaag catgcgccc tgcgcggga 2160
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tccctgagag tgaagtccag caggagcgca gacgcccccg cgtaccagca gggccagaac 2460
cagctctata acgagctcaa tctaggacga agagaggagt acgatgtttt ggacaagaga 2520

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cgtggccggg accctgagat ggggggaaag ccgagaagga agaaccctca ggaaggcctg 2580
tacaatgaac tgcagaaaga taagatggcg gaggcctaca gtgagattgg gatgaaaggc 2640
gagcgccgga ggggcaaggg gcacgatggc ctttaccagg gtctcagtac agccaccaag 2700
gacacctacg acgcccctca catgcaggcc ctgcccctc gcaactagg aagcggagct 2760
actaacttca gcctgctgaa gcaggctgga gacgtggagg agaaccctgg accttctaga 2820
atggacatgg cggatgagcc actcaatgga agccacacat ggctatccat tccatttgac 2880
ctcaatggct ctgtgtgttc aaccaacacc tcaaaccaga cagagccgta ctatgaactg 2940
acaagcaatg cagtoctcac attcatctat tttgtggtct gcatcattgg gttgtgtggc 3000
aacacacttg tcaatttatgt catcctccgc tatgccaaaga tgaagaccat caccacatt 3060
tacatctca acctggccat cgcagatgag ctcttcacgc tgggtctgcc tttcttggt 3120
atgcagggtg ctctgggtcca ctggcccttt ggcaaggcca tttgccgggt ggtcatgact 3180
gtggatggca tcaatcagtt caccagcatc ttctgacctg cagtcagtag catcgaccga 3240
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atgacacca tggctgtgtg gggagtctct ctgctggta tcttgcccat catgatata 3360
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cgagtgggct cctctaagag gaagaagtct gagaagaagg tcaccogaat ggtgtccatc 3600
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SEQ ID NO: 68 moltype = DNA length = 3930
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 organism = synthetic construct

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SEQ ID NO: 69 moltype = DNA length = 3930
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 organism = synthetic construct

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FEATURE           Location/Qualifiers
source           1..16
                  mol_type = protein
                  organism = synthetic construct

```

```

SEQUENCE: 71
LEWMGWIKPN NGLANY 16

```

```

SEQ ID NO: 72      moltype = AA  length = 9
FEATURE           Location/Qualifiers
source           1..9
                  mol_type = protein
                  organism = synthetic construct

```

```

SEQUENCE: 72
SEATTEFDY 9

```

```

SEQ ID NO: 73      moltype = AA  length = 9
FEATURE           Location/Qualifiers
source           1..9
                  mol_type = protein
                  organism = synthetic construct

```

```

SEQUENCE: 73
SEWTTEFDY 9

```

```

SEQ ID NO: 74      moltype = AA  length = 9
FEATURE           Location/Qualifiers
source           1..9
                  mol_type = protein
                  organism = synthetic construct

```

```

SEQUENCE: 74
SEFTTEFDY 9

```

```

SEQ ID NO: 75      moltype = AA  length = 9
FEATURE           Location/Qualifiers
source           1..9
                  mol_type = protein
                  organism = synthetic construct

```

```

SEQUENCE: 75
SEGTTEFDY 9

```

```

SEQ ID NO: 76      moltype = AA  length = 6
FEATURE           Location/Qualifiers
source           1..6
                  mol_type = protein
                  organism = synthetic construct

```

```

SEQUENCE: 76
ESVDSY 6

```

```

SEQ ID NO: 77      moltype = AA  length = 12
FEATURE           Location/Qualifiers
source           1..12
                  mol_type = protein
                  organism = synthetic construct

```

```

SEQUENCE: 77
PKLLIYRAST RE 12

```

```

SEQ ID NO: 78      moltype = AA  length = 6
FEATURE           Location/Qualifiers
source           1..6
                  mol_type = protein
                  organism = synthetic construct

```

-continued

SEQUENCE: 78
SKEDPL 6

SEQ ID NO: 79 moltype = AA length = 118
FEATURE Location/Qualifiers
source 1..118
mol_type = protein
organism = synthetic construct

SEQUENCE: 79
QVQLVQSGAE VKKPGASVKV SCKASGYIFT AYTMHWVRQA PGQGLEWMGW IKPNNGLAN Y 60
AQKFQGRVTM TRDTSISTAY MELSLRSD TAVYYCARSE ITTEFDYWGQ GTLVTVSS 118

SEQ ID NO: 80 moltype = AA length = 118
FEATURE Location/Qualifiers
source 1..118
mol_type = protein
organism = synthetic construct

SEQUENCE: 80
QVQLVQSGAE VKKPGASVKV SCKASGYIFT AYTMHWVRQA PGQGLEWMGW IKPNNGLAN Y 60
AQKFQGRVTM TRDTSISTAY MELSLRSD TAVYYCARSE ATTEFDYWGQ GTLVTVSS 118

SEQ ID NO: 81 moltype = AA length = 118
FEATURE Location/Qualifiers
source 1..118
mol_type = protein
organism = synthetic construct

SEQUENCE: 81
QVQLVQSGAE VKKPGASVKV SCKASGYIFT AYTMHWVRQA PGQGLEWMGW IKPNNGLAN Y 60
AQKFQGRVTM TRDTSISTAY MELSLRSD TAVYYCARSE WTTEFDYWGQ GTLVTVSS 118

SEQ ID NO: 82 moltype = AA length = 118
FEATURE Location/Qualifiers
source 1..118
mol_type = protein
organism = synthetic construct

SEQUENCE: 82
QVQLVQSGAE VKKPGASVKV SCKASGYIFT AYTMHWVRQA PGQGLEWMGW IKPNNGLAN Y 60
AQKFQGRVTM TRDTSISTAY MELSLRSD TAVYYCARSE FTTEFDYWGQ GTLVTVSS 118

SEQ ID NO: 83 moltype = AA length = 118
FEATURE Location/Qualifiers
source 1..118
mol_type = protein
organism = synthetic construct

SEQUENCE: 83
QVQLVQSGAE VKKPGASVKV SCKASGYIFT AYTMHWVRQA PGQGLEWMGW IKPNNGLAN Y 60
AQKFQGRVTM TRDTSISTAY MELSLRSD TAVYYCARSE GTTEFDYWGQ GTLVTVSS 118

SEQ ID NO: 84 moltype = AA length = 111
FEATURE Location/Qualifiers
source 1..111
mol_type = protein
organism = synthetic construct

SEQUENCE: 84
DIVLTQSPDS LAVSLGERAT INCKSSESVD SYANSFMHWY QKPGQPPKL LIYRASTRES 60
GVPDRFSGSG SRTDFTLTIS SLQAEDVAVY YCQSKEDPL TFGGGTKVEI K 111

SEQ ID NO: 85 moltype = DNA length = 354
FEATURE Location/Qualifiers
source 1..354
mol_type = other DNA
organism = synthetic construct

SEQUENCE: 85
caggtgcagc tegtgcagag cggcgccgag gtgaagaagc cggcgcttc cgtgaagggtg 60
tcctgtaagg cctctggcta catcttcacc gcctacacca tgcactgggt gcgccaggcc 120
ccgggacagg ggctggagtg gatgggctgg atcaagccta acaacggtct ggccaactac 180
ggcgagaagt tccagggcgc cgtgaccatg actcgggaca ccagcatctc taccgcgtac 240
atggagctgt cccgcctgcg ctccgatgac acggccgtgt actactgcgc ccgctcggag 300
atcaccaccg agttcgacta ttggggccag ggcaccctgg tgaccgtgtc gtcc 354

SEQ ID NO: 86 moltype = DNA length = 354
FEATURE Location/Qualifiers
source 1..354
mol_type = other DNA
organism = synthetic construct

SEQUENCE: 86

-continued

```

caggtgcagc tcgtgcagag cggcgccgag gtgaagaagc cggcgcttc cgtgaagggtg 60
tcctgtaagg cctctggcta catcttcacc gcctacacca tgcactgggt gcgccaggcc 120
cgggacagc ggctggagtg gatgggctgg atcaagccta acaacggtct ggccaactac 180
gcgcagaagt tccagggccg cgtgaccatg actcgggaca ccagcatctc taccgcgtac 240
atggagctgt cccgcctgcg ctccgatgac acggccgtgt actactgcgc ccgctcggag 300
gcaccaccg agttcgacta ttggggccag ggcaccctgg tgaccgtgtc gtcc 354

```

```

SEQ ID NO: 87          moltype = DNA length = 354
FEATURE               Location/Qualifiers
source                1..354
                      mol_type = other DNA
                      organism = synthetic construct

```

```

SEQUENCE: 87
caggtgcagc tcgtgcagag cggcgccgag gtgaagaagc cggcgcttc cgtgaagggtg 60
tcctgtaagg cctctggcta catcttcacc gcctacacca tgcactgggt gcgccaggcc 120
cgggacagc ggctggagtg gatgggctgg atcaagccta acaacggtct ggccaactac 180
gcgcagaagt tccagggccg cgtgaccatg actcgggaca ccagcatctc taccgcgtac 240
atggagctgt cccgcctgcg ctccgatgac acggccgtgt actactgcgc ccgctcggag 300
tgaccaccg agttcgacta ttggggccag ggcaccctgg tgaccgtgtc gtcc 354

```

```

SEQ ID NO: 88          moltype = DNA length = 354
FEATURE               Location/Qualifiers
source                1..354
                      mol_type = other DNA
                      organism = synthetic construct

```

```

SEQUENCE: 88
caggtgcagc tcgtgcagag cggcgccgag gtgaagaagc cggcgcttc cgtgaagggtg 60
tcctgtaagg cctctggcta catcttcacc gcctacacca tgcactgggt gcgccaggcc 120
cgggacagc ggctggagtg gatgggctgg atcaagccta acaacggtct ggccaactac 180
gcgcagaagt tccagggccg cgtgaccatg actcgggaca ccagcatctc taccgcgtac 240
atggagctgt cccgcctgcg ctccgatgac acggccgtgt actactgcgc ccgctcggag 300
ttcaccaccg agttcgacta ttggggccag ggcaccctgg tgaccgtgtc gtcc 354

```

```

SEQ ID NO: 89          moltype = DNA length = 354
FEATURE               Location/Qualifiers
source                1..354
                      mol_type = other DNA
                      organism = synthetic construct

```

```

SEQUENCE: 89
caggtgcagc tcgtgcagag cggcgccgag gtgaagaagc cggcgcttc cgtgaagggtg 60
tcctgtaagg cctctggcta catcttcacc gcctacacca tgcactgggt gcgccaggcc 120
cgggacagc ggctggagtg gatgggctgg atcaagccta acaacggtct ggccaactac 180
gcgcagaagt tccagggccg cgtgaccatg actcgggaca ccagcatctc taccgcgtac 240
atggagctgt cccgcctgcg ctccgatgac acggccgtgt actactgcgc ccgctcggag 300
ggcaccaccg agttcgacta ttggggccag ggcaccctgg tgaccgtgtc gtcc 354

```

```

SEQ ID NO: 90          moltype = DNA length = 333
FEATURE               Location/Qualifiers
source                1..333
                      mol_type = other DNA
                      organism = synthetic construct

```

```

SEQUENCE: 90
gacatcgtgc tgaccagag ccttgacagc ctggccgtgt ccctgggcca gcgcgccacc 60
attaactgca agagctccga gagtgtcgat agctacgcca actccttcac gcactggtag 120
cagcagaagc cgggccagcc acccaagctg ctgatctaca gggcttcac ccgcgagagc 180
ggcgccccg acaggttttc aggtttctggc tccgcaccg acttcaccct gacctctct 240
tctctgcagg ccgaggatgt ggcgggtgac tactgtcaac agtccaagga ggaccgctt 300
accttcggcg gcgcaccaa ggtggagatc aag 333

```

```

SEQ ID NO: 91          moltype = AA length = 244
FEATURE               Location/Qualifiers
source                1..244
                      mol_type = protein
                      organism = synthetic construct

```

```

SEQUENCE: 91
QVQLVQSGAE VKKPGASVKV SCKASGYIFT AYTMHWVRQA PGQGLEWMGW IKPNNGLAN 60
AQKFPQGRVTM TRDTSISTAY MELSLRLSDD TAVYYCARSE ITTEFDYWQ GTLVTVSSGG 120
GGSGGGGSGG GGSDIVLTQS PDSLAVSLGE RATINCKSSE SVDSYANSFM HWYQQKPGQP 180
PKLLIYRAST RESGVPDRFS GSGSRTDFTL TISSLQAEDV AVYYCQQSKE DPLTFGGGTK 240
VEIK 244

```

```

SEQ ID NO: 92          moltype = AA length = 244
FEATURE               Location/Qualifiers
source                1..244
                      mol_type = protein
                      organism = synthetic construct

```

-continued

SEQUENCE: 92
 QVQLVQSGAE VKKPGASVKV SCKASGYIFT AYTMHWVRQA PGQGLEWMGW IKPNNGLANY 60
 AQKFQGRVTM TRDTSISTAY MELSLRLSDD TAVYYCARSE ATTEFDYWGQ GTLVTVSSGG 120
 GSGGGGGSGG GGSDIVLTQS PDSLAVSLGE RATINCKSSE SVDSYANSFM HWYQQKPGQP 180
 PKLLIYRAST RESGVPDRFS GSGSRTDFTL TISSLQAEDV AVYYCQQSKE DPLTFGGGTK 240
 VEIK 244

SEQ ID NO: 93 moltype = AA length = 244
 FEATURE Location/Qualifiers
 source 1..244
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 93
 QVQLVQSGAE VKKPGASVKV SCKASGYIFT AYTMHWVRQA PGQGLEWMGW IKPNNGLANY 60
 AQKFQGRVTM TRDTSISTAY MELSLRLSDD TAVYYCARSE WTTEFDYWGQ GTLVTVSSGG 120
 GSGGGGGSGG GGSDIVLTQS PDSLAVSLGE RATINCKSSE SVDSYANSFM HWYQQKPGQP 180
 PKLLIYRAST RESGVPDRFS GSGSRTDFTL TISSLQAEDV AVYYCQQSKE DPLTFGGGTK 240
 VEIK 244

SEQ ID NO: 94 moltype = AA length = 244
 FEATURE Location/Qualifiers
 source 1..244
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 94
 QVQLVQSGAE VKKPGASVKV SCKASGYIFT AYTMHWVRQA PGQGLEWMGW IKPNNGLANY 60
 AQKFQGRVTM TRDTSISTAY MELSLRLSDD TAVYYCARSE FTTEFDYWGQ GTLVTVSSGG 120
 GSGGGGGSGG GGSDIVLTQS PDSLAVSLGE RATINCKSSE SVDSYANSFM HWYQQKPGQP 180
 PKLLIYRAST RESGVPDRFS GSGSRTDFTL TISSLQAEDV AVYYCQQSKE DPLTFGGGTK 240
 VEIK 244

SEQ ID NO: 95 moltype = AA length = 244
 FEATURE Location/Qualifiers
 source 1..244
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 95
 QVQLVQSGAE VKKPGASVKV SCKASGYIFT AYTMHWVRQA PGQGLEWMGW IKPNNGLANY 60
 AQKFQGRVTM TRDTSISTAY MELSLRLSDD TAVYYCARSE GTTEFDYWGQ GTLVTVSSGG 120
 GSGGGGGSGG GGSDIVLTQS PDSLAVSLGE RATINCKSSE SVDSYANSFM HWYQQKPGQP 180
 PKLLIYRAST RESGVPDRFS GSGSRTDFTL TISSLQAEDV AVYYCQQSKE DPLTFGGGTK 240
 VEIK 244

SEQ ID NO: 96 moltype = DNA length = 732
 FEATURE Location/Qualifiers
 source 1..732
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 96
 caggtgcagc tcgtgcagag cggcgccgag gtgaagaagc ccggcgcttc cgtgaaggtg 60
 tcctgtaagg cctctggcta catcttcacc gcctacacca tgcactgggt gcgccaggcc 120
 ccgggacagg gctctggagt gatgggctgg atcaagccta acaacggctc ggccaactac 180
 gcgcagaagt tccaggcccg cgtgaccatg actcgggaca ccagcatctc taccgcgtac 240
 atggagctgt cccgcctgcg ctccgatgac acggccgtgt actactgcgc ccgctcggag 300
 atcaccaccg agttcgacta ttggggccag ggcaccctgg tgaccgtgtc gtcgggtgga 360
 ggcggttcag gcggaggtgg ctctggcggt ggcggatcgg acatcgtgct gaccagagc 420
 cctgacagcc tggcctgtgc cctgggagag cgcgccacca ttaactgcaa gagctccgag 480
 agtgtcgata gctacgcca ctccttcatt cactggatcc agcagaagcc cggccagcca 540
 cccaagctgc tgatctacag ggtctccacc cgcgagagcg gcgtcccca cagggtttca 600
 ggtctctggc cccgcaccga cttcaccctg accatctctt ctctgcagcg cgaggatgtg 660
 gcgggtgact actgtcaaca gtccaaggag gaccgcgcta ccttcggcgg cggcaccagg 720
 gtggagatca ag 732

SEQ ID NO: 97 moltype = DNA length = 732
 FEATURE Location/Qualifiers
 source 1..732
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 97
 caggtgcagc tcgtgcagag cggcgccgag gtgaagaagc ccggcgcttc cgtgaaggtg 60
 tcctgtaagg cctctggcta catcttcacc gcctacacca tgcactgggt gcgccaggcc 120
 ccgggacagg gctctggagt gatgggctgg atcaagccta acaacggctc ggccaactac 180
 gcgcagaagt tccaggcccg cgtgaccatg actcgggaca ccagcatctc taccgcgtac 240
 atggagctgt cccgcctgcg ctccgatgac acggccgtgt actactgcgc ccgctcggag 300
 gccaccaccg agttcgacta ttggggccag ggcaccctgg tgaccgtgtc gtcgggtgga 360
 ggcggttcag gcggaggtgg ctctggcggt ggcggatcgg acatcgtgct gaccagagc 420

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```

cctgacagcc tggccgtgtc cctgggcccag cgcgccacca ttaactgcaa gagctccgag 480
agtgtcgata gctacgccaa ctcttccatg cactgggtacc agcagaagcc cggccagcca 540
cccaagctgc tgatctacag ggcttccacc cgcgagagcg gcgtcccga cagggtttca 600
ggttctggct cccgcaccga ctccaccctg accatctctt ctctgcaggc cgaggatgtg 660
gcggtgtact actgtcaaca gtccaaggag gaccgcgcta ccttcggcgg cggcaccaaag 720
gtggagatca ag 732

```

```

SEQ ID NO: 98      moltype = DNA length = 732
FEATURE           Location/Qualifiers
source            1..732
                  mol_type = other DNA
                  organism = synthetic construct

```

```

SEQUENCE: 98
caggtgcagc tcgtgcagag cggcgcccag gtgaagaagc cggcgcttc cgtgaagggt 60
tcctgtaagg cctctggcta catcttcacc gcctacacca tgcactgggt gcgccaggcc 120
ccgggacagg ggctggagtg gatgggctgg atcaagccta acaacggtct ggccaactac 180
gcgcagaagt tccagggccg cgtgaccatg actcgggaca ccagcatctc taccgcgtac 240
atggagctgt cccgcctgcg ctccgatgac acggccgtgt actactgcgc ccgctcggag 300
tggaccaccg agttcgacta ttggggccag ggcaccctgg tgaccgtgtc gtccggtgga 360
ggcggttcag cggagagtggt ctctggcggg ggcggatcgg acatcgtgct gaccagagc 420
cctgacagcc tggccgtgtc cctgggcccag cgcgccacca ttaactgcaa gagctccgag 480
agtgtcgata gctacgccaa ctcttccatg cactgggtacc agcagaagcc cggccagcca 540
cccaagctgc tgatctacag ggcttccacc cgcgagagcg gcgtcccga cagggtttca 600
ggttctggct cccgcaccga ctccaccctg accatctctt ctctgcaggc cgaggatgtg 660
gcggtgtact actgtcaaca gtccaaggag gaccgcgcta ccttcggcgg cggcaccaaag 720
gtggagatca ag 732

```

```

SEQ ID NO: 99      moltype = DNA length = 732
FEATURE           Location/Qualifiers
source            1..732
                  mol_type = other DNA
                  organism = synthetic construct

```

```

SEQUENCE: 99
caggtgcagc tcgtgcagag cggcgcccag gtgaagaagc cggcgcttc cgtgaagggt 60
tcctgtaagg cctctggcta catcttcacc gcctacacca tgcactgggt gcgccaggcc 120
ccgggacagg ggctggagtg gatgggctgg atcaagccta acaacggtct ggccaactac 180
gcgcagaagt tccagggccg cgtgaccatg actcgggaca ccagcatctc taccgcgtac 240
atggagctgt cccgcctgcg ctccgatgac acggccgtgt actactgcgc ccgctcggag 300
tccaccaccg agttcgacta ttggggccag ggcaccctgg tgaccgtgtc gtccggtgga 360
ggcggttcag cggagagtggt ctctggcggg ggcggatcgg acatcgtgct gaccagagc 420
cctgacagcc tggccgtgtc cctgggcccag cgcgccacca ttaactgcaa gagctccgag 480
agtgtcgata gctacgccaa ctcttccatg cactgggtacc agcagaagcc cggccagcca 540
cccaagctgc tgatctacag ggcttccacc cgcgagagcg gcgtcccga cagggtttca 600
ggttctggct cccgcaccga ctccaccctg accatctctt ctctgcaggc cgaggatgtg 660
gcggtgtact actgtcaaca gtccaaggag gaccgcgcta ccttcggcgg cggcaccaaag 720
gtggagatca ag 732

```

```

SEQ ID NO: 100     moltype = DNA length = 732
FEATURE           Location/Qualifiers
source            1..732
                  mol_type = other DNA
                  organism = synthetic construct

```

```

SEQUENCE: 100
caggtgcagc tcgtgcagag cggcgcccag gtgaagaagc cggcgcttc cgtgaagggt 60
tcctgtaagg cctctggcta catcttcacc gcctacacca tgcactgggt gcgccaggcc 120
ccgggacagg ggctggagtg gatgggctgg atcaagccta acaacggtct ggccaactac 180
gcgcagaagt tccagggccg cgtgaccatg actcgggaca ccagcatctc taccgcgtac 240
atggagctgt cccgcctgcg ctccgatgac acggccgtgt actactgcgc ccgctcggag 300
ggcaccaccg agttcgacta ttggggccag ggcaccctgg tgaccgtgtc gtccggtgga 360
ggcggttcag cggagagtggt ctctggcggg ggcggatcgg acatcgtgct gaccagagc 420
cctgacagcc tggccgtgtc cctgggcccag cgcgccacca ttaactgcaa gagctccgag 480
agtgtcgata gctacgccaa ctcttccatg cactgggtacc agcagaagcc cggccagcca 540
cccaagctgc tgatctacag ggcttccacc cgcgagagcg gcgtcccga cagggtttca 600
ggttctggct cccgcaccga ctccaccctg accatctctt ctctgcaggc cgaggatgtg 660
gcggtgtact actgtcaaca gtccaaggag gaccgcgcta ccttcggcgg cggcaccaaag 720
gtggagatca ag 732

```

```

SEQ ID NO: 101     moltype = AA length = 472
FEATURE           Location/Qualifiers
source            1..472
                  mol_type = protein
                  organism = synthetic construct

```

```

SEQUENCE: 101
QVQLVQSGAE VKKPGASVKV SCKASGYIFT AYTMHWVRQA PGQGLEWMGW IKPNNGLAN 60
AQKFQGRVTM TRDTSISTAY MELSLRLSDD TAVYYCARSE ITTEFDYWGQ GTLVTVSSGG 120
GSGSGGSGSG GGSIVLTQS PDSLAVSLGE RATINCKSSE SVDSYANSFM HWYQKPGQP 180

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PKLLIYRAST	RESGVPDRFS	SGSRTDFTL	TISSLQAEDV	AVYYCQKSKE	DPLTFGGGTK	240
VEIKASTTTP	APRPPTPAPT	IASQPLSLRP	EACRPAAGGA	VHTRGLDFAC	DFWVLVVVG	300
VLACYSLLVT	VAFIIFWVRS	KRSRLHSDY	MNMTPRRPGP	TRKHYQPYAP	PRDFAAYRSL	360
RVKFSRSADA	PAYQQGQNQL	YNELNLGRRE	EYDVLDKRRG	RDPEMGGKPR	RKNPQEGLYN	420
ELQKDKMAEA	YSEIGMKGER	RRGKGHDGLY	QGLSTATKDT	YDALHMQALP	PR	472

SEQ ID NO: 102 moltype = AA length = 472
 FEATURE Location/Qualifiers
 source 1..472
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 102

QVQLVQSGAE	VKKPGASVKV	SCKASGYIFT	AYTMHWVRQA	PGQGLEWMGW	IKPNNGNLANY	60
AQKFQGRVTM	TRDTSISTAY	MELSLRLSDD	TAVYYCARSE	ATTEFDYWGQ	GTLVTVSSGG	120
GGSGGGGSGG	GGSDIVLTQS	PDSLAVSLGE	RATINCKSSE	SVDSYANSFM	HWYQQKPGQP	180
PKLLIYRAST	RESGVPDRFS	SGSRTDFTL	TISSLQAEDV	AVYYCQKSKE	DPLTFGGGTK	240
VEIKASTTTP	APRPPTPAPT	IASQPLSLRP	EACRPAAGGA	VHTRGLDFAC	DFWVLVVVG	300
VLACYSLLVT	VAFIIFWVRS	KRSRLHSDY	MNMTPRRPGP	TRKHYQPYAP	PRDFAAYRSL	360
RVKFSRSADA	PAYQQGQNQL	YNELNLGRRE	EYDVLDKRRG	RDPEMGGKPR	RKNPQEGLYN	420
ELQKDKMAEA	YSEIGMKGER	RRGKGHDGLY	QGLSTATKDT	YDALHMQALP	PR	472

SEQ ID NO: 103 moltype = AA length = 472
 FEATURE Location/Qualifiers
 source 1..472
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 103

QVQLVQSGAE	VKKPGASVKV	SCKASGYIFT	AYTMHWVRQA	PGQGLEWMGW	IKPNNGNLANY	60
AQKFQGRVTM	TRDTSISTAY	MELSLRLSDD	TAVYYCARSE	WTTEFDYWGQ	GTLVTVSSGG	120
GGSGGGGSGG	GGSDIVLTQS	PDSLAVSLGE	RATINCKSSE	SVDSYANSFM	HWYQQKPGQP	180
PKLLIYRAST	RESGVPDRFS	SGSRTDFTL	TISSLQAEDV	AVYYCQKSKE	DPLTFGGGTK	240
VEIKASTTTP	APRPPTPAPT	IASQPLSLRP	EACRPAAGGA	VHTRGLDFAC	DFWVLVVVG	300
VLACYSLLVT	VAFIIFWVRS	KRSRLHSDY	MNMTPRRPGP	TRKHYQPYAP	PRDFAAYRSL	360
RVKFSRSADA	PAYQQGQNQL	YNELNLGRRE	EYDVLDKRRG	RDPEMGGKPR	RKNPQEGLYN	420
ELQKDKMAEA	YSEIGMKGER	RRGKGHDGLY	QGLSTATKDT	YDALHMQALP	PR	472

SEQ ID NO: 104 moltype = AA length = 472
 FEATURE Location/Qualifiers
 source 1..472
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 104

QVQLVQSGAE	VKKPGASVKV	SCKASGYIFT	AYTMHWVRQA	PGQGLEWMGW	IKPNNGNLANY	60
AQKFQGRVTM	TRDTSISTAY	MELSLRLSDD	TAVYYCARSE	FTTEFDYWGQ	GTLVTVSSGG	120
GGSGGGGSGG	GGSDIVLTQS	PDSLAVSLGE	RATINCKSSE	SVDSYANSFM	HWYQQKPGQP	180
PKLLIYRAST	RESGVPDRFS	SGSRTDFTL	TISSLQAEDV	AVYYCQKSKE	DPLTFGGGTK	240
VEIKASTTTP	APRPPTPAPT	IASQPLSLRP	EACRPAAGGA	VHTRGLDFAC	DFWVLVVVG	300
VLACYSLLVT	VAFIIFWVRS	KRSRLHSDY	MNMTPRRPGP	TRKHYQPYAP	PRDFAAYRSL	360
RVKFSRSADA	PAYQQGQNQL	YNELNLGRRE	EYDVLDKRRG	RDPEMGGKPR	RKNPQEGLYN	420
ELQKDKMAEA	YSEIGMKGER	RRGKGHDGLY	QGLSTATKDT	YDALHMQALP	PR	472

SEQ ID NO: 105 moltype = AA length = 472
 FEATURE Location/Qualifiers
 source 1..472
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 105

QVQLVQSGAE	VKKPGASVKV	SCKASGYIFT	AYTMHWVRQA	PGQGLEWMGW	IKPNNGNLANY	60
AQKFQGRVTM	TRDTSISTAY	MELSLRLSDD	TAVYYCARSE	GTTEFDYWGQ	GTLVTVSSGG	120
GGSGGGGSGG	GGSDIVLTQS	PDSLAVSLGE	RATINCKSSE	SVDSYANSFM	HWYQQKPGQP	180
PKLLIYRAST	RESGVPDRFS	SGSRTDFTL	TISSLQAEDV	AVYYCQKSKE	DPLTFGGGTK	240
VEIKASTTTP	APRPPTPAPT	IASQPLSLRP	EACRPAAGGA	VHTRGLDFAC	DFWVLVVVG	300
VLACYSLLVT	VAFIIFWVRS	KRSRLHSDY	MNMTPRRPGP	TRKHYQPYAP	PRDFAAYRSL	360
RVKFSRSADA	PAYQQGQNQL	YNELNLGRRE	EYDVLDKRRG	RDPEMGGKPR	RKNPQEGLYN	420
ELQKDKMAEA	YSEIGMKGER	RRGKGHDGLY	QGLSTATKDT	YDALHMQALP	PR	472

SEQ ID NO: 106 moltype = AA length = 881
 FEATURE Location/Qualifiers
 source 1..881
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 106

GSEQKLISEE	DLGSQVQLVQ	SGAEVKKPGA	SVKVSCKASG	YIFTAYTMHW	VRQAPQGGL	60
WMGWIKPNNG	LANYAQKFQ	RVMTMTRDTSI	STAYMELSLR	RSDDTAVYYC	ARSEITTEFD	120
YWGQGTLLTV	SSGGGSGGG	LTQSPDSLAV	SLGERATINC	KSESSEVDSYA		180
NSFMHWYQQK	PGQPKLLIY	RASTRESGVP	DRFSGSGSRT	DFTLTISSLQ	AEDVAVYYCQ	240

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QSKEDPLTFG	GGTKVEIKAS	TTTPAPRPPT	PAPTIASQPL	SLRPEACRPA	AGGAVHTRGL	300
DFACDFWVLV	VVGGLVACYS	LLVTVAFIIF	WVRSKRSRL	HSDYMNMTPR	RPGPTRKHYQ	360
PYAPPRDFAA	YRSLRVKFSR	SADAPAYQQG	QNQLYNELNL	GRREEYDVL	KRRGRDPEMG	420
GKPRRKNPQE	GLYNELQDK	MAEAYSEIGM	KGERRRGKGH	DGLYQGLSTA	TKDTYDALHM	480
QALPRTSGS	GATNFSLKQ	AGDVEENPGP	SRMDMADEPL	NGSHTWLSIP	FDLNGSVVST	540
NTSNQTEPY	DLTSNAVLTF	IYFVVCIIGL	CGNTLVIYVI	LRYAKMKTIT	NIYILNLAIA	600
DELFLMLGLPF	LAMQVALVHW	PFGKAICRVV	MTVDGINQFT	SIFCLTVMSI	DRYLAVVHPI	660
KSAKWRPRT	AKMITMAVWG	VSLLVILPIM	IYAGLRSNQW	GRSSCTINWP	GESGAWYTF	720
IYTFILGFL	VPLTIICLCY	LFIIIKVKSS	GIRVGSSKRK	KSEKKVTRMV	SIVVAVFIFC	780
WLPFYIFNVS	SVSMAISPTP	ALKGMFDFVV	VLTYANSCAN	PILYAFSLDN	FKKSPQNVLC	840
LVKVSGTDDG	ERSDSKQDKS	RLNETTETQR	TLLNGDLQTS	I		881

SEQ ID NO: 107 moltype = AA length = 881
FEATURE Location/Qualifiers
source 1..881
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 107

GSEQKLISEE	DLGSQVQLVQ	SGAEVKKPGA	SVKVSCKASG	YIFTAYTMHW	VRQAPGGGLE	60
WMGWIKPNNG	LANYAQKFGQ	RVTMTRDTSI	STAYMELSRL	RSDDTAVYYC	ARSEATTEFD	120
YWGQGTLLTV	SSGGGSGGGG	GSGGGGSDIV	LTQSPDSLAV	SLGERATINC	KSSESVDSYA	180
NSFMHWYQQK	PGQPPKLLIY	RASTRESGVP	DRFSGSGSRT	DFTLTISLQ	AEDVAVYYCQ	240
QSKEDPLTFG	GGTKVEIKAS	TTTPAPRPPT	PAPTIASQPL	SLRPEACRPA	AGGAVHTRGL	300
DFACDFWVLV	VVGGLVACYS	LLVTVAFIIF	WVRSKRSRL	HSDYMNMTPR	RPGPTRKHYQ	360
PYAPPRDFAA	YRSLRVKFSR	SADAPAYQQG	QNQLYNELNL	GRREEYDVL	KRRGRDPEMG	420
GKPRRKNPQE	GLYNELQDK	MAEAYSEIGM	KGERRRGKGH	DGLYQGLSTA	TKDTYDALHM	480
QALPRTSGS	GATNFSLKQ	AGDVEENPGP	SRMDMADEPL	NGSHTWLSIP	FDLNGSVVST	540
NTSNQTEPY	DLTSNAVLTF	IYFVVCIIGL	CGNTLVIYVI	LRYAKMKTIT	NIYILNLAIA	600
DELFLMLGLPF	LAMQVALVHW	PFGKAICRVV	MTVDGINQFT	SIFCLTVMSI	DRYLAVVHPI	660
KSAKWRPRT	AKMITMAVWG	VSLLVILPIM	IYAGLRSNQW	GRSSCTINWP	GESGAWYTF	720
IYTFILGFL	VPLTIICLCY	LFIIIKVKSS	GIRVGSSKRK	KSEKKVTRMV	SIVVAVFIFC	780
WLPFYIFNVS	SVSMAISPTP	ALKGMFDFVV	VLTYANSCAN	PILYAFSLDN	FKKSPQNVLC	840
LVKVSGTDDG	ERSDSKQDKS	RLNETTETQR	TLLNGDLQTS	I		881

SEQ ID NO: 108 moltype = AA length = 881
FEATURE Location/Qualifiers
source 1..881
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 108

GSEQKLISEE	DLGSQVQLVQ	SGAEVKKPGA	SVKVSCKASG	YIFTAYTMHW	VRQAPGGGLE	60
WMGWIKPNNG	LANYAQKFGQ	RVTMTRDTSI	STAYMELSRL	RSDDTAVYYC	ARSEWTTEFD	120
YWGQGTLLTV	SSGGGSGGGG	GSGGGGSDIV	LTQSPDSLAV	SLGERATINC	KSSESVDSYA	180
NSFMHWYQQK	PGQPPKLLIY	RASTRESGVP	DRFSGSGSRT	DFTLTISLQ	AEDVAVYYCQ	240
QSKEDPLTFG	GGTKVEIKAS	TTTPAPRPPT	PAPTIASQPL	SLRPEACRPA	AGGAVHTRGL	300
DFACDFWVLV	VVGGLVACYS	LLVTVAFIIF	WVRSKRSRL	HSDYMNMTPR	RPGPTRKHYQ	360
PYAPPRDFAA	YRSLRVKFSR	SADAPAYQQG	QNQLYNELNL	GRREEYDVL	KRRGRDPEMG	420
GKPRRKNPQE	GLYNELQDK	MAEAYSEIGM	KGERRRGKGH	DGLYQGLSTA	TKDTYDALHM	480
QALPRTSGS	GATNFSLKQ	AGDVEENPGP	SRMDMADEPL	NGSHTWLSIP	FDLNGSVVST	540
NTSNQTEPY	DLTSNAVLTF	IYFVVCIIGL	CGNTLVIYVI	LRYAKMKTIT	NIYILNLAIA	600
DELFLMLGLPF	LAMQVALVHW	PFGKAICRVV	MTVDGINQFT	SIFCLTVMSI	DRYLAVVHPI	660
KSAKWRPRT	AKMITMAVWG	VSLLVILPIM	IYAGLRSNQW	GRSSCTINWP	GESGAWYTF	720
IYTFILGFL	VPLTIICLCY	LFIIIKVKSS	GIRVGSSKRK	KSEKKVTRMV	SIVVAVFIFC	780
WLPFYIFNVS	SVSMAISPTP	ALKGMFDFVV	VLTYANSCAN	PILYAFSLDN	FKKSPQNVLC	840
LVKVSGTDDG	ERSDSKQDKS	RLNETTETQR	TLLNGDLQTS	I		881

SEQ ID NO: 109 moltype = AA length = 881
FEATURE Location/Qualifiers
source 1..881
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 109

GSEQKLISEE	DLGSQVQLVQ	SGAEVKKPGA	SVKVSCKASG	YIFTAYTMHW	VRQAPGGGLE	60
WMGWIKPNNG	LANYAQKFGQ	RVTMTRDTSI	STAYMELSRL	RSDDTAVYYC	ARSEFTTEFD	120
YWGQGTLLTV	SSGGGSGGGG	GSGGGGSDIV	LTQSPDSLAV	SLGERATINC	KSSESVDSYA	180
NSFMHWYQQK	PGQPPKLLIY	RASTRESGVP	DRFSGSGSRT	DFTLTISLQ	AEDVAVYYCQ	240
QSKEDPLTFG	GGTKVEIKAS	TTTPAPRPPT	PAPTIASQPL	SLRPEACRPA	AGGAVHTRGL	300
DFACDFWVLV	VVGGLVACYS	LLVTVAFIIF	WVRSKRSRL	HSDYMNMTPR	RPGPTRKHYQ	360
PYAPPRDFAA	YRSLRVKFSR	SADAPAYQQG	QNQLYNELNL	GRREEYDVL	KRRGRDPEMG	420
GKPRRKNPQE	GLYNELQDK	MAEAYSEIGM	KGERRRGKGH	DGLYQGLSTA	TKDTYDALHM	480
QALPRTSGS	GATNFSLKQ	AGDVEENPGP	SRMDMADEPL	NGSHTWLSIP	FDLNGSVVST	540
NTSNQTEPY	DLTSNAVLTF	IYFVVCIIGL	CGNTLVIYVI	LRYAKMKTIT	NIYILNLAIA	600
DELFLMLGLPF	LAMQVALVHW	PFGKAICRVV	MTVDGINQFT	SIFCLTVMSI	DRYLAVVHPI	660
KSAKWRPRT	AKMITMAVWG	VSLLVILPIM	IYAGLRSNQW	GRSSCTINWP	GESGAWYTF	720
IYTFILGFL	VPLTIICLCY	LFIIIKVKSS	GIRVGSSKRK	KSEKKVTRMV	SIVVAVFIFC	780
WLPFYIFNVS	SVSMAISPTP	ALKGMFDFVV	VLTYANSCAN	PILYAFSLDN	FKKSPQNVLC	840

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LVKVSQTDDG ERSDSKQDKS RLNETTETQR TLLNGDLQTS I 881

SEQ ID NO: 110 moltype = AA length = 881
FEATURE Location/Qualifiers
source 1..881
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 110
GSEQKLISEE DLGSQVQLVQ SGAEVKKPGA SVKVSCKASG YIFTAYTMHW VRQAPQGGL 60
WMGWIKPNNG LANYAQKFGQ RVTMTRDTSI STAYMELSRL RSDDTAVYYC ARSEGTEFD 120
YWQGGLTVTV SSGGGGSGGG GSGGGGSDIV LTQSPDSLAV SLGERATINC KSESVDVSYA 180
NSFMHWYQQK PGQPPKLLIY RASTRESGVP DRFSGSGSRT DFTLTISLQ AEDVAVYYCQ 240
QSKEDPLTFG GGTKEIKAS TTPAPRPPT PPTIASQPL SLRPEACRPA AGGAVHTRGL 300
DFACDFWLVV VVGVLACYS LLVTVAFIIF WVRSKRSRL HSDYMNMTPR RGPTRKHYQ 360
PYAPPRDFAA YRSLRVKFSR SADAPAYQQG QNQLYNELNL GRREEYDVL KRRGRDPEMG 420
GKPRRNPKQE GLYNELQKDK MAEAYSEIGM KGERRRGKGH DGLYQGLSTA TKDLYDALHM 480
QALPPRTSGS GATNFSLLKQ AGDVEENPGP SRMDMADEPL NGSHTWLSIP FDLNGSVVST 540
NTSNQTEPHY DLTSNAVLTF IYFVVCIIGL CGNTLVIYVI LRYAKMKIT NIYILNLAIA 600
DELFLMLGLPF LAMQVALVHW PFGKAICRVV MTVDGINQFT SIFCLTVMSI DRYLAVVHPI 660
KSAKWRRPRT AKMITMAVWG VSLLVILPIM IYAGLRSNQW GRSSCTINWP GESGAWYTF 720
IYTFILGFL VPLTIICLCY LFIIKVKSS GIRVSSSKRK KSEKKVTRMV SIVVAVFIFC 780
WLPFYIFNVS SVSMAISPTP ALKGMDFDVV VLTANSNCAN PILYAFSLDN FKKSFPQNVLC 840
LVKVSQTDDG ERSDSKQDKS RLNETTETQR TLLNGDLQTS I 881

SEQ ID NO: 111 moltype = DNA length = 1416
FEATURE Location/Qualifiers
source 1..1416
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 111
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ccgggacagg ggctggagtg gatgggctgg atcaagccta acaacggtct ggccaactac 180
gcgcagaagt tccagggccg cgtgacctg actcgggaca ccagcatctc taccgcgtac 240
atggagctgt cccgcctgcg ctccgatgac acggccgtgt actactgcgc ccgctcggag 300
atcaccaccg agttcgacta ttggggccag ggcaccctgt tgaccgtgtc gtccggtgga 360
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cctgacagcc tggccgtgtc cctggcgag cgcgccacca ttaactgcaa gagctccgag 480
agtgtcgata gctacgccaa ctcttcctat cactggtacc agcagaagcc cggccagcca 540
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cgggaccctg agatgggggg aaagccgaga aggaagaacc ctacggaagg cctgtacaat 1260
gaactgcaga aagataagat ggcgagggcc tacagtgaga ttgggatgaa aggcgagcgc 1320
cggaggggca agggggcacga tggcctttac cagggtctca gtacagccac caaggacacc 1380
tacgacgccc ttcacatgca ggcctcgcgc cctcgc 1416

SEQ ID NO: 112 moltype = DNA length = 1416
FEATURE Location/Qualifiers
source 1..1416
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 112
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ccgggacagg ggctggagtg gatgggctgg atcaagccta acaacggtct ggccaactac 180
gcgcagaagt tccagggccg cgtgacctg actcgggaca ccagcatctc taccgcgtac 240
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gccaccaccg agttcgacta ttggggccag ggcaccctgt tgaccgtgtc gtccggtgga 360
ggcgggtcag gcggaggtgg cctctggcgtt ggccgatcgg acatcgtgtc gaccagagc 420
cctgacagcc tggccgtgtc cctggcgag cgcgccacca ttaactgcaa gagctccgag 480
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ggttctggct cccgcaccga ctccaccctg acctctctt ctctgcaggc cgaggatgtg 660
gcggtgtact actgtcaaca gtccaaggag gacccgctta ccttcggcgg cggcaccagg 720
gtggagatca aggtacagc gaccactccg gcgcgcgcgc caccgactcc ggccccaact 780
atcgcagacc agccctgtc cgtgagggcg gaagcatgcc gccctgcgc cggaggtgct 840
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gtcctggcctt gctatagcctt gctagtaaca gtggccttta ttattttctg ggtgaggagt 960
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cgggaccctg agatgggggg aaagccgaga aggaagaacc ctcaggaagg cctgtacaat 1260
gaactgcaga aagataagat ggcggaggcc tacagtgaga ttgggatgaa aggcgagcgc 1320
cggaggggca agggggcacga tggcctttac cagggtctca gtacagccac caaggacacc 1380
tacgacgccc ttcacatgca ggccttgccc cctcgc 1416

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SEQ ID NO: 113      moltype = DNA length = 1416
FEATURE            Location/Qualifiers
source             1..1416
                   mol_type = other DNA
                   organism = synthetic construct

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SEQUENCE: 113
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tcctgttaagg cctctggcta catcttcacc gcctacacca tgcactgggt gcgccaggcc 120
ccgggacagg ggctggagtg gatgggctgg atcaagccta acaacggtct ggccaactac 180
gcgcagaagt tccagggccg cgtgaccatg actcgggaca ccagcatctc taccgcgtac 240
atggagctgt cccgcctgcg ctccgatgac acggccgtgt actactgcgc ccgctcggag 300
tggaccaccg agttcgacta ttggggccag ggcaccctgg tgaccgtgtc gtccggtgga 360
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cctgacagcc tggccgtgtc cctgggcgag cgcgccacca ttaactgcaa gagctccgag 480
agtgtcgata gctacgccaa ctcttcctg cactggtacc agcagaagcc cggccagcca 540
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aagaggagca ggctcctgca cagtgaactac atgaacatga ctccccgccg ccccgggccc 1020
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cggaggggca agggggcacga tggcctttac cagggtctca gtacagccac caaggacacc 1380
tacgacgccc ttcacatgca ggccttgccc cctcgc 1416

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SEQ ID NO: 114      moltype = DNA length = 1416
FEATURE            Location/Qualifiers
source             1..1416
                   mol_type = other DNA
                   organism = synthetic construct

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SEQUENCE: 114
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ttcaccaccg agttcgacta ttggggccag ggcaccctgg tgaccgtgtc gtccggtgga 360
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gaactgcaga aagataagat ggcggaggcc tacagtgaga ttgggatgaa aggcgagcgc 1320
cggaggggca agggggcacga tggcctttac cagggtctca gtacagccac caaggacacc 1380
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SEQ ID NO: 115      moltype = DNA length = 1416
FEATURE            Location/Qualifiers
source             1..1416
                   mol_type = other DNA

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organism = synthetic construct

SEQUENCE: 115
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 FEATURE Location/Qualifiers
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 mol_type = other DNA
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ctccaaacca gtatctaa 3918

```

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SEQ ID NO: 120      moltype = DNA length = 3918
FEATURE             Location/Qualifiers
source               1..3918
                    mol_type = other DNA
                    organism = synthetic construct

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SEQUENCE: 120
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ctccaaacca gtatctaa 3918

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```

SEQ ID NO: 121      moltype = AA length = 15
FEATURE            Location/Qualifiers
source              1..15
                    mol_type = protein
                    organism = synthetic construct

```

```

SEQUENCE: 121
GGGSGGGGS GGGGS 15

```

```

SEQ ID NO: 122      moltype = AA length = 42
FEATURE            Location/Qualifiers
source              1..42
                    mol_type = protein
                    organism = synthetic construct

```

```

SEQUENCE: 122
KRGRKLLLYI FKQPFMRPVQ TTQEEDGCSC RFPEEEEGGC EL 42

```

```

SEQ ID NO: 123      moltype = AA length = 41
FEATURE            Location/Qualifiers
source              1..41
                    mol_type = protein
                    organism = synthetic construct

```

```

SEQUENCE: 123
RSKRSRLLS DYMNMTPRRP GPTRKHYQPY APPRDFAAAYR S 41

```

```

SEQ ID NO: 124      moltype = AA length = 42
FEATURE            Location/Qualifiers
source              1..42
                    mol_type = protein
                    organism = synthetic construct

```

```

SEQUENCE: 124
ALYLLRRDQR LPPDAHKPPG GGSFRTPIQE EQADAHSTLA KI 42

```

```

SEQ ID NO: 125      moltype = AA length = 112
FEATURE            Location/Qualifiers
source              1..112
                    mol_type = protein
                    organism = synthetic construct

```

```

SEQUENCE: 125
RVKFSRSADA PAYKQGQNQL YNELNLGRRE EYDVLDKRRG RDPEMGGKPR RKNPQEGLYN 60
ELQKDKMAEA YSEIGMKGER RRGKGDGLY QGLSTATKDT YDALHMQALP PR 112

```

```

SEQ ID NO: 126      moltype = AA length = 45
FEATURE            Location/Qualifiers
source              1..45
                    mol_type = protein
                    organism = synthetic construct

```

```

SEQUENCE: 126
TTTPAPRPPT PAPTIASQPL SLRPEACRPA AGGAVHTRGL DFACD 45

```

```

SEQ ID NO: 127      moltype = AA length = 47
FEATURE            Location/Qualifiers
source              1..47
                    mol_type = protein
                    organism = synthetic construct

```

```

SEQUENCE: 127
RAAAIEVMYP PPYLDNEKSN GTIIHVKGKH LCPSPLFPGP SKPKDPK 47

```

```

SEQ ID NO: 128      moltype = AA length = 11
FEATURE            Location/Qualifiers
source              1..11
                    mol_type = protein
                    organism = synthetic construct

```

```

SEQUENCE: 128
SKYGPPCPSC P 11

```


SEQ ID NO: 129	moltype = AA length = 24					
FEATURE	Location/Qualifiers					
source	1..24					
	mol_type = protein					
	organism = synthetic construct					
SEQUENCE: 129						
IYIWAPLAGT	CGVLLLSLVI	TYLC	24			
SEQ ID NO: 130	moltype = AA length = 24					
FEATURE	Location/Qualifiers					
source	1..24					
	mol_type = protein					
	organism = synthetic construct					
SEQUENCE: 130						
FWVLVVVGGV	LACYSLLVTV	AFII	24			
SEQ ID NO: 131	moltype = AA length = 24					
FEATURE	Location/Qualifiers					
source	1..24					
	mol_type = protein					
	organism = synthetic construct					
SEQUENCE: 131						
FWLPIGCAAF	VVVCILGCIL	ICWL	24			
SEQ ID NO: 132	moltype = AA length = 25					
FEATURE	Location/Qualifiers					
source	1..25					
	mol_type = protein					
	organism = synthetic construct					
SEQUENCE: 132						
LGWLTVVLLA	VAACVLLLTS	AQLGL	25			
SEQ ID NO: 133	moltype = AA length = 1145					
FEATURE	Location/Qualifiers					
source	1..1145					
	mol_type = protein					
	organism = Homo sapiens					
SEQUENCE: 133						
YNLDVRGARS	FSPPRAGRHF	GYRVLQVGNG	VIVGAPGEGN	STGSLYQCQS	GTGHCLPVTL	60
RGSNYTSKYL	GMTLATDPTD	GSILACDPGL	SRTCDQNTYL	SGLCYLFRQN	LQGPMLQGRP	120
RGQEICKGNV	DLVFLFDGSM	SLQPDEFQKI	LDFMKDVMMK	LSNTSYQFAA	VQFSTSYKTE	180
PDFS DYVKWK	DPDALLKHVK	HMLLLTNTFG	AINYVATEVF	REELGARPD	TKVLIITDGT	240
EATDSGNIDA	AKDIIRYIIG	IGKHFQTKES	QETLHKFASK	PASEFVKILD	TFEKLDLFLT	300
ELQKKIYVIE	GTSKQDLTSF	NMELSSSGIS	ADLSRGHAVV	GAVGAKDWAG	GFLDLKADLQ	360
DDTFIGNEPL	TPEVRAGYLG	YTVTWLPSRQ	KTSLLASGAP	RYQHMGRVLL	FQEPQGGGHW	420
SQVQTIHGTD	IGSYFGGELC	GVDVDQDGET	ELLIGAPLFL	YGEQGRGGRV	IYQRRQLGFE	480
EVSELQGDGP	YPLGRFGAEI	TALTDINGDG	LVDVAVGAPL	EEQGAVYIFN	GRHGGLSPQP	540
SQRIEGTQVL	SGIQWFGRSI	HGVKDLGEGD	LADVAVGAES	QMIVLSSRPV	VDVMTLMSFS	600
PAEPIPVHEVE	CSYSTSNMKM	EGVNITICFQ	IKSLYPQFQG	RLVANLTYTL	QLDGHRTRRR	660
PIPKAGHSVA	LQMMFNTLVN	SWGDSVELH	ANVTCNNEDS	DLLEDNSATT	IIPILYPINI	720
QKDIPIPILR	PSLHSETWEI	PFEKNCGEDK	KCEANLRVSF	SPARSRALRL	TAFASLSVEL	780
SLSNLEEDAY	WVQDLHLFPP	GLSFRKVEML	KPHSQIPVSC	EELPEESRLL	SRALSCNVSS	840
LIQDQEDSTL	YVSFTPKGPK	IHQVKHMYQV	RIQPSIHHDN	IPTLEAVVGV	PQPPSEGPIT	900
HQWSVQMEPP	VPCHYEDLER	LPDAAEPCLP	GALPRCPVVF	RQEILVQVIG	TLELVGEIEA	1020
SSMFLSCSSL	SISFNSSKHF	HLYGSNASLA	QVMKMYDVVY	EKQMLYLVVL	SGGGGLLLLL	1080
LIFIVLYKVG	PFKRNLEKEM	EAGRGVPNGI	PAEDSEQLAS	GQEAGDPGCL	KPLHEKDSSES	1140
GGGKD						1145
SEQ ID NO: 134	moltype = AA length = 181					
FEATURE	Location/Qualifiers					
source	1..181					
	mol_type = protein					
	organism = Homo sapiens					
SEQUENCE: 134						
VDLVFLPDGS	MSLQPDEFQK	ILDFMKDVMK	KLSNTSYQFA	AVQFSTSYKT	EFDFS DYVKW	60
KDPDALLKHV	KHMLLLTNTF	GAINYVATEV	FREELGARPD	ATKVLIIITD	GEATDSGNID	120
AAADKIIRYII	GIGKHFQTK	SQETLHKFAS	KPASEFVKIL	DTFEKLDLFL	TELQKKIYVI	180
E						181
SEQ ID NO: 135	moltype = AA length = 369					
FEATURE	Location/Qualifiers					
source	1..369					
	mol_type = protein					
	organism = Homo sapiens					

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SEQUENCE: 135
MDMADEPLNG SHTWLSIPFD LNGSVVSTNT SNQTEPYDYL TSNVLTFIY FVVCIIIGLCG 60
NTLVIYVILR YAKMKTITNI YILNLAIAD LFMGLPLFLA MQVALVHWPF GKAICRVVMT 120
VDGINQFTSI FCLTVMSIDR YLAVVHPIKS AKWRRPRTAK MITMAVWGS LLVILPIMIY 180
AGLRSNQWGR SSCTINWPGE SGAWYTGFIY YTFILGFLVP LTIICLCYLF IIKVKSSGI 240
RVGSSKRKKS EKKVTRMVIS VVAVFIFCWL PFYIFNVSSV SMAISPTPAL KGMFDFVVVL 300
TYANSCANPI LYAFSLDNFK KSPQNVLCV KVSQTDDGER SDSKQDKSRL NETTETQRTL 360
LNGDLQTSI 369

SEQ ID NO: 136 moltype = AA length = 314
FEATURE Location/Qualifiers
source 1..314
mol_type = protein
organism = Homo sapiens

SEQUENCE: 136
MDMADEPLNG SHTWLSIPFD LNGSVVSTNT SNQTEPYDYL TSNVLTFIY FVVCIIIGLCG 60
NTLVIYVILR YAKMKTITNI YILNLAIAD LFMGLPLFLA MQVALVHWPF GKAICRVVMT 120
VDGINQFTSI FCLTVMSIDR YLAVVHPIKS AKWRRPRTAK MITMAVWGS LLVILPIMIY 180
AGLRSNQWGR SSCTINWPGE SGAWYTGFIY YTFILGFLVP LTIICLCYLF IIKVKSSGI 240
RVGSSKRKKS EKKVTRMVIS VVAVFIFCWL PFYIFNVSSV SMAISPTPAL KGMFDFVVVL 300
TYANSCANPI LYAF 314

SEQ ID NO: 137 moltype = DNA length = 1110
FEATURE Location/Qualifiers
source 1..1110
mol_type = genomic DNA
organism = Homo sapiens

SEQUENCE: 137
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acaagcaatg cagtccctcac attcatctat tttgtggtct gcatcattgg gttgtgtggc 180
aacacacttg tcatattatgt catcctccgc tatgccaaga tgaagaccat caccaacatt 240
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aagagcttcc agaattgctt ctgcttggtc aaggtgagcg gcacagatga tggggagcgg 1020
agtgacagta agcaggacaa atcccggctg aatgagacca cggagaccca gaggaccctc 1080
ctcaatggag acctccaaac cagtatctaa 1110

SEQ ID NO: 138 moltype = DNA length = 936
FEATURE Location/Qualifiers
source 1..936
mol_type = genomic DNA
organism = Homo sapiens

SEQUENCE: 138
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acaagcaatg cagtccctcac attcatctat tttgtggtct gcatcattgg gttgtgtggc 180
aacacacttg tcatattatgt catcctccgc tatgccaaga tgaagaccat caccaacatt 240
tacatcctca acctggccat cgcagatgag ctcttcctgc tgggtctgcc tttcttggtc 300
atgcagggtg ctctgggtcca ctggcccttt ggcaaggcca tttgccgggt ggtcatgact 360
gtggatggca tcaatcagtt caccagcctc ttctgcttga cagtcattgag catcgaccga 420
tacctggctg tgggtccacc catcaagtgc gccaaagtga ggagaccccg gacggccaag 480
atgatcacca tgggtgtgtg gggagtctct ctgctggtca tcttgcccat catgatatat 540
gctgggtctc ggagcaacca gtgggggaga agcagctgca ccatcaactg gccaggtgaa 600
tctggggctt ggtacacagg gtccatcctc tacactttca tctgggggtt cctgggtacc 660
ctcaccatca tctgtctttg ctacctgttc attatcatca aggtgaagtc ctctggaatc 720
cgagtgggct cctctaagag gaagaagtct gagaagaagg tcacccgaat ggtgtccatc 780
gtggtggctg tctcatcctt ctgctgggtt cctctctaca tattaacagt ttcttccgtc 840
tccatggcca tcagcccccac cccagccctt aaaggcattg ttgactttgt ggtggtctct 900
acctatgcta acagctgtgc caaccctatc ctatat 936

SEQ ID NO: 139 moltype = AA length = 22
FEATURE Location/Qualifiers
source 1..22
mol_type = protein
organism = synthetic construct

-continued

SEQUENCE: 139
GSGATNFSLL KQAGDVEENP GP 22

SEQ ID NO: 140 moltype = AA length = 23
FEATURE Location/Qualifiers
source 1..23
mol_type = protein
organism = synthetic construct

SEQUENCE: 140
GSGQCTNYAL LKLAGDVESN PGP 23

SEQ ID NO: 141 moltype = AA length = 21
FEATURE Location/Qualifiers
source 1..21
mol_type = protein
organism = synthetic construct

SEQUENCE: 141
GSGEGRGSLL TCGDVEENPG P 21

SEQ ID NO: 142 moltype = AA length = 25
FEATURE Location/Qualifiers
source 1..25
mol_type = protein
organism = synthetic construct

SEQUENCE: 142
GSGVKQTLNF DLLKLAGDVE SNPGP 25

SEQ ID NO: 143 moltype = AA length = 40
FEATURE Location/Qualifiers
source 1..40
mol_type = protein
organism = synthetic construct

VARIANT 1..40
note = This sequence may encompass 1-10 GGGS repeating units

SEQUENCE: 143
GGSGGGSGG GSGGGSGGS GGGSGGGSGG GSGGGSGGS 40

SEQ ID NO: 144 moltype = DNA length = 28
FEATURE Location/Qualifiers
source 1..28
mol_type = other DNA
organism = synthetic construct

SEQUENCE: 144
ctgtgccacc gcgggctcct acgtgagc 28

SEQ ID NO: 145 moltype = DNA length = 18
FEATURE Location/Qualifiers
source 1..18
mol_type = other DNA
organism = synthetic construct

SEQUENCE: 145
tagtacacgg cgctgtcc 18

SEQ ID NO: 146 moltype = DNA length = 25
FEATURE Location/Qualifiers
source 1..25
mol_type = other DNA
organism = synthetic construct

SEQUENCE: 146
cgtgagccct gcggattatt ggggc 25

SEQ ID NO: 147 moltype = DNA length = 18
FEATURE Location/Qualifiers
source 1..18
mol_type = other DNA
organism = synthetic construct

SEQUENCE: 147
taggagccgt aggtggca 18

SEQ ID NO: 148 moltype = DNA length = 23
FEATURE Location/Qualifiers
source 1..23
mol_type = other DNA
organism = synthetic construct

SEQUENCE: 148
ctgtgccacc gtcggctcct acg 23

-continued

SEQ ID NO: 149	moltype = DNA length = 18	
FEATURE	Location/Qualifiers	
source	1..18	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 149		
tagtacacgg cgctgtcc		18
SEQ ID NO: 150	moltype = DNA length = 22	
FEATURE	Location/Qualifiers	
source	1..22	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 150		
ctgtgccacc ctcggctcct ac		22
SEQ ID NO: 151	moltype = DNA length = 22	
FEATURE	Location/Qualifiers	
source	1..22	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 151		
ctgtgccacc atcggctcct ac		22
SEQ ID NO: 152	moltype = DNA length = 30	
FEATURE	Location/Qualifiers	
source	1..30	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 152		
ctgtgccacc atgggctcct acgtgagccc		30
SEQ ID NO: 153	moltype = DNA length = 22	
FEATURE	Location/Qualifiers	
source	1..22	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 153		
ccgctcggag gccaccaccg ag		22
SEQ ID NO: 154	moltype = DNA length = 18	
FEATURE	Location/Qualifiers	
source	1..18	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 154		
gcgcagtagt acacggcc		18
SEQ ID NO: 155	moltype = DNA length = 22	
FEATURE	Location/Qualifiers	
source	1..22	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 155		
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SEQ ID NO: 156	moltype = DNA length = 18	
FEATURE	Location/Qualifiers	
source	1..18	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 156		
cgggcgcagt agtacacg		18
SEQ ID NO: 157	moltype = DNA length = 18	
FEATURE	Location/Qualifiers	
source	1..18	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 157		
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SEQ ID NO: 158	moltype = DNA length = 25	
FEATURE	Location/Qualifiers	
source	1..25	

-continued

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mol_type = other DNA
organism = synthetic construct

SEQUENCE: 158
ccgctcggag ggcaccaccg agttc 25

SEQ ID NO: 159      moltype = DNA length = 23
FEATURE            Location/Qualifiers
source             1..23
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 159
ccgctcggag ttcaccaccg agt 23

SEQ ID NO: 160      moltype = DNA length = 25
FEATURE            Location/Qualifiers
source             1..25
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 160
ccgctcggag tggaccaccg agttc 25

SEQ ID NO: 161      moltype = DNA length = 1179
FEATURE            Location/Qualifiers
source             1..1179
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 161
ggctccggtg cccgtcagtg ggcagagcgc acatcgccca cagtccccga gaagtgggg 60
ggaggggtcg gcaattgaac cgtgacctag agaaggtggc gcggggtaaa ctgggaaagt 120
gatgtcgtgt actggctccg cctttttccc gaggttgggg gagaaccgta tataagtgc 180
gtagtcgccc tgaacgttct ttttcgcaac gggtttgccg ccagaacaca ggtaagtgcc 240
gtgtgtggtt ccccggggcc tggcctcttt acgggttatg gcccttgctg gccttgaatt 300
acttccacct ggctgcagta cgtgattctt gatcccgagc ttcgggttgg aagtgggtgg 360
gagagttcga ggccttgccg ttaaggagcc ccttcgcctc gtgcttgagt tgaggcctgg 420
cctgggcgct ggggcgcgcg cgtgcgaatc tggtggcacc ttcgcgcctg tctcgtgct 480
ttcgataagt ctctagccat ttaaaathtt tgatgacctg ctgcgacgct tttttcttg 540
caagatagtc ttgtaaatgc gggccaagat ctgcacactg gtatttcggt ttttggggcc 600
gcgggcggcg acggggcccg tgcgtcccag cgcacatggt cggcgaggcg gggcctgcga 660
gcgcggccac cgagaatcgg acgggggtag tctcaagctg gccggcctgc tctggtgcct 720
ggctcgcgcg cgcctgtgat cgcccgcgcc tgggcggcaa ggctggcccc gtcggcacca 780
gttgctgtag cggaagatg gccgcttccc ggcctgctg caggagagctc aaaatggagg 840
acgcggcgct cgggagagcg ggcgggtgag tcacccacac aaaggaaaag ggcctttccg 900
tcctcagccg tcgcttcctg tgactccacg gagtaccggg cgccgtccag gcacctcgat 960
tagttctcga gcttttggag tacgtcgtct ttaggttggg gggagggggt ttatgcgatg 1020
gagtttcccc acactgagtg ggtggagact gaagttaggc cagcttgcca cttgatgtaa 1080
ttctccttgg aatttgccct ttttgagttt ggatcttggt tcatttctca gcctcagaca 1140
tggtttcaaa gtttttttct tccatttcag gtgtcgtga 1179

SEQ ID NO: 162      moltype = AA length = 21
FEATURE            Location/Qualifiers
source             1..21
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 162
MALPVTALLL PLALLHAAR P 21

SEQ ID NO: 163      moltype = DNA length = 63
FEATURE            Location/Qualifiers
source             1..63
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 163
atggccttac cagtgaaccg cttgctcctg ccgctggcct tgctgctcca cgccgccagg 60
ccg 63

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1. A chimeric antigen receptor (CAR), comprising:
 - (a) an extracellular binding domain;
 - (b) a costimulatory domain; and
 - (c) an activation domain,
 wherein the extracellular binding domain comprises a heavy chain variable (V_H) region comprising complementary determining 1 (HC CDR1), complementary determining 2 (HC CDR2), and complementary determining 3 (HC CDR3) regions; and a light chain variable (V_L) region comprising complementary determining 1 (LC CDR1), complementary determining 2 (LC CDR2), and complementary determining 3 (LC CDR3) regions, wherein:
 - (i) the V_H region comprises HC CDR1 and HC CDR2 regions comprising amino acid sequences set forth in SEQ ID NOs: 1 and 2, respectively, and an HC CDR3 region comprising an amino acid sequence set forth in any one of SEQ ID NOs: 3-8; and the V_L region comprises LC CDR1, LC CDR2, and LC CDR3 regions comprising amino acid sequences set forth in SEQ ID NOs: 9-11, respectively, or
 - (ii) the V_H region comprises HC CDR1 and HC CDR2 regions comprising amino acid sequences set forth in SEQ ID NOs: 70 and 71, respectively, and an HC CDR3 region comprising an amino acid sequence set forth in any one of SEQ ID NOs: 72-75; and the V_L region comprises LC CDR1, LC CDR2, and LC CDR3 regions comprising amino acid sequences set forth in SEQ ID NOs: 76-78, respectively.
2. The CAR of claim 1, wherein the V_H region comprises the amino acid sequences set forth in (i).
3. The CAR of claim 1, wherein the HC CDR3 comprises an amino acid sequence set forth in SEQ ID NO: 5 or SEQ ID NO: 8.
4. The CAR of claim 1, which comprises V_H and V_L region amino acid sequences selected from the group consisting of:
 - (a) SEQ ID NOs: 13 and 19, respectively;
 - (b) SEQ ID NOs: 14 and 19, respectively;
 - (c) SEQ ID NOs: 15 and 19, respectively;
 - (d) SEQ ID NOs: 16 and 19, respectively;
 - (e) SEQ ID NOs: 17 and 19, respectively; and
 - (f) SEQ ID NOs: 18 and 19, respectively;
5. The CAR of claim 1, wherein the extracellular domain is a single chain antibody fragment (scFv).
6. The CAR of claim 5, wherein scFv comprises a (GxS)_n linker between the V_H and V_L regions.
7. The CAR of claim 5, wherein the scFv comprises an amino acid sequence set forth in any one of SEQ ID NOs: 29-34.
8. The CAR of claim 1, wherein the V_H region comprises the amino acid sequences set forth in (ii).
9. The CAR of claim 8, wherein the HC CDR3 comprises an amino acid sequence set forth in SEQ ID NO: 74 or 75.
10. The CAR of claim 8, which comprises V_H and V_L region amino acid sequences selected from the group consisting of:
 - (a) SEQ ID NOs: 80 and 84, respectively;
 - (b) SEQ ID NOs: 81 and 84, respectively;
 - (c) SEQ ID NOs: 82 and 84, respectively; and
 - (d) SEQ ID NOs: 83 and 84, respectively.
11. The CAR of claim 8, wherein the extracellular domain is an scFv.
12. The CAR of claim 11, wherein the scFv comprises a (GxS)_n linker between the V_H and V_L regions.
13. The CAR of claim 11, wherein the scFv comprises an amino acid sequence set forth in any one of SEQ ID NOs: 92-95.
14. The CAR of claim 1, wherein the costimulatory domain is from 4-1BB (CD 137), CD28, OX40, ICOS, GITR, CD27, CD30, CD40, DAP 10, DAP12, BAFTR, HVEM, ICAM-1, lymphocyte function-associated antigen-1 (LFA-1), CD2, CDS, CD7, CD287, LIGHT, NKG2C, NKG2D, SLAMF7, NKp80, NKp30, NKp44, NKp46, CD 160, B7-H3, a ligand that specifically binds with CD83 or a combination thereof; optionally wherein the costimulatory domain comprises an amino acid sequence set forth in any one of SEQ ID NOs: 122-124.
15. The CAR of claim 14, wherein the costimulatory domain comprises 4-1BB or CD28.
16. The CAR of claim 1, wherein the extracellular binding domain comprises costimulatory domains from 4-1BB and CD28, or CD-28 and OX-40.
17. The CAR of claim 1, wherein the activation domain is from CD3 zeta, common FcR gamma (FCER1G), Fc gamma RIla, FcR beta (Fc epsilon 1b), CD3 gamma, CD3 delta, CD3epsilon, CD5, CD22, CD79a, CD79b, CD278 (ICOS), FcεRI, or CD66d.
18. The CAR of claim 17, wherein the activation domain is from CD3 zeta, optionally comprising the amino acid sequence of SEQ ID NO: 125.
19. The CAR of claim 1, further comprising a hinge domain, a transmembrane domain, or a combination thereof, which optionally is located between the extracellular binding domain and the costimulatory domain.
20. The CAR of claim 19 comprising a hinge domain from CD8 alpha, CD28, or IgG4; optionally wherein the hinge domain comprises an amino sequence set forth in any one of SEQ ID NOs: 126-128.
21. The CAR of claim 19, comprising a transmembrane domain from a cell surface receptor selected from the group consisting of an alpha, beta or zeta chain of a T cell receptor, CD8 alpha, CD28, ICOS, GITR, CD3 epsilon, CD45, CD4, CD5, CD8, CD9, CD16, CD22, CD33, CD37, CD64, CD80, CD86, CD134, CD137, CD154, CD271, TNFRSF19, Killer Cell Immunoglobulin-Like Receptor (KIR), and a combination thereof.
22. The CAR of claim 21, wherein the transmembrane domain is from CD8 alpha, CD28, ICOS, and GITR; optionally wherein the transmembrane domain comprises an amino acid sequence set forth in any one SEQ ID NOs: 129-131.
23. The CAR of claim 1, further comprising a signal peptide located at the N-terminus of a CAR precursor protein; optionally wherein the signal peptide is a CD8 alpha chain signal peptide.
24. The CAR of claim 1, further comprising a second extracellular antigen binding domain.
25. The CAR of claim 23, wherein the second extracellular antigen binding domain comprises a second scFv.
26. The CAR of claim 24, wherein the second extracellular antigen binding domain comprises an ICAM-1 binding domain.
27. The CAR of claim 26, wherein the ICAM-1 binding domain comprises an αL subunit I domain of human lymphocyte function-associated antigen-1 (LFA-1); optionally

wherein the ICAM-1 binding domain comprises an α L subunit I domain of human lymphocyte function-associated antigen-1 (LFA-1).

28. The CAR of claim **1**, wherein the CAR comprises an amino acid sequence set forth in any one of SEQ ID NOs: 43-48 or 102-105.

29-36. (canceled)

37. A population of immune cells expressing the CAR of claim **1**.

38. (canceled)

39. The population of immune cells of claim **37**, wherein the immune cells are T cells, natural killer (NK) cells, tumor infiltrating lymphocytes, dendritic cells, macrophages, B cells, neutrophils, eosinophils, basophils, mast cells, myeloid-derived suppressor cells, stem cells, precursors thereof, subtypes thereof, or a combination thereof; optionally wherein the immune cell is a human immune cell.

40. The population of immune cells of claim **39**, wherein the immune cells are T cells.

41. The population of immune cells of claim **40**, wherein the T cells express SSTR2.

42. The population of immune cells of claim **37**, further comprising a second population of immune cells, wherein the second population of immune cells expresses a second CAR or another polypeptide of interest.

43. (canceled)

44. The population of immune cells of claim **42**, wherein the second CAR comprises an ICAM-1 binding domain; optionally wherein the ICAM-1 binding domain comprises an α L subunit I domain of human LFA-1.

45. A cell therapy-based method of treating cancer, comprising administering to a subject in need thereof the population of immune cells of claim **37**.

46-66. (canceled)

67. A method for treating cancer and monitoring CAR-T cell distribution in a patient, comprising:
incubating the population of CAR-T cells of claim **41** with a radioactive label that binds to SSTR2,
intravenously infusing the labeled CAR-T cells into a patient in an amount of 10^4 - 10^8 cells/kg patient, and
detecting the labeled CAR-T cell distribution by PET/CT imaging,
wherein the labeled CAR-T cells are infiltrated into cancer cells to kill the cancer cells.

68-70. (canceled)

71. A method for treating cancer and monitoring CAR-T cell distribution in a patient, comprising:
intravenously infusing the population of CAR-T cells of claim **41** into a patient, wherein the CAR-T cells express or have been transduced to express at least 100,000 molecules of SSTR2 per T cell
injecting into the patient a radioactive label that binds to SSTR2 at least one hour prior to PET/CT imaging, and
detecting the labeled CAR-T cell distribution by PET/CT imaging,
wherein the labeled CAR-T cells are infiltrated into cancer cells to kill the cancer cells.

72. (canceled)

73. A method of producing a population of genetically engineered immune cells, the method comprising:
(a) providing a population of immune cells; and
(b) introducing into the immune cells a nucleic acid coding for the CAR claim **1** to produce a population of genetically engineered immune cells.

74-78. (canceled)

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